

Baghdad Proposal

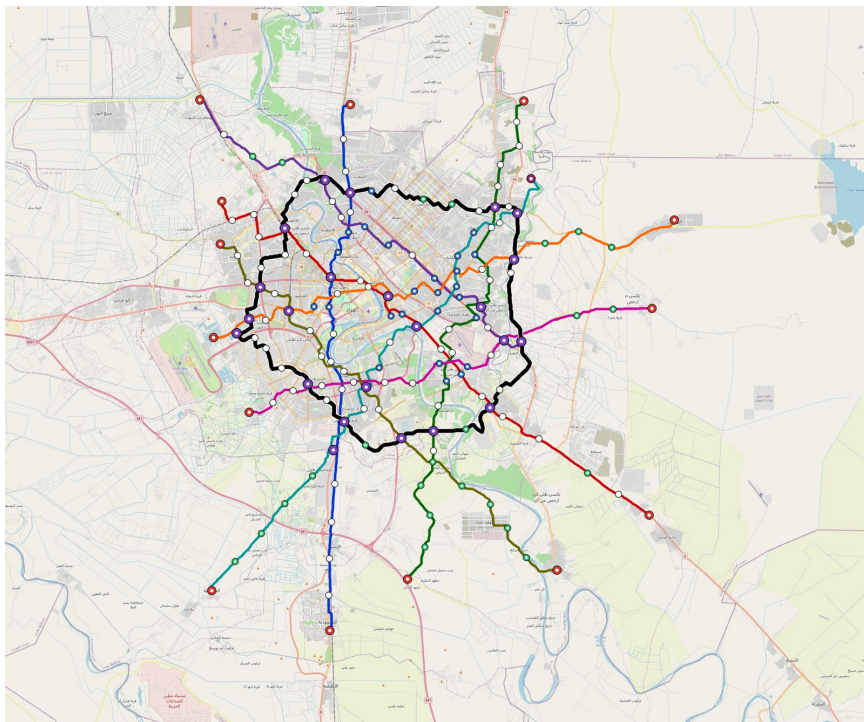
Urban railway and Iraqi manufacturing
Future national development

OpenSourceRail · Planning baseline 2026-10-04

Complete proposal, detailed schedules and technical evidence annexes

Prepared for the prospective Iraqi sponsor, Baghdad authorities, operating organisation and financing partners.

Baghdad finance only. National expansion is a separate future development programme. Construction and operating approval remain pending.



Baghdad planning network

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Baghdad Proposal

Source: cities/catalogue/west-asia/Iraq/Baghdad/proposal/BAGHDAD-PROPOSAL.md

OpenSourceRail proposes an owner led feasibility and front end engineering programme for Baghdad, with Iraqi train manufacture and local infrastructure delivery. This proposal brings the Baghdad network, railway systems, operating organisation, delivery evidence and financing together, and sets out a separate path for future national development. It is addressed to the prospective Iraqi public sponsor, Baghdad authorities, operating organisation and financing partners; no appointment or financing commitment is asserted.

The current planning network is 9 lines, 516.5 km of double track route, 182 stations and 831 six car trainsets. Baghdad capital, including one manufacturing plant and its EPC, is USD 7.880 billion equivalent. The direct government capital contribution is 25%. Imports are financed 50% government USD cash and 50% proposed Chinese USD credit; all remaining capital cash, bonds and bank debt are IQD.

The immediate decision proposed is to establish a sponsor, commission survey and demand work, develop the first operable line and plant packages, qualify suppliers and obtain executable financing terms. Construction and operating release require the recorded physical and approval gates. The current resource constrained plan reaches first line revenue in month 41 and full operation in month 83 after financial close. That long schedule is a material design and delivery problem to resolve; this proposal does not substitute a five year promise.

How to read the proposal

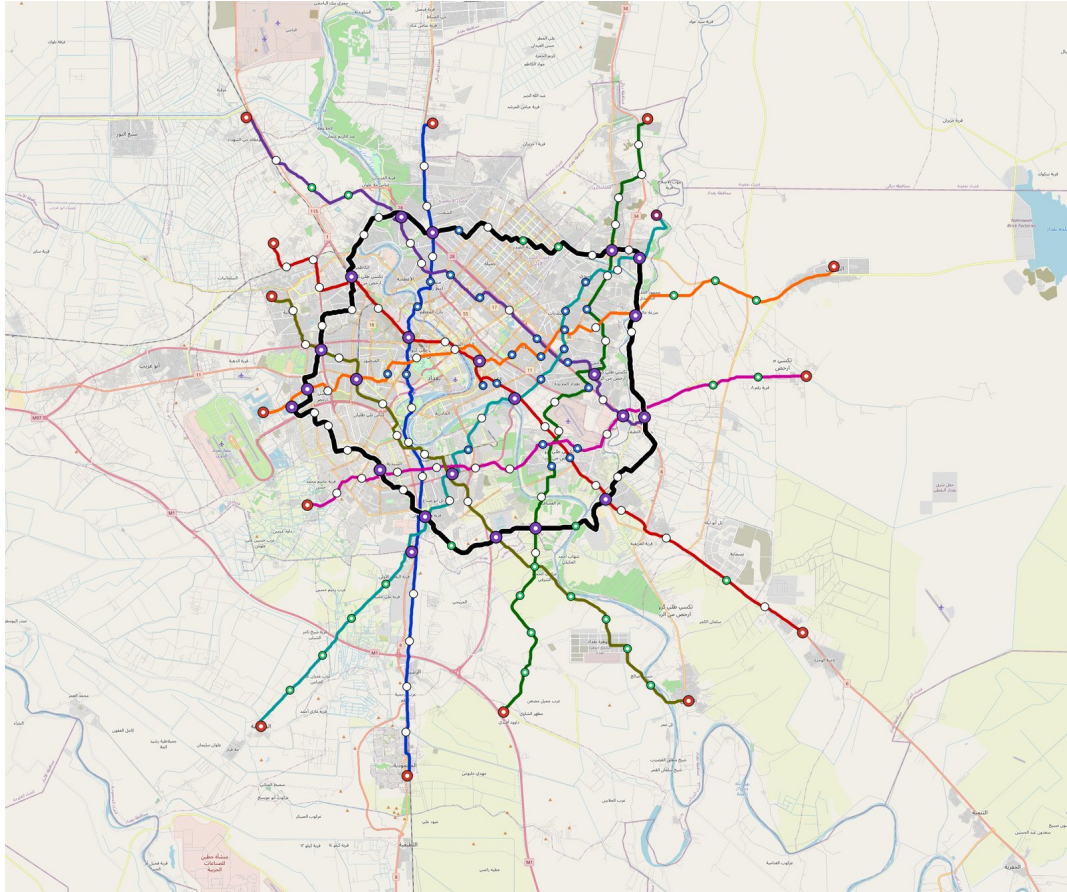
The first part states the integrated proposal and the decisions it needs. The next part prints the detailed network registers and six month financing schedules. The technical appendices reproduce every current Baghdad Markdown report, the national brief and selected shared standards. The supporting ZIP preserves the full controlled Baghdad files, every Baghdad financing spreadsheet, the operations payload, national city design inputs and the cited shared references. The source inventory and manifest identify exact file bytes.

Baghdad is the only financed city. National expansion is a future strategic option with its own budgets and approvals. Procurement origin, loan currency and reporting currency are different measures. USD equivalents use the historical model anchor of IQD 1,300/USD; this is not a current execution quote. All prices, demand, debt terms and rights proceeds remain planning assumptions. Older generic foreign turnkey examples reproduced in source appendices are separate from the historical 148 km comparison and the scheduled Baghdad financing cases.

Baghdad network and population access

The design retains a planning population of 9,780,429. The 46.4% anchor weighted coverage proxy represents 4,538,119 residents under the model. It is not measured population within an 800 m walking network, nor the number of unique passengers. River crossings, actual entrances, walking barriers and feeders need a surveyed population and access model. A larger network provides a scope to evaluate; it does not prove higher coverage than another proposal.

Line	Shape	Route km	Stations	Peak fleet	Total fleet	Opening month
line-1	radial	53.822	19	86	95	41
line-2	radial	56.994	21	93	103	46
line-3	radial	55.819	21	98	108	52
line-4	radial	44.443	16	77	85	57
line-5	radial	53.737	18	89	98	62
line-6	radial	57.516	19	100	111	69
line-7	radial	43.502	16	76	84	73
line-8	radial	51.366	18	90	100	79
line-9	ring	99.317	34	42	47	83



Baghdad network

Line names are controlled design identifiers. Public station names, route brands and final termini require owner approval. Chainages, coordinates, interchange platforms and fleet roles are printed in the network registers and retained without replacing the established layout.

The service concept operates 05:30 to 02:00 with a three minute protected peak headway. The current scheduled journeys and fleet sizing are capacity led; accepted junction/authority capacity, ridership, a timetable, station crowding and degraded recovery need operating review. The 831 trainsets comprise 751 peak, spare and cold reserve roles documented in the annex.

Trains and imported component strategy

The Baghdad profile is metro 6car: 6 cars, 111 m body length, 720 passengers at nominal planning load including 120 seats, and 960 at short duration crush load. Each train has 1,350 kWh nameplate / 1,080 kWh usable LFP battery capacity, 12 traction controllers, 3,600 kW peak traction and a 50 C design ambient. These are reference profiles requiring supplier and physical qualification, not delivered fleet performance.

The local industrial scope is train assembly, body modules, fit out, wiring, coatings, inspection, testing and maintenance. Imported scope includes bogies, batteries, windows, doors, solar equipment and manufacturing tooling within existing capital allowances. Chinese supplier origin and export lender eligibility need evidence for every financed item; the entire imported basket is currently an unqualified scenario. Candidate CRRC equipment remains subject to competitive supplier selection, interface and safety qualification. There is no established CRRC partnership, quotation or endorsement.

Shared LM3 fabrication and first article documentation is reference process evidence for a three car platform. It does not qualify Baghdad's six car consist. The national programme should qualify the shared modules and then validate each consist and its interfaces, rather than treating a shared drawing as an accepted Baghdad train.

Civil infrastructure and stations

The civil screen contains 2,312 segments: 428.3 km at-grade, 75.8 km elevated, 12.4 km bridge. Route kilometres describe double track corridors; they are not track kilometres or a measurement of all sidings. Station products, 23 interchange complexes and 19 junction records are bound to the generated layout.

The proposed civil programme starts with survey control, utilities, property and access, geotechnical investigation, flood/drainage levels, alignment and station fit. Desktop soil inputs contain 7,287 sample locations and 65 missing profiles; they do not supply foundation bearing capacity, groundwater or deep stratigraphy. Structural calculations, spans, erection,

movements and independent checking must follow route specific evidence. Generated alignment exports have unfitted curves, placeholder vertical profiles and undesigned cant.

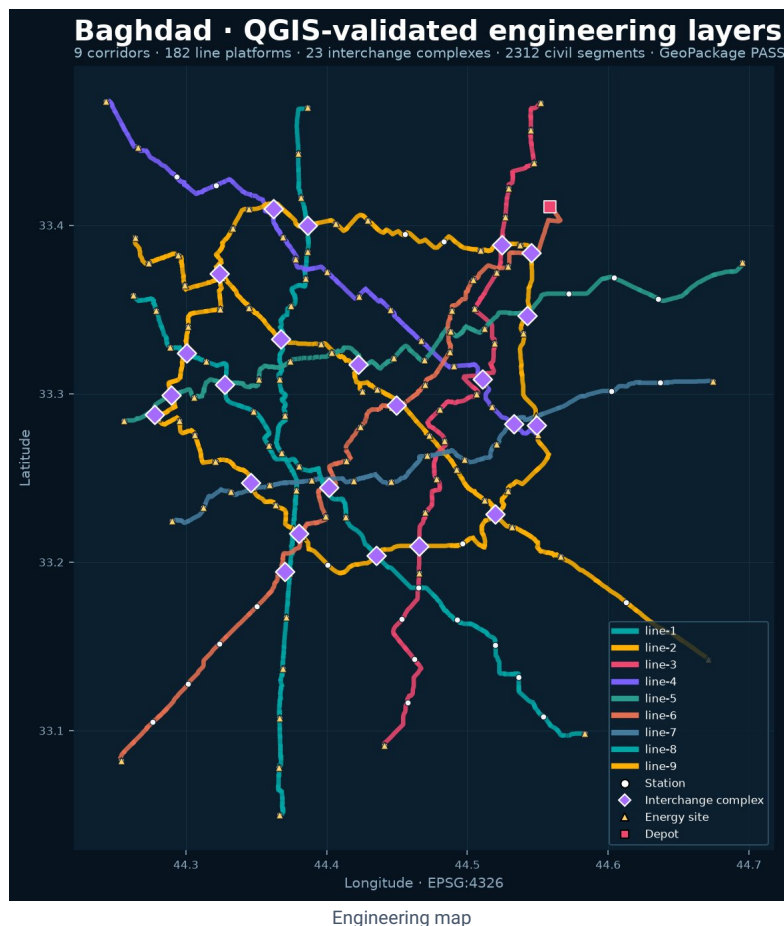
Stations require accessible approaches, platforms, passenger information, fire and evacuation design, fare equipment, retail and advertising layouts, security, sanitation and maintenance access. Platform and station access standards must be checked against the final six car envelope and passenger demand. Equipment and architecture references do not establish installed compliance.

The current USD 8m depot allowance remains unreconciled to physical stabling, workshops, power, fire and security. Workshop bays cannot be counted as overnight train parking. The current policy proposes two revenue trains at selected powered stations and line local storage for remaining fleet; usable tracks, charging, protected morning release, evening repositioning and repeated day replay remain open. The depot and stabling appendices retain these failures explicitly.

Energy and desert operation

The current duty model schedules 3,952 one way journeys and about 217,090 train km per day, with 2,053.8 GWh annual traction demand. It assigns 52.1 MW station/depot PV, 354.0 MWh site storage, 316.0 MW connected charging and 1018.6 MW dedicated solar. The design's zero annual residual grid/PPA import is an energy accounting result; it does not establish uninterrupted operation or installed grid independence.

Storage endurance, adverse weather, PV land, heat/dust derating, losses, supplier fire separation, actual charging duty, protection and backup import need a time resolved operating appraisal. Snapshot solver passes and grid only diagnostics do not close these gates. Battery protection and cabin/battery thermal separation must be qualified at the declared ambient and duty. No claimed unlimited battery autonomy or accepted solar islanding is used to close financing.



Train control and operational safety

The current architecture reference is RFC 0033, TACS runtime and committed resource control. RFC 0032's separate authority/protection prototype is superseded. The design retains committed resource ownership in the interlocking, ordered consensus decisions, onboard movement authority, automatic train protection, station departure gates and final brake requests. OCC supplies service intentions and observations; ERP and supervision cannot bypass railway authority or clear protection latches.

The reference uses three static consensus voters and two logical protection channels. Hardware placement, independent power/network domains, sensing, brake outputs and common cause analysis need assessment. Two software channels do not establish independent safety hardware. Lost required permissions, stale/invalid evidence and missing physical proving retain restrictive protection. The runtime is a synthetic process reference and not a deployed Baghdad control system or safety certificate. Existing Baghdad timetable simulations do not constitute RFC 0033 physical acceptance.

Iraqi manufacture and local economic benefit

The Baghdad plant is sized from 4,986 vehicle modules. Its base allowance is USD 303.505m plus USD 21.245m EPC, counted once outside city CAPEX. The current plant construction assumption is 390 working days before train production. Factory siting, freight access, utilities, tooling, staff, production rate, quality capacity and six car qualification require their own approved business and delivery plan.

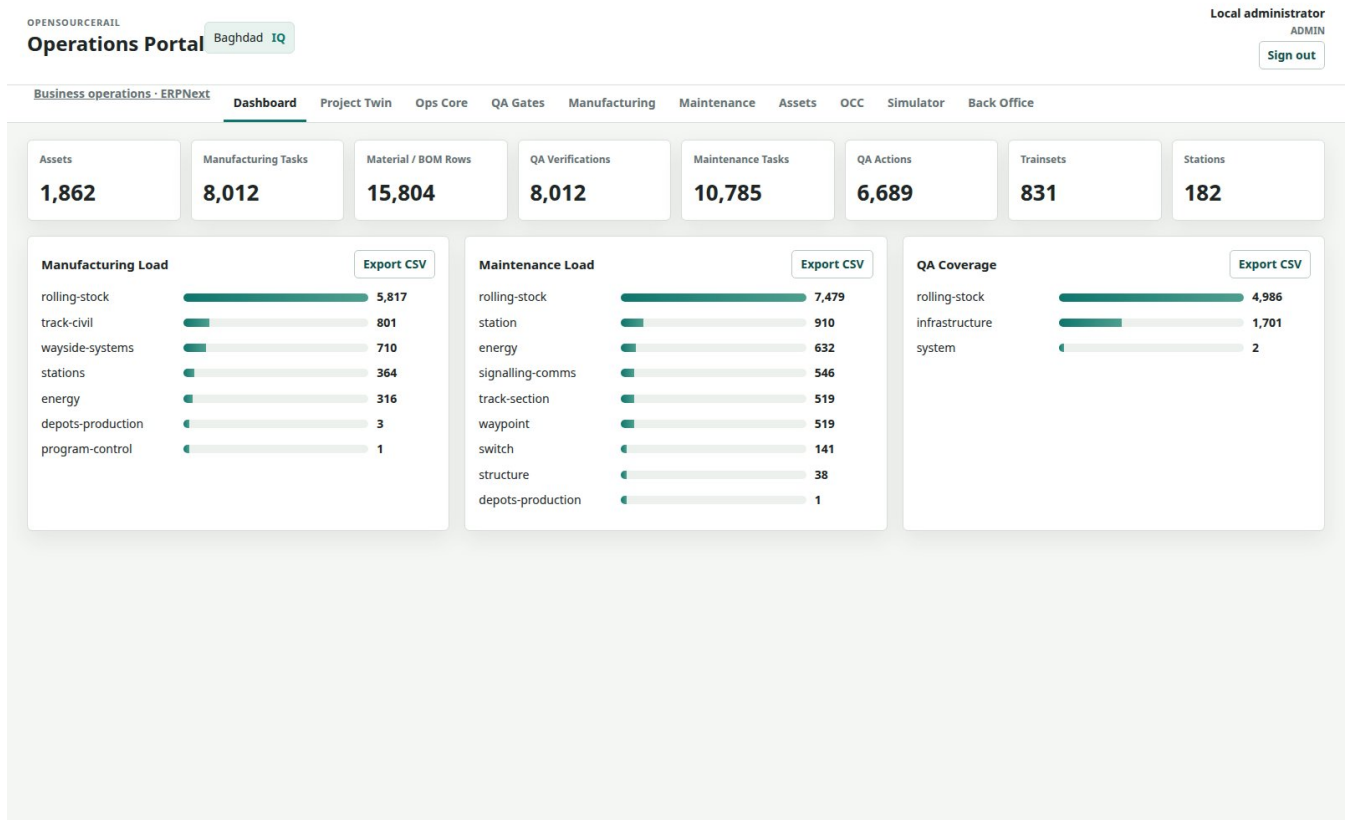
Baghdad procurement assigns USD 6.087bn equivalent locally. This is potential local expenditure, not payroll, GDP added or a guaranteed Iraqi content ratio. Local train and infrastructure work can retain skills, supplier income, repair capacity and spares knowledge. The plant appraisal covers capital financing only; manufacturing income, factory OPEX and commercial margins need a separate business case. No multiplier, tax recovery, construction job count or future national plant profit is booked as project cash without evidence. Exported components retain foreign exchange and supply chain exposure.

Operating organisation and digital management

The funded operating allowance is 2,350 indicative full time equivalents, including OCC/remote assistance, fleet maintenance, infrastructure/energy, stations, passenger service, administration and training. Annual labour allowance is IQD 19.503bn. It is a planning FTE and cost model rather than an accepted shift, leave or legal duty roster. Construction and factory headcount await measured work hours, crew mixes, wages and throughput.

Operating group	Indicative FTE
admin training	33
fleet maintenance	535
infrastructure energy	404
occ remote assist	426
passenger service	231
station platform	721

The Baghdad operating package contains 1,862 assets, 8,012 manufacturing/verification tasks, 15,804 material/procurement rows, 10,785 maintenance tasks and 6,689 QA actions. These are generated planning records. Actual purchase orders, execution, measurements and accountable release evidence remain distinct. The project twin, ERPNext/Frappe integration, supervision, QR identities, maintenance and advisory AI support business work; they do not issue movement or safety release authority.



Baghdad operations dashboard

Delivery and commissioning sequence

First obtain survey and demand inputs, freeze a viable first line and plant scope, and reconcile depot and energy duties. Qualify long lead components and the first six car train, then deliver infrastructure, energy, station systems and trained operating staff in accepted phases. Each line needs its own operating and safety acceptance before fare revenue is realised.

The city-sized plant becomes available after 18 months from NTP, followed by first-article qualification and finite six-car production cells. All 831 trains remain in scope. The calculated full fleet finishes alongside the overall infrastructure programme. Civil work and invoice milestones are rephased within the existing crew lanes and dependency graph. Line 1's infrastructure moves from day 219 to 682, reducing its fleet wait from 553 to 90 working days. Resources, durations and opening dates are preserved. Survey, land, utility, permit and contract approval remain necessary before deferring work. See the [physical factory sizing, crews, test paths and capital reconciliation](#).

Current capital milestones span 84 months, based on 260 working days/year and 30 pre NTP working days. Conditional first/full network opening is month 41/83, including the separate three-month line commissioning allowance. Opening-weighted demand and the 25% fixed / 75% variable OPEX proxy require a surveyed phase-specific plan. Factory sizing and its explicit capital increase are included in the new monthly and six-month financing schedules; rates, physical qualification and delivery risk remain open.

The factory's final fleet margin is only 20 working days; test-path throughput margin is 2.71%. The [frozen-resource delivery and financing study](#) separates civil productivity from investment timing: a 1.0 cycle multiplier preserves the complete baseline schedule, faster cycles retain the rephased start floors, and earliest construction is a separate comparison. Removing spending delays at unchanged productivity must not be called a productivity financing penalty.

Recovery options price extra structural/electrical/composite or coordinated production shifts, hiring/training, supplier expediting and testing, with unchanged cell counts and indexed incremental payroll/nonlabour costs. Testing alone does not improve the 75% availability opening dates; upstream production still limits them. A separate temporary first-article facility sensitivity adds USD 37.450m capital plus support staffing and tests first opening in month 36; permanent acceptance paths, first-article qualification and full line fleets remain required. These options are unquoted deterministic comparisons, not adopted delivery commitments.

Shift compression now excludes fixed curing/bonding/inspection holds and the additional 60-day first-article qualification. The [industrial qualification and funding-gate package](#) adds a metric temporary-site layout, cell tooling/transfer interfaces, quantity-based RFQs reconciled inside the existing USD 35m direct allowance, and ten ERPNext evidence tasks with source-bound measurement templates and authenticated independent result verification. Named owners, measurements, quotations and signatures remain pending.

Funding interruptions halt procurement/construction/production and defer invoices, adding local remobilisation and carrying/prolongation costs. Recovered domestic placement and delayed export credit are conditional on re-placement; permanent refusal has no opening or debt-clearance date, and is never filled by an automatic government or gap-loan replacement. The separate first-section study uses 14.034 km and five actual line-1 stations, 16 already-planned six-car trains at six-minute headways, independent turnbacks/charging/maintenance and USD 29.96m extra capital. Conditional section service is month 36, with surveyed demand and physical acceptance still required; the full 831-train baseline remains unchanged. Earlier small-section fares alone do not establish better finance: the study includes advanced invoices, additional support costs, phase demand deducted from later full-line receipts, and native-currency reserves/debt.

Combined delay-cost cases add extended staffing, supervision, carrying, storage/insurance and construction prolongation allowances without buying baseline crew-months twice. The financial downside ladder tests 30% fewer paid trips, 25% lower retail/advertising receipts, 5% annual invoice escalation, 7% rail OPEX growth, no assumed green/grant/rights enhancements, core rates two percentage points higher and an 8% IQD gap facility limited to IQD 4tn. The joint case leaves IQD 26.862tn cumulative uncovered cash and IQD 4.000tn terminal gap debt. Uncovered cash is a missing funding requirement; it is not an additional government contribution or secured credit. Reported repayment in such a case is conditional on filling that gap. The [physical qualification register](#) remains entirely not demonstrated; no model run supplies measured production/civil evidence or lender commitments.

Scenario	First/full month	Peak IQD gap tn	Interest/fees USD eq bn	Debt cleared month
Baseline	41/83	4.461	5.076	303
Faster civil; retained starts	41/83	4.465	5.078	303
Earliest civil; original cycles	41/83	4.818	5.211	304
Earliest civil; faster cycles	41/82	5.139	5.332	306
75% availability, costed	42/90	4.728	5.156	306
75% availability + test shift	42/90	4.735	5.159	307
75% availability + all stages	40/83	4.542	5.114	304
Temporary first-article facility	36/83	4.422	5.103	302
Combined delays, costed	55/106	5.308	5.291	316
Joint downside	55/106	4.000	15.384	Unfunded

OPENSOURCERAIL

Operations Portal

Baghdad IQ

Local administrator
ADMIN
Sign out

Business operations - ERPNext

Dashboards

Project Twin

Ops Core

QA Gates

Manufacturing

Maintenance

Assets

OCC

Simulator

Back Office

Programme days

1683

Critical work

1074

Order candidates

15804

Pre-NTP actions

49

Planned CAPEX

\$7,555,743,753

ERP submitted / draft POs

Snapshot unavailable

Critical Path and Project Baseline

Export Work

Revision

twin-c0718527dea001ed

Status

planning-digital-twin-not-construction-release

Scheduler

earliest-feasible ready-task dispatch with finite-resource lanes followed by CPM forward/backward pass

Critical tasks

1074

External gates

None unresolved in template graph

ERP task progress

Select or export the matching operating baseline

ERP observation

No unambiguous matching snapshot

Explore city operating twins and ERP execution ...

Schedule-linked Cash Requirements

Export Cashflow

Month	Project days	Requirement	Local	Imported	Cumulative
0	-30--9	\$3,040,063	\$2,584,054	\$456,009	\$3,040,063
1	-8-13	\$7,738,343	\$6,577,591	\$1,160,751	\$10,778,406
2	14-34	\$28,673,323	\$24,372,324	\$4,300,998	\$39,451,729

Baghdad project twin

Baghdad capital and procurement origin

Capital scope	USD eq m	Imported USD m	Local USD eq m
civil	3,894.477	584.172	3,310.306
stations	907.900	181.580	726.320
depots	8.000	2.000	6.000
rolling stock	1,396.080	488.628	907.452
solar plant	814.869	366.691	448.178
signalling	25.826	12.913	12.913
charging microgrid	67.600	27.040	40.560
epc overhead	440.992	66.149	374.843
Baghdad plant	303.505	60.701	242.804
Plant EPC	21.245	3.187	18.059
Total Baghdad capital	7,880.494	1,793.060	6,087.434

The city and plant total includes the plant EPC once. Budgets are unquoted planning estimates; land, utilities, taxes/duties, escalation, contingency and accepted depot/site scope require closure. Loan principal repaid later is a financing cashflow, not additional construction CAPEX. Origin shares do not establish citizenship of vendors, employment or lender eligibility.

Baghdad financing in USD and Iraqi dinars

Capital source	Currency	Native amount	USD equivalent m
bank credit	IQD	1,629,498,072,786	1,253.460
chinese export credit	USD	896,529,966	896.530
domestic bonds	IQD	4,888,494,218,359	3,760.380
government import cash	USD	896,529,966	896.530
government local cash	IQD	1,395,671,459,457	1,073.593

Government cash totals USD 1.970bn equivalent, exactly 25% of capital. Its USD 896.530m import cash is inside that limit. The same amount of Chinese USD debt covers the other half of imports. Remaining capital sources are IQD. Combined USD capital funding is USD 1.793bn (22.75%); IQD is 77.25%. Only Chinese credit is USD debt, while government also needs USD cash for downpayments.

Facility	Currency	Rate assumption	Grace months per draw	Amortisation months	Arrangement fee
chinese export credit	USD	5%	48	180	1.0%
domestic bonds	IQD	8%	24	180	0.5%
bank credit	IQD	9%	0	84	1.0%

Bond and bank capital split the residual after government and Chinese credit 75:25. Interest, fees and reserve deposits are funded separately in the cash forecast. A direct 25% capital contribution does not cap sovereign bond liabilities, guarantees or all lifetime public risk. Ministry of Finance issuance powers, sponsor structure, investor mandates, IQD placement capacity, draw availability and creditor consent for city/plant pooling need confirmation. Chinese supplier and lender qualification remains pending. Terms are sensitivities, not loan offers or an approved appropriation.

The capital source table above is the reference allocation. The conditional blended/early repayment case below replaces part of those ordinary bonds with green bonds and adds a climate grant that reduces domestic borrowing. It is a separate capital mix with the same total uses, government cash and Chinese credit; the two tables must not be added together.

Conditional blended capital source	Currency	Native amount	USD equivalent m
government import cash	USD	896,529,966	896.530
government local cash	IQD	1,395,671,459,457	1,073.593
Chinese export credit	USD	896,529,966	896.530
Ordinary IQD bonds	IQD	3,652,455,446,154	2,809.581
Green IQD bonds	IQD	1,211,663,772,206	932.049
Bank capital credit	IQD	1,621,373,072,787	1,247.210
Conditional climate capital grant	IQD	32,500,000,000	25.000

The conditional capital grant replaces USD 25m equivalent of domestic borrowing. Additional development rights and new local operating receipts enter later project cash and are not counted as construction capital a second time. Supplemental gap draws pay financing/OPEX/reserve cash needs and are separate from both capital tables.

Early deficits and additional financing

The independent flat price reconstruction requires USD 6.168bn gross early additional cash and retains USD 3.941bn later. Their difference is USD 2226.818m net nominal deficit before pricing gap finance. These figures cannot be added to construction capital as though they were new infrastructure. Later revenue cannot fund an earlier payment without a priced and available facility.

The candidate replaces eligible ordinary capital bonds with IQD green debt at 4% plus arrangement and 0.5% annual guarantee charges; it also tests an uncommitted USD 25m equivalent climate grant, USD 300m equivalent net development rights and USD 25m equivalent annual additional net local receipts. A green label alone changes no coupon. Grants replace eligible domestic capital borrowing; guarantees enhance credit rather than provide cash. Supplemental IQD credit at 2% and a 0.5% draw fee is capped at IQD 13tn outstanding, not treated as proven market capacity. All these conditional sources require legal, donor, investor and valuation evidence.

Additional routes to qualify include climate/renewable energy grants or concessional finance, guaranteed IQD on lending, phased green bonds/sukuk, development rights and station land leases, telecom/fibre leases, naming rights, sponsorship, employer travel contracts and carefully priced concessions. Carbon receipts remain unbooked contingent upside. Upfront lease receipts cannot be added while the same future rents remain in revenue. Keep net proceeds after costs and any transferred liabilities; foreign development finance needs confirmed IQD on lending or a priced hedge to preserve the currency structure.

Financing and pricing sensitivity	Peak gap IQD tn	Uncovered IQD tn	Terminal gap IQD tn
blended candidate	8.074	0.000	5.153
commercial gap credit	13.000	18.472	13.000
concessional gap credit	10.320	0.000	8.083
fare 5pct flat costs elastic	4.033	0.000	0.000
fare 5pct flat costs fixed demand	3.321	0.000	0.000
fare 5pct opex 5pct	4.362	0.000	0.000
fare 5pct opex 5pct income 2pct	5.501	0.000	0.000
fare 5pct opex 5pct rents indexed	4.166	0.000	0.000
fare 5pct opex 7pct	5.723	0.000	0.000
fixed fare 5pct opex	13.000	9.457	13.000
green concessional only	9.386	0.000	7.455
green label only	0.000	8.019	0.000
unfunded reference	0.000	8.019	0.000
variable fare 5pct opex 5pct	4.144	0.000	0.000
zero cost liquidity bound	8.019	0.000	2.895

Rail, property, energy and industrial financing redesign

The [twelve-case financing redesign](#) tests opening-linked principal, longer civil amortisation, a 15-year total insured green tenor and USD 1bn-equivalent total station rights replacing USD 300m, plus three retained viaduct-rental variants. Separate rail, solar, factory and developer monthly/native-currency accounts cancel PPA payments, train invoices, plant capacity fees and rights transfers on consolidation. The rental variants add a fifth borrower with zero extra operating-gap cap and no unapproved transfers to rail. Six-month placement envelopes and loan-vintage dates accompany every borrower. Fares/kiosks/advertising remain included; no future national order, surplus-power sale or unawarded climate grant services the integrated case.

The integrated sensitivity leaves peak aggregate IQD liquidity of IQD 8.450tn, cumulative missing funding IQD 0.070tn, and resource NPV USD -2.945bn, or USD -3.412bn after unverified public-land opportunity cost. Company cash balances and an assumed credit cap cannot establish bankability. Longer grace still pays interest, and changing finance terms leaves the core unlevered NPV unchanged.

Government remains 25% of the original rail/energy/factory capital. Import cash/Chinese credit still split 50:50 in USD; other credit, equity, fares, rights and payments are IQD. Private property construction is an additional explicitly priced scope, with its own IQD debt and equity. Optional indexed availability payments add public obligations outside the 25% capital limit; resource NPV cancels those government transfers. The combined downside retains delayed openings, weaker fare/nonfare demand, inflation and restricted financing, and adds weaker property sales/rights: IQD 48.418tn remains missing. Fifteen station candidates and six 90-day evidence workstreams have no invented title, valuation, investor term or named owner.

Iraqi mixed joint-stock holding and staged ordinary equity

The [eighteen-case ordinary-equity study](#) proposes a holding company with 100%-owned rail, energy, factory and station-development businesses, including retained viaduct premises. It replaces the prior subsidiary private equity rather than adding it twice. Government's original USD 1.970bn-equivalent cash remains 25% of original capital; converting it into shares creates zero new money. Same-price private subscriptions of USD 500m/1bn/2bn equivalent leave government ownership of 79.8%/66.3%/49.6%. IQD subscriptions, paid share registers, premium/dilution and fees are explicit. A secondary government share sale gives cash to the seller and zero to the company.

The \$1bn primary sensitivity requests six founder calls and a conditional later issue at month 60, with 2% fees. Monthly paid state capital constrains actual settlement; listing at that date is unproven. Against a matched tax-stressed holding reference, peak IQD liquidity falls from 5.455tn to 2.631tn. Domestic principal waits for full-network opening. Subscription failure cannot silently expand agreed six-month capital-credit placements or draw operating rescue for construction: the next unfunded invoice is withheld and no opening is reported. Whole-programme subscription delay and joint downside have separate cash/debt ledgers. Government's USD cash and Chinese USD credit remain the import split; other money remains IQD.

The \$1bn case's illustrative private equity return is 3.00% nominal IRR, below the assumed 15% hurdle, with first dividend at month 335. Dividends require profit, cleared debt, intact reserves and no missing cash; shares have no guaranteed redemption. Tax, depreciation, property inventory, factory impairment, retained earnings and internal-charge eliminations reconcile in pro-forma accounts. The base tax sensitivity avoids assumed group loss relief; a separate aggregate-profit proxy is explicitly unqualified. Resource NPV after land remains USD -3.412bn; changing financing does not create additional resources or financial feasibility.

Current Iraqi corporate/admission rules require an accepted mandate, legal/capital approvals, eligible audited business/accounts and actual investors. The 2019 liabilities restriction also needs entity-level assessment: the consolidated screening proxy flags 215 months in the \$1bn sensitivity. The structure therefore needs capitalisation/legal resolution before execution. Incorporated status, listing approval, public-land equity valuation and independent audit are all absent. Six additional source-bound ERP evidence packages cover legal/tax, mandate, subscriptions, accounts, governance and admission. Open licences are retained; no exclusive design valuation, national train order or uncontracted export income is assumed.

Retained premises beneath suitable viaducts

The [commercial-space register](#), [open unit plan](#), [pilot and rental model](#) links 1,143 elevated segments / 75.8105 km to civil chainage, planning coordinates and existing track assets. A preliminary 25 m bay screen reserves approaches, support inspection zones and independent access, suggesting 128,070 m². Confirmed eligible area remains zero: surveyed height/footprint, ownership, street/utility access, fire/flood/impact protection and station-sale overlap are unaccepted. The 200,000 m² illustration has 71,930 m² unmapped and supplies no integrated lease cash. A 30-unit draft pilot and reusable independent 30 m² enclosure have no invented tenant, permit or live contract.

The medium 100,000 m² scenario adds USD 93.532m equivalent of indexed fit-out capital. Leasing follows civil/rail readiness, rent-free periods, occupancy ramp, tenant turnover and arrears. Occupied/vacant maintenance, insurance, 12-year refurbishment, depreciation and tax are visible. Security deposits are liabilities backed by restricted cash; they supply no revenue, capital or dividends. Existing kiosks are unchanged and station buildings assumed sold cannot also generate rent. Additional under-viaduct land/rights cost is unverified; no free land value is asserted.

Before-tax incremental rental resource NPV is USD 17.244m, with the lower-rent/prolonged-vacancy case USD -35.427m. In the wholly owned \$1bn holding case, medium rents give private nominal IRR 3.22%, first dividend month 323 and peak IQD liquidity 2.646tn. Recurring rent adds value only after its resources/costs; it does not close the multibillion-dollar financial deficit.

The independent lender-policy comparison allows annual dividends with at least 1.30 historical/projected debt coverage, positive profit, completed capital, intact operating/tax/forward-service/renewal reserves and conservative leverage. These terms are unapproved and do not produce earlier distributions under the current tested assumptions: the non-rental first dividend remains month 335. A separately labelled final-cash diagnostic leaves actual accounts unchanged: proportional unrestricted terminal cash raises the original \$1bn private IRR from 3.00% to 4.22%, with no guaranteed redemption, unvalued asset sale or quoted share-exit price.

Each pilot unit carries lease/meter/inspection/use fields, civil-parent links and five common physical hazard paths. Tenant fire, flooding, impact, blocked bearing/inspection access and utility faults can affect piers, spans and railway operation; controller redundancy provides no mitigation for these hazards. Six additional native ERP evidence drafts require actual site, civil/fire, market, quote, maintenance/lease and finance evidence before occupancy or construction release.

Passenger fares and other operating income

Base model average paid trip yield is IQD 1,317. At full steady operation, existing fares contribute IQD 547.335bn/year, shops/kiosks IQD 12.984bn and advertising IQD 20.175bn. These already reduce the funding gap. Rental occupancy is 88%, advertising 85%; prices, collection and customer demand are not verified Baghdad lease quotations. Commercial figures are gross receipts, with dedicated concession costs requiring appraisal.

Low capacity use assumes 1,138,320 paid trips/day, not unique travellers or a demand survey. The proposed fare design should evaluate concession funding, student and low income access, commuter caps, transfer integration, peak/off peak tiers and collection costs before adopting a tariff. Neither a fare increase nor future kiosk rent places the early financing by itself.

The base full network operating allowance totals USD 180.567m/year equivalent, before the separate inflation sensitivities. Existing maintenance includes a battery renewal reserve on a 12 year reference cycle and fixed asset renewal allowances; it is not an accepted lifecycle replacement plan. Do not add the same battery reserve as new CAPEX. Condition based renewal quantities, inflation, dedicated concession costs, tax and supplier maintenance terms need qualification.

Base steady annual OPEX	USD equivalent m
civil station depot maintenance	96.208
labour	15.002
residual grid energy	0.000
rolling stock maintenance including battery renewal reserve	55.843
signalling maintenance	1.291
solar plant maintenance	12.223

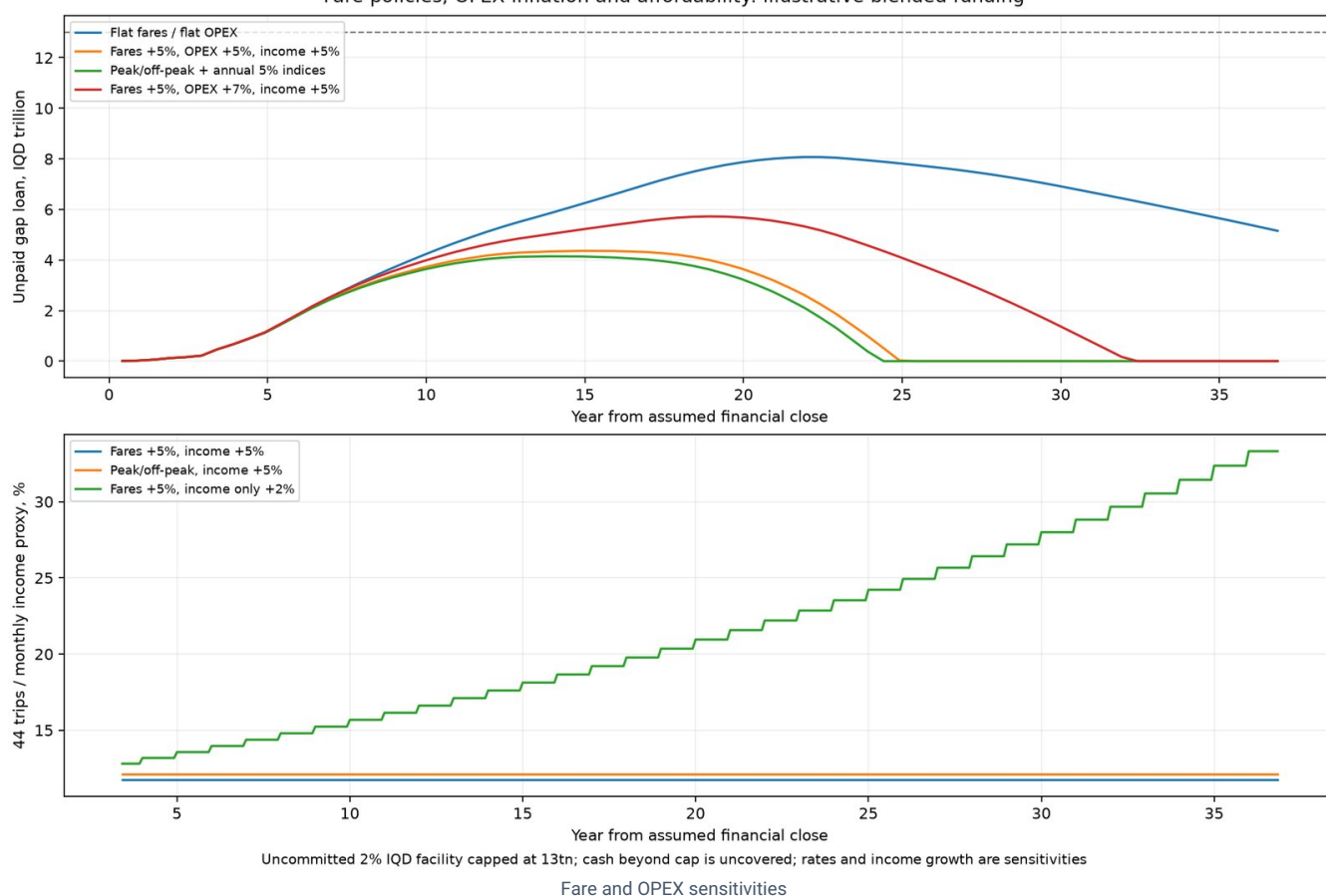
Annual ticket increases and OPEX inflation

The requested paired sensitivity indexes fares, OPEX and income 5% annually from financial close. No tickets are sold before opening. Nominal average fares are IQD 1,525 at first opening and IQD 1,765 at full opening. With 5% income growth, 44 trips remain 11.7% of the income proxy. With only 2% income growth that burden reaches 13.96% at full opening and the assumed real price elasticity reduces paid trips.

Variable pricing tests 40% of baseline trips at 1.25 times the standard fare and 60% at 0.90 times it, with separate demand response and the same capacity limit. If OPEX grows 7% while fares/income grow 5%, the model leaves IQD 0.000tn uncovered cash and IQD 0.000tn terminal unpaid gap debt. This sensitivity still needs additional financing before later surpluses; repayment remains conditional on that funding being available. Existing rent is flat unless the rental indexation sensitivity is chosen; new net rights/receipt targets are held nominal.

The paired case's unlevered NPV is USD -3.299bn at 13.4% nominal discount, excluding new grant/rights/net income targets and with un-escalated capital. Paying debt under a nominal model is not evidence of positive discounted project value. CAPEX escalation, renewal inflation, future FX, floating rates and surveyed demand remain material appraisal work.

Fare policies, OPEX inflation and affordability: illustrative blended funding



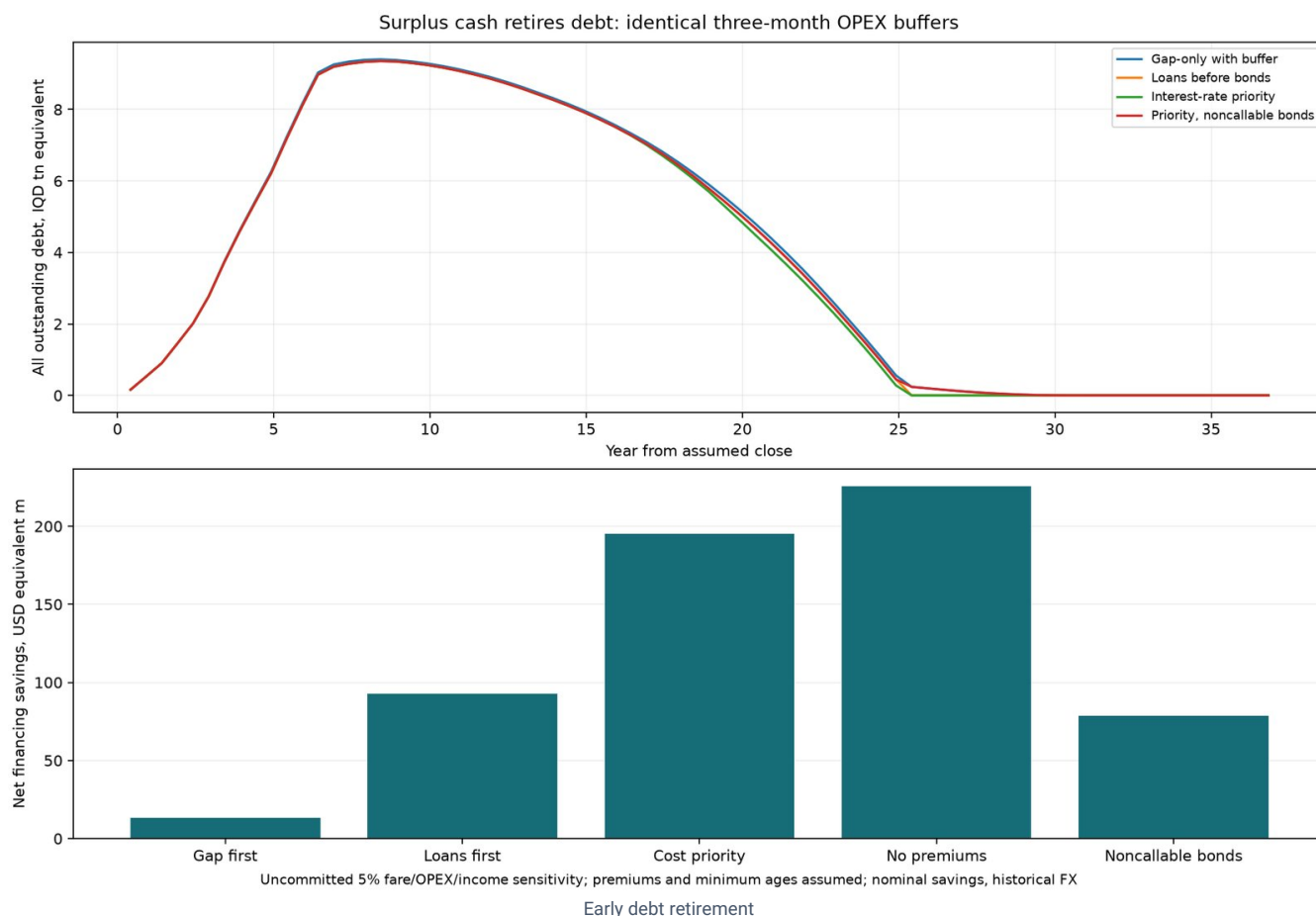
Surplus cash and early debt retirement

Surplus first pays OPEX, scheduled principal, interest and fees, then debt service reserves and three months of current OPEX. Voluntary payments cannot be funded by a new gap draw or uncovered cash in the same month. All comparisons below use the same buffers and 5% fare/OPEX/income inputs. Scheduled and voluntary principal are deducted once from native balances.

Repayment policy	All debt cleared month	Net saving vs buffered gap only USD eq m	Premium USD eq m
cost priority	303	195.029	23.681
cost priority noncallable bonds	371	78.731	5.175
cost priority zero premium	302	225.425	0.000
gap only buffered	371	0.000	0.000
gap then core	305	13.320	1.896
loans then bonds	304	92.811	7.121

Cost priority retires bank credit, ordinary bonds, Chinese credit, green bonds, then cheaper gap credit. Under assumed contractual rights it saves USD 195.029m equivalent after premiums and clears debt in month 303 versus 371 for buffered gap only. Loans first saves USD 92.811m. All retain IQD 0.340tn operating buffer at the horizon.

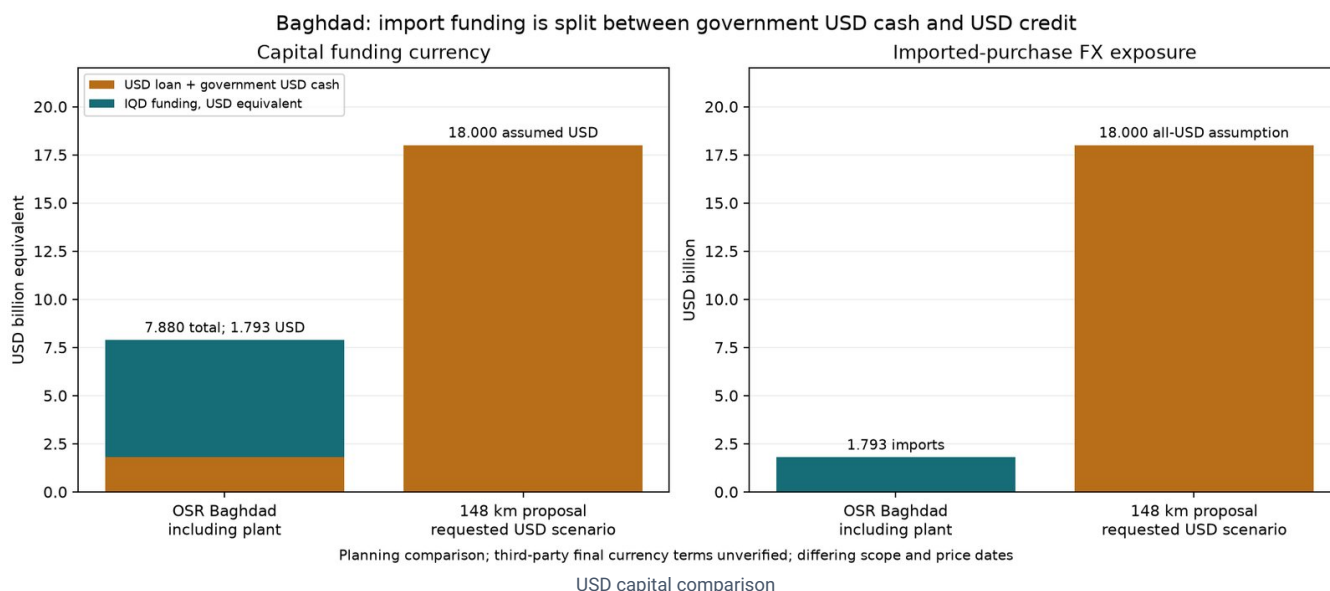
Premiums assume 1% bank/Chinese/green and 2% ordinary bonds, with minimum draw ages 6/12/24 months respectively. Eligible vintages are oldest first, retaining instalments and shortening maturity. Calls, notice, compensation, tax and market buyback prices need actual terms. The noncallable case makes no early bond payments; cost ordering is a heuristic rather than a globally best solution. Savings are nominal finance costs, not principal savings or present value wealth. Month numbers run from financial close, with no calendar commencement date assumed.



Historical Baghdad metro comparison

Measure	OpenSourceRail Baghdad plus plant	Historical proposal or requested comparator
Route km	516.5	148
Lines / stations	9 / 182	7 / 64
Capital USD equivalent bn	7.880	18.000 reported
Capital USD equivalent m / route km	15.26	121.62
USD capital funding bn	1.793	18.000 requested all USD scenario
Chinese USD debt bn	0.897	Final debt and government split unverified

The July 2024 reported estimate is a historical 148 km, USD 18bn scope. An entirely USD foreign loan/government cash basis is the requested comparator, not a verified financing contract. Under that assumption Baghdad's USD capital requirement is 90.0% lower. Distinct scope, price date, tunnelling/structures, land, utilities, qualification and schedule prevent a like for like bid saving claim. Third party fares, actual financing and comparable population access are not established. Its Iraqi labour share cannot be assumed zero.



Future national development

Baghdad can establish manufacturing, maintenance, training, procurement and digital delivery capacity that later Iraqi cities can reuse. The existing catalogue contains 18 cities representing 29,491,199 planning residents, 3,650 trainsets and 15,996 vehicle modules across several standard families. These are proposed urban networks; national intercity rail and freight connections are a future feasibility topic, without routes, budgets, revenues or approvals in these totals.

Future catalogue city	Planning population	Fleet family	Route km	Trainsets	City CAPEX USD eq m
Baghdad	9,780,429	metro-6car	516.5	831	7,555.744
Basra	3,955,000	metro-6car	305.4	450	3,114.674
Sulaymaniyah	2,150,000	metro-4car	119.8	129	1,026.507
Erbil	1,952,000	metro-4car	138.1	212	1,020.570
Mosul	1,940,000	metro-4car	207.3	256	1,926.740
Kirkuk	1,780,000	metro-4car	143.9	179	1,167.046
Najaf	1,540,000	metro-4car	165.1	229	1,521.180
Karbala	1,390,000	metro-4car	168.5	198	1,373.564
Nasiriyah	705,000	light-metro-3car	51.1	147	538.681
Hillah	700,000	light-metro-3car	57.9	125	461.207
Amarah	660,000	light-metro-3car	46.6	101	433.015
Ramadi	525,000	light-metro-3car	48.0	104	430.330
Baqubah	470,000	light-metro-3car	60.9	130	479.224
Diwaniyah	440,000	light-metro-3car	49.8	106	434.542
Kut	410,000	light-metro-3car	46.4	101	426.622
Samawah	373,770	light-metro-3car	50.4	108	415.341
Duhok	360,000	light-metro-3car	57.3	122	518.450
Fallujah	360,000	light-metro-3car	55.8	122	452.393

All city capital totals sum to USD 23.296bn. Adding one shared factory at USD 303.505m and its EPC at USD 21.245m produces USD 23.621bn equivalent nationally. Baghdad already contains this same plant and EPC. The additional city capital beyond the Baghdad scope is therefore USD 15.740bn, with no second plant added. Imported/local procurement in the generic national origin model is USD 5.562bn / USD 18.058bn; this is procurement composition, not a national loan programme.

No national factory expansion or replacement, intercity connection, research/training institution, shared governance or additional capital acceleration is priced. Sizing by the largest city's module order is not proof of annual production capacity. The 18 city aggregate is not a five year delivery commitment. Future orders require a throughput/renewal study, scheduled allocation and separate appropriations; no national revenue or profit services Baghdad debt in this proposal.

National industrial and institutional programme

Create common interface standards and supplier quality processes; qualify body, interior and infrastructure suppliers in Iraq; develop local module assembly, battery service, tools, spares and repair capability. A shared apprenticeship and technician programme should cover fabrication, concrete/precast, rail installation, power electronics, charging, software, accessibility and accountable inspection. Set competency outcomes and measured hours before assigning job numbers or training budgets.

Establish a national asset and evidence framework with separate city ownership, budgets, installed configuration and operating approval. Shared procurement and maintenance standards may reduce duplicated development, but each city still needs its own survey, demand, fiscal powers, financial close and safety acceptance. National development should coordinate with existing railways, freight access and universities where feasible; institutional appointments and agreements remain proposed.

Conditional national sequence

Stage	Proposed work	Decision before investment
Baghdad foundation	Sponsor, surveyed first line, plant and six car qualification	Accepted scope, executable finance and physical evidence
Demonstration and repeatability	Prove operating and industrial outcomes; maintain Mosul and Samawah as planning examples	Evidence of safe service, actual cost, production and affordable demand
Next city feasibility	Evaluate regional demand, access, land, energy and local implementation capacity	City specific business case and legal/financial approval
Broader urban rollout	Allocate shared factory output by accepted opening dates; qualify family differences	Priced capacity expansion, renewal and approved city finance
Intercity and national services	Study railway interfaces, freight/logistics, training and shared standards	New scope, environmental approvals and independently funded appraisal

Mosul and Samawah are useful reference packages; their inclusion here is not a selected construction order. Basra, Erbil, Sulaymaniyah and the other cities require transparent prioritisation from actual need and readiness. Future national borrowing shares and USD/IQD exposure must be recalculated from each qualified import basket. Baghdad's 25% contribution and 50:50 import split are not silently imposed on future city budgets.

Ownership and decisions requested

The proposed public sponsor should establish the legal project vehicle and accountable budget authority. The city/operator should own demand, fares, service and operating competence. Manufacturing management should own supplier interfaces, tooling, work hours and first article evidence. Independent survey, civil, energy and safety assessors should control their acceptance evidence. Lenders and placement advisers should validate rights, currency, timing, fees and investor capacity. These are proposed functions, not appointments.

Commission a phased Baghdad feasibility package that closes the route, station, depot and energy scope, independently verifies demand and affordable fares, prices capital and lifecycle obligations, and obtains term sheets for the USD import and IQD domestic funding. Review the long production schedule and fund any acceleration explicitly. Present the accepted first phase to the sponsor and competent authorities before construction. Maintain the national chapter as a separately approved development framework.

Current Baghdad deployment gates

Gate	Status	Responsible function	Closure action
soil-desktop-inputs	closed	civil/geotechnical designer	Sample current station and civil geometry from the pinned soilDB source and retain means, uncertainty and nodata.
soil-coverage	open	geotechnical investigator	Fill mapped coverage gaps through local records and targeted sampling; keep estimated and measured records distinct.
control-processing	open	survey team	Receive survey control observations; process and check the project CRS, datum and residuals.
ground-model	open	survey team	Receive and inspect the terrain, point cloud and ground-model deliveries against independent control.
surveyed-alignment	open	alignment designer	Replace or confirm generated geometry with checked survey alignment and platform reconciliation.
route-station-fit	open	civil/station designer	Resolve utilities, land/access, flood levels, station access and construction staging against the current route.
drainage-ground	open	civil/geotechnical designer	Use the soil investigation plan to collect local geotechnics and groundwater, size foundations/treatment, and check drainage with local rainfall and surveyed levels.
structural-release	open	structural designer and checker	Complete route-specific support/span, foundation, movement and erection calculations and close checking comments.
model-timing-comparison	closed	simulation engineer	Reconcile current native reference journey times with SUMO using the same scenario dwells and route.
full-service-validation	closed	simulation engineer	Run current nominal and degraded full-service cases against the current scenario and simulator; retain failures and exact provenance.
operating-release	open	operator	Complete conflict-aware capacity and degraded-operation review and sign the bound operating evidence.
morning-fleet-allocation	open	service planner	Reconcile revenue fleet with required morning departures on each line; retain two station berths and same-line overflow storage.
continuous-stabling-replay	open	simulation engineer	Run the selected line-local station/depot candidate across consecutive evenings and synchronised morning starts.
stabling-physical-fit	open	track/station designer	Locate usable station and line-local depot tracks, shared charging, isolation, inspection access and protected morning release slots.
solar-storage-endurance	open	energy designer	Bind declared station/ROW/dedicated PV to site storage and actual charging duty, reconcile conversion losses and prove replenishment across adverse weather; specify residual backup duty explicitly.
depot-placement-and-budget	open	depot and cost designer	Place the declared PV/storage inventory within the controlled site layout and reconcile itemised installed costs and renewal scope with existing allowances.
workforce-duty-and-workload	open	city operating lead	Reconcile the funded role plan with named shift/leave cover, measured inspection and cleaning workloads, simultaneous service-depot arrivals and actual local employment inputs.

3 checks are closed and 14 deployment gates remain open. Manufacturing, system certification and donor/lender decisions have additional distinct gates. The Baghdad package is complete as a planning example and has operational release false. Synthetic simulation, generated registers and documentary completeness are not physical acceptance.

Detailed schedules and source appendices

The [detailed engineering plan](#) and [component register](#) expand Baghdad's six-car mechanical interfaces, civil works, Iraqi slab manufacture, missing parts, onboard power/wiring, software host allocation and ERP handover. The register carries 69 reference part rows and all 60 Rust software allocations, with unknown prices and supplier identities left open. It corrects the ST6 seat layout and electronics interface errors; it does not release shop drawings or add unpriced components to the accepted finance totals. Local ERP recovery and planning-contract checks are recorded separately from production readiness.

The following proposal annex prints every station, interchange, fleet role, energy site and the complete cost priority six month draw/repayment schedule. Civil segment chainages and junction details are in the attached registers and design. Full monthly

and alternative case ledgers remain in the supporting archive and repository. Technical annexes reproduce the city survey, ground, alignment, depot, stabling, delivery, deployment, finance and acceptance reports, followed by the current shared architecture and engineering references.

[Source inventory](#) and [publication manifest](#) identify exact inputs and outputs. [Supporting data archive](#) includes the controlled files and complete operations payload. The editable proposal, PDF and appendix source list can be regenerated with the repository's proposal builder.

External instrument and historical sources are retained from the financing baseline. Source retrieval attempts on 3 October 2026 for China Exim, the GCF Iraq page and the NIC notice returned a timeout or access denial; this proposal does not claim a new source verification or a new lending commitment. Detailed financing reports retain the original source URLs and their stated evidence limits.

Baghdad detailed schedules

Source: cities/catalogue/west-asia/Iraq/Baghdad/proposal/DETAILED-SCHEDULES.md

Station register

Coordinates and chainages are planning inputs. IDs are controlled; anchor names remain in the UTF-8 CSV rather than being adopted as public station names. Latitude/longitude are WGS84 degrees.

Station ID	Line	Chainage km	Latitude	Longitude	Platform m
line-1-2500-0775-s000000	line-1	0.000	33.049995	44.366651	121.0
line-1-2343-0771-s003515	line-1	3.515	33.078249	44.365791	121.0
line-1-2180-0774-s007014	line-1	7.014	33.107584	44.366436	121.0
line-1-2017-0784-s010529	line-1	10.529	33.136918	44.368585	121.0
line-1-1848-0797-s014034	line-1	14.034	33.167333	44.371379	121.0
line-1-1702-0808-s017045	line-1	17.045	33.193608	44.373743	121.0
line-1-1554-0820-s020402	line-1	20.402	33.220243	44.376321	121.0
line-1-1429-0829-s023054	line-1	23.054	33.242738	44.378255	121.0
line-1-1306-0780-s026078	line-1	26.078	33.264874	44.367726	121.0
line-1-1182-0792-s029092	line-1	29.092	33.287190	44.370304	121.0
line-1-1063-0773-s032095	line-1	32.095	33.308606	44.366221	121.0
line-1-0940-0787-s035101	line-1	35.101	33.330742	44.369230	121.0
line-1-0821-0812-s038113	line-1	38.113	33.352158	44.374602	121.0
line-1-0731-0859-s041121	line-1	41.121	33.368355	44.384702	121.0
line-1-0643-0866-s043055	line-1	43.055	33.384192	44.386206	121.0
line-1-0557-0867-s044984	line-1	44.984	33.399669	44.386421	121.0
line-1-0463-0845-s047129	line-1	47.129	33.416586	44.381694	121.0
line-1-0319-0836-s050133	line-1	50.133	33.442501	44.379760	121.0
line-1-0166-0866-s053822	line-1	53.822	33.470036	44.386206	121.0
line-2-1987-2192-s000000	line-2	0.000	33.142317	44.671157	121.0
line-2-1895-2056-s003494	line-2	3.494	33.158874	44.641931	121.0
line-2-1800-1920-s007001	line-2	7.001	33.175971	44.612706	117.0
line-2-1648-1703-s012612	line-2	12.612	33.203326	44.566074	121.0
line-2-1550-1543-s017049	line-2	17.049	33.220962	44.531690	121.0
line-2-1509-1487-s018636	line-2	18.636	33.228341	44.519656	121.0
line-2-1461-1452-s020064	line-2	20.064	33.236979	44.512135	121.0
line-2-1361-1359-s023069	line-2	23.069	33.254976	44.492150	121.0
line-2-1249-1269-s026078	line-2	26.078	33.275132	44.472809	121.0
line-2-1141-1176-s029095	line-2	29.095	33.294569	44.452824	121.0
line-2-1095-1097-s031056	line-2	31.056	33.302847	44.435847	121.0
line-2-1103-1047-s033035	line-2	33.035	33.301407	44.425102	121.0
line-2-1022-1022-s035123	line-2	35.123	33.315985	44.419730	121.0
line-2-0947-0911-s038129	line-2	38.129	33.329482	44.395877	121.0
line-2-0923-0772-s041136	line-2	41.136	33.333801	44.366006	121.0
line-2-0828-0672-s044160	line-2	44.160	33.350898	44.344517	121.0
line-2-0728-0584-s047170	line-2	47.169	33.368895	44.325606	121.0
line-2-0753-0460-s050189	line-2	50.189	33.364395	44.298959	121.0
line-2-0654-0439-s052456	line-2	52.456	33.382212	44.294446	121.0
line-2-0679-0342-s054719	line-2	54.719	33.377713	44.273602	121.0
line-2-0595-0296-s056994	line-2	56.994	33.392830	44.263716	121.0
line-3-2271-1120-s000000	line-3	0.000	33.091207	44.440790	121.0
line-3-2129-1197-s003513	line-3	3.513	33.116762	44.457337	117.0
line-3-1987-1220-s007007	line-3	7.007	33.142317	44.462279	117.0
line-3-1854-1179-s010503	line-3	10.503	33.166253	44.453469	117.0
line-3-1702-1235-s014007	line-3	14.007	33.193608	44.465503	121.0
line-3-1613-1236-s015796	line-3	15.796	33.209625	44.465718	121.0
line-3-1502-1256-s018214	line-3	18.214	33.229601	44.470015	121.0
line-3-1391-1293-s021226	line-3	21.226	33.249577	44.477967	121.0
line-3-1265-1319-s024226	line-3	24.226	33.272253	44.483554	121.0
line-3-1162-1305-s027239	line-3	27.239	33.290789	44.480545	121.0
line-3-1111-1426-s030244	line-3	30.244	33.299968	44.506548	121.0
line-3-1046-1445-s033249	line-3	33.249	33.311665	44.510631	121.0
line-3-0943-1486-s036258	line-3	36.258	33.330202	44.519441	121.0
line-3-0830-1419-s039265	line-3	39.265	33.350538	44.505043	121.0

Station ID	Line	Chainage km	Latitude	Longitude	Platform m
line-3-0711-1492-s042273	line-3	42.273	33.371954	44.520731	121.0
line-3-0620-1509-s044279	line-3	44.279	33.388331	44.524384	121.0
line-3-0528-1521-s046268	line-3	46.268	33.404888	44.526963	121.0
line-3-0433-1532-s048276	line-3	48.276	33.421985	44.529326	121.0
line-3-0350-1616-s051291	line-3	51.291	33.436922	44.547378	121.0
line-3-0242-1604-s053550	line-3	53.550	33.456358	44.544799	121.0
line-3-0151-1637-s055819	line-3	55.819	33.472735	44.551890	121.0
line-4-0146-0199-s000000	line-4	0.000	33.473635	44.242872	121.0
line-4-0298-0305-s004261	line-4	4.261	33.446280	44.265650	121.0
line-4-0395-0433-s007765	line-4	7.765	33.428823	44.293157	117.0
line-4-0423-0565-s011263	line-4	11.263	33.423784	44.321523	117.0
line-4-0515-0750-s016457	line-4	16.457	33.407227	44.361279	121.0
line-4-0594-0786-s018376	line-4	18.376	33.393010	44.369015	121.0
line-4-0666-0827-s020297	line-4	20.297	33.380053	44.377826	121.0
line-4-0708-0930-s023314	line-4	23.314	33.372494	44.399960	121.0
line-4-0789-1036-s026330	line-4	26.330	33.357917	44.422739	121.0
line-4-0834-1141-s029342	line-4	29.342	33.349818	44.445303	121.0
line-4-0936-1241-s032367	line-4	32.367	33.331462	44.466792	121.0
line-4-1020-1350-s035373	line-4	35.373	33.316345	44.490216	121.0
line-4-1080-1447-s038373	line-4	38.373	33.305547	44.511060	121.0
line-4-1154-1476-s040395	line-4	40.395	33.292229	44.517292	121.0
line-4-1217-1551-s042429	line-4	42.429	33.280891	44.533409	121.0
line-4-1246-1628-s044443	line-4	44.443	33.275672	44.549956	121.0
line-5-1199-0260-s000000	line-5	0.000	33.284131	44.255980	121.0
line-5-1170-0351-s002077	line-5	2.077	33.289350	44.275536	121.0
line-5-1116-0417-s003985	line-5	3.985	33.299068	44.289719	121.0
line-5-1122-0491-s006033	line-5	6.033	33.297988	44.305621	121.0
line-5-1077-0591-s009034	line-5	9.034	33.306086	44.327110	121.0
line-5-1062-0705-s012053	line-5	12.053	33.308786	44.351608	121.0
line-5-1004-0809-s015076	line-5	15.076	33.319224	44.373957	121.0
line-5-0974-0947-s018085	line-5	18.085	33.324623	44.403613	121.0
line-5-1008-1046-s020829	line-5	20.829	33.318504	44.424888	121.0
line-5-0993-1151-s023847	line-5	23.847	33.321204	44.447452	121.0
line-5-1000-1254-s026861	line-5	26.861	33.319944	44.469586	121.0
line-5-0904-1340-s029889	line-5	29.889	33.337221	44.488067	121.0
line-5-0895-1451-s032907	line-5	32.907	33.338840	44.511920	121.0
line-5-0853-1593-s036330	line-5	36.331	33.346399	44.542435	121.0
line-5-0781-1730-s039913	line-5	39.913	33.359356	44.571876	117.0
line-5-0729-1880-s043410	line-5	43.410	33.368715	44.604110	117.0
line-5-0798-2027-s046922	line-5	46.922	33.356297	44.635699	117.0
line-5-0677-2304-s053737	line-5	53.737	33.378073	44.695225	121.0
line-6-2322-0251-s000000	line-6	0.000	33.082029	44.254046	121.0
line-6-2194-0355-s003511	line-6	3.511	33.105064	44.276395	117.0
line-6-2069-0472-s007016	line-6	7.016	33.127560	44.301538	117.0
line-6-1936-0575-s010529	line-6	10.529	33.151496	44.323672	117.0
line-6-1813-0699-s014028	line-6	14.028	33.173631	44.350319	117.0
line-6-1694-0774-s017029	line-6	17.029	33.195047	44.366436	121.0
line-6-1580-0848-s020342	line-6	20.342	33.215563	44.382338	121.0
line-6-1515-0927-s023073	line-6	23.073	33.227261	44.399315	121.0
line-6-1417-0951-s026078	line-6	26.078	33.244898	44.404473	121.0
line-6-1332-0995-s029084	line-6	29.084	33.260195	44.413928	121.0
line-6-1219-1042-s032087	line-6	32.087	33.280531	44.424028	121.0
line-6-1156-1145-s035101	line-6	35.101	33.291869	44.446162	121.0
line-6-1081-1255-s038128	line-6	38.128	33.305367	44.469801	121.0
line-6-0973-1337-s041141	line-6	41.141	33.324803	44.487422	121.0
line-6-0837-1343-s044142	line-6	44.142	33.349278	44.488711	121.0
line-6-0734-1414-s047145	line-6	47.145	33.367815	44.503969	121.0
line-6-0691-1529-s050169	line-6	50.169	33.375553	44.528682	121.0
line-6-0647-1607-s052578	line-6	52.578	33.383472	44.545444	121.0
line-6-0495-1669-s057516	line-6	57.516	33.410827	44.558767	121.0
line-7-1069-2206-s000000	line-7	0.000	33.307526	44.674166	121.0
line-7-1073-2033-s003493	line-7	3.493	33.306806	44.636989	117.0
line-7-1102-1872-s007005	line-7	7.005	33.301587	44.602391	117.0

Station ID	Line	Chainage km	Latitude	Longitude	Platform m
line-7-1185-1622-s012693	line-7	12.693	33.286650	44.548667	121.0
line-7-1206-1546-s014599	line-7	14.599	33.282871	44.532335	121.0
line-7-1277-1491-s016522	line-7	16.522	33.270093	44.520516	121.0
line-7-1328-1386-s019526	line-7	19.526	33.260915	44.497952	121.0
line-7-1314-1261-s022537	line-7	22.537	33.263434	44.471090	121.0
line-7-1401-1143-s025930	line-7	25.930	33.247777	44.445732	121.0
line-7-1398-1020-s028935	line-7	28.935	33.248317	44.419300	121.0
line-7-1396-0881-s031947	line-7	31.947	33.248677	44.389430	121.0
line-7-1412-0741-s034958	line-7	34.958	33.245798	44.359345	121.0
line-7-1420-0682-s036232	line-7	36.232	33.244358	44.346666	121.0
line-7-1434-0612-s037982	line-7	37.982	33.241839	44.331623	121.0
line-7-1487-0523-s040748	line-7	40.748	33.232300	44.312498	121.0
line-7-1531-0418-s043502	line-7	43.502	33.224382	44.289934	121.0
line-8-2231-1783-s000000	line-8	0.000	33.098405	44.583265	121.0
line-8-2175-1647-s003504	line-8	3.504	33.108484	44.554039	117.0
line-8-2045-1565-s007016	line-8	7.016	33.131879	44.536418	117.0
line-8-1940-1488-s010527	line-8	10.527	33.150776	44.519871	117.0
line-8-1857-1362-s014032	line-8	14.032	33.165713	44.492794	117.0
line-8-1751-1233-s017689	line-8	17.689	33.184789	44.465073	117.0
line-8-1645-1094-s021347	line-8	21.347	33.203866	44.435203	121.0
line-8-1518-0993-s024893	line-8	24.893	33.226721	44.413498	121.0
line-8-1422-0924-s027911	line-8	27.911	33.243998	44.398670	121.0
line-8-1349-0838-s030923	line-8	30.924	33.257136	44.380189	121.0
line-8-1281-0738-s033926	line-8	33.926	33.269373	44.358700	121.0
line-8-1168-0688-s036929	line-8	36.929	33.289710	44.347955	121.0
line-8-1085-0596-s039937	line-8	39.937	33.304647	44.328185	121.0
line-8-1004-0531-s042945	line-8	42.945	33.319224	44.314217	121.0
line-8-0976-0467-s044485	line-8	44.485	33.324263	44.300463	121.0
line-8-0957-0411-s045951	line-8	45.951	33.327682	44.288429	121.0
line-8-0835-0366-s048955	line-8	48.955	33.349638	44.278759	121.0
line-8-0785-0290-s051366	line-8	51.366	33.358637	44.262427	121.0
line-9-0565-0620-s003116	line-9	3.116	33.398229	44.333342	121.0
line-9-0502-0673-s005049	line-9	5.049	33.409567	44.344732	121.0
line-9-0487-0760-s006980	line-9	6.980	33.412266	44.363428	121.0
line-9-0557-0867-s010067	line-9	10.067	33.399669	44.386421	121.0
line-9-0549-0960-s012159	line-9	12.159	33.401109	44.406407	121.0
line-9-0538-1065-s015166	line-9	15.166	33.403088	44.428971	121.0
line-9-0585-1190-s018666	line-9	18.666	33.394630	44.455832	117.0
line-9-0608-1317-s022187	line-9	22.187	33.390491	44.483124	117.0
line-9-0637-1439-s025194	line-9	25.194	33.385272	44.509341	121.0
line-9-0620-1509-s026894	line-9	26.895	33.388331	44.524384	121.0
line-9-0619-1569-s028209	line-9	28.209	33.388511	44.537278	121.0
line-9-0647-1607-s029412	line-9	29.412	33.383472	44.545444	121.0
line-9-0853-1593-s033809	line-9	33.809	33.346399	44.542435	121.0
line-9-0911-1579-s035217	line-9	35.217	33.335961	44.539427	121.0
line-9-1185-1622-s041363	line-9	41.363	33.286650	44.548667	121.0
line-9-1246-1628-s042633	line-9	42.633	33.275672	44.549956	121.0
line-9-1430-1529-s048432	line-9	48.432	33.242558	44.528682	121.0
line-9-1509-1487-s050681	line-9	50.681	33.228341	44.519656	121.0
line-9-1605-1380-s055450	line-9	55.450	33.211064	44.496662	117.0
line-9-1613-1236-s058396	line-9	58.396	33.209625	44.465718	121.0
line-9-1645-1094-s061501	line-9	61.501	33.203866	44.435203	121.0
line-9-1675-0934-s065467	line-9	65.467	33.198467	44.400819	117.0
line-9-1580-0848-s068248	line-9	68.248	33.215563	44.382338	121.0
line-9-1477-0761-s071522	line-9	71.522	33.234100	44.363642	121.0
line-9-1389-0675-s074529	line-9	74.529	33.249937	44.345162	121.0
line-9-1335-0562-s077540	line-9	77.540	33.259655	44.320878	121.0
line-9-1246-0493-s080552	line-9	80.552	33.275672	44.306051	121.0
line-9-1200-0443-s082619	line-9	82.619	33.283951	44.295306	121.0
line-9-1190-0372-s084695	line-9	84.695	33.285750	44.280048	121.0
line-9-1116-0417-s087021	line-9	87.021	33.299068	44.289719	121.0
line-9-0976-0467-s090748	line-9	90.748	33.324263	44.300463	121.0
line-9-0889-0471-s092604	line-9	92.604	33.339920	44.301323	121.0

Station ID	Line	Chainage km	Latitude	Longitude	Platform m
line-9-0832-0577-s095621	line-9	95.621	33.350178	44.324102	121.0
line-9-0700-0568-s098647	line-9	98.647	33.373934	44.322168	121.0

Interchange register

Interchange	Lines	Latitude	Longitude
interchange-000	line-1, line-6	33.194327	44.370089
interchange-001	line-1, line-2	33.332272	44.367618
interchange-002	line-1, line-9	33.399669	44.386421
interchange-003	line-2, line-9	33.228341	44.519656
interchange-004	line-2, line-6	33.293219	44.449493
interchange-005	line-2, line-5	33.317244	44.422309
interchange-006	line-2, line-9	33.371414	44.323887
interchange-007	line-3, line-9	33.209625	44.465718
interchange-008	line-3, line-4	33.308606	44.510846
interchange-009	line-3, line-9	33.388331	44.524384
interchange-010	line-4, line-9	33.409747	44.362353
interchange-011	line-4, line-7	33.281881	44.532872
interchange-012	line-4, line-7, line-9	33.281161	44.549312
interchange-013	line-5, line-9	33.287550	44.277792
interchange-014	line-5, line-9	33.299068	44.289719
interchange-015	line-5, line-8	33.305367	44.327648
interchange-016	line-5, line-9	33.346399	44.542435
interchange-017	line-1, line-6, line-9	33.217123	44.380333
interchange-018	line-6, line-8	33.244448	44.401571
interchange-019	line-6, line-9	33.383472	44.545444
interchange-020	line-7, line-9	33.247148	44.345914
interchange-021	line-8, line-9	33.203866	44.435203
interchange-022	line-8, line-9	33.324263	44.300463

Fleet roles

Line	Peak	Rotation	Spare	Cold reserve	Total
line-1	86	0	8	1	95
line-2	93	0	9	1	103
line-3	98	0	9	1	108
line-4	77	0	7	1	85
line-5	89	0	8	1	98
line-6	100	0	10	1	111
line-7	76	0	7	1	84
line-8	90	0	9	1	100
line-9	42	0	4	1	47

Energy site register

Site station	Tier	PV kW	Storage kWh	Grid import kW	Charger kW
line-1-2500-0775-s000000	terminal	300.0	2000.0	3200.0	2000.0
line-1-2343-0771-s003515	standard	300.0	2000.0	2000.0	2000.0
line-1-2180-0774-s007014	standard	300.0	2000.0	2000.0	2000.0
line-1-2017-0784-s010529	standard	300.0	2000.0	2000.0	2000.0
line-1-1848-0797-s014034	standard	300.0	2000.0	2000.0	2000.0
line-1-1702-0808-s017045	interchange	300.0	2000.0	2400.0	2000.0
line-1-1554-0820-s020402	interchange	300.0	2000.0	2400.0	2000.0
line-1-1429-0829-s023054	standard	300.0	2000.0	2000.0	2000.0
line-1-1306-0780-s026078	standard	300.0	2000.0	2000.0	2000.0
line-1-1182-0792-s029092	standard	300.0	2000.0	2000.0	2000.0
line-1-1063-0773-s032095	major	300.0	2000.0	2400.0	2000.0
line-1-0940-0787-s035101	interchange	300.0	2000.0	2400.0	2000.0
line-1-0821-0812-s038113	major	300.0	2000.0	2400.0	2000.0
line-1-0731-0859-s041121	major	300.0	2000.0	2400.0	2000.0
line-1-0643-0866-s043055	standard	300.0	2000.0	2000.0	2000.0
line-1-0557-0867-s044984	interchange	300.0	2000.0	2400.0	2000.0
line-1-0463-0845-s047129	standard	300.0	2000.0	2000.0	2000.0
line-1-0319-0836-s050133	standard	300.0	2000.0	2000.0	2000.0
line-1-0166-0866-s053822	terminal	300.0	2000.0	3200.0	2000.0
line-2-1987-2192-s000000	terminal	300.0	2000.0	3200.0	2000.0
line-2-1895-2056-s003494	standard	300.0	2000.0	2000.0	2000.0
line-2-1648-1703-s012612	standard	300.0	2000.0	2000.0	2000.0
line-2-1550-1543-s017049	standard	300.0	2000.0	2000.0	2000.0
line-2-1509-1487-s018636	interchange	300.0	2000.0	2400.0	2000.0
line-2-1461-1452-s020064	standard	300.0	2000.0	2000.0	2000.0
line-2-1361-1359-s023069	major	300.0	2000.0	2400.0	2000.0
line-2-1249-1269-s026078	standard	300.0	2000.0	2000.0	2000.0
line-2-1141-1176-s029095	interchange	300.0	2000.0	2400.0	2000.0
line-2-1095-1097-s031056	major	300.0	2000.0	2400.0	2000.0
line-2-1103-1047-s033035	major	300.0	2000.0	2400.0	2000.0
line-2-1022-1022-s035123	interchange	300.0	2000.0	2400.0	2000.0
line-2-0947-0911-s038129	standard	300.0	2000.0	2000.0	2000.0
line-2-0923-0772-s041136	interchange	300.0	2000.0	2400.0	2000.0
line-2-0828-0672-s044160	standard	300.0	2000.0	2000.0	2000.0
line-2-0728-0584-s047170	interchange	300.0	2000.0	2400.0	2000.0
line-2-0753-0460-s050189	standard	300.0	2000.0	2000.0	2000.0
line-2-0654-0439-s052456	standard	300.0	2000.0	2000.0	2000.0
line-2-0679-0342-s054719	standard	300.0	2000.0	2000.0	2000.0
line-2-0595-0296-s056994	terminal	300.0	2000.0	3200.0	2000.0
line-3-2271-1120-s000000	terminal	300.0	2000.0	3200.0	2000.0
line-3-1702-1235-s014007	standard	300.0	2000.0	2000.0	2000.0
line-3-1613-1236-s015796	interchange	300.0	2000.0	2400.0	2000.0
line-3-1502-1256-s018214	standard	300.0	2000.0	2000.0	2000.0
line-3-1391-1293-s021226	major	300.0	2000.0	2400.0	2000.0
line-3-1265-1319-s024226	standard	300.0	2000.0	2000.0	2000.0
line-3-1162-1305-s027239	standard	300.0	2000.0	2000.0	2000.0
line-3-1111-1426-s030244	major	300.0	2000.0	2400.0	2000.0
line-3-1046-1445-s033249	interchange	300.0	2000.0	2400.0	2000.0
line-3-0943-1486-s036258	standard	300.0	2000.0	2000.0	2000.0
line-3-0830-1419-s039265	major	300.0	2000.0	2400.0	2000.0
line-3-0711-1492-s042273	standard	300.0	2000.0	2000.0	2000.0
line-3-0620-1509-s044279	interchange	300.0	2000.0	2400.0	2000.0
line-3-0528-1521-s046268	standard	300.0	2000.0	2000.0	2000.0
line-3-0433-1532-s048276	standard	300.0	2000.0	2000.0	2000.0
line-3-0350-1616-s051291	standard	300.0	2000.0	2000.0	2000.0
line-3-0242-1604-s053550	standard	300.0	2000.0	2000.0	2000.0
line-3-0151-1637-s055819	terminal	300.0	2000.0	3200.0	2000.0
line-4-0146-0199-s000000	terminal	300.0	2000.0	3200.0	2000.0
line-4-0298-0305-s004261	standard	300.0	2000.0	2000.0	2000.0
line-4-0515-0750-s016457	interchange	300.0	2000.0	2400.0	2000.0
line-4-0594-0786-s018376	standard	300.0	2000.0	2000.0	2000.0

Site station	Tier	PV kW	Storage kWh	Grid import kW	Charger kW
line-4-0666-0827-s020297	standard	300.0	2000.0	2000.0	2000.0
line-4-0708-0930-s023314	major	300.0	2000.0	2400.0	2000.0
line-4-0789-1036-s026330	major	300.0	2000.0	2400.0	2000.0
line-4-0834-1141-s029342	standard	300.0	2000.0	2000.0	2000.0
line-4-0936-1241-s032367	major	300.0	2000.0	2400.0	2000.0
line-4-1020-1350-s035373	major	300.0	2000.0	2400.0	2000.0
line-4-1080-1447-s038373	interchange	300.0	2000.0	2400.0	2000.0
line-4-1154-1476-s040395	standard	300.0	2000.0	2000.0	2000.0
line-4-1217-1551-s042429	interchange	300.0	2000.0	2400.0	2000.0
line-4-1246-1628-s044443	terminal	300.0	2000.0	3200.0	2000.0
line-5-1199-0260-s000000	terminal	300.0	2000.0	3200.0	2000.0
line-5-1170-0351-s002077	interchange	300.0	2000.0	2400.0	2000.0
line-5-1116-0417-s003985	interchange	300.0	2000.0	2400.0	2000.0
line-5-1122-0491-s006033	standard	300.0	2000.0	2000.0	2000.0
line-5-1077-0591-s009034	interchange	300.0	2000.0	2400.0	2000.0
line-5-1062-0705-s012053	major	300.0	2000.0	2400.0	2000.0
line-5-1004-0809-s015076	standard	300.0	2000.0	2000.0	2000.0
line-5-0974-0947-s018085	standard	300.0	2000.0	2000.0	2000.0
line-5-1008-1046-s020829	interchange	300.0	2000.0	2400.0	2000.0
line-5-0993-1151-s023847	major	300.0	2000.0	2400.0	2000.0
line-5-1000-1254-s026861	major	300.0	2000.0	2400.0	2000.0
line-5-0904-1340-s029889	major	300.0	2000.0	2400.0	2000.0
line-5-0895-1451-s032907	standard	300.0	2000.0	2000.0	2000.0
line-5-0853-1593-s036330	interchange	300.0	2000.0	2400.0	2000.0
line-5-0677-2304-s053737	terminal	300.0	2000.0	3200.0	2000.0
line-6-2322-0251-s000000	terminal	300.0	2000.0	3200.0	2000.0
line-6-1694-0774-s017029	interchange	300.0	2000.0	2400.0	2000.0
line-6-1580-0848-s020342	interchange	300.0	2000.0	2400.0	2000.0
line-6-1515-0927-s023073	standard	300.0	2000.0	2000.0	2000.0
line-6-1417-0951-s026078	interchange	300.0	2000.0	2400.0	2000.0
line-6-1332-0995-s029084	major	300.0	2000.0	2400.0	2000.0
line-6-1219-1042-s032087	standard	300.0	2000.0	2000.0	2000.0
line-6-1156-1145-s035101	interchange	300.0	2000.0	2400.0	2000.0
line-6-1081-1255-s038128	major	300.0	2000.0	2400.0	2000.0
line-6-0973-1337-s041141	major	300.0	2000.0	2400.0	2000.0
line-6-0837-1343-s044142	major	300.0	2000.0	2400.0	2000.0
line-6-0734-1414-s047145	major	300.0	2000.0	2400.0	2000.0
line-6-0691-1529-s050169	standard	300.0	2000.0	2000.0	2000.0
line-6-0647-1607-s052578	interchange	300.0	2000.0	2400.0	2000.0
line-6-0495-1669-s057516	depot-main	5000.0	40000.0	12000.0	2000.0
line-7-1069-2206-s000000	terminal	300.0	2000.0	3200.0	2000.0
line-7-1185-1622-s012693	interchange	300.0	2000.0	2400.0	2000.0
line-7-1206-1546-s014599	interchange	300.0	2000.0	2400.0	2000.0
line-7-1277-1491-s016522	standard	300.0	2000.0	2000.0	2000.0
line-7-1328-1386-s019526	major	300.0	2000.0	2400.0	2000.0
line-7-1314-1261-s022537	major	300.0	2000.0	2400.0	2000.0
line-7-1401-1143-s025930	standard	300.0	2000.0	2000.0	2000.0
line-7-1398-1020-s028935	standard	300.0	2000.0	2000.0	2000.0
line-7-1396-0881-s031947	standard	300.0	2000.0	2000.0	2000.0
line-7-1412-0741-s034958	standard	300.0	2000.0	2000.0	2000.0
line-7-1420-0682-s036232	interchange	300.0	2000.0	2400.0	2000.0
line-7-1434-0612-s037982	standard	300.0	2000.0	2000.0	2000.0
line-7-1487-0523-s040748	standard	300.0	2000.0	2000.0	2000.0
line-7-1531-0418-s043502	terminal	300.0	2000.0	3200.0	2000.0
line-8-2231-1783-s000000	terminal	300.0	2000.0	3200.0	2000.0
line-8-1645-1094-s021347	interchange	300.0	2000.0	2400.0	2000.0
line-8-1518-0993-s024893	standard	300.0	2000.0	2000.0	2000.0
line-8-1422-0924-s027911	interchange	300.0	2000.0	2400.0	2000.0
line-8-1349-0838-s030923	standard	300.0	2000.0	2000.0	2000.0
line-8-1281-0738-s033926	standard	300.0	2000.0	2000.0	2000.0
line-8-1168-0688-s036929	standard	300.0	2000.0	2000.0	2000.0
line-8-1085-0596-s039937	interchange	300.0	2000.0	2400.0	2000.0
line-8-1004-0531-s042945	standard	300.0	2000.0	2000.0	2000.0

Site station	Tier	PV kW	Storage kWh	Grid import kW	Charger kW
line-8-0976-0467-s044485	interchange	300.0	2000.0	2400.0	2000.0
line-8-0957-0411-s045951	standard	300.0	2000.0	2000.0	2000.0
line-8-0835-0366-s048955	standard	300.0	2000.0	2000.0	2000.0
line-8-0785-0290-s051366	terminal	300.0	2000.0	3200.0	2000.0
line-9-0565-0620-s003116	standard	300.0	2000.0	2000.0	2000.0
line-9-0502-0673-s005049	standard	300.0	2000.0	2000.0	2000.0
line-9-0487-0760-s006980	interchange	300.0	2000.0	2400.0	2000.0
line-9-0557-0867-s010067	interchange	300.0	2000.0	2400.0	2000.0
line-9-0549-0960-s012159	major	300.0	2000.0	2400.0	2000.0
line-9-0538-1065-s015166	standard	300.0	2000.0	2000.0	2000.0
line-9-0637-1439-s025194	standard	300.0	2000.0	2000.0	2000.0
line-9-0620-1509-s026894	interchange	300.0	2000.0	2400.0	2000.0
line-9-0619-1569-s028209	standard	300.0	2000.0	2000.0	2000.0
line-9-0647-1607-s029412	interchange	300.0	2000.0	2400.0	2000.0
line-9-0853-1593-s033809	interchange	300.0	2000.0	2400.0	2000.0
line-9-0911-1579-s035217	standard	300.0	2000.0	2000.0	2000.0
line-9-1185-1622-s041363	interchange	300.0	2000.0	2400.0	2000.0
line-9-1246-1628-s042633	interchange	300.0	2000.0	2400.0	2000.0
line-9-1430-1529-s048432	standard	300.0	2000.0	2000.0	2000.0
line-9-1509-1487-s050681	interchange	300.0	2000.0	2400.0	2000.0
line-9-1613-1236-s058396	interchange	300.0	2000.0	2400.0	2000.0
line-9-1645-1094-s061501	interchange	300.0	2000.0	2400.0	2000.0
line-9-1580-0848-s068248	interchange	300.0	2000.0	2400.0	2000.0
line-9-1477-0761-s071522	standard	300.0	2000.0	2000.0	2000.0
line-9-1389-0675-s074529	interchange	300.0	2000.0	2400.0	2000.0
line-9-1335-0562-s077540	standard	300.0	2000.0	2000.0	2000.0
line-9-1246-0493-s080552	standard	300.0	2000.0	2000.0	2000.0
line-9-1200-0443-s082619	standard	300.0	2000.0	2000.0	2000.0
line-9-1190-0372-s084695	interchange	300.0	2000.0	2400.0	2000.0
line-9-1116-0417-s087021	interchange	300.0	2000.0	2400.0	2000.0
line-9-0976-0467-s090748	interchange	300.0	2000.0	2400.0	2000.0
line-9-0889-0471-s092604	standard	300.0	2000.0	2000.0	2000.0
line-9-0832-0577-s095621	standard	300.0	2000.0	2000.0	2000.0
line-9-0700-0568-s098647	interchange	300.0	2000.0	2400.0	2000.0

Six month capital and supplemental draw requirements

Cost priority with identical three month operating buffers. Settlement is monthly; six month rows are envelopes, not advance placements. Unrounded CSV values reconcile; displayed amounts are rounded. Tranches continue through the entire model horizon.

Months	Gov USD m	Gov IQD bn	China USD m	Ordinary bond IQD bn	Green IQD bn	Bank IQD bn	Gap IQD bn
0 to 5	12.63	35.03	12.63	103.44	0.00	34.48	4.86
6 to 11	31.51	77.51	31.51	235.84	0.00	78.61	17.02
12 to 17	26.67	80.20	26.67	232.45	0.00	77.48	37.22
18 to 23	42.52	111.55	42.52	332.12	1.79	111.30	61.05
24 to 29	60.40	103.18	60.40	287.02	61.71	116.24	71.32
30 to 35	90.90	127.12	90.90	321.38	139.09	153.49	96.39
36 to 41	92.85	127.88	92.85	316.88	148.90	155.26	293.29
42 to 47	92.43	126.92	92.43	313.81	149.00	154.27	210.57
48 to 53	86.76	109.79	86.76	264.79	148.43	137.74	233.49
54 to 59	85.84	108.48	85.84	259.89	148.58	136.16	235.77
60 to 65	93.99	130.42	93.99	325.48	148.24	157.90	311.69
66 to 71	89.94	123.89	89.94	306.92	144.30	150.41	318.26
72 to 77	80.04	119.69	80.04	313.28	109.90	141.06	321.22
78 to 83	10.04	14.02	10.04	39.17	11.73	16.97	289.34
84 to 89	0.00	0.00	0.00	0.00	0.00	0.00	250.55
90 to 95	0.00	0.00	0.00	0.00	0.00	0.00	236.08
96 to 101	0.00	0.00	0.00	0.00	0.00	0.00	221.02
102 to 107	0.00	0.00	0.00	0.00	0.00	0.00	205.47
108 to 113	0.00	0.00	0.00	0.00	0.00	0.00	181.69
114 to 119	0.00	0.00	0.00	0.00	0.00	0.00	166.81

Months	Gov USD m	Gov IQD bn	China USD m	Ordinary bond IQD bn	Green IQD bn	Bank IQD bn	Gap IQD bn
120 to 125	0.00	0.00	0.00	0.00	0.00	0.00	150.08
126 to 131	0.00	0.00	0.00	0.00	0.00	0.00	130.59
132 to 137	0.00	0.00	0.00	0.00	0.00	0.00	109.08
138 to 143	0.00	0.00	0.00	0.00	0.00	0.00	94.03
144 to 149	0.00	0.00	0.00	0.00	0.00	0.00	69.68
150 to 155	0.00	0.00	0.00	0.00	0.00	0.00	50.62
156 to 161	0.00	0.00	0.00	0.00	0.00	0.00	28.09
162 to 167	0.00	0.00	0.00	0.00	0.00	0.00	24.87
168 to 173	0.00	0.00	0.00	0.00	0.00	0.00	18.81
174 to 179	0.00	0.00	0.00	0.00	0.00	0.00	13.33
180 to 185	0.00	0.00	0.00	0.00	0.00	0.00	5.48
186 to 191	0.00	0.00	0.00	0.00	0.00	0.00	0.00
192 to 197	0.00	0.00	0.00	0.00	0.00	0.00	2.77
198 to 203	0.00	0.00	0.00	0.00	0.00	0.00	0.00
204 to 209	0.00	0.00	0.00	0.00	0.00	0.00	0.00
210 to 215	0.00	0.00	0.00	0.00	0.00	0.00	0.00
216 to 221	0.00	0.00	0.00	0.00	0.00	0.00	0.00
222 to 227	0.00	0.00	0.00	0.00	0.00	0.00	0.00
228 to 233	0.00	0.00	0.00	0.00	0.00	0.00	0.00
234 to 239	0.00	0.00	0.00	0.00	0.00	0.00	0.00
240 to 245	0.00	0.00	0.00	0.00	0.00	0.00	0.00
246 to 251	0.00	0.00	0.00	0.00	0.00	0.00	0.00
252 to 257	0.00	0.00	0.00	0.00	0.00	0.00	0.00
258 to 263	0.00	0.00	0.00	0.00	0.00	0.00	0.00
264 to 269	0.00	0.00	0.00	0.00	0.00	0.00	0.00
270 to 275	0.00	0.00	0.00	0.00	0.00	0.00	0.00
276 to 281	0.00	0.00	0.00	0.00	0.00	0.00	0.00
282 to 287	0.00	0.00	0.00	0.00	0.00	0.00	0.00
288 to 293	0.00	0.00	0.00	0.00	0.00	0.00	0.00
294 to 299	0.00	0.00	0.00	0.00	0.00	0.00	0.00
300 to 305	0.00	0.00	0.00	0.00	0.00	0.00	0.00
306 to 311	0.00	0.00	0.00	0.00	0.00	0.00	0.00
312 to 317	0.00	0.00	0.00	0.00	0.00	0.00	0.00
318 to 323	0.00	0.00	0.00	0.00	0.00	0.00	0.00
324 to 329	0.00	0.00	0.00	0.00	0.00	0.00	0.00
330 to 335	0.00	0.00	0.00	0.00	0.00	0.00	0.00
336 to 341	0.00	0.00	0.00	0.00	0.00	0.00	0.00
342 to 347	0.00	0.00	0.00	0.00	0.00	0.00	0.00
348 to 353	0.00	0.00	0.00	0.00	0.00	0.00	0.00
354 to 359	0.00	0.00	0.00	0.00	0.00	0.00	0.00
360 to 365	0.00	0.00	0.00	0.00	0.00	0.00	0.00
366 to 371	0.00	0.00	0.00	0.00	0.00	0.00	0.00
372 to 377	0.00	0.00	0.00	0.00	0.00	0.00	0.00
378 to 383	0.00	0.00	0.00	0.00	0.00	0.00	0.00
384 to 389	0.00	0.00	0.00	0.00	0.00	0.00	0.00
390 to 395	0.00	0.00	0.00	0.00	0.00	0.00	0.00
396 to 401	0.00	0.00	0.00	0.00	0.00	0.00	0.00
402 to 407	0.00	0.00	0.00	0.00	0.00	0.00	0.00
408 to 413	0.00	0.00	0.00	0.00	0.00	0.00	0.00
414 to 419	0.00	0.00	0.00	0.00	0.00	0.00	0.00
420 to 425	0.00	0.00	0.00	0.00	0.00	0.00	0.00
426 to 431	0.00	0.00	0.00	0.00	0.00	0.00	0.00
432 to 437	0.00	0.00	0.00	0.00	0.00	0.00	0.00
438 to 442	0.00	0.00	0.00	0.00	0.00	0.00	0.00

Six month early repayments and closing balances

Early principal is additional to the scheduled principal separately retained in the full ledger. Outstanding debt uses the final month of each period. USD equivalent aggregates convert Chinese native debt at the historical IQD 1,300 anchor. Premiums and operating buffers are distinct from principal.

Months	Core early principal USD eq m	Premium USD eq m	Gap principal IQD bn	All debt closing IQD tn eq	Cash closing IQD tn	OPEX buffer IQD bn
0 to 5	0.00	0.00	0.00	0.158	0.000	0.000
6 to 11	0.00	0.00	0.00	0.528	0.000	0.000
12 to 17	0.00	0.00	0.00	0.902	0.000	0.000
18 to 23	0.00	0.00	0.00	1.450	0.000	0.000
24 to 29	27.79	0.28	0.00	2.011	0.000	0.000
30 to 35	36.56	0.37	0.00	2.766	0.000	0.000
36 to 41	32.31	0.32	0.00	3.723	0.000	22.808
42 to 47	19.61	0.20	0.00	4.598	0.000	29.124
48 to 53	14.65	0.15	0.00	5.414	0.000	37.533
54 to 59	15.16	0.15	0.00	6.212	0.000	43.005
60 to 65	0.00	0.00	0.00	7.186	0.000	51.780
66 to 71	0.00	0.00	0.00	8.113	0.000	59.283
72 to 77	0.00	0.00	0.00	8.972	0.000	68.209
78 to 83	0.00	0.00	0.00	9.194	0.000	78.643
84 to 89	0.00	0.00	0.00	9.279	0.000	82.575
90 to 95	0.00	0.00	0.00	9.332	0.000	82.575
96 to 101	0.00	0.00	0.00	9.352	0.000	86.704
102 to 107	0.00	0.00	0.00	9.341	0.000	86.704
108 to 113	0.00	0.00	0.00	9.299	0.000	91.039
114 to 119	0.00	0.00	0.00	9.242	0.000	91.039
120 to 125	0.00	0.00	0.00	9.169	0.000	95.591
126 to 131	0.00	0.00	0.00	9.081	0.000	95.591
132 to 137	0.00	0.00	0.00	8.978	0.000	100.370
138 to 143	0.00	0.00	0.00	8.865	0.000	100.370
144 to 149	0.00	0.00	0.00	8.734	0.000	105.389
150 to 155	0.00	0.00	0.00	8.592	0.000	105.389
156 to 161	0.00	0.00	0.00	8.436	0.000	110.658
162 to 167	0.00	0.00	0.00	8.278	0.000	110.658
168 to 173	0.00	0.00	0.00	8.109	0.000	116.191
174 to 179	0.00	0.00	0.00	7.927	0.000	116.191
180 to 185	1.36	0.03	0.00	7.729	0.000	122.001
186 to 191	3.23	0.06	0.00	7.516	0.000	122.001
192 to 197	22.43	0.45	0.00	7.279	0.000	128.101
198 to 203	44.44	0.89	0.00	7.016	0.000	128.101
204 to 209	80.99	1.62	0.00	6.722	0.000	134.506
210 to 215	117.72	2.35	0.00	6.406	0.000	134.506
216 to 221	156.14	3.12	0.00	6.066	0.000	141.231
222 to 227	192.01	3.84	0.00	5.707	0.000	141.231
228 to 233	249.40	3.73	0.00	5.305	0.000	148.293
234 to 239	284.55	2.85	0.00	4.900	0.000	148.293
240 to 245	308.15	3.08	0.00	4.487	0.000	155.707
246 to 251	20.19	0.20	383.74	4.077	0.000	155.707
252 to 257	0.00	0.00	421.44	3.655	0.000	163.493
258 to 263	0.00	0.00	433.52	3.222	0.000	163.493
264 to 269	0.00	0.00	451.49	2.770	0.000	171.667
270 to 275	0.00	0.00	464.27	2.306	0.000	171.667
276 to 281	0.00	0.00	483.23	1.823	0.000	180.251
282 to 287	0.00	0.00	496.74	1.326	0.000	180.251
288 to 293	0.00	0.00	516.75	0.809	0.000	189.263
294 to 299	0.00	0.00	531.02	0.278	0.000	189.263
300 to 305	0.00	0.00	278.35	0.000	0.273	198.726
306 to 311	0.00	0.00	0.00	0.000	0.835	198.726
312 to 317	0.00	0.00	0.00	0.000	1.414	208.663
318 to 323	0.00	0.00	0.00	0.000	2.003	208.663
324 to 329	0.00	0.00	0.00	0.000	2.609	219.096
330 to 335	0.00	0.00	0.00	0.000	3.225	219.096
336 to 341	0.00	0.00	0.00	0.000	3.860	230.051
342 to 347	0.00	0.00	0.00	0.000	4.505	230.051
348 to 353	0.00	0.00	0.00	0.000	5.170	241.553
354 to 359	0.00	0.00	0.00	0.000	5.846	241.553
360 to 365	0.00	0.00	0.00	0.000	6.542	253.631
366 to 371	0.00	0.00	0.00	0.000	7.251	253.631

Months	Core early principal USD eq m	Premium USD eq m	Gap principal IQD bn	All debt closing IQD tn eq	Cash closing IQD tn	OPEX buffer IQD bn
372 to 377	0.00	0.00	0.00	0.000	7.980	266.312
378 to 383	0.00	0.00	0.00	0.000	8.722	266.312
384 to 389	0.00	0.00	0.00	0.000	9.486	279.628
390 to 395	0.00	0.00	0.00	0.000	10.264	279.628
396 to 401	0.00	0.00	0.00	0.000	11.065	293.609
402 to 407	0.00	0.00	0.00	0.000	11.880	293.609
408 to 413	0.00	0.00	0.00	0.000	12.719	308.290
414 to 419	0.00	0.00	0.00	0.000	13.573	308.290
420 to 425	0.00	0.00	0.00	0.000	14.452	323.704
426 to 431	0.00	0.00	0.00	0.000	15.347	323.704
432 to 437	0.00	0.00	0.00	0.000	16.269	339.889
438 to 442	0.00	0.00	0.00	0.000	17.051	339.889

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Baghdad detailed engineering and Iraqi manufacturing plan

Source: cities/catalogue/west-asia/Iraq/Baghdad/DETAILED-ENGINEERING.md

Baghdad uses a six-car, 111 m metro family. The next design stage retains ballastless slab track and expands the civil, mechanical, electronics and software packages into controlled interfaces, parts and manufacturing operations. The [component register](#), [parts CSV](#) and [source-bound JSON](#) contain the calculated city quantities and open supplier positions. These are engineering references; production drawings, surveys, qualified suppliers and operational approvals remain required.

The current city allocation is 831 trainsets, 4,986 cars, 182 stations and 158 energy sites. The two-bogie-per-car reference gives 9,972 bogies and 19,944 wheelsets. Wheelsets are children of complete bogies, not extra purchases. Doors, cooling loops, fixtures, harnesses and equipment repeats are reference allocations that must pass the six-car family review. The detailed three-car LM3 manufacturing package is useful process evidence; it is not a released Baghdad production package and cannot be doubled to obtain qualified six-car interfaces.

Reference architecture and drawing package

The mechanical load path is wheelset to suspension/axlebox to bogie frame to secondary suspension/pivot to car underframe to coupler and body structure. Cladding attaches to the structural frame through controlled service rails and retention features; FRP panels do not replace the primary crash structure. Keep longitudinal crash loads, recovery loads, passenger fixture loads and equipment retention as separate calculation cases with a common drawing revision.

The electrical path is protected charging contact to traction battery/HV junction to inverters/motors and protected auxiliary DC/DC branches. Battery, inverter and motor cooling circuits require a coordinated heat balance. The nominal traction bus is 675 V and normal maximum 740 V; “800 V class” describes a component class, not the normal bus voltage. Coordination must include abnormal/transient limits, insulation monitoring, HV interlocks, precharge, discharge, isolation and emergency disconnect.

Controlled drawing	Required content	Reference data available	Release owner / remaining evidence
BG-M-001 Trainset arrangement	Six-car positions, couplers, gangways, swept and evacuation envelopes	111 m train envelope; six cars; five intercar interfaces; two ends	Rolling-stock authority: exact end/coupler lengths, curve sweep and tolerance stack
BG-M-002 Car structure	Underframe/side/roof members, joints, datums, lift/recovery points	18.5 m average module allocation; geometry is not a cut list	Structures: stress/fatigue/crash, weld map, material thicknesses, distortion and NDT
BG-M-003 Bogie interface	Pivots, spring/air attachments, wheel profile, brake/motor clearances	One powered and one trailer bogie per car reference	Supplier + dynamics authority: axle loads, suspension rates, hunting, curves and brake duty
BG-M-004 Battery and HV bay	Racks, crash retention, isolation, vent paths and removable pack access	225 kWh gross allocation per car; 1,350 gross / 1,080 usable per train	HV/battery authority: actual pack mass/dimensions, fault current, thermal propagation and retention
BG-M-005 Cooling and HVAC	Chiller/radiator/pump, hoses, expansion/bleed, ducts and condensate	Hot-desert 50 °C design ambient	Thermal authority: traction/aux duty, degraded operation, dust and heat rejection
BG-M-006 Door/glazing	Openings, leaves, locks, seals, glazing retention and maintenance access	Two double-leaf openings per side/car reference	Body + accessibility authority: family-specific layout, loading, escape and obstruction
BG-M-007 Harness and connector	Wire/terminal IDs, cavity views, cut lengths, routing, clamps and separation	HV/LV/data service zones; reference host counts	Electrical authority: current/voltage drop, insulation, EMC, connector and supplier pin freeze
BG-M-008 Interior/accessibility	Seats, rails, ramps/steps, lighting, fire barriers and service panels	720 nominal passengers, 120 seats, 960 crush per train	Interior authority: fixture loads, fire/smoke, accessibility and passenger evacuation
BG-C-001 ST6 and slipform section	Concrete geometry, cage, inserts, rail seats, drainage and transitions	Six-metre panel geometry and 20 seats	Track/civil authority: reinforcement, support stiffness, shrinkage, fasteners and actual geotechnical zones
BG-C-002 Viaduct and bridge interfaces	Pi20/Pi25, foundations, bearings, joints, CWR and walkway interfaces	Standard 20/25 m spans; bridge-specific exceptions	Structures: permanent/variable/thermal/seismic/braking, fatigue, scour and foundation release
BG-E-001 End electronics cabinet	Rails, protection, output chain, harness, thermal and enclosure	Corrected T-OBS capacity and exposed Pico bench pins	Electronics authority: actual vendor BOM, schematic, HAL, EMC and injected faults

Use train longitudinal datum X, track centre Y and surveyed rail datum Z for integration. Each supplier drawing must state its own mating datum and tolerance contribution. The track reference gauge is 1,435 mm between the running faces; rail-head width and inclination are needed to convert this into rail-centre/baseplate coordinates. Existing simplified CAD rail-line positions must not be used as drilling coordinates. A platform step/gap or bogie pivot offset copied from LM3 is an unresolved six-car interface until reviewed.

Mechanical manufacturing and missing kit content

Local work can cover steel structure fabrication, service rails, FRP moulding/trim, interiors, mounts, battery enclosures, harness manufacture, pipework and final integration. Initially import complete qualified bogies, motor/inverter packages, cells/qualified packs, safety glazing, doors and electronics where local process and product qualification is absent. A later local-content decision needs supplier drawings/licences, process capability, acceptance tests and economics; local fabrication is not evidence that a safety-critical design is accepted.

Work package	Make/buy and production route	Child content that must be explicit	Verification / hold point
Car frame	Iraqi cut/form/machine/fixture/weld	Plates/sections, brackets, joint preparation, isolation, drains and finish	Released datum/fixture, WPS, weld/NDT, dimensional and load inspection
FRP skin	Iraqi mould/layout/cure/trim/fit	Resin/fibre/core, inserts, retention clips, edge seals and fire-compatible finish	Laminate/fire/smoke, cure coupon, insert pullout and panel retention
Bogie	Complete supplier package first	Wheels/axles, bearings, suspension, dampers, brakes, grounding and guards	Family axle-load/dynamics/brake qualification and complete serial records
Battery/HV	Import qualified core; locally integrate after approval	Rack, fuses, contactors, precharge, disconnect, HVIL, BMS, sensors, vent/containment and service lift	Supplier duty, propagation, isolation, fault-current and retention tests
Thermal	Local formed tube/hose/brackets with sourced pumps/chillers	Fluid, filters, manifolds, isolation valves, drains, bleed/expansion and leak detection	Pressure/cleanliness, heat balance at hot ambient and failure response
Brake	Supplier-controlled complete design	Compressor, dryer, reservoir, valve blocks, parking brake, hoses and sensors	Pressure/leak, stop distance, adhesion/degraded modes and rescue
Doors/glass	Qualified sourced units with local frame/fitting	Lock/detection/obstruction, emergency release, drainage, seals, adhesive and retainers	Opening alignment, force/leak/retention and interlock to safety host
Wiring	Iraqi formboard/cut/crimp/test	Cable, contacts, mating halves, seals, glands, shielding, clamps and labels	Calibrated crimp/pull, cavity checks, continuity and insulation
Final vehicle	Local installation and configuration	Torque evidence, fluids, firmware, mass/CG, identification and handover records	Weighed family vehicle, static brake/door tests, route acceptance and traceability

The existing small-component standard, factory release, tooling, inspection and mass-closure modules remain useful. Their product IDs and drawing seeds must be cross-referenced to a Baghdad family-specific EBOM/MBOM before ERP Work Orders are released. No exact bolt torque, weld size, laminate layup or heat-exchanger area is invented here. Those values must close the applicable load, thermal or supplier calculation.

Slab track decision

Choose ballastless slab track as the reference permanent way. Use continuous slipforming for long, open, machine-accessible at-grade sections; use Iraqi-made OSR-ST6 single-track precast panels in constrained streets, utility/transition zones, short possessions, station/depot interfaces and replaceable sections. On standard elevated Pi20/Pi25 structures, use direct-fixation plinths above the structural flange with a thin alignment layer only where needed. Floating track is a location-specific vibration treatment, not a blanket addition.

Slab track is selected for the design package because it supports controlled alignment, modular installation and local concrete work. Lifecycle value must be tested against settlement risk, access, reparability, construction traffic, drainage and noise. It is not inherently cheaper on weak or uninvestigated ground. Keep ballasted/special transition arrangements available as engineered exceptions where life-cycle, geotechnical or depot requirements justify them.

ST6 single-track reference	Value / quantity
Dimensions	6,000 × 2,900 mm; 250 mm base; two 380 × 160 mm plinths
Double track	Two independently manufactured rows on 3,500 mm centres
Concrete, base and plinths	5.0796 m³ per panel; excludes service-trough envelope, bedding and steel
Bare concrete planning mass	12.699 t at 2,500 kg/m³; steel, inserts and lifting rig add weight
Reinforcement allowance	761.94 kg at 150 kg/m³; not a released bar schedule
Rail seats	20 per panel; 10 per rail, at $X = 300 + 600 \times i$ mm, $i = 0 \dots 9$
Repeated-module seat pitch	600 mm including the panel joint; maximum reference spacing remains 650 mm
Complete seat child allocation	One pad/baseplate kit, two clips, two lateral insulators, two anchor assemblies and one adjustment kit; supplier pattern to freeze
Bedding illustration	30 mm over 17.4 m² = 0.522 m³; actual support/grout design required
Lifting illustration	Four engineered anchors; supplier load sharing, strength, angle and rigging required

The previous CAD placed the first baseplate centre at the concrete edge, extending its 260 mm footprint beyond the panel. The corrected layout centres seats inside the module and keeps uniform pitch across repeated joints. Continuous-run planning quantities still use the existing 650 mm basis; precast method zones need their own 20-seat-per-panel schedule. Do not apply a single rounded per-km seat allowance as a fabrication order for every six-metre panel.

A complete track seat needs the matched rail, baseplate, resilient pad/layer, clips, insulators, anchors, shim/adjustment and tested fastening system. The legacy sleeper/clip CAD remains a ballasted alternative; it is not the default slab-track MBOM.

Source proprietary/safety-critical elastic parts and qualified anchors initially; manufacture concrete/cages, ducts/covers, moulds, jigs and appropriate metalwork locally. Local clip or pad manufacture requires licensing/material/process/fatigue and insulation qualification, not just matching the rendered shape.

Pandrol's published direct-fixation system provides an example of adjustable, resilient fastening for slab structures with system-specific performance and testing. This supports the architecture; it does not select Pandrol, approve an equivalent, transfer the vendor stiffness/adjustment values to OSR or establish Iraq pricing. [Pandrol technical sheet](#).

Iraqi slab manufacturing line

Add a dedicated slab line alongside the shared Baghdad plant or a route-adjacent civil yard after a haulage/capacity comparison. Do not consume the same Pi-beam moulds, crane shifts or curing spaces twice in the production programme. A mould row must record mould ID, cycle, available shifts, crane allocation, rejected pieces and actual release strength. The previous 24-hour mould-cycle target remains a trial target; use the existing 48-hour planning basis until hot-weather trials substantiate an alternative.

- 1 Release panel/cage/insert drawings, supplier fastener pattern, concrete mix/exposure class and prototype inspection plan.
- 2 Qualify Iraqi cement, aggregate, water and admixtures for the site sulfate/chloride exposure. Check shrinkage, creep and curing rather than assuming a higher concrete grade solves durability.
- 3 Build surveyed reusable steel moulds, cage/weld jigs and insert/rail-seat gauges. Fit engineered lifting sockets, bonding/drain sleeves and unique panel identity.
- 4 Inspect cage cover, insert positions, cleanliness and mould dimensions before casting. Record supplier/batch and mixing conditions.
- 5 Place/vibrate the approved mix or use qualified self-compacting concrete. Protect from heat/wind, control moisture and record maturity against strength tests.
- 6 Release lifting only at the calculated strength. Survey rail-seat datums, flatness, geometry and cracks; record repair/rejection decisions against panel serials.
- 7 Transport with designed support points and route clearance; inspect after transport. Install on accepted formation/bed, grout under the released procedure and survey rail alignment.
- 8 Install the matched fastening system, complete CWR welding/stress management, verify gauge/cant/height and inspect bonding/insulation/drainage. Record the as-installed rail/fastener/panel configuration.
- 9 Perform proof loading/ride/geometry and monitored settlement trials before series deployment. Retain replaceability and maintenance-access demonstrations.

The line requires batching/aggregate handling, controlled curing, slab moulds, cage jigs, insert fixtures, maturity/strength testing, survey gear, lifting frames, certified rigging and loading/storage space. Iraqi labour can fabricate cages/moulds and operate casting, survey, installation and maintenance; specialist training, qualified lifting/welding and independent inspection need funded work packages. No construction-job count or local-content percentage is asserted without a priced resource and origin schedule.

A quantity example for one kilometre of double track wholly assigned to ST6 is 334 full panels = $2 \times \text{ceil}(1,000/6)$, 6,680 seats and 1,696.5864 m³ of base/plinth concrete. These cover 1,002 m per track with end trimming/interface treatment unresolved. It is an example, not Baghdad's method allocation. If the surveyed constrained length is L on each of the two tracks, use $2 \times \text{ceil}(L/6)$ and separately schedule short end pieces; do not infer L from all at-grade kilometres.

Civil interfaces that need completion

Interface	Required design information	Acceptance and cost implication
Formation/slab	Ground zones, groundwater, modulus, settlement, sulfate exposure, utility trenches	Settlement and transition design; preparation/ground improvement measured by zone
Drainage/flood	Catchments, design storm, outlet levels, maintenance access, debris and scour	Cable routes separated from wet channels; outfalls/attenuation/pumps counted where required
Viaduct/slab/CWR	Thermal range, neutral rail temperature, braking, deck movement and restraint	Rail–structure interaction analysis; no automatic expansion device at every deck gap
Bearings/continuity	Unit lengths, pier stiffness, seismic/thermal restraints, replacement/lift access	Supplier bearing envelope and certified reactions; quantity schedule by actual support
River/highway bridges	Survey/navigation, scour, traffic/diversion, foundations and special spans	Separate bridge design; standard Pi spans do not release major crossings
Elevated evacuation	Walkway/barrier loads, clear width, station access and rescue route	Outer 1 m cassette reference checked against obstruction and fire/evacuation rules
Station boarding	Six-car swept envelope, floor height, tolerance and accessibility	Surveyed step/gap and dwell/evacuation capacity; LM3 offsets cannot be copied blindly
Depot/stabling	831-train circulation and storage, workshop equipment, wash/drainage/recovery	Existing USD8m depot allowance needs detailed scope and pricing; workshop bays are not parking
Plant/logistics	Crane charts, rig loads, haul route, curing, power/water/waste and laydown	12.699 t concrete panel is not the certified shipping/lift weight; steel/rigging included

Keep the existing construction-method and foundation catalogs. Their “actual length and cost required” gates prevent generic deep-pile lengths from being adopted as surveyed Baghdad foundations. Release each geotechnical zone and map it to the alignment segments before recalculating quantities and installed costs.

Onboard electronics corrections

The revised [T-OBS bench design](#) and [integration JSON](#) close erroneous interface assumptions and add missing power, watchdog, relay-feedback, harness and external-controller blocks. They do not release a production board or pretend that a driver exists because its crate appears in the workspace.

Issue identified	Corrected design treatment
HDR-60-24 treated as 24 V to 5/12 V	It is AC/high-voltage DC input to 24 V output; use appropriately protected DC/DC branches onboard
USB isolator used as SPI / Pico device-to-device link	Use directionally correct SPI isolation with one controller and one peripheral
TPS3701 called dual heartbeat watchdog	Voltage detector only; separate channel heartbeat watchdogs and hardware inhibition
Pico GP30–39, ADC at GP14–17, native CAN/RMII	Exposed Pico pins and supported functions only; custom chip pinout separately frozen; external CAN/Ethernet controllers
ATECC608B driven through SPI	Selected breakout uses I2C; provisioning and exact variant confirmed
Digital ultrasonic echo sampled through ADS1115	Timed digital capture with voltage-level conversion; raw piezo requires an AFE/driver design
Radar said to ship M12 / USB-C	AWR1843BOOST EVM uses vendor J3 CAN, micro USB and 5 V barrel; add qualified adapter/harness
HAP claimed Gigabit/12 V PoE	Select HAP TX 100BASE-TX; vendor 9–18 V power separately; T1 interface differs
Ultrasonic face hidden behind radome	Separate acoustic apertures and tested radar/optical windows
Under-20-W host and 1-A input protection	71.5 W simultaneous output sizing envelope; 79.44 W at 90% efficiency; 99.31 W with 25% margin; 100 W reference capacity
Two relays asserted “always safe”	Require end-to-end polarity, feedback, welded-contact/driver faults and common-cause analysis

Capacity is not measured average power. Do not inflate train OPEX from a 100 W converter nameplate or retain a 40 W allowance as if it covers the updated candidate assembly. Production sensor and firmware selection must provide a measured duty cycle for the auxiliary energy/thermal model. Exact wire gauge, fuse, supply voltage range, bus topology, PCB layout, heat spreader and relay remain design-owned.

Primary specifications checked 3 October 2026: [Mean Well](#), [TI voltage detector](#), [TI radar](#), [Livox](#), [Raspberry Pi carrier](#), [Pico 2](#), [Microchip secure element](#). Candidate lifecycle/price and rail suitability are not implied by a working datasheet link.

Software and hardware integration work

The component register verifies all 60 Rust workspace packages against deployment/components.toml. Existing control, diagnostics, AFC, CBM, station, supervision and identity software is retained. RFC0033 TACS runtime/resource control remains a process reference; it is not installed Baghdad protection equipment. A logical redundant process cannot substitute for independent power, I/O, sensing and qualified timing.

Work package	Required binding / acceptance
Device HAL	Board-revision ADC/timer/PIO, external CAN/Ethernet, GPIO defaults, watchdog and relay feedback; mock tests plus actual bench traces
Sensor ingress	Vendor framing, calibration, timestamps/sequence/CRC, data age, malformed frames, packet loss and common sensor faults

Work package	Required binding / acceptance
Protection output	Complete boot/reset/timeout/permission truth table and hardware watchdog inhibition; braking and traction supplier interfaces
Runtime deployment	Qualified host composition, task/WCET budgets, fault containment, resource authority, committed resource state and degraded recovery
Clock/network	Actual timestamp accuracy, redundant routes/switches, PTP loss, CAN termination, bandwidth and shared-path failure analysis
Target images	Reproducible signed build, configuration/board hash, driver versions, SBOM, secure boot/update/rollback and release evidence
Operations integration	Event provenance, CBM/Issue triage, maintenance states, trusted operator decisions and reporting; ERP/FUXA remain outside the safety authority
Ticketing/revenue	Actual validators/payment/acquirer contracts, offline/replay/refund behaviour, Arabic UX, tariff versions and daily settlement reconciliation
Cybersecurity/service	Key provisioning/revocation, account/role access, logs/storage limits, patch ownership and restore drills

The image cookbook explicitly records that production SD images are absent. Leave that fact visible. Add real driver/board work against each package and evidence gate; do not create empty drivers or announce “complete onboard firmware” from successful host-library tests. Hardware-in-loop must test a full protected output path using representative I/O and faults.

ERP recheck and detail handover

The city component compiler, city configuration validator and supervision asset validator pass for all 266 cities; supervision covers 249,114 equipment records. This proves reusable planning/asset contracts. Baghdad’s enabled reusable profile currently contains the induction training programme; it does not automatically create production, purchasing, quality or maintenance documents.

The local ERP MariaDB service was repeatedly failing on 3 October 2026 because it could not create temporary InnoDB files under /tmp. The compose definition now gives the database a writable, bounded 512 MiB temporary filesystem with mode 1777. The database was recreated against its existing persistent volume; it became healthy and the ERP ping API returned HTTP 200. No city assets, Work Orders, stock movements or loan records were fabricated during this repair. Health does not replace backup/restore or load testing.

ERP handover	Existing capability	Inputs needed to activate the detailed package
Engineering parts / Item / BOM	Native Items, nested BOMs and version exposure/disposition	Exact family part/revision, UOM, make/buy, supplier alternatives, child content and approved production BOM
Production	Draft Work Orders and native execution/Job Cards	Approved BOM, city source/WIP/FG warehouses, workstations, routing, actual hours, dated capacity and operator permissions
Procurement / stock	Project material requests, purchasing, receipts, replenishment	Landed freight/duty/tax, lead time, MOQ, safety stock, serial/batch and warehouse availability; no invented stock
Quality	Draft native Quality Inspections and evidence references	Released characteristics, sampling and limits, gauges/calibration, inspector, receipt line and retained actual results
Maintenance / CBM	Asset Maintenance, Issue-driven repair preview and controlled draft repair	Commissioned serial asset, approved interval/team/date, spares and handback acceptance
Training	Reusable programme and native events	Competence criteria, assessor, actual attendance and role authorisation
Finance	Native currency/accounting and project budgets; proposal cashflow remains a separate planning model	Iraqi legal company, fiscal year, IQD chart/accounts, USD-import clearing, actual bank/loan/bond terms, treasury schedule and approvals
Lifecycle	Version/evidence/disposition and physical identity templates	Real HTTPS resolver, physical serial binding and verified evidence before printable deployment labels

Use the 69 engineering rows as reference requirements first. They are not automatically posted as live ERP Items or added to existing priced city materials; complete assemblies and their child content need a nested MBOM to avoid double counting. The generic ERP manufacturing/quality/maintenance contracts should remain disabled until actual approved inputs exist.

For Baghdad finance, keep the previously selected government share 25%, imported component cash/Chinese-credit split 50/50 and all other cash/debt/bonds in IQD. Confirm company currency IQD, actual FX conversion dates and USD exposures when live treasury accounting is configured. Imported procurement is not automatically all Chinese-origin credit-eligible; rails, fastening hardware, safety electronics and tooling need actual origin/loan-eligibility schedules. Foreign climate money is not IQD financing unless an on-lender or a priced hedge provides that currency structure.

Cost and release reconciliation

The parts register does not reprice individual civil or vehicle components. The separate [city-sized factory calculation](#) now replaces the four-slot production assumption with physical six-car cells, crews, tooling and test paths, with facility readiness at 18 months from NTP. Its larger physical plant allowance and EPC are included once in the Baghdad funding programme. It adds no second plant or future national-city debt. Existing civil and vehicle allowances already contain planning permanent-way and equipment scope; the component breakdowns clarify what a complete package must contain. Unknown detailed unit costs remain null/open, not zero or a claim that all omissions are covered. The depot, civil manufacturing line, full control-electronics integration, spares, qualification and measured auxiliary duty need RFQs and scope reconciliation before the current financing model becomes a tender-grade model.

For each priced assembly, mark child lines as included, separately payable or excluded; bind quantities, installed rates and origin to one revision. Compare the detail subtotal with its current allowance, update CAPEX/import split/EPC only for an accepted scope delta, then regenerate the Baghdad cashflow, tranche and debt sweep outputs together. A prototype/localisation plan must not be treated as an agreed Chinese loan, grant or bond subscription.

Release sequence: surveyed alignment/ground/utility zones to family and track interface freeze to priced complete BOM/MBOM to local manufacturing trials to hardware/software integration and fault tests to independent review to approved construction/production documents to field acceptance. This makes missing components actionable without marking them as manufactured or qualified.

Baghdad – Urban Rail Network

Source: [cities/catalogue/west-asia/Iraq/Baghdad/README.md](https://cities.catalogue/west-asia/Iraq/Baghdad/README.md)

Country: IQ · Population: 9,780,429 · [National brief](#)

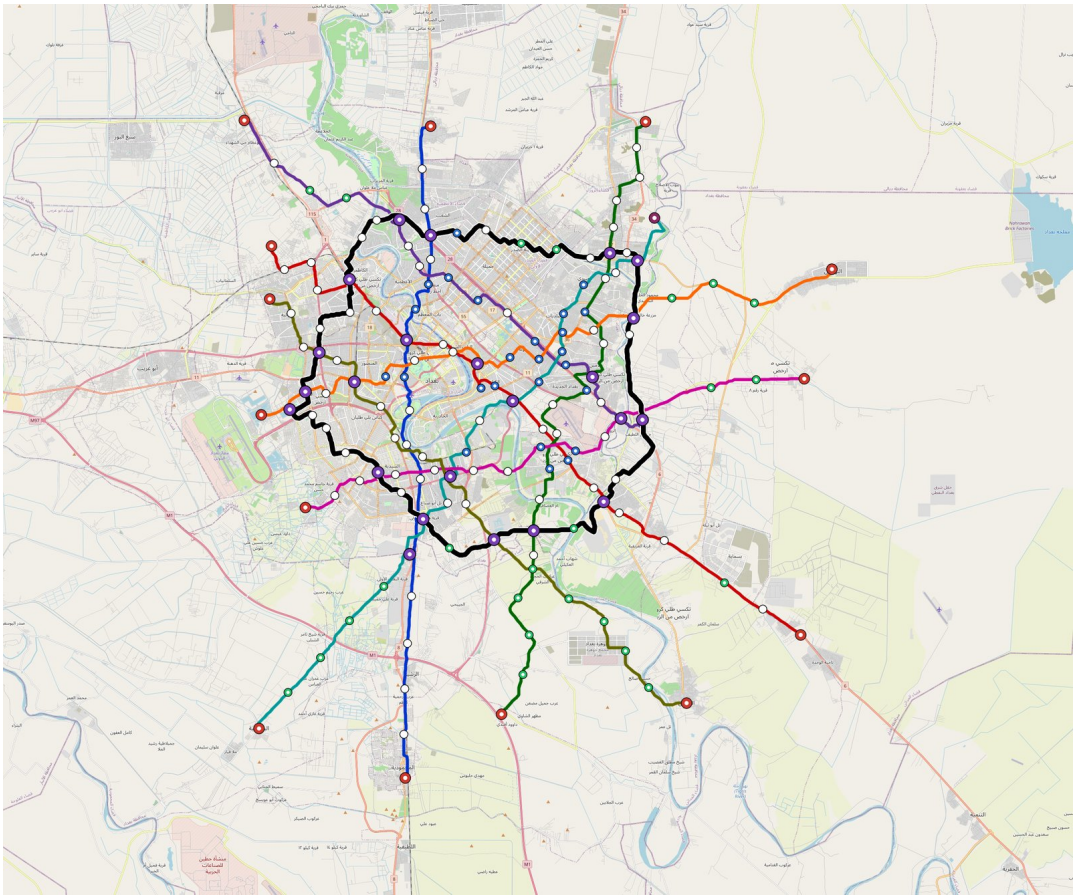
This page contains only Baghdad-specific results. Shared routing, service, energy, civil, cost, finance, QA and validation methods are defined once in the [deployment planning reference](#).

Important

Foreign-capital advantage: against the default equivalent foreign-turnkey sensitivity, this local plan avoids \$11.87 bn (87.3%) of external capital and \$14.60 bn of external interest. Capital plus saved interest totals \$26.47 bn. See the common reference for interpretation and limitations.

Auto-planned by the OpenSourceRail design pipeline from the controlled city catalogue, source-locked geospatial inputs and shared templates.

Network



Baghdad rail network on OpenStreetMap

Local measure	Value
Lines / unique stations / interchanges	9 / 182 / 23
Route length	516.5 km double track
Coverage / transfer reachability	46.4% / 44%
Estimated station catchment	4,538,119 residents
Service span / peak headway	05:30–02:00 / 3 min
Fleet	831 × 6-car metro - 6car trainsets (751 peak revenue)
Peak network throughput	259,200 passengers/hour
Practical service capacity	2,276,640 passenger-trips/day
Annual paid-trip planning range	415.5–664.8 M

Lines

Line	Length	Stations	Trainsets	Termini
line-1	53.8 km	19	95	S Outer N Outer
line-2	57.0 km	21	103	SE Outer NW Mid
line-3	55.8 km	21	108	S Outer NE Outer
line-4	44.4 km	16	85	NW Outer E Mid
line-5	53.7 km	18	98	W Mid E Outer
line-6	57.5 km	19	111	SW Outer NE Mid
line-7	43.5 km	16	84	E Outer SW Mid
line-8	51.4 km	18	100	SE Outer NW Mid
line-9	99.3 km	34	47	NW Mid NW Mid
Total	516.5 km	182 unique	831	

Energy

Local measure	Value
Scheduled service	3,952 one-way journeys / 217,090 train-km/day
Annual traction demand	2,053.8 GWh
Station/depot PV / storage	52.1 MW / 354.0 MWh
Aggregate charging power	316.0 MW
Dedicated solar plant	1,018.6 MW
Residual grid/PPA import	0.0 GWh/yr
Worst powered-stop gap	line-8: 21.3 km / 344 kWh
Lowest traversal charging margin	line-5: 304 kWh

Capital And Funding

Local CAPEX bucket	Planning value
Civil works	\$3.89 bn
Stations	\$908 M
Depots	\$8.0 M
Rolling stock	\$1.40 bn
Dedicated solar plant	\$815 M
Residual train control	\$26 M
Charging microgrids	\$68 M
EPC / project services	\$441 M
Total city programme	\$7.56 bn

Iraq funding

Proposed facilities and appropriations remain uncommitted. The conditional ledger calculates government capital and the extra support required for fees, interest, reserves and cash shortfalls; additional support is not a funding commitment.

Capital source	Planning USD equivalent
bank credit	\$1.20 bn
chinese export credit	\$865 M
domestic bonds	\$3.60 bn
government	\$1.89 bn

The procurement schedule requires 84 calendar months of capital cash under an assumed 260-working-day year. The resource-constrained full-network rollout needs review before a construction commitment.

Peak annual government cash: \$971 M. This includes support required under the low capacity-use case; it is not a funded appropriation.

Chinese export buyer credit is allocated within existing imported budgets for solar equipment, bogies, batteries, windows and doors. City CAPEX excludes manufacturing tooling; the Baghdad-only programme separately funds one plant for Baghdad. IQD bonds assume a proposed Ministry of Finance programme; municipal borrowing authority is pending legal review.

The model includes actual scheduled draws, native-currency principal/interest, fees, revenue ramps, operating/debt support, reserve movements and downside cases. Short bullet bonds have explicit redemptions without assumed refinancing.

See [funding model](#), [monthly cashflow](#), [annual cashflow](#) and [Baghdad-only funding programme](#). Government capital is 25% of total CAPEX, including USD cash for half the imports; the other half uses proposed Chinese USD credit. The remaining

government capital, bonds, bank credit and local cash are IQD. Full import-basket loan eligibility is an unqualified scenario assumption. Additional support is a separate unfunded requirement if public cash is capped at that contribution.

Conditional phased-opening sensitivity: first / last line revenue starts in month 41 / 83 from financial close. City additional support is \$5.74 bn, excluding the separately financed plant. Fleet-weighted revenue, independent ramps and a 25% fixed OPEX allowance require a validated phase-specific operating plan. Actual plant, depot and line acceptance remain pending.

Annual operating allowance: \$181 M; demand remains capacity-led.

Local Evidence

Package	Current status	Evidence
Finance	pass	summary.json
Native simulation + degraded cases	pass	validation-summary.json
SUMO timetable	pass	summary.json
Independent OSR/SUMO running-time cross-check	pass; junction/authority gate open	operations-crosscheck.md
GIS package	pass	summary.json
Solar/storage snapshot (operating duty unverified)	pass; 0 findings; 69 grid-only diagnostics	summary.json
Operations, QA and maintenance	1,862 assets / 10,785 tasks	baghdad-operations-manifest.json

Local Files And Regeneration

File	Local role
design.toml	Authoritative generated city design
baghdad.toml	Expanded simulator scenario
baghdad.corridor.geojson	GIS corridor and stations
baghdad.design-quality.yaml	Coverage, source and civil-quality gates

```
tools/automation/regenerate-city.sh baghdad
```

Baghdad Planning OSR-ALN Package

Source: [cities/catalogue/west-asia/Iraq/Baghdad/engineering/alignment/README.md](https://github.com/OpenSourceRail/Baghdad/blob/master/cities/catalogue/west-asia/Iraq/Baghdad/engineering/alignment/README.md)

Deterministic alignment exports for every line in the current generated network.

File	Design line	Length	Stations
baghdad-line1.aln.toml	line-1	53,822.4 m	19
baghdad-line2.aln.toml	line-2	56,994.2 m	21
baghdad-line3.aln.toml	line-3	55,819.3 m	21
baghdad-line4.aln.toml	line-4	44,443.3 m	16
baghdad-line5.aln.toml	line-5	53,737.2 m	18
baghdad-line6.aln.toml	line-6	57,516.3 m	19
baghdad-line7.aln.toml	line-7	43,501.9 m	16
baghdad-line8.aln.toml	line-8	51,366.4 m	18
baghdad-line9.aln.toml	line-9	99,316.5 m	34

Status

These files are planning-only and not for construction. Horizontal control comes from the current WGS84 corridor and is projected to the local UTM zone. Circular curves and transitions are not fitted, the vertical profile is a zero-datum placeholder, and cant has not been designed. Survey, curve-fit, vertical-profile, cant, geotechnical, utility, property, drainage, and structural release gates therefore remain open.

Each OSR-ALN file records the SHA-256 hashes of `../.. / design.toml` and the city corridor GeoJSON used to generate it.

Baghdad organisation and design work

Source: cities/catalogue/west-asia/Iraq/Baghdad/engineering/delivery/README.md

2350 indicative operating FTE, including management; annual labour allowance \$15,002,400. Group totals reconcile to the existing finance model. Individual role shares are editable planning allocations, not measured workload or an accepted duty roster.

GoA 4 control and remote assistance; local station batteries supply train top-ups. Two revenue trainsets per selected station with same-line depot overflow. Shared domestic design/software/factory capability serves city teams.

City operating organisation

Function	FTE	Reports to	Skills	Tasks
City accountable lead	1	Local owner board	Local operating governance; budget and emergency accountability	Own city delivery and operations; coordinate shared domestic capabilities
Engineering authority	4	City director	Systems integration; configuration and design checking	Own interfaces and technical concessions; commission competent discipline checks
Quality and safety assurance	7	Local owner board (independent escalation)	Independent inspection; hazard review; traceability	Audit evidence; raise stop-work/hold points independently of production
Training and competence assessors	6	City director	Practical assessment; instruction; competence records	Train local recruits and apprentices; witness competence before unsupervised work
Procurement and stores	8	City director	BOM traceability; repairable spares; supplier quality	Purchase controlled common parts; manage local repair loops and critical stock
Finance and people administration	7	City director	Payroll; cost control; roster and employment administration	Maintain local pay/leave inputs; reconcile labour budget and workload
Fleet workshop supervision	54	Chief engineer	Isolation authority; work planning; calibrated tooling	Allocate same-line workshop slots; control defect and release records
Mechanical technicians	187	Workshop lead	Wheels/brakes; joint inspection; controlled fastener assembly	Inspect bogies and brakes; replace modules; record torque/preload evidence
Electrical and battery technicians	160	Workshop lead	HV isolation; LFP diagnostics; controls and harness repair	Diagnose packs, HVAC and controllers; verify protective interlocks
Vehicle cleaning and finish repair	134	Workshop lead	GFRP/EPDM compatibility; coating preparation; controlled washing	Turnaround clean; inspect seals; repair finish damage under qualified process
Infrastructure work supervision	41	Chief engineer	Track access; lifting and electrical isolation coordination	Plan possessions; check civil and energy defect closure
Track and civil technicians	161	Infrastructure lead	Survey checks; track geometry; bearing/joint inspection	Inspect settlement, track, deck joints, drainage and walkways
Solar and station storage technicians	141	Infrastructure lead	PV DC safety; storage BMS; charger and protection testing	Maintain PV/storage/top-up equipment; measure soiling and energy balance
Wayside and communications technicians	61	Infrastructure lead	Fail-restrictive controls; networking; configuration control	Maintain point/crossing equipment where installed; test communications and recovery
Shift leadership and degraded-operation authority	65	City director	GoA 4 incident command; dispatch competence	Own service start; coordinate recovery and same-line depot releases
Line dispatch and control	170	OCC lead	Timetable supervision; protected movement authority	Monitor headways and station berths; authorise line-local recovery
Remote assistance	191	OCC lead	Vehicle fault diagnosis; passenger emergency response	Resolve remote-assist requests; coordinate on-site responders
Passenger service and fare support	127	City director	Accessible communication; fare-system support	Resolve passenger queries; verify fare equipment and complaints
Station cleaning and waste	104	Station lead	Product compatibility; wet-floor control; waste segregation	Clean measured public areas; record defects and wash-water disposal
Station team coordination	109	OCC lead	Accessible evacuation; platform protection	Allocate mobile teams; inspect opening and closing readiness
Platform and accessibility assistance	612	Station lead	Accessibility assistance; crowd management; first aid	Assist boarding; manage obstructions and platform incidents

Recruit and apprentice locally; assess practical task competence, isolate/hold/release authority and refresher needs. Training duration follows the demonstrated skills gap, not a universal weeks-to-qualification claim.

Quality and safety staff can escalate holds directly to the local owner board. Functions can be combined where competence and cover permit. Management allocations are already inside the operating totals.

Roster closure: named duty and leave roster including service-depot turnaround crews; measured cleaning/inspection person-minutes and simultaneous arrivals; country/local pay and actual grade costs; local employment/rest requirements and night emergency cover.

Construction and domestic manufacturing work

Concurrent crews/workstations below come from the existing dated work orders, with inclusive finish days. Worker headcount awaits an explicit crew mix; named skills are not assumed to mean one person each. Factory capacity is shared across city orders. Peaks from different work centres need not occur together.

Work centre	Peak resources (unknown if unsized)	Peak task slots	First peak day	Resource-days	Skills/tasks
Baghdad six-car body installation	unknown	3	601	unknown	body-cell lead; composite fitter; quality technician / rs-25-clip-on-body
Baghdad six-car composite kits	unknown	10	597	unknown	composite cell lead; laminator; quality technician; trim technician / rs-23-moulded-body-modules
Baghdad six-car electrical integration	unknown	20	625	unknown	battery technician; electrical lead; quality technician; software technician / rs-30-traction-battery-control
Baghdad six-car fitout and static test	unknown	16	644	unknown	HVAC technician; door technician; fit-out lead; quality technician / rs-40-fitout-static-test
Baghdad six-car structural assembly	unknown	24	580	unknown	bogie technician; brake technician; fabrication lead; weld inspector / rs-20-carbody-bogie
Baghdad trainset acceptance bays	unknown	13	659	unknown	commissioning lead; quality manager; software technician; test operator / rs-50-dynamic-commissioning
Baghdad trainset kitting	unknown	4	556	unknown	production planner; quality technician; stores clerk / rs-10-material-kit
corridor civil crew	2	2	5	3308	civil supervisor; drainage technician; permit coordinator; plant operator; quality technician; safety officer; survey lead / trk-10-survey-utility, trk-20-subgrade-drainage
depot works	unknown	1	638	unknown	depot operations lead; depot supervisor; electrical supervisor; quality technician; systems technician; tooling technician / dep-10-building-plant, dep-20-charging-stabling
energy crew	unknown	2	498	unknown	PV technician; battery technician; commissioning lead; electrical supervisor; quality technician; software technician / en-10-energy-build, en-20-energy-commission
project controls	unknown	1	0	unknown	production planner; project controls lead; quality manager; safety assurance lead / sys-00-baseline-freeze
station civil crew	unknown	2	467	unknown	accessibility lead; quality technician; station civil supervisor; steelwork lead / st-10-civil-structure
station systems crew	unknown	2	593	unknown	electrical supervisor; fare-system technician; quality technician; systems technician / st-20-mep-passenger-systems
track crew	2	2	252	2678	quality technician; rail welder; survey technician; track supervisor / trk-30-trackform-rail

Work centre	Peak resources (unknown if unsized)	Peak task slots	First peak day	Resource-days	Skills/tasks
track/switch crew	unknown	2	594	unknown	quality technician; survey technician; switch technician; systems technician; track supervisor / sw-10-turnout-build, sw-20-install-proof
wayside bench	unknown	3	472	unknown	quality technician; software technician; systems technician / wpt-10-cabinet-bench
wayside systems crew	unknown	3	655	unknown	quality technician; radio technician; software technician; systems supervisor / wpt-20-install-commission

Civil and mechanical refinements

Climate: hot-desert; 28 °C reference, 50 °C upper ambient. Cold ambient: missing.

Free movement $\alpha L \Delta T$. 8/10/12 microstrain/K are declared sensitivity cases, not measured concrete properties. Reference is the preset annual mean, not installation temperature; sunlit component temperatures and gradients require separate inputs.

Each elevated segment is separate; 25 m catalogue span, up to 4 spans/unit from the shared production/continuity templates, two tracks and two webs. Rounded spans are a procurement screen, not surveyed support positions. Deck gaps count the existing catalogue unit interfaces; abutment details remain separate.

Elevated segment	Longest unit m	Bearings	Link slabs	Hot movement mm at 8 / 10 / 12 μ strain/K
line-1 388.3–428.3 m	50.0	12	1	8.8 / 11.0 / 13.2
line-1 488.3–528.3 m	50.0	12	1	8.8 / 11.0 / 13.2
line-1 588.3–676.6 m	100.0	20	3	17.6 / 22.0 / 26.4
line-1 696.6–736.6 m	50.0	12	1	8.8 / 11.0 / 13.2
line-1 956.6–1016.6 m	75.0	16	2	13.2 / 16.5 / 19.8
line-1 1336.6–1356.6 m	25.0	8	0	4.4 / 5.5 / 6.6
line-1 1829.7–1849.7 m	25.0	8	0	4.4 / 5.5 / 6.6
line-1 1909.7–1949.7 m	50.0	12	1	8.8 / 11.0 / 13.2
line-1 2086.3–2146.3 m	75.0	16	2	13.2 / 16.5 / 19.8
line-1 2274.6–2334.6 m	75.0	16	2	13.2 / 16.5 / 19.8
line-1 3034.6–3114.6 m	100.0	20	3	17.6 / 22.0 / 26.4
line-1 3474.6–3514.6 m	50.0	12	1	8.8 / 11.0 / 13.2
line-1 3602.8–3651.1 m	50.0	12	1	8.8 / 11.0 / 13.2
line-1 4444.3–4464.3 m	25.0	8	0	4.4 / 5.5 / 6.6
line-1 4512.5–4532.5 m	25.0	8	0	4.4 / 5.5 / 6.6
line-1 4689.1–4709.1 m	25.0	8	0	4.4 / 5.5 / 6.6
line-1 5029.1–5049.1 m	25.0	8	0	4.4 / 5.5 / 6.6
line-1 5237.4–5297.4 m	75.0	16	2	13.2 / 16.5 / 19.8
line-1 5905.7–5925.7 m	25.0	8	0	4.4 / 5.5 / 6.6
line-1 6425.7–6485.7 m	75.0	16	2	13.2 / 16.5 / 19.8
line-1 7014.0–7074.0 m	75.0	16	2	13.2 / 16.5 / 19.8
line-1 7154.0–7222.3 m	75.0	16	2	13.2 / 16.5 / 19.8
line-1 7242.3–7302.3 m	75.0	16	2	13.2 / 16.5 / 19.8
line-1 7522.3–7582.3 m	75.0	16	2	13.2 / 16.5 / 19.8
line-1 7642.3–7670.5 m	50.0	12	1	8.8 / 11.0 / 13.2
line-1 7690.5–7770.5 m	100.0	20	3	17.6 / 22.0 / 26.4
line-1 8058.8–8118.8 m	75.0	16	2	13.2 / 16.5 / 19.8
line-1 8327.1–8347.1 m	25.0	8	0	4.4 / 5.5 / 6.6
line-1 8503.7–8543.7 m	50.0	12	1	8.8 / 11.0 / 13.2
line-1 8623.7–8643.7 m	25.0	8	0	4.4 / 5.5 / 6.6
line-1 8892.0–9028.5 m	100.0	32	4	17.6 / 22.0 / 26.4
line-1 9205.1–9225.1 m	25.0	8	0	4.4 / 5.5 / 6.6
line-1 9309.9–9358.2 m	50.0	12	1	8.8 / 11.0 / 13.2
line-1 9458.2–9554.8 m	100.0	20	3	17.6 / 22.0 / 26.4
line-1 9859.7–9987.9 m	100.0	32	4	17.6 / 22.0 / 26.4
line-1 10187.9–10256.2 m	75.0	16	2	13.2 / 16.5 / 19.8
line-1 10276.2–10364.5 m	100.0	20	3	17.6 / 22.0 / 26.4
line-1 10412.8–10441.1 m	50.0	12	1	8.8 / 11.0 / 13.2

Elevated segment	Longest unit m	Bearings	Link slabs	Hot movement mm at 8 / 10 / 12 µstrain/K
line-1 10609.4–10709.4 m	100.0	20	3	17.6 / 22.0 / 26.4
line-1 10769.4–10857.6 m	100.0	20	3	17.6 / 22.0 / 26.4
line-1 11145.9–11205.9 m	75.0	16	2	13.2 / 16.5 / 19.8
line-1 11474.2–11662.5 m	100.0	40	6	17.6 / 22.0 / 26.4
line-1 11830.8–11850.8 m	25.0	8	0	4.4 / 5.5 / 6.6
line-1 12259.1–12299.1 m	50.0	12	1	8.8 / 11.0 / 13.2
line-1 14301.9–14321.9 m	25.0	8	0	4.4 / 5.5 / 6.6
line-1 15206.8–15275.0 m	75.0	16	2	13.2 / 16.5 / 19.8
line-1 15375.0–15475.0 m	100.0	20	3	17.6 / 22.0 / 26.4
line-1 15851.6–15891.6 m	50.0	12	1	8.8 / 11.0 / 13.2
line-1 16059.9–16079.9 m	25.0	8	0	4.4 / 5.5 / 6.6
line-1 16736.5–16804.8 m	75.0	16	2	13.2 / 16.5 / 19.8
line-1 16824.8–16844.8 m	25.0	8	0	4.4 / 5.5 / 6.6
line-1 17084.8–17104.8 m	25.0	8	0	4.4 / 5.5 / 6.6
line-1 17144.8–17193.0 m	50.0	12	1	8.8 / 11.0 / 13.2
line-1 17854.5–17874.5 m	25.0	8	0	4.4 / 5.5 / 6.6
line-1 17974.5–18034.5 m	75.0	16	2	13.2 / 16.5 / 19.8
line-1 18182.7–18202.7 m	25.0	8	0	4.4 / 5.5 / 6.6
line-1 18251.0–18299.3 m	50.0	12	1	8.8 / 11.0 / 13.2
line-1 18905.6–18953.9 m	50.0	12	1	8.8 / 11.0 / 13.2
line-1 19627.0–19647.0 m	25.0	8	0	4.4 / 5.5 / 6.6
line-1 19735.3–19815.3 m	100.0	20	3	17.6 / 22.0 / 26.4
line-1 20108.4–20128.4 m	25.0	8	0	4.4 / 5.5 / 6.6
line-1 20148.4–20225.0 m	100.0	20	3	17.6 / 22.0 / 26.4
line-1 20245.0–20293.3 m	50.0	12	1	8.8 / 11.0 / 13.2
line-1 20313.3–20401.6 m	100.0	20	3	17.6 / 22.0 / 26.4
line-1 20598.1–20666.4 m	75.0	16	2	13.2 / 16.5 / 19.8
line-1 20694.7–20803.0 m	100.0	28	3	17.6 / 22.0 / 26.4
line-1 20843.0–20903.0 m	75.0	16	2	13.2 / 16.5 / 19.8
line-1 20983.0–21011.3 m	50.0	12	1	8.8 / 11.0 / 13.2
line-1 21296.1–21384.4 m	100.0	20	3	17.6 / 22.0 / 26.4
line-1 21424.4–21452.7 m	50.0	12	1	8.8 / 11.0 / 13.2
line-1 21672.7–21781.0 m	100.0	28	3	17.6 / 22.0 / 26.4
line-1 21881.0–21961.0 m	100.0	20	3	17.6 / 22.0 / 26.4
line-1 22029.3–22129.3 m	100.0	20	3	17.6 / 22.0 / 26.4
line-1 22149.3–22229.3 m	100.0	20	3	17.6 / 22.0 / 26.4
line-1 22477.5–22565.8 m	100.0	20	3	17.6 / 22.0 / 26.4
line-1 22645.8–22705.8 m	75.0	16	2	13.2 / 16.5 / 19.8
line-1 22765.8–22785.8 m	25.0	8	0	4.4 / 5.5 / 6.6
line-1 22825.8–22894.1 m	75.0	16	2	13.2 / 16.5 / 19.8
line-1 22954.1–23054.1 m	100.0	20	3	17.6 / 22.0 / 26.4
line-1 23074.1–23102.4 m	50.0	12	1	8.8 / 11.0 / 13.2
line-1 23122.4–23202.4 m	100.0	20	3	17.6 / 22.0 / 26.4
line-1 23262.4–23310.7 m	50.0	12	1	8.8 / 11.0 / 13.2
line-1 23419.0–23507.2 m	100.0	20	3	17.6 / 22.0 / 26.4
line-1 23607.2–23655.5 m	50.0	12	1	8.8 / 11.0 / 13.2
line-1 23683.8–23723.8 m	50.0	12	1	8.8 / 11.0 / 13.2
line-1 23860.4–23928.7 m	75.0	16	2	13.2 / 16.5 / 19.8
line-1 23948.7–23997.0 m	50.0	12	1	8.8 / 11.0 / 13.2
line-1 24093.5–24141.8 m	50.0	12	1	8.8 / 11.0 / 13.2
line-1 24548.1–24652.9 m	100.0	28	3	17.6 / 22.0 / 26.4
line-1 24837.8–24857.8 m	25.0	8	0	4.4 / 5.5 / 6.6
line-1 24877.8–24946.1 m	75.0	16	2	13.2 / 16.5 / 19.8
line-1 25224.1–25348.9 m	100.0	28	3	17.6 / 22.0 / 26.4
line-1 25924.9–25953.2 m	50.0	12	1	8.8 / 11.0 / 13.2
line-1 26472.6–26500.9 m	50.0	12	1	8.8 / 11.0 / 13.2
line-1 26577.4–26625.7 m	50.0	12	1	8.8 / 11.0 / 13.2
line-1 26790.6–26810.6 m	25.0	8	0	4.4 / 5.5 / 6.6
line-1 26878.9–26907.1 m	50.0	12	1	8.8 / 11.0 / 13.2
line-1 26955.4–26975.4 m	25.0	8	0	4.4 / 5.5 / 6.6
line-1 27120.3–27216.9 m	100.0	20	3	17.6 / 22.0 / 26.4
line-1 27365.1–27501.7 m	100.0	32	4	17.6 / 22.0 / 26.4

Elevated segment	Longest unit m	Bearings	Link slabs	Hot movement mm at 8 / 10 / 12 µstrain/K
line-1 27570.0–27746.6 m	100.0	40	6	17.6 / 22.0 / 26.4
line-1 27906.6–27934.8 m	50.0	12	1	8.8 / 11.0 / 13.2
line-1 28099.7–28156.3 m	75.0	16	2	13.2 / 16.5 / 19.8
line-1 28176.3–28196.3 m	25.0	8	0	4.4 / 5.5 / 6.6
line-1 28224.5–28281.1 m	75.0	16	2	13.2 / 16.5 / 19.8
line-1 28357.7–28406.0 m	50.0	12	1	8.8 / 11.0 / 13.2
line-1 28466.0–28494.3 m	50.0	12	1	8.8 / 11.0 / 13.2
line-1 28804.0–28824.0 m	25.0	8	0	4.4 / 5.5 / 6.6
line-1 28964.0–29004.0 m	50.0	12	1	8.8 / 11.0 / 13.2
line-1 29525.4–29585.4 m	75.0	16	2	13.2 / 16.5 / 19.8
line-1 29935.1–29983.4 m	50.0	12	1	8.8 / 11.0 / 13.2
line-1 30039.9–30068.2 m	50.0	12	1	8.8 / 11.0 / 13.2
line-1 30502.8–30531.1 m	50.0	12	1	8.8 / 11.0 / 13.2
line-1 30659.4–30699.4 m	50.0	12	1	8.8 / 11.0 / 13.2
line-1 30784.2–30840.8 m	75.0	16	2	13.2 / 16.5 / 19.8
line-1 31070.5–31098.8 m	50.0	12	1	8.8 / 11.0 / 13.2
line-1 31468.5–31488.5 m	25.0	8	0	4.4 / 5.5 / 6.6
line-1 32307.9–32327.9 m	25.0	8	0	4.4 / 5.5 / 6.6
line-1 32874.2–32942.4 m	75.0	16	2	13.2 / 16.5 / 19.8
line-1 32982.4–33010.7 m	50.0	12	1	8.8 / 11.0 / 13.2
line-1 33263.9–33292.1 m	50.0	12	1	8.8 / 11.0 / 13.2
line-1 33621.8–33641.8 m	25.0	8	0	4.4 / 5.5 / 6.6
line-1 33758.4–33963.3 m	100.0	48	6	17.6 / 22.0 / 26.4
line-1 34079.8–34184.7 m	100.0	28	3	17.6 / 22.0 / 26.4
line-1 34554.4–34662.7 m	100.0	28	3	17.6 / 22.0 / 26.4
line-1 34895.8–34915.8 m	25.0	8	0	4.4 / 5.5 / 6.6
line-1 34955.8–35012.4 m	75.0	16	2	13.2 / 16.5 / 19.8
line-1 35786.9–35855.2 m	75.0	16	2	13.2 / 16.5 / 19.8
line-1 35951.8–35971.8 m	25.0	8	0	4.4 / 5.5 / 6.6
line-1 36048.4–36076.7 m	50.0	12	1	8.8 / 11.0 / 13.2
line-1 36193.2–36261.5 m	75.0	16	2	13.2 / 16.5 / 19.8
line-1 36672.6–36805.8 m	100.0	32	4	17.6 / 22.0 / 26.4
line-1 36825.8–36854.1 m	50.0	12	1	8.8 / 11.0 / 13.2
line-1 36902.3–36922.3 m	25.0	8	0	4.4 / 5.5 / 6.6
line-1 37018.9–37075.5 m	75.0	16	2	13.2 / 16.5 / 19.8
line-1 37160.3–37216.9 m	75.0	16	2	13.2 / 16.5 / 19.8
line-1 37818.3–37846.6 m	50.0	12	1	8.8 / 11.0 / 13.2
line-1 37894.9–37943.2 m	50.0	12	1	8.8 / 11.0 / 13.2
line-1 37971.5–38056.3 m	100.0	20	3	17.6 / 22.0 / 26.4
line-1 38112.9–38132.9 m	25.0	8	0	4.4 / 5.5 / 6.6
line-1 38735.7–38812.3 m	100.0	20	3	17.6 / 22.0 / 26.4
line-1 38880.6–38928.9 m	50.0	12	1	8.8 / 11.0 / 13.2
line-1 38948.9–38988.9 m	50.0	12	1	8.8 / 11.0 / 13.2
line-1 39017.1–39057.1 m	50.0	12	1	8.8 / 11.0 / 13.2
line-1 39238.6–39258.6 m	25.0	8	0	4.4 / 5.5 / 6.6
line-1 39423.4–39471.7 m	50.0	12	1	8.8 / 11.0 / 13.2
line-1 39994.5–40062.8 m	75.0	16	2	13.2 / 16.5 / 19.8
line-1 40151.1–40199.4 m	50.0	12	1	8.8 / 11.0 / 13.2
line-1 40396.0–40416.0 m	25.0	8	0	4.4 / 5.5 / 6.6
line-1 40532.5–40572.5 m	50.0	12	1	8.8 / 11.0 / 13.2
line-1 40780.8–40840.8 m	75.0	16	2	13.2 / 16.5 / 19.8
line-1 41100.8–41120.8 m	25.0	8	0	4.4 / 5.5 / 6.6
line-1 41420.8–41440.8 m	25.0	8	0	4.4 / 5.5 / 6.6
line-1 41489.1–41509.1 m	25.0	8	0	4.4 / 5.5 / 6.6
line-1 41634.0–41674.0 m	50.0	12	1	8.8 / 11.0 / 13.2
line-1 41807.1–41835.4 m	50.0	12	1	8.8 / 11.0 / 13.2
line-1 41855.4–41883.7 m	50.0	12	1	8.8 / 11.0 / 13.2
line-1 42236.8–42305.1 m	75.0	16	2	13.2 / 16.5 / 19.8
line-1 42701.6–42721.6 m	25.0	8	0	4.4 / 5.5 / 6.6
line-1 42769.9–42878.2 m	100.0	28	3	17.6 / 22.0 / 26.4
line-1 42898.2–43006.5 m	100.0	28	3	17.6 / 22.0 / 26.4
line-1 43547.9–43607.9 m	75.0	16	2	13.2 / 16.5 / 19.8

Elevated segment	Longest unit m	Bearings	Link slabs	Hot movement mm at 8 / 10 / 12 µstrain/K
line-1 43667.9–43727.9 m	75.0	16	2	13.2 / 16.5 / 19.8
line-1 43796.2–43816.2 m	25.0	8	0	4.4 / 5.5 / 6.6
line-1 43896.2–43956.2 m	75.0	16	2	13.2 / 16.5 / 19.8
line-1 44565.9–44594.2 m	50.0	12	1	8.8 / 11.0 / 13.2
line-1 44923.9–44983.9 m	75.0	16	2	13.2 / 16.5 / 19.8
line-1 45240.5–45373.6 m	100.0	32	4	17.6 / 22.0 / 26.4
line-1 45638.5–45666.7 m	50.0	12	1	8.8 / 11.0 / 13.2
line-1 45686.7–45706.7 m	25.0	8	0	4.4 / 5.5 / 6.6
line-1 45863.3–45931.6 m	75.0	16	2	13.2 / 16.5 / 19.8
line-1 46076.5–46153.0 m	100.0	20	3	17.6 / 22.0 / 26.4
line-1 46181.3–46229.6 m	50.0	12	1	8.8 / 11.0 / 13.2
line-1 46992.4–47020.7 m	50.0	12	1	8.8 / 11.0 / 13.2
line-1 47109.0–47129.0 m	25.0	8	0	4.4 / 5.5 / 6.6
line-1 47393.9–47470.4 m	100.0	20	3	17.6 / 22.0 / 26.4
line-1 47703.6–47723.6 m	25.0	8	0	4.4 / 5.5 / 6.6
line-1 47931.8–47951.8 m	25.0	8	0	4.4 / 5.5 / 6.6
line-1 48885.0–48905.0 m	25.0	8	0	4.4 / 5.5 / 6.6
line-2 2228.8–2285.3 m	75.0	16	2	13.2 / 16.5 / 19.8
line-2 2418.5–2475.0 m	75.0	16	2	13.2 / 16.5 / 19.8
line-2 2579.9–2656.5 m	100.0	20	3	17.6 / 22.0 / 26.4
line-2 4649.3–4705.8 m	75.0	16	2	13.2 / 16.5 / 19.8
line-2 6133.8–6162.1 m	50.0	12	1	8.8 / 11.0 / 13.2
line-2 6694.6–6722.9 m	50.0	12	1	8.8 / 11.0 / 13.2
line-2 9444.2–9472.5 m	50.0	12	1	8.8 / 11.0 / 13.2
line-2 9625.6–9673.9 m	50.0	12	1	8.8 / 11.0 / 13.2
line-2 12450.4–12506.9 m	75.0	16	2	13.2 / 16.5 / 19.8
line-2 12924.9–12973.2 m	50.0	12	1	8.8 / 11.0 / 13.2
line-2 13001.5–13049.8 m	50.0	12	1	8.8 / 11.0 / 13.2
line-2 13106.4–13214.6 m	100.0	28	3	17.6 / 22.0 / 26.4
line-2 13311.2–13339.5 m	50.0	12	1	8.8 / 11.0 / 13.2
line-2 13359.5–13407.8 m	50.0	12	1	8.8 / 11.0 / 13.2
line-2 13447.8–13467.8 m	25.0	8	0	4.4 / 5.5 / 6.6
line-2 13496.1–13516.1 m	25.0	8	0	4.4 / 5.5 / 6.6
line-2 13644.3–13692.6 m	50.0	12	1	8.8 / 11.0 / 13.2
line-2 14170.6–14190.6 m	25.0	8	0	4.4 / 5.5 / 6.6
line-2 14360.3–14445.2 m	100.0	20	3	17.6 / 22.0 / 26.4
line-2 14858.3–14934.9 m	100.0	20	3	17.6 / 22.0 / 26.4
line-2 14954.9–15011.5 m	75.0	16	2	13.2 / 16.5 / 19.8
line-2 15144.6–15286.0 m	100.0	32	4	17.6 / 22.0 / 26.4
line-2 15487.4–15544.0 m	75.0	16	2	13.2 / 16.5 / 19.8
line-2 15592.3–15648.9 m	75.0	16	2	13.2 / 16.5 / 19.8
line-2 15773.7–15822.0 m	50.0	12	1	8.8 / 11.0 / 13.2
line-2 15946.8–15975.1 m	50.0	12	1	8.8 / 11.0 / 13.2
line-2 16023.4–16051.7 m	50.0	12	1	8.8 / 11.0 / 13.2
line-2 16120.0–16168.3 m	50.0	12	1	8.8 / 11.0 / 13.2
line-2 16341.4–16369.7 m	50.0	12	1	8.8 / 11.0 / 13.2
line-2 16699.4–16727.7 m	50.0	12	1	8.8 / 11.0 / 13.2
line-2 16747.7–16796.0 m	50.0	12	1	8.8 / 11.0 / 13.2
line-2 16844.2–16864.2 m	25.0	8	0	4.4 / 5.5 / 6.6
line-2 17157.4–17177.4 m	25.0	8	0	4.4 / 5.5 / 6.6
line-2 19130.2–19215.0 m	100.0	20	3	17.6 / 22.0 / 26.4
line-2 19255.0–19283.3 m	50.0	12	1	8.8 / 11.0 / 13.2
line-2 19339.9–19416.5 m	100.0	20	3	17.6 / 22.0 / 26.4
line-2 19484.7–19513.0 m	50.0	12	1	8.8 / 11.0 / 13.2
line-2 19714.4–19742.7 m	50.0	12	1	8.8 / 11.0 / 13.2
line-2 19762.7–19802.7 m	50.0	12	1	8.8 / 11.0 / 13.2
line-2 19911.0–20007.6 m	100.0	20	3	17.6 / 22.0 / 26.4
line-2 20302.1–20350.4 m	50.0	12	1	8.8 / 11.0 / 13.2
line-2 20580.1–20608.4 m	50.0	12	1	8.8 / 11.0 / 13.2
line-2 20809.8–20858.1 m	50.0	12	1	8.8 / 11.0 / 13.2
line-2 20954.7–20983.0 m	50.0	12	1	8.8 / 11.0 / 13.2
line-2 21264.4–21292.7 m	50.0	12	1	8.8 / 11.0 / 13.2

Elevated segment	Longest unit m	Bearings	Link slabs	Hot movement mm at 8 / 10 / 12 µstrain/K
line-2 21425.8–21474.1 m	50.0	12	1	8.8 / 11.0 / 13.2
line-2 21542.4–21570.7 m	50.0	12	1	8.8 / 11.0 / 13.2
line-2 21618.9–21667.2 m	50.0	12	1	8.8 / 11.0 / 13.2
line-2 21687.2–21743.8 m	75.0	16	2	13.2 / 16.5 / 19.8
line-2 21812.1–21945.2 m	100.0	32	4	17.6 / 22.0 / 26.4
line-2 22263.2–22311.5 m	50.0	12	1	8.8 / 11.0 / 13.2
line-2 23157.2–23205.5 m	50.0	12	1	8.8 / 11.0 / 13.2
line-2 23282.0–23310.3 m	50.0	12	1	8.8 / 11.0 / 13.2
line-2 23338.6–23386.9 m	50.0	12	1	8.8 / 11.0 / 13.2
line-2 23588.3–23636.6 m	50.0	12	1	8.8 / 11.0 / 13.2
line-2 23656.6–23713.2 m	75.0	16	2	13.2 / 16.5 / 19.8
line-2 23741.4–23818.0 m	100.0	20	3	17.6 / 22.0 / 26.4
line-2 24084.3–24169.1 m	100.0	20	3	17.6 / 22.0 / 26.4
line-2 24480.3–24500.3 m	25.0	8	0	4.4 / 5.5 / 6.6
line-2 24596.8–24665.1 m	75.0	16	2	13.2 / 16.5 / 19.8
line-2 24798.3–24826.5 m	50.0	12	1	8.8 / 11.0 / 13.2
line-2 25226.0–25282.5 m	75.0	16	2	13.2 / 16.5 / 19.8
line-2 25993.1–26049.6 m	75.0	16	2	13.2 / 16.5 / 19.8
line-2 26106.2–26191.1 m	100.0	20	3	17.6 / 22.0 / 26.4
line-2 26364.2–26392.5 m	50.0	12	1	8.8 / 11.0 / 13.2
line-2 26432.5–26549.0 m	100.0	28	3	17.6 / 22.0 / 26.4
line-2 26577.3–26605.6 m	50.0	12	1	8.8 / 11.0 / 13.2
line-2 27429.3–27562.4 m	100.0	32	4	17.6 / 22.0 / 26.4
line-2 28171.5–28219.8 m	50.0	12	1	8.8 / 11.0 / 13.2
line-2 29123.5–29180.1 m	75.0	16	2	13.2 / 16.5 / 19.8
line-2 29304.9–29381.5 m	100.0	20	3	17.6 / 22.0 / 26.4
line-2 30122.3–30142.3 m	25.0	8	0	4.4 / 5.5 / 6.6
line-2 30258.9–30287.2 m	50.0	12	1	8.8 / 11.0 / 13.2
line-2 30343.7–30392.0 m	50.0	12	1	8.8 / 11.0 / 13.2
line-2 30488.6–30516.9 m	50.0	12	1	8.8 / 11.0 / 13.2
line-2 30556.9–30613.4 m	75.0	16	2	13.2 / 16.5 / 19.8
line-2 30951.4–30979.7 m	50.0	12	1	8.8 / 11.0 / 13.2
line-2 31008.0–31144.6 m	100.0	32	4	17.6 / 22.0 / 26.4
line-2 31172.9–31201.1 m	50.0	12	1	8.8 / 11.0 / 13.2
line-2 31539.1–31567.4 m	50.0	12	1	8.8 / 11.0 / 13.2
line-2 31644.0–31672.3 m	50.0	12	1	8.8 / 11.0 / 13.2
line-2 31785.4–31813.7 m	50.0	12	1	8.8 / 11.0 / 13.2
line-2 32938.0–32966.2 m	50.0	12	1	8.8 / 11.0 / 13.2
line-2 33131.1–33159.4 m	50.0	12	1	8.8 / 11.0 / 13.2
line-2 34262.2–34338.8 m	100.0	20	3	17.6 / 22.0 / 26.4
line-2 34367.1–34407.1 m	50.0	12	1	8.8 / 11.0 / 13.2
line-2 34745.1–34765.1 m	25.0	8	0	4.4 / 5.5 / 6.6
line-2 34921.6–35179.6 m	100.0	56	8	17.6 / 22.0 / 26.4
line-2 35855.6–35875.6 m	25.0	8	0	4.4 / 5.5 / 6.6
line-2 35895.6–35915.6 m	25.0	8	0	4.4 / 5.5 / 6.6
line-2 36097.0–36153.6 m	75.0	16	2	13.2 / 16.5 / 19.8
line-2 36266.7–36383.3 m	100.0	28	3	17.6 / 22.0 / 26.4
line-2 36423.3–36463.3 m	50.0	12	1	8.8 / 11.0 / 13.2
line-2 38052.7–38129.2 m	100.0	20	3	17.6 / 22.0 / 26.4
line-2 38349.2–38457.5 m	100.0	28	3	17.6 / 22.0 / 26.4
line-2 38725.8–38785.8 m	75.0	16	2	13.2 / 16.5 / 19.8
line-2 39918.9–39938.9 m	25.0	8	0	4.4 / 5.5 / 6.6
line-2 40118.9–41346.0 m	100.0	252	37	17.6 / 22.0 / 26.4
line-2 41366.0–41406.0 m	50.0	12	1	8.8 / 11.0 / 13.2
line-2 41434.3–41454.3 m	25.0	8	0	4.4 / 5.5 / 6.6
line-2 41833.7–41862.0 m	50.0	12	1	8.8 / 11.0 / 13.2
line-2 42176.6–42204.9 m	50.0	12	1	8.8 / 11.0 / 13.2
line-2 42700.8–42785.7 m	100.0	20	3	17.6 / 22.0 / 26.4
line-2 42814.0–42898.8 m	100.0	20	3	17.6 / 22.0 / 26.4
line-2 43201.7–43230.0 m	50.0	12	1	8.8 / 11.0 / 13.2
line-2 43544.5–43564.5 m	25.0	8	0	4.4 / 5.5 / 6.6
line-2 43612.8–43689.4 m	100.0	20	3	17.6 / 22.0 / 26.4

Elevated segment	Longest unit m	Bearings	Link slabs	Hot movement mm at 8 / 10 / 12 µstrain/K
line-2 43717.7–43745.9 m	50.0	12	1	8.8 / 11.0 / 13.2
line-2 43794.2–43850.8 m	75.0	16	2	13.2 / 16.5 / 19.8
line-2 43907.4–43927.4 m	25.0	8	0	4.4 / 5.5 / 6.6
line-2 43947.4–43987.4 m	50.0	12	1	8.8 / 11.0 / 13.2
line-2 44132.2–44160.5 m	50.0	12	1	8.8 / 11.0 / 13.2
line-2 44268.8–44297.1 m	50.0	12	1	8.8 / 11.0 / 13.2
line-2 44365.3–44393.6 m	50.0	12	1	8.8 / 11.0 / 13.2
line-2 44490.2–44518.5 m	50.0	12	1	8.8 / 11.0 / 13.2
line-2 44606.8–44683.3 m	100.0	20	3	17.6 / 22.0 / 26.4
line-2 44816.5–44844.8 m	50.0	12	1	8.8 / 11.0 / 13.2
line-2 44873.0–44901.3 m	50.0	12	1	8.8 / 11.0 / 13.2
line-2 45224.2–45252.5 m	50.0	12	1	8.8 / 11.0 / 13.2
line-2 45433.9–45510.4 m	100.0	20	3	17.6 / 22.0 / 26.4
line-2 45918.1–45946.4 m	50.0	12	1	8.8 / 11.0 / 13.2
line-2 46003.0–46031.3 m	50.0	12	1	8.8 / 11.0 / 13.2
line-2 46059.6–46136.1 m	100.0	20	3	17.6 / 22.0 / 26.4
line-2 46665.2–46693.5 m	50.0	12	1	8.8 / 11.0 / 13.2
line-2 46798.4–46826.7 m	50.0	12	1	8.8 / 11.0 / 13.2
line-2 46903.2–46931.5 m	50.0	12	1	8.8 / 11.0 / 13.2
line-2 46979.8–47008.1 m	50.0	12	1	8.8 / 11.0 / 13.2
line-2 47547.5–47567.5 m	25.0	8	0	4.4 / 5.5 / 6.6
line-2 47635.8–47655.8 m	25.0	8	0	4.4 / 5.5 / 6.6
line-2 47704.1–47965.5 m	100.0	56	8	17.6 / 22.0 / 26.4
line-2 48182.1–48202.1 m	25.0	8	0	4.4 / 5.5 / 6.6
line-2 49013.2–49041.5 m	50.0	12	1	8.8 / 11.0 / 13.2
line-2 49856.0–49884.3 m	50.0	12	1	8.8 / 11.0 / 13.2
line-2 49932.6–50009.2 m	100.0	20	3	17.6 / 22.0 / 26.4
line-2 50189.2–50217.5 m	50.0	12	1	8.8 / 11.0 / 13.2
line-2 50385.7–50445.7 m	75.0	16	2	13.2 / 16.5 / 19.8
line-2 50782.3–50850.6 m	75.0	16	2	13.2 / 16.5 / 19.8
line-2 51585.1–51605.1 m	25.0	8	0	4.4 / 5.5 / 6.6
line-2 51801.7–51850.0 m	50.0	12	1	8.8 / 11.0 / 13.2
line-2 51990.0–52098.3 m	100.0	28	3	17.6 / 22.0 / 26.4
line-2 52263.1–52283.1 m	25.0	8	0	4.4 / 5.5 / 6.6
line-2 52476.3–52536.3 m	75.0	16	2	13.2 / 16.5 / 19.8
line-2 52842.5–52862.5 m	25.0	8	0	4.4 / 5.5 / 6.6
line-2 53115.7–53135.7 m	25.0	8	0	4.4 / 5.5 / 6.6
line-2 53437.1–53457.1 m	25.0	8	0	4.4 / 5.5 / 6.6
line-2 53525.4–53610.2 m	100.0	20	3	17.6 / 22.0 / 26.4
line-2 53658.5–53686.8 m	50.0	12	1	8.8 / 11.0 / 13.2
line-2 53959.9–54008.2 m	50.0	12	1	8.8 / 11.0 / 13.2
line-2 54048.2–54076.5 m	50.0	12	1	8.8 / 11.0 / 13.2
line-2 54096.5–54144.8 m	50.0	12	1	8.8 / 11.0 / 13.2
line-2 54417.9–54437.9 m	25.0	8	0	4.4 / 5.5 / 6.6
line-3 6748.7–6777.0 m	50.0	12	1	8.8 / 11.0 / 13.2
line-3 7995.2–8043.5 m	50.0	12	1	8.8 / 11.0 / 13.2
line-3 8100.0–8128.3 m	50.0	12	1	8.8 / 11.0 / 13.2
line-3 8689.2–8717.4 m	50.0	12	1	8.8 / 11.0 / 13.2
line-3 9278.3–9306.6 m	50.0	12	1	8.8 / 11.0 / 13.2
line-3 18009.5–18089.5 m	100.0	20	3	17.6 / 22.0 / 26.4
line-3 18214.4–18291.0 m	100.0	20	3	17.6 / 22.0 / 26.4
line-3 18347.5–18375.8 m	50.0	12	1	8.8 / 11.0 / 13.2
line-3 18424.1–18492.4 m	75.0	16	2	13.2 / 16.5 / 19.8
line-3 18653.8–18682.1 m	50.0	12	1	8.8 / 11.0 / 13.2
line-3 19160.1–19220.1 m	75.0	16	2	13.2 / 16.5 / 19.8
line-3 19628.4–19648.4 m	25.0	8	0	4.4 / 5.5 / 6.6
line-3 20261.5–20289.8 m	50.0	12	1	8.8 / 11.0 / 13.2
line-3 20766.4–20786.4 m	25.0	8	0	4.4 / 5.5 / 6.6
line-3 21129.2–21149.2 m	25.0	8	0	4.4 / 5.5 / 6.6
line-3 21197.5–21282.3 m	100.0	20	3	17.6 / 22.0 / 26.4
line-3 21447.2–21467.2 m	25.0	8	0	4.4 / 5.5 / 6.6
line-3 21603.8–21623.8 m	25.0	8	0	4.4 / 5.5 / 6.6

Elevated segment	Longest unit m	Bearings	Link slabs	Hot movement mm at 8 / 10 / 12 µstrain/K
line-3 21963.8–22003.8 m	50.0	12	1	8.8 / 11.0 / 13.2
line-3 22292.0–22352.0 m	75.0	16	2	13.2 / 16.5 / 19.8
line-3 22380.3–22428.6 m	50.0	12	1	8.8 / 11.0 / 13.2
line-3 22448.6–22516.9 m	75.0	16	2	13.2 / 16.5 / 19.8
line-3 22733.5–22761.7 m	50.0	12	1	8.8 / 11.0 / 13.2
line-3 22901.7–22930.0 m	50.0	12	1	8.8 / 11.0 / 13.2
line-3 22986.6–23128.0 m	100.0	32	4	17.6 / 22.0 / 26.4
line-3 24442.8–24511.1 m	75.0	16	2	13.2 / 16.5 / 19.8
line-3 24539.4–24567.7 m	50.0	12	1	8.8 / 11.0 / 13.2
line-3 25213.4–25241.6 m	50.0	12	1	8.8 / 11.0 / 13.2
line-3 25656.2–25712.8 m	75.0	16	2	13.2 / 16.5 / 19.8
line-3 25741.1–25761.1 m	25.0	8	0	4.4 / 5.5 / 6.6
line-3 25905.9–25982.5 m	100.0	20	3	17.6 / 22.0 / 26.4
line-3 26002.5–26050.8 m	50.0	12	1	8.8 / 11.0 / 13.2
line-3 26107.3–26135.6 m	50.0	12	1	8.8 / 11.0 / 13.2
line-3 26192.2–26220.5 m	50.0	12	1	8.8 / 11.0 / 13.2
line-3 26673.0–26701.3 m	50.0	12	1	8.8 / 11.0 / 13.2
line-3 27493.3–27549.8 m	75.0	16	2	13.2 / 16.5 / 19.8
line-3 27663.0–27747.8 m	100.0	20	3	17.6 / 22.0 / 26.4
line-3 27852.7–27881.0 m	50.0	12	1	8.8 / 11.0 / 13.2
line-3 28174.1–28202.4 m	50.0	12	1	8.8 / 11.0 / 13.2
line-3 28222.4–28242.4 m	25.0	8	0	4.4 / 5.5 / 6.6
line-3 28475.5–28495.5 m	25.0	8	0	4.4 / 5.5 / 6.6
line-3 28925.2–28965.2 m	50.0	12	1	8.8 / 11.0 / 13.2
line-3 29729.5–29757.8 m	50.0	12	1	8.8 / 11.0 / 13.2
line-3 30884.9–30913.2 m	50.0	12	1	8.8 / 11.0 / 13.2
line-3 31046.3–31074.6 m	50.0	12	1	8.8 / 11.0 / 13.2
line-3 33566.0–33586.0 m	25.0	8	0	4.4 / 5.5 / 6.6
line-3 33614.2–33634.2 m	25.0	8	0	4.4 / 5.5 / 6.6
line-3 34613.6–34641.9 m	50.0	12	1	8.8 / 11.0 / 13.2
line-3 34670.2–34690.2 m	25.0	8	0	4.4 / 5.5 / 6.6
line-3 36017.9–36037.9 m	25.0	8	0	4.4 / 5.5 / 6.6
line-3 39128.7–39197.0 m	75.0	16	2	13.2 / 16.5 / 19.8
line-3 39849.5–39877.8 m	50.0	12	1	8.8 / 11.0 / 13.2
line-3 40147.5–40195.8 m	50.0	12	1	8.8 / 11.0 / 13.2
line-3 40224.1–40272.4 m	50.0	12	1	8.8 / 11.0 / 13.2
line-3 40320.7–40397.2 m	100.0	20	3	17.6 / 22.0 / 26.4
line-3 40610.4–40678.6 m	75.0	16	2	13.2 / 16.5 / 19.8
line-3 42040.3–42088.6 m	50.0	12	1	8.8 / 11.0 / 13.2
line-3 42796.3–42836.3 m	50.0	12	1	8.8 / 11.0 / 13.2
line-3 43069.4–43109.4 m	50.0	12	1	8.8 / 11.0 / 13.2
line-3 43654.3–43702.6 m	50.0	12	1	8.8 / 11.0 / 13.2
line-3 44367.4–44407.4 m	50.0	12	1	8.8 / 11.0 / 13.2
line-3 44495.7–44600.6 m	100.0	28	3	17.6 / 22.0 / 26.4
line-3 44868.8–44937.1 m	75.0	16	2	13.2 / 16.5 / 19.8
line-3 44965.4–44985.4 m	25.0	8	0	4.4 / 5.5 / 6.6
line-3 46011.7–46031.7 m	25.0	8	0	4.4 / 5.5 / 6.6
line-3 47073.1–47213.1 m	100.0	32	4	17.6 / 22.0 / 26.4
line-3 47509.7–47634.5 m	100.0	28	3	17.6 / 22.0 / 26.4
line-3 49118.8–49138.8 m	25.0	8	0	4.4 / 5.5 / 6.6
line-3 49287.1–49347.1 m	75.0	16	2	13.2 / 16.5 / 19.8
line-3 49467.1–49487.1 m	25.0	8	0	4.4 / 5.5 / 6.6
line-3 50571.3–50591.3 m	25.0	8	0	4.4 / 5.5 / 6.6
line-3 50696.2–50716.2 m	25.0	8	0	4.4 / 5.5 / 6.6
line-3 51182.5–51222.5 m	50.0	12	1	8.8 / 11.0 / 13.2
line-3 51735.6–51763.9 m	50.0	12	1	8.8 / 11.0 / 13.2
line-3 52545.3–52565.3 m	25.0	8	0	4.4 / 5.5 / 6.6
line-3 52585.3–52605.3 m	25.0	8	0	4.4 / 5.5 / 6.6
line-3 53050.2–53110.2 m	75.0	16	2	13.2 / 16.5 / 19.8
line-3 54330.2–54390.2 m	75.0	16	2	13.2 / 16.5 / 19.8
line-3 54450.2–54510.2 m	75.0	16	2	13.2 / 16.5 / 19.8
line-3 54878.4–54918.4 m	50.0	12	1	8.8 / 11.0 / 13.2

Elevated segment	Longest unit m	Bearings	Link slabs	Hot movement mm at 8 / 10 / 12 µstrain/K
line-4 0.0–80.0 m	100.0	20	3	17.6 / 22.0 / 26.4
line-4 140.0–180.0 m	50.0	12	1	8.8 / 11.0 / 13.2
line-4 220.0–296.6 m	100.0	20	3	17.6 / 22.0 / 26.4
line-4 421.4–489.7 m	75.0	16	2	13.2 / 16.5 / 19.8
line-4 546.3–622.8 m	100.0	20	3	17.6 / 22.0 / 26.4
line-4 642.8–731.1 m	100.0	20	3	17.6 / 22.0 / 26.4
line-4 1097.4–1117.4 m	25.0	8	0	4.4 / 5.5 / 6.6
line-4 2119.7–2167.9 m	50.0	12	1	8.8 / 11.0 / 13.2
line-4 2292.8–2369.4 m	100.0	20	3	17.6 / 22.0 / 26.4
line-4 2389.4–2437.6 m	50.0	12	1	8.8 / 11.0 / 13.2
line-4 2514.2–2562.5 m	50.0	12	1	8.8 / 11.0 / 13.2
line-4 2880.5–2908.8 m	50.0	12	1	8.8 / 11.0 / 13.2
line-4 3524.8–3544.8 m	25.0	8	0	4.4 / 5.5 / 6.6
line-4 3621.3–3641.3 m	25.0	8	0	4.4 / 5.5 / 6.6
line-4 4184.2–4212.4 m	50.0	12	1	8.8 / 11.0 / 13.2
line-4 4547.0–4567.0 m	25.0	8	0	4.4 / 5.5 / 6.6
line-4 4927.0–4947.0 m	25.0	8	0	4.4 / 5.5 / 6.6
line-4 7308.9–7337.2 m	50.0	12	1	8.8 / 11.0 / 13.2
line-4 7393.8–7413.8 m	25.0	8	0	4.4 / 5.5 / 6.6
line-4 7918.0–7946.3 m	50.0	12	1	8.8 / 11.0 / 13.2
line-4 8002.9–8031.2 m	50.0	12	1	8.8 / 11.0 / 13.2
line-4 8385.7–8414.0 m	50.0	12	1	8.8 / 11.0 / 13.2
line-4 8502.3–8550.6 m	50.0	12	1	8.8 / 11.0 / 13.2
line-4 8827.1–8855.4 m	50.0	12	1	8.8 / 11.0 / 13.2
line-4 9759.1–9864.0 m	100.0	28	3	17.6 / 22.0 / 26.4
line-4 10040.5–10088.8 m	50.0	12	1	8.8 / 11.0 / 13.2
line-4 10273.7–10302.0 m	50.0	12	1	8.8 / 11.0 / 13.2
line-4 10864.8–10913.1 m	50.0	12	1	8.8 / 11.0 / 13.2
line-4 10981.4–11001.4 m	25.0	8	0	4.4 / 5.5 / 6.6
line-4 11174.5–11214.5 m	50.0	12	1	8.8 / 11.0 / 13.2
line-4 11359.4–11387.6 m	50.0	12	1	8.8 / 11.0 / 13.2
line-4 11657.3–11830.5 m	100.0	36	5	17.6 / 22.0 / 26.4
line-4 12148.5–12176.8 m	50.0	12	1	8.8 / 11.0 / 13.2
line-4 12389.9–12418.2 m	50.0	12	1	8.8 / 11.0 / 13.2
line-4 13479.8–13528.1 m	50.0	12	1	8.8 / 11.0 / 13.2
line-4 14071.0–14131.0 m	75.0	16	2	13.2 / 16.5 / 19.8
line-4 14151.0–14199.3 m	50.0	12	1	8.8 / 11.0 / 13.2
line-4 15454.6–15482.9 m	50.0	12	1	8.8 / 11.0 / 13.2
line-4 15651.2–15671.2 m	25.0	8	0	4.4 / 5.5 / 6.6
line-4 15711.2–15739.5 m	50.0	12	1	8.8 / 11.0 / 13.2
line-4 15759.5–15807.8 m	50.0	12	1	8.8 / 11.0 / 13.2
line-4 16702.3–16730.6 m	50.0	12	1	8.8 / 11.0 / 13.2
line-4 17050.6–17118.9 m	75.0	16	2	13.2 / 16.5 / 19.8
line-4 17223.8–17292.0 m	75.0	16	2	13.2 / 16.5 / 19.8
line-4 17340.3–17368.6 m	50.0	12	1	8.8 / 11.0 / 13.2
line-4 17388.6–17416.9 m	50.0	12	1	8.8 / 11.0 / 13.2
line-4 17581.7–17630.0 m	50.0	12	1	8.8 / 11.0 / 13.2
line-4 17658.3–17686.6 m	50.0	12	1	8.8 / 11.0 / 13.2
line-4 17714.9–17783.2 m	75.0	16	2	13.2 / 16.5 / 19.8
line-4 17859.7–17944.6 m	100.0	20	3	17.6 / 22.0 / 26.4
line-4 17992.9–18069.4 m	100.0	20	3	17.6 / 22.0 / 26.4
line-4 18194.3–18250.9 m	75.0	16	2	13.2 / 16.5 / 19.8
line-4 18319.2–18347.4 m	50.0	12	1	8.8 / 11.0 / 13.2
line-4 18597.1–18653.7 m	75.0	16	2	13.2 / 16.5 / 19.8
line-4 18722.0–18778.6 m	75.0	16	2	13.2 / 16.5 / 19.8
line-4 18963.4–19048.3 m	100.0	20	3	17.6 / 22.0 / 26.4
line-4 19459.4–19507.7 m	50.0	12	1	8.8 / 11.0 / 13.2
line-4 20208.5–20236.8 m	50.0	12	1	8.8 / 11.0 / 13.2
line-4 21321.1–21369.3 m	50.0	12	1	8.8 / 11.0 / 13.2
line-4 21425.9–21454.2 m	50.0	12	1	8.8 / 11.0 / 13.2
line-4 21482.5–21539.0 m	75.0	16	2	13.2 / 16.5 / 19.8
line-4 21599.0–21619.0 m	25.0	8	0	4.4 / 5.5 / 6.6

Elevated segment	Longest unit m	Bearings	Link slabs	Hot movement mm at 8 / 10 / 12 µstrain/K
line-4 21679.0–21727.3 m	50.0	12	1	8.8 / 11.0 / 13.2
line-4 21932.2–21960.5 m	50.0	12	1	8.8 / 11.0 / 13.2
line-4 23064.7–23121.3 m	75.0	16	2	13.2 / 16.5 / 19.8
line-4 23141.3–23237.9 m	100.0	20	3	17.6 / 22.0 / 26.4
line-4 23266.2–23371.0 m	100.0	28	3	17.6 / 22.0 / 26.4
line-4 23524.1–23580.7 m	75.0	16	2	13.2 / 16.5 / 19.8
line-4 23960.1–23988.4 m	50.0	12	1	8.8 / 11.0 / 13.2
line-4 24056.7–24113.3 m	75.0	16	2	13.2 / 16.5 / 19.8
line-4 24141.5–24169.8 m	50.0	12	1	8.8 / 11.0 / 13.2
line-4 26794.0–26822.3 m	50.0	12	1	8.8 / 11.0 / 13.2
line-4 26927.1–26975.4 m	50.0	12	1	8.8 / 11.0 / 13.2
line-4 27334.8–27391.4 m	75.0	16	2	13.2 / 16.5 / 19.8
line-4 27609.4–27666.0 m	75.0	16	2	13.2 / 16.5 / 19.8
line-4 28319.9–28339.9 m	25.0	8	0	4.4 / 5.5 / 6.6
line-4 28473.1–28501.3 m	50.0	12	1	8.8 / 11.0 / 13.2
line-4 29058.7–29228.5 m	100.0	36	5	17.6 / 22.0 / 26.4
line-4 29454.7–29483.0 m	50.0	12	1	8.8 / 11.0 / 13.2
line-4 29559.6–29587.9 m	50.0	12	1	8.8 / 11.0 / 13.2
line-4 29672.7–29701.0 m	50.0	12	1	8.8 / 11.0 / 13.2
line-4 30087.3–30115.6 m	50.0	12	1	8.8 / 11.0 / 13.2
line-4 30143.8–30200.4 m	75.0	16	2	13.2 / 16.5 / 19.8
line-4 30390.1–30446.7 m	75.0	16	2	13.2 / 16.5 / 19.8
line-4 30789.5–30886.1 m	100.0	20	3	17.6 / 22.0 / 26.4
line-4 31039.2–31067.5 m	50.0	12	1	8.8 / 11.0 / 13.2
line-4 31208.9–31237.2 m	50.0	12	1	8.8 / 11.0 / 13.2
line-4 31305.5–31382.1 m	100.0	20	3	17.6 / 22.0 / 26.4
line-4 31495.2–31523.5 m	50.0	12	1	8.8 / 11.0 / 13.2
line-4 31798.1–31826.3 m	50.0	12	1	8.8 / 11.0 / 13.2
line-4 31854.6–31996.0 m	100.0	32	4	17.6 / 22.0 / 26.4
line-4 32726.6–32754.9 m	50.0	12	1	8.8 / 11.0 / 13.2
line-4 33210.8–33239.1 m	50.0	12	1	8.8 / 11.0 / 13.2
line-4 33267.4–33315.7 m	50.0	12	1	8.8 / 11.0 / 13.2
line-4 34490.5–34558.8 m	75.0	16	2	13.2 / 16.5 / 19.8
line-4 34901.6–34949.9 m	50.0	12	1	8.8 / 11.0 / 13.2
line-4 35167.9–35264.5 m	100.0	20	3	17.6 / 22.0 / 26.4
line-4 36397.6–36437.6 m	50.0	12	1	8.8 / 11.0 / 13.2
line-4 36645.9–36674.2 m	50.0	12	1	8.8 / 11.0 / 13.2
line-4 36754.2–36794.2 m	50.0	12	1	8.8 / 11.0 / 13.2
line-4 37300.5–38469.6 m	100.0	236	35	17.6 / 22.0 / 26.4
line-4 41871.7–41956.6 m	100.0	20	3	17.6 / 22.0 / 26.4
line-4 42013.1–42041.4 m	50.0	12	1	8.8 / 11.0 / 13.2
line-4 42098.0–42126.3 m	50.0	12	1	8.8 / 11.0 / 13.2
line-4 42792.0–42812.0 m	25.0	8	0	4.4 / 5.5 / 6.6
line-4 43259.7–43279.7 m	25.0	8	0	4.4 / 5.5 / 6.6
line-4 43504.5–43524.5 m	25.0	8	0	4.4 / 5.5 / 6.6
line-5 0.0–20.0 m	25.0	8	0	4.4 / 5.5 / 6.6
line-5 48.3–156.6 m	100.0	28	3	17.6 / 22.0 / 26.4
line-5 926.3–986.3 m	75.0	16	2	13.2 / 16.5 / 19.8
line-5 2479.7–2539.7 m	75.0	16	2	13.2 / 16.5 / 19.8
line-5 3828.2–3848.2 m	25.0	8	0	4.4 / 5.5 / 6.6
line-5 4004.8–4033.0 m	50.0	12	1	8.8 / 11.0 / 13.2
line-5 4514.5–4542.7 m	50.0	12	1	8.8 / 11.0 / 13.2
line-5 4611.0–4639.3 m	50.0	12	1	8.8 / 11.0 / 13.2
line-5 5595.3–5615.3 m	25.0	8	0	4.4 / 5.5 / 6.6
line-5 5985.0–6033.3 m	50.0	12	1	8.8 / 11.0 / 13.2
line-5 6161.6–6189.8 m	50.0	12	1	8.8 / 11.0 / 13.2
line-5 6326.4–6346.4 m	25.0	8	0	4.4 / 5.5 / 6.6
line-5 6526.4–6623.0 m	100.0	20	3	17.6 / 22.0 / 26.4
line-5 6876.1–6924.4 m	50.0	12	1	8.8 / 11.0 / 13.2
line-5 6944.4–6992.7 m	50.0	12	1	8.8 / 11.0 / 13.2
line-5 7052.7–7072.7 m	25.0	8	0	4.4 / 5.5 / 6.6
line-5 8238.4–8258.4 m	25.0	8	0	4.4 / 5.5 / 6.6

Elevated segment	Longest unit m	Bearings	Link slabs	Hot movement mm at 8 / 10 / 12 µstrain/K
line-5 8394.9–8414.9 m	25.0	8	0	4.4 / 5.5 / 6.6
line-5 8531.5–8551.5 m	25.0	8	0	4.4 / 5.5 / 6.6
line-5 8599.8–8628.1 m	50.0	12	1	8.8 / 11.0 / 13.2
line-5 8812.9–8901.2 m	100.0	20	3	17.6 / 22.0 / 26.4
line-5 9034.4–9062.6 m	50.0	12	1	8.8 / 11.0 / 13.2
line-5 9299.2–9319.2 m	25.0	8	0	4.4 / 5.5 / 6.6
line-5 9439.2–9479.2 m	50.0	12	1	8.8 / 11.0 / 13.2
line-5 9667.5–9687.5 m	25.0	8	0	4.4 / 5.5 / 6.6
line-5 9895.8–9955.8 m	75.0	16	2	13.2 / 16.5 / 19.8
line-5 10144.1–10184.1 m	50.0	12	1	8.8 / 11.0 / 13.2
line-5 10224.1–10304.1 m	100.0	20	3	17.6 / 22.0 / 26.4
line-5 10906.9–10926.9 m	25.0	8	0	4.4 / 5.5 / 6.6
line-5 11235.2–11275.2 m	50.0	12	1	8.8 / 11.0 / 13.2
line-5 11873.2–11893.2 m	25.0	8	0	4.4 / 5.5 / 6.6
line-5 11953.2–11973.2 m	25.0	8	0	4.4 / 5.5 / 6.6
line-5 12121.5–12161.5 m	50.0	12	1	8.8 / 11.0 / 13.2
line-5 12646.3–12706.3 m	75.0	16	2	13.2 / 16.5 / 19.8
line-5 13080.9–13129.2 m	50.0	12	1	8.8 / 11.0 / 13.2
line-5 13317.4–13365.7 m	50.0	12	1	8.8 / 11.0 / 13.2
line-5 13474.0–13494.0 m	25.0	8	0	4.4 / 5.5 / 6.6
line-5 13542.3–13562.3 m	25.0	8	0	4.4 / 5.5 / 6.6
line-5 13582.3–13622.3 m	50.0	12	1	8.8 / 11.0 / 13.2
line-5 13682.3–13722.3 m	50.0	12	1	8.8 / 11.0 / 13.2
line-5 13858.9–13938.9 m	100.0	20	3	17.6 / 22.0 / 26.4
line-5 13958.9–14038.9 m	100.0	20	3	17.6 / 22.0 / 26.4
line-5 14078.9–14158.9 m	100.0	20	3	17.6 / 22.0 / 26.4
line-5 14198.9–14287.1 m	100.0	20	3	17.6 / 22.0 / 26.4
line-5 14335.4–14383.7 m	50.0	12	1	8.8 / 11.0 / 13.2
line-5 14576.9–14605.1 m	50.0	12	1	8.8 / 11.0 / 13.2
line-5 14750.0–14770.0 m	25.0	8	0	4.4 / 5.5 / 6.6
line-5 14883.1–14903.1 m	25.0	8	0	4.4 / 5.5 / 6.6
line-5 15019.7–15048.0 m	50.0	12	1	8.8 / 11.0 / 13.2
line-5 15104.5–15161.1 m	75.0	16	2	13.2 / 16.5 / 19.8
line-5 15181.1–15237.7 m	75.0	16	2	13.2 / 16.5 / 19.8
line-5 15434.3–15494.3 m	75.0	16	2	13.2 / 16.5 / 19.8
line-5 15514.3–15562.5 m	50.0	12	1	8.8 / 11.0 / 13.2
line-5 15622.5–15722.5 m	100.0	20	3	17.6 / 22.0 / 26.4
line-5 15770.8–15879.1 m	100.0	28	3	17.6 / 22.0 / 26.4
line-5 16107.4–16127.4 m	25.0	8	0	4.4 / 5.5 / 6.6
line-5 16187.4–16267.4 m	100.0	20	3	17.6 / 22.0 / 26.4
line-5 16487.4–16515.7 m	50.0	12	1	8.8 / 11.0 / 13.2
line-5 16615.7–16635.7 m	25.0	8	0	4.4 / 5.5 / 6.6
line-5 16972.2–17052.2 m	100.0	20	3	17.6 / 22.0 / 26.4
line-5 17923.4–18008.2 m	100.0	20	3	17.6 / 22.0 / 26.4
line-5 18313.1–18333.1 m	25.0	8	0	4.4 / 5.5 / 6.6
line-5 18353.1–18373.1 m	25.0	8	0	4.4 / 5.5 / 6.6
line-5 18393.1–18517.9 m	100.0	28	3	17.6 / 22.0 / 26.4
line-5 18942.8–18982.8 m	50.0	12	1	8.8 / 11.0 / 13.2
line-5 19935.3–19963.6 m	50.0	12	1	8.8 / 11.0 / 13.2
line-5 20209.9–20238.2 m	50.0	12	1	8.8 / 11.0 / 13.2
line-5 20258.2–20286.5 m	50.0	12	1	8.8 / 11.0 / 13.2
line-5 20567.9–20644.4 m	100.0	20	3	17.6 / 22.0 / 26.4
line-5 21477.0–21517.0 m	50.0	12	1	8.8 / 11.0 / 13.2
line-5 21803.3–21879.8 m	100.0	20	3	17.6 / 22.0 / 26.4
line-5 21956.4–21984.7 m	50.0	12	1	8.8 / 11.0 / 13.2
line-5 22146.1–22194.4 m	50.0	12	1	8.8 / 11.0 / 13.2
line-5 22876.7–22953.2 m	100.0	20	3	17.6 / 22.0 / 26.4
line-5 23066.4–23151.2 m	100.0	20	3	17.6 / 22.0 / 26.4
line-5 23227.8–23247.8 m	25.0	8	0	4.4 / 5.5 / 6.6
line-5 23432.6–23460.9 m	50.0	12	1	8.8 / 11.0 / 13.2
line-5 23529.2–23557.5 m	50.0	12	1	8.8 / 11.0 / 13.2
line-5 23585.8–23625.8 m	50.0	12	1	8.8 / 11.0 / 13.2

Elevated segment	Longest unit m	Bearings	Link slabs	Hot movement mm at 8 / 10 / 12 µstrain/K
line-5 24461.2–24489.4 m	50.0	12	1	8.8 / 11.0 / 13.2
line-5 25296.6–25353.1 m	75.0	16	2	13.2 / 16.5 / 19.8
line-5 25373.1–25449.7 m	100.0	20	3	17.6 / 22.0 / 26.4
line-5 26042.2–26110.5 m	75.0	16	2	13.2 / 16.5 / 19.8
line-5 26138.8–26195.4 m	75.0	16	2	13.2 / 16.5 / 19.8
line-5 28711.8–28740.1 m	50.0	12	1	8.8 / 11.0 / 13.2
line-5 28788.4–28808.4 m	25.0	8	0	4.4 / 5.5 / 6.6
line-5 29412.7–29441.0 m	50.0	12	1	8.8 / 11.0 / 13.2
line-5 29489.2–29517.5 m	50.0	12	1	8.8 / 11.0 / 13.2
line-5 29840.4–29965.2 m	100.0	28	3	17.6 / 22.0 / 26.4
line-5 30078.4–30134.9 m	75.0	16	2	13.2 / 16.5 / 19.8
line-5 30191.5–30239.8 m	50.0	12	1	8.8 / 11.0 / 13.2
line-5 30582.6–30602.6 m	25.0	8	0	4.4 / 5.5 / 6.6
line-5 30650.9–30679.2 m	50.0	12	1	8.8 / 11.0 / 13.2
line-5 30735.8–30764.0 m	50.0	12	1	8.8 / 11.0 / 13.2
line-5 30792.3–30848.9 m	75.0	16	2	13.2 / 16.5 / 19.8
line-5 31030.3–31126.9 m	100.0	20	3	17.6 / 22.0 / 26.4
line-5 31203.5–31251.7 m	50.0	12	1	8.8 / 11.0 / 13.2
line-5 31521.4–31541.4 m	25.0	8	0	4.4 / 5.5 / 6.6
line-5 31686.3–31714.6 m	50.0	12	1	8.8 / 11.0 / 13.2
line-5 31851.2–31879.4 m	50.0	12	1	8.8 / 11.0 / 13.2
line-5 33217.7–33246.0 m	50.0	12	1	8.8 / 11.0 / 13.2
line-5 34947.6–34975.9 m	50.0	12	1	8.8 / 11.0 / 13.2
line-5 35395.9–35415.9 m	25.0	8	0	4.4 / 5.5 / 6.6
line-5 35715.9–35775.9 m	75.0	16	2	13.2 / 16.5 / 19.8
line-5 36020.8–36097.3 m	100.0	20	3	17.6 / 22.0 / 26.4
line-5 42687.2–42735.5 m	50.0	12	1	8.8 / 11.0 / 13.2
line-5 42932.0–42952.0 m	25.0	8	0	4.4 / 5.5 / 6.6
line-5 45461.4–45489.7 m	50.0	12	1	8.8 / 11.0 / 13.2
line-5 46342.2–46362.2 m	25.0	8	0	4.4 / 5.5 / 6.6
line-5 49023.3–49071.6 m	50.0	12	1	8.8 / 11.0 / 13.2
line-5 51318.9–51347.2 m	50.0	12	1	8.8 / 11.0 / 13.2
line-5 51395.5–51415.5 m	25.0	8	0	4.4 / 5.5 / 6.6
line-5 52614.9–52654.9 m	50.0	12	1	8.8 / 11.0 / 13.2
line-6 0.0–173.1 m	100.0	36	5	17.6 / 22.0 / 26.4
line-6 664.3–749.1 m	100.0	20	3	17.6 / 22.0 / 26.4
line-6 1552.8–1581.1 m	50.0	12	1	8.8 / 11.0 / 13.2
line-6 1657.6–1685.9 m	50.0	12	1	8.8 / 11.0 / 13.2
line-6 2982.2–3010.4 m	50.0	12	1	8.8 / 11.0 / 13.2
line-6 3038.7–3067.0 m	50.0	12	1	8.8 / 11.0 / 13.2
line-6 4117.0–4145.2 m	50.0	12	1	8.8 / 11.0 / 13.2
line-6 4278.4–4306.7 m	50.0	12	1	8.8 / 11.0 / 13.2
line-6 4617.8–4646.1 m	50.0	12	1	8.8 / 11.0 / 13.2
line-6 4807.5–4835.8 m	50.0	12	1	8.8 / 11.0 / 13.2
line-6 4864.1–4892.3 m	50.0	12	1	8.8 / 11.0 / 13.2
line-6 11462.3–11490.6 m	50.0	12	1	8.8 / 11.0 / 13.2
line-6 13745.1–13773.4 m	50.0	12	1	8.8 / 11.0 / 13.2
line-6 15723.6–15751.8 m	50.0	12	1	8.8 / 11.0 / 13.2
line-6 15820.1–15840.1 m	25.0	8	0	4.4 / 5.5 / 6.6
line-6 15888.4–15908.4 m	25.0	8	0	4.4 / 5.5 / 6.6
line-6 16599.5–17760.4 m	100.0	236	35	17.6 / 22.0 / 26.4
line-6 18373.5–18461.8 m	100.0	20	3	17.6 / 22.0 / 26.4
line-6 18481.8–18521.8 m	50.0	12	1	8.8 / 11.0 / 13.2
line-6 18641.8–18718.4 m	100.0	20	3	17.6 / 22.0 / 26.4
line-6 20490.3–20546.9 m	75.0	16	2	13.2 / 16.5 / 19.8
line-6 21000.0–21028.3 m	50.0	12	1	8.8 / 11.0 / 13.2
line-6 21346.3–21366.3 m	25.0	8	0	4.4 / 5.5 / 6.6
line-6 21602.9–21631.2 m	50.0	12	1	8.8 / 11.0 / 13.2
line-6 21927.7–21956.0 m	50.0	12	1	8.8 / 11.0 / 13.2
line-6 22445.7–22514.0 m	75.0	16	2	13.2 / 16.5 / 19.8
line-6 22582.3–22650.6 m	75.0	16	2	13.2 / 16.5 / 19.8
line-6 23664.5–23761.1 m	100.0	20	3	17.6 / 22.0 / 26.4

Elevated segment	Longest unit m	Bearings	Link slabs	Hot movement mm at 8 / 10 / 12 µstrain/K
line-6 24917.1–24973.6 m	75.0	16	2	13.2 / 16.5 / 19.8
line-6 25335.1–25355.1 m	25.0	8	0	4.4 / 5.5 / 6.6
line-6 25403.4–25423.4 m	25.0	8	0	4.4 / 5.5 / 6.6
line-6 25471.6–25499.9 m	50.0	12	1	8.8 / 11.0 / 13.2
line-6 25616.5–25656.5 m	50.0	12	1	8.8 / 11.0 / 13.2
line-6 25793.1–25813.1 m	25.0	8	0	4.4 / 5.5 / 6.6
line-6 26254.5–26274.5 m	25.0	8	0	4.4 / 5.5 / 6.6
line-6 26371.1–26391.1 m	25.0	8	0	4.4 / 5.5 / 6.6
line-6 27255.3–27275.3 m	25.0	8	0	4.4 / 5.5 / 6.6
line-6 27653.3–27681.6 m	50.0	12	1	8.8 / 11.0 / 13.2
line-6 27778.2–27846.4 m	75.0	16	2	13.2 / 16.5 / 19.8
line-6 28585.9–28614.1 m	50.0	12	1	8.8 / 11.0 / 13.2
line-6 28702.4–28770.7 m	75.0	16	2	13.2 / 16.5 / 19.8
line-6 29083.8–29123.8 m	50.0	12	1	8.8 / 11.0 / 13.2
line-6 29317.0–29345.3 m	50.0	12	1	8.8 / 11.0 / 13.2
line-6 29385.3–29461.8 m	100.0	20	3	17.6 / 22.0 / 26.4
line-6 29646.7–29715.0 m	75.0	16	2	13.2 / 16.5 / 19.8
line-6 30109.5–30129.5 m	25.0	8	0	4.4 / 5.5 / 6.6
line-6 30289.5–30329.5 m	50.0	12	1	8.8 / 11.0 / 13.2
line-6 30349.5–30417.8 m	75.0	16	2	13.2 / 16.5 / 19.8
line-6 30695.8–30724.1 m	50.0	12	1	8.8 / 11.0 / 13.2
line-6 30808.9–30865.5 m	75.0	16	2	13.2 / 16.5 / 19.8
line-6 31849.2–31905.8 m	75.0	16	2	13.2 / 16.5 / 19.8
line-6 32256.9–32285.2 m	50.0	12	1	8.8 / 11.0 / 13.2
line-6 32639.7–32668.0 m	50.0	12	1	8.8 / 11.0 / 13.2
line-6 32688.0–32748.0 m	75.0	16	2	13.2 / 16.5 / 19.8
line-6 32852.9–32912.9 m	75.0	16	2	13.2 / 16.5 / 19.8
line-6 32952.9–32992.9 m	50.0	12	1	8.8 / 11.0 / 13.2
line-6 34117.1–34278.5 m	100.0	36	5	17.6 / 22.0 / 26.4
line-6 34463.4–34511.7 m	50.0	12	1	8.8 / 11.0 / 13.2
line-6 34848.2–34936.5 m	100.0	20	3	17.6 / 22.0 / 26.4
line-6 35013.1–35101.4 m	100.0	20	3	17.6 / 22.0 / 26.4
line-6 35129.7–35158.0 m	50.0	12	1	8.8 / 11.0 / 13.2
line-6 35274.5–35294.5 m	25.0	8	0	4.4 / 5.5 / 6.6
line-6 35314.5–36656.2 m	100.0	272	40	17.6 / 22.0 / 26.4
line-6 37047.3–37075.6 m	50.0	12	1	8.8 / 11.0 / 13.2
line-6 37220.5–37240.5 m	25.0	8	0	4.4 / 5.5 / 6.6
line-6 37490.2–37546.7 m	75.0	16	2	13.2 / 16.5 / 19.8
line-6 37575.0–37595.0 m	25.0	8	0	4.4 / 5.5 / 6.6
line-6 37615.0–37671.6 m	75.0	16	2	13.2 / 16.5 / 19.8
line-6 37719.9–37768.1 m	50.0	12	1	8.8 / 11.0 / 13.2
line-6 38582.1–38630.4 m	50.0	12	1	8.8 / 11.0 / 13.2
line-6 38687.0–38707.0 m	25.0	8	0	4.4 / 5.5 / 6.6
line-6 38985.0–39013.2 m	50.0	12	1	8.8 / 11.0 / 13.2
line-6 39549.2–39597.5 m	50.0	12	1	8.8 / 11.0 / 13.2
line-6 41313.7–41413.7 m	100.0	20	3	17.6 / 22.0 / 26.4
line-6 41562.0–41590.3 m	50.0	12	1	8.8 / 11.0 / 13.2
line-6 41876.6–41904.9 m	50.0	12	1	8.8 / 11.0 / 13.2
line-6 43632.6–43700.8 m	75.0	16	2	13.2 / 16.5 / 19.8
line-6 44447.1–44515.4 m	75.0	16	2	13.2 / 16.5 / 19.8
line-6 44543.7–44583.7 m	50.0	12	1	8.8 / 11.0 / 13.2
line-6 45103.1–45131.4 m	50.0	12	1	8.8 / 11.0 / 13.2
line-6 46150.8–46170.8 m	25.0	8	0	4.4 / 5.5 / 6.6
line-6 46190.8–46219.1 m	50.0	12	1	8.8 / 11.0 / 13.2
line-6 46335.6–46363.9 m	50.0	12	1	8.8 / 11.0 / 13.2
line-6 46875.1–46971.6 m	100.0	20	3	17.6 / 22.0 / 26.4
line-6 47434.5–47482.7 m	50.0	12	1	8.8 / 11.0 / 13.2
line-6 49387.3–49427.3 m	50.0	12	1	8.8 / 11.0 / 13.2
line-6 51290.1–51310.1 m	25.0	8	0	4.4 / 5.5 / 6.6
line-6 52598.4–52618.4 m	25.0	8	0	4.4 / 5.5 / 6.6
line-7 0.0–20.0 m	25.0	8	0	4.4 / 5.5 / 6.6
line-7 3153.1–3173.1 m	25.0	8	0	4.4 / 5.5 / 6.6

Elevated segment	Longest unit m	Bearings	Link slabs	Hot movement mm at 8 / 10 / 12 µstrain/K
line-7 3493.1–3513.1 m	25.0	8	0	4.4 / 5.5 / 6.6
line-7 5093.1–5113.1 m	25.0	8	0	4.4 / 5.5 / 6.6
line-7 9310.8–9339.1 m	50.0	12	1	8.8 / 11.0 / 13.2
line-7 11202.2–11222.2 m	25.0	8	0	4.4 / 5.5 / 6.6
line-7 12467.8–12487.8 m	25.0	8	0	4.4 / 5.5 / 6.6
line-7 12516.1–12564.4 m	50.0	12	1	8.8 / 11.0 / 13.2
line-7 12644.4–12672.7 m	50.0	12	1	8.8 / 11.0 / 13.2
line-7 13002.4–13022.4 m	25.0	8	0	4.4 / 5.5 / 6.6
line-7 13227.2–13315.5 m	100.0	20	3	17.6 / 22.0 / 26.4
line-7 13432.1–13452.1 m	25.0	8	0	4.4 / 5.5 / 6.6
line-7 13548.7–13605.2 m	75.0	16	2	13.2 / 16.5 / 19.8
line-7 13918.4–13938.4 m	25.0	8	0	4.4 / 5.5 / 6.6
line-7 14046.7–15398.0 m	100.0	276	41	17.6 / 22.0 / 26.4
line-7 16493.4–16521.7 m	50.0	12	1	8.8 / 11.0 / 13.2
line-7 16578.3–16646.6 m	75.0	16	2	13.2 / 16.5 / 19.8
line-7 17281.1–17309.4 m	50.0	12	1	8.8 / 11.0 / 13.2
line-7 17807.4–17827.4 m	25.0	8	0	4.4 / 5.5 / 6.6
line-7 18445.4–18465.4 m	25.0	8	0	4.4 / 5.5 / 6.6
line-7 20335.3–20363.6 m	50.0	12	1	8.8 / 11.0 / 13.2
line-7 21121.0–21149.3 m	50.0	12	1	8.8 / 11.0 / 13.2
line-7 21237.6–21257.6 m	25.0	8	0	4.4 / 5.5 / 6.6
line-7 21759.0–21787.3 m	50.0	12	1	8.8 / 11.0 / 13.2
line-7 22023.9–22052.1 m	50.0	12	1	8.8 / 11.0 / 13.2
line-7 22280.4–22340.4 m	75.0	16	2	13.2 / 16.5 / 19.8
line-7 22468.7–22497.0 m	50.0	12	1	8.8 / 11.0 / 13.2
line-7 22517.0–22537.0 m	25.0	8	0	4.4 / 5.5 / 6.6
line-7 22565.3–22585.3 m	25.0	8	0	4.4 / 5.5 / 6.6
line-7 22701.8–22761.8 m	75.0	16	2	13.2 / 16.5 / 19.8
line-7 23404.7–23424.7 m	25.0	8	0	4.4 / 5.5 / 6.6
line-7 23707.5–23735.8 m	50.0	12	1	8.8 / 11.0 / 13.2
line-7 23877.2–23897.2 m	25.0	8	0	4.4 / 5.5 / 6.6
line-7 24579.5–24599.5 m	25.0	8	0	4.4 / 5.5 / 6.6
line-7 24970.6–25050.6 m	100.0	20	3	17.6 / 22.0 / 26.4
line-7 25445.2–25521.7 m	100.0	20	3	17.6 / 22.0 / 26.4
line-7 25978.3–26034.9 m	75.0	16	2	13.2 / 16.5 / 19.8
line-7 26091.5–26148.0 m	75.0	16	2	13.2 / 16.5 / 19.8
line-7 26281.2–26301.2 m	25.0	8	0	4.4 / 5.5 / 6.6
line-7 27066.8–27095.1 m	50.0	12	1	8.8 / 11.0 / 13.2
line-7 27344.8–27373.1 m	50.0	12	1	8.8 / 11.0 / 13.2
line-7 27393.1–27558.0 m	100.0	36	5	17.6 / 22.0 / 26.4
line-7 27794.5–27902.8 m	100.0	28	3	17.6 / 22.0 / 26.4
line-7 27971.1–28039.4 m	75.0	16	2	13.2 / 16.5 / 19.8
line-7 28167.7–28196.0 m	50.0	12	1	8.8 / 11.0 / 13.2
line-7 28244.2–28272.5 m	50.0	12	1	8.8 / 11.0 / 13.2
line-7 28340.8–28429.1 m	100.0	20	3	17.6 / 22.0 / 26.4
line-7 28537.4–28557.4 m	25.0	8	0	4.4 / 5.5 / 6.6
line-7 28585.7–28614.0 m	50.0	12	1	8.8 / 11.0 / 13.2
line-7 28682.2–28730.5 m	50.0	12	1	8.8 / 11.0 / 13.2
line-7 28790.5–28858.8 m	75.0	16	2	13.2 / 16.5 / 19.8
line-7 28995.4–29103.7 m	100.0	28	3	17.6 / 22.0 / 26.4
line-7 29288.5–29308.5 m	25.0	8	0	4.4 / 5.5 / 6.6
line-7 29976.2–29996.2 m	25.0	8	0	4.4 / 5.5 / 6.6
line-7 30981.1–31001.1 m	25.0	8	0	4.4 / 5.5 / 6.6
line-7 31525.9–31574.2 m	50.0	12	1	8.8 / 11.0 / 13.2
line-7 31642.5–31662.5 m	25.0	8	0	4.4 / 5.5 / 6.6
line-7 31879.0–31947.3 m	75.0	16	2	13.2 / 16.5 / 19.8
line-7 32015.6–32055.6 m	50.0	12	1	8.8 / 11.0 / 13.2
line-7 32143.9–32183.9 m	50.0	12	1	8.8 / 11.0 / 13.2
line-7 32737.0–32797.0 m	75.0	16	2	13.2 / 16.5 / 19.8
line-7 32877.0–32925.3 m	50.0	12	1	8.8 / 11.0 / 13.2
line-7 33521.9–33550.2 m	50.0	12	1	8.8 / 11.0 / 13.2
line-7 33978.5–33998.5 m	25.0	8	0	4.4 / 5.5 / 6.6

Elevated segment	Longest unit m	Bearings	Link slabs	Hot movement mm at 8 / 10 / 12 µstrain/K
line-7 34046.7–34066.7 m	25.0	8	0	4.4 / 5.5 / 6.6
line-7 34135.0–34203.3 m	75.0	16	2	13.2 / 16.5 / 19.8
line-7 34399.9–34468.2 m	75.0	16	2	13.2 / 16.5 / 19.8
line-7 34684.7–34713.0 m	50.0	12	1	8.8 / 11.0 / 13.2
line-7 34801.3–34861.3 m	75.0	16	2	13.2 / 16.5 / 19.8
line-7 34889.6–34937.9 m	50.0	12	1	8.8 / 11.0 / 13.2
line-7 35159.3–35179.3 m	25.0	8	0	4.4 / 5.5 / 6.6
line-7 35415.9–35484.1 m	75.0	16	2	13.2 / 16.5 / 19.8
line-7 36052.4–36072.4 m	25.0	8	0	4.4 / 5.5 / 6.6
line-7 36490.4–36518.7 m	50.0	12	1	8.8 / 11.0 / 13.2
line-7 37591.3–37647.8 m	75.0	16	2	13.2 / 16.5 / 19.8
line-7 38038.9–38067.2 m	50.0	12	1	8.8 / 11.0 / 13.2
line-7 38843.2–38871.5 m	50.0	12	1	8.8 / 11.0 / 13.2
line-7 40526.3–40554.6 m	50.0	12	1	8.8 / 11.0 / 13.2
line-7 40671.2–40699.4 m	50.0	12	1	8.8 / 11.0 / 13.2
line-7 40796.0–40836.0 m	50.0	12	1	8.8 / 11.0 / 13.2
line-7 40884.3–40980.9 m	100.0	20	3	17.6 / 22.0 / 26.4
line-7 41000.9–41077.4 m	100.0	20	3	17.6 / 22.0 / 26.4
line-7 42241.1–42337.7 m	100.0	20	3	17.6 / 22.0 / 26.4
line-7 42800.5–42828.8 m	50.0	12	1	8.8 / 11.0 / 13.2
line-7 42908.8–42948.8 m	50.0	12	1	8.8 / 11.0 / 13.2
line-7 42988.8–43008.8 m	25.0	8	0	4.4 / 5.5 / 6.6
line-7 43137.1–43157.1 m	25.0	8	0	4.4 / 5.5 / 6.6
line-7 43177.1–43197.1 m	25.0	8	0	4.4 / 5.5 / 6.6
line-8 1198.0–1298.0 m	100.0	20	3	17.6 / 22.0 / 26.4
line-8 2029.1–2057.4 m	50.0	12	1	8.8 / 11.0 / 13.2
line-8 2114.0–2170.5 m	75.0	16	2	13.2 / 16.5 / 19.8
line-8 2661.7–2746.5 m	100.0	20	3	17.6 / 22.0 / 26.4
line-8 2843.1–2871.4 m	50.0	12	1	8.8 / 11.0 / 13.2
line-8 3370.8–3390.8 m	25.0	8	0	4.4 / 5.5 / 6.6
line-8 3943.3–3971.6 m	50.0	12	1	8.8 / 11.0 / 13.2
line-8 4402.7–4431.0 m	50.0	12	1	8.8 / 11.0 / 13.2
line-8 4471.0–4499.3 m	50.0	12	1	8.8 / 11.0 / 13.2
line-8 4519.3–4615.9 m	100.0	20	3	17.6 / 22.0 / 26.4
line-8 6159.0–6199.0 m	50.0	12	1	8.8 / 11.0 / 13.2
line-8 6259.0–6279.0 m	25.0	8	0	4.4 / 5.5 / 6.6
line-8 6339.0–6379.0 m	50.0	12	1	8.8 / 11.0 / 13.2
line-8 7483.8–7523.8 m	50.0	12	1	8.8 / 11.0 / 13.2
line-8 7723.8–7803.8 m	100.0	20	3	17.6 / 22.0 / 26.4
line-8 8223.8–8283.8 m	75.0	16	2	13.2 / 16.5 / 19.8
line-8 10971.5–11019.8 m	50.0	12	1	8.8 / 11.0 / 13.2
line-8 11684.1–11732.3 m	50.0	12	1	8.8 / 11.0 / 13.2
line-8 11760.6–11780.6 m	25.0	8	0	4.4 / 5.5 / 6.6
line-8 12002.1–12050.3 m	50.0	12	1	8.8 / 11.0 / 13.2
line-8 12146.9–12175.2 m	50.0	12	1	8.8 / 11.0 / 13.2
line-8 12561.5–12609.7 m	50.0	12	1	8.8 / 11.0 / 13.2
line-8 12629.7–12678.0 m	50.0	12	1	8.8 / 11.0 / 13.2
line-8 12706.3–12726.3 m	25.0	8	0	4.4 / 5.5 / 6.6
line-8 12996.0–13044.3 m	50.0	12	1	8.8 / 11.0 / 13.2
line-8 13112.6–13152.6 m	50.0	12	1	8.8 / 11.0 / 13.2
line-8 13765.7–13822.3 m	75.0	16	2	13.2 / 16.5 / 19.8
line-8 13878.9–13907.1 m	50.0	12	1	8.8 / 11.0 / 13.2
line-8 14012.0–14052.0 m	50.0	12	1	8.8 / 11.0 / 13.2
line-8 14248.6–14305.1 m	75.0	16	2	13.2 / 16.5 / 19.8
line-8 14325.1–14353.4 m	50.0	12	1	8.8 / 11.0 / 13.2
line-8 14551.4–14599.7 m	50.0	12	1	8.8 / 11.0 / 13.2
line-8 14668.0–14696.3 m	50.0	12	1	8.8 / 11.0 / 13.2
line-8 14999.1–15027.4 m	50.0	12	1	8.8 / 11.0 / 13.2
line-8 15084.0–15124.0 m	50.0	12	1	8.8 / 11.0 / 13.2
line-8 15144.0–15240.5 m	100.0	20	3	17.6 / 22.0 / 26.4
line-8 15317.1–15365.4 m	50.0	12	1	8.8 / 11.0 / 13.2
line-8 15425.4–15510.2 m	100.0	20	3	17.6 / 22.0 / 26.4

Elevated segment	Longest unit m	Bearings	Link slabs	Hot movement mm at 8 / 10 / 12 µstrain/K
line-8 15595.1–15623.4 m	50.0	12	1	8.8 / 11.0 / 13.2
line-8 16102.8–16131.1 m	50.0	12	1	8.8 / 11.0 / 13.2
line-8 16405.6–16433.9 m	50.0	12	1	8.8 / 11.0 / 13.2
line-8 16462.2–16490.5 m	50.0	12	1	8.8 / 11.0 / 13.2
line-8 16575.3–16603.6 m	50.0	12	1	8.8 / 11.0 / 13.2
line-8 17549.3–17569.3 m	25.0	8	0	4.4 / 5.5 / 6.6
line-8 17669.3–17689.3 m	25.0	8	0	4.4 / 5.5 / 6.6
line-8 20555.5–20583.8 m	50.0	12	1	8.8 / 11.0 / 13.2
line-8 22429.1–22457.4 m	50.0	12	1	8.8 / 11.0 / 13.2
line-8 25333.0–25453.0 m	100.0	28	3	17.6 / 22.0 / 26.4
line-8 25481.3–25501.3 m	25.0	8	0	4.4 / 5.5 / 6.6
line-8 25617.9–25637.9 m	25.0	8	0	4.4 / 5.5 / 6.6
line-8 25746.2–25774.4 m	50.0	12	1	8.8 / 11.0 / 13.2
line-8 25959.3–25979.3 m	25.0	8	0	4.4 / 5.5 / 6.6
line-8 26007.6–26027.6 m	25.0	8	0	4.4 / 5.5 / 6.6
line-8 26140.7–26180.7 m	50.0	12	1	8.8 / 11.0 / 13.2
line-8 26313.9–26342.1 m	50.0	12	1	8.8 / 11.0 / 13.2
line-8 26370.4–26418.7 m	50.0	12	1	8.8 / 11.0 / 13.2
line-8 26895.3–28240.4 m	100.0	272	40	17.6 / 22.0 / 26.4
line-8 28308.7–28356.9 m	50.0	12	1	8.8 / 11.0 / 13.2
line-8 28698.4–28738.4 m	50.0	12	1	8.8 / 11.0 / 13.2
line-8 28834.9–28863.2 m	50.0	12	1	8.8 / 11.0 / 13.2
line-8 29324.6–29472.9 m	100.0	32	4	17.6 / 22.0 / 26.4
line-8 29629.5–29697.8 m	75.0	16	2	13.2 / 16.5 / 19.8
line-8 29717.8–29737.8 m	25.0	8	0	4.4 / 5.5 / 6.6
line-8 29834.3–29862.6 m	50.0	12	1	8.8 / 11.0 / 13.2
line-8 29890.9–29919.2 m	50.0	12	1	8.8 / 11.0 / 13.2
line-8 30252.3–30292.3 m	50.0	12	1	8.8 / 11.0 / 13.2
line-8 31607.7–31647.7 m	50.0	12	1	8.8 / 11.0 / 13.2
line-8 32518.8–32567.1 m	50.0	12	1	8.8 / 11.0 / 13.2
line-8 32627.1–32667.1 m	50.0	12	1	8.8 / 11.0 / 13.2
line-8 33238.3–33286.5 m	50.0	12	1	8.8 / 11.0 / 13.2
line-8 33721.1–33749.4 m	50.0	12	1	8.8 / 11.0 / 13.2
line-8 33857.7–33877.7 m	25.0	8	0	4.4 / 5.5 / 6.6
line-8 34212.2–34232.2 m	25.0	8	0	4.4 / 5.5 / 6.6
line-8 34365.4–34413.6 m	50.0	12	1	8.8 / 11.0 / 13.2
line-8 34897.9–34926.2 m	50.0	12	1	8.8 / 11.0 / 13.2
line-8 34954.5–34982.8 m	50.0	12	1	8.8 / 11.0 / 13.2
line-8 35051.1–35107.6 m	75.0	16	2	13.2 / 16.5 / 19.8
line-8 35127.6–35155.9 m	50.0	12	1	8.8 / 11.0 / 13.2
line-8 35204.2–35252.5 m	50.0	12	1	8.8 / 11.0 / 13.2
line-8 35675.3–35695.3 m	25.0	8	0	4.4 / 5.5 / 6.6
line-8 35751.9–35800.2 m	50.0	12	1	8.8 / 11.0 / 13.2
line-8 36259.6–36279.6 m	25.0	8	0	4.4 / 5.5 / 6.6
line-8 36384.4–36412.7 m	50.0	12	1	8.8 / 11.0 / 13.2
line-8 36650.7–36679.0 m	50.0	12	1	8.8 / 11.0 / 13.2
line-8 36860.4–36880.4 m	25.0	8	0	4.4 / 5.5 / 6.6
line-8 36900.4–36928.7 m	50.0	12	1	8.8 / 11.0 / 13.2
line-8 36977.0–37045.3 m	75.0	16	2	13.2 / 16.5 / 19.8
line-8 37286.7–37315.0 m	50.0	12	1	8.8 / 11.0 / 13.2
line-8 37628.1–37648.1 m	25.0	8	0	4.4 / 5.5 / 6.6
line-8 38086.1–38106.1 m	25.0	8	0	4.4 / 5.5 / 6.6
line-8 38134.4–38154.4 m	25.0	8	0	4.4 / 5.5 / 6.6
line-8 38262.7–38339.2 m	100.0	20	3	17.6 / 22.0 / 26.4
line-8 39080.1–39208.4 m	100.0	32	4	17.6 / 22.0 / 26.4
line-8 39816.6–39836.6 m	25.0	8	0	4.4 / 5.5 / 6.6
line-8 39896.6–40004.9 m	100.0	28	3	17.6 / 22.0 / 26.4
line-8 41118.1–41158.1 m	50.0	12	1	8.8 / 11.0 / 13.2
line-8 41551.2–41571.2 m	25.0	8	0	4.4 / 5.5 / 6.6
line-8 41719.5–41739.5 m	25.0	8	0	4.4 / 5.5 / 6.6
line-8 41767.8–41796.0 m	50.0	12	1	8.8 / 11.0 / 13.2
line-8 41980.9–42000.9 m	25.0	8	0	4.4 / 5.5 / 6.6

Elevated segment	Longest unit m	Bearings	Link slabs	Hot movement mm at 8 / 10 / 12 µstrain/K
line-8 42362.3–42382.3 m	25.0	8	0	4.4 / 5.5 / 6.6
line-8 43150.0–43286.6 m	100.0	32	4	17.6 / 22.0 / 26.4
line-8 43479.7–43536.3 m	75.0	16	2	13.2 / 16.5 / 19.8
line-8 43681.1–43701.1 m	25.0	8	0	4.4 / 5.5 / 6.6
line-8 43817.7–43857.7 m	50.0	12	1	8.8 / 11.0 / 13.2
line-8 43954.3–44002.6 m	50.0	12	1	8.8 / 11.0 / 13.2
line-8 44110.8–44139.1 m	50.0	12	1	8.8 / 11.0 / 13.2
line-8 44275.7–44380.6 m	100.0	28	3	17.6 / 22.0 / 26.4
line-8 44505.4–44533.7 m	50.0	12	1	8.8 / 11.0 / 13.2
line-8 44980.0–45008.2 m	50.0	12	1	8.8 / 11.0 / 13.2
line-8 45156.5–45184.8 m	50.0	12	1	8.8 / 11.0 / 13.2
line-8 45373.1–45421.4 m	50.0	12	1	8.8 / 11.0 / 13.2
line-8 45501.4–45529.7 m	50.0	12	1	8.8 / 11.0 / 13.2
line-8 45774.5–45814.5 m	50.0	12	1	8.8 / 11.0 / 13.2
line-8 45882.8–46019.4 m	100.0	32	4	17.6 / 22.0 / 26.4
line-8 46144.2–46192.5 m	50.0	12	1	8.8 / 11.0 / 13.2
line-8 46369.1–46409.1 m	50.0	12	1	8.8 / 11.0 / 13.2
line-8 46449.1–46533.9 m	100.0	20	3	17.6 / 22.0 / 26.4
line-8 46795.4–46871.9 m	100.0	20	3	17.6 / 22.0 / 26.4
line-8 47651.3–47679.6 m	50.0	12	1	8.8 / 11.0 / 13.2
line-8 48793.6–48821.9 m	50.0	12	1	8.8 / 11.0 / 13.2
line-8 48983.3–49003.3 m	25.0	8	0	4.4 / 5.5 / 6.6
line-8 49091.6–49119.9 m	50.0	12	1	8.8 / 11.0 / 13.2
line-8 49324.7–49344.7 m	25.0	8	0	4.4 / 5.5 / 6.6
line-8 49421.3–49441.3 m	25.0	8	0	4.4 / 5.5 / 6.6
line-8 49469.6–49489.6 m	25.0	8	0	4.4 / 5.5 / 6.6
line-8 49517.9–49626.1 m	100.0	28	3	17.6 / 22.0 / 26.4
line-8 49646.1–49714.4 m	75.0	16	2	13.2 / 16.5 / 19.8
line-8 49839.3–49867.6 m	50.0	12	1	8.8 / 11.0 / 13.2
line-8 49887.6–49927.6 m	50.0	12	1	8.8 / 11.0 / 13.2
line-8 50084.1–50144.1 m	75.0	16	2	13.2 / 16.5 / 19.8
line-8 50204.1–50272.4 m	75.0	16	2	13.2 / 16.5 / 19.8
line-8 50569.0–50609.0 m	50.0	12	1	8.8 / 11.0 / 13.2
line-8 51221.5–51249.8 m	50.0	12	1	8.8 / 11.0 / 13.2
line-9 0.0–48.3 m	50.0	12	1	8.8 / 11.0 / 13.2
line-9 124.9–201.4 m	100.0	20	3	17.6 / 22.0 / 26.4
line-9 439.4–536.0 m	100.0	20	3	17.6 / 22.0 / 26.4
line-9 556.0–584.3 m	50.0	12	1	8.8 / 11.0 / 13.2
line-9 1096.8–1153.4 m	75.0	16	2	13.2 / 16.5 / 19.8
line-9 1249.9–1306.5 m	75.0	16	2	13.2 / 16.5 / 19.8
line-9 1411.4–1439.7 m	50.0	12	1	8.8 / 11.0 / 13.2
line-9 1669.4–1689.4 m	25.0	8	0	4.4 / 5.5 / 6.6
line-9 2052.2–2092.2 m	50.0	12	1	8.8 / 11.0 / 13.2
line-9 2346.8–2403.3 m	75.0	16	2	13.2 / 16.5 / 19.8
line-9 4749.3–4769.3 m	25.0	8	0	4.4 / 5.5 / 6.6
line-9 4789.3–4929.3 m	100.0	32	4	17.6 / 22.0 / 26.4
line-9 5702.4–5742.4 m	50.0	12	1	8.8 / 11.0 / 13.2
line-9 6035.5–6063.8 m	50.0	12	1	8.8 / 11.0 / 13.2
line-9 6120.4–6180.4 m	75.0	16	2	13.2 / 16.5 / 19.8
line-9 6438.4–7682.1 m	100.0	252	37	17.6 / 22.0 / 26.4
line-9 7795.2–7823.5 m	50.0	12	1	8.8 / 11.0 / 13.2
line-9 8013.2–8041.5 m	50.0	12	1	8.8 / 11.0 / 13.2
line-9 8126.3–8154.6 m	50.0	12	1	8.8 / 11.0 / 13.2
line-9 9030.0–9106.6 m	100.0	20	3	17.6 / 22.0 / 26.4
line-9 9219.7–9248.0 m	50.0	12	1	8.8 / 11.0 / 13.2
line-9 9276.3–9304.5 m	50.0	12	1	8.8 / 11.0 / 13.2
line-9 9526.0–10645.4 m	100.0	228	33	17.6 / 22.0 / 26.4
line-9 11471.7–11511.7 m	50.0	12	1	8.8 / 11.0 / 13.2
line-9 12131.1–12187.6 m	75.0	16	2	13.2 / 16.5 / 19.8
line-9 12914.7–12943.0 m	50.0	12	1	8.8 / 11.0 / 13.2
line-9 12991.3–13019.6 m	50.0	12	1	8.8 / 11.0 / 13.2
line-9 14517.2–14545.5 m	50.0	12	1	8.8 / 11.0 / 13.2

Elevated segment	Longest unit m	Bearings	Link slabs	Hot movement mm at 8 / 10 / 12 µstrain/K
line-9 14730.4–14758.7 m	50.0	12	1	8.8 / 11.0 / 13.2
line-9 15118.1–15146.4 m	50.0	12	1	8.8 / 11.0 / 13.2
line-9 15356.1–15432.6 m	100.0	20	3	17.6 / 22.0 / 26.4
line-9 15634.1–15718.9 m	100.0	20	3	17.6 / 22.0 / 26.4
line-9 15747.2–15803.8 m	75.0	16	2	13.2 / 16.5 / 19.8
line-9 15860.3–15888.6 m	50.0	12	1	8.8 / 11.0 / 13.2
line-9 15945.2–15993.5 m	50.0	12	1	8.8 / 11.0 / 13.2
line-9 16619.2–16647.4 m	50.0	12	1	8.8 / 11.0 / 13.2
line-9 22804.3–22832.6 m	50.0	12	1	8.8 / 11.0 / 13.2
line-9 23410.0–23438.3 m	50.0	12	1	8.8 / 11.0 / 13.2
line-9 24022.5–24042.5 m	25.0	8	0	4.4 / 5.5 / 6.6
line-9 24210.8–24279.1 m	75.0	16	2	13.2 / 16.5 / 19.8
line-9 24399.1–24427.4 m	50.0	12	1	8.8 / 11.0 / 13.2
line-9 24595.7–24623.9 m	50.0	12	1	8.8 / 11.0 / 13.2
line-9 24672.2–24732.2 m	75.0	16	2	13.2 / 16.5 / 19.8
line-9 25057.1–25125.4 m	75.0	16	2	13.2 / 16.5 / 19.8
line-9 25193.6–25253.6 m	75.0	16	2	13.2 / 16.5 / 19.8
line-9 25293.6–25341.9 m	50.0	12	1	8.8 / 11.0 / 13.2
line-9 25421.9–25490.2 m	75.0	16	2	13.2 / 16.5 / 19.8
line-9 25686.8–25706.8 m	25.0	8	0	4.4 / 5.5 / 6.6
line-9 26309.6–26337.9 m	50.0	12	1	8.8 / 11.0 / 13.2
line-9 26386.2–27455.9 m	100.0	216	32	17.6 / 22.0 / 26.4
line-9 28577.3–28617.3 m	50.0	12	1	8.8 / 11.0 / 13.2
line-9 28845.6–29931.9 m	100.0	220	33	17.6 / 22.0 / 26.4
line-9 31353.3–31449.9 m	100.0	20	3	17.6 / 22.0 / 26.4
line-9 31518.2–31634.7 m	100.0	28	3	17.6 / 22.0 / 26.4
line-9 32857.6–32885.9 m	50.0	12	1	8.8 / 11.0 / 13.2
line-9 33250.7–34469.5 m	100.0	248	36	17.6 / 22.0 / 26.4
line-9 37331.2–37359.5 m	50.0	12	1	8.8 / 11.0 / 13.2
line-9 37780.9–37877.5 m	100.0	20	3	17.6 / 22.0 / 26.4
line-9 37905.8–37945.8 m	50.0	12	1	8.8 / 11.0 / 13.2
line-9 37965.8–38054.0 m	100.0	20	3	17.6 / 22.0 / 26.4
line-9 41295.1–41315.1 m	25.0	8	0	4.4 / 5.5 / 6.6
line-9 43262.8–43291.1 m	50.0	12	1	8.8 / 11.0 / 13.2
line-9 44178.2–44254.8 m	100.0	20	3	17.6 / 22.0 / 26.4
line-9 44456.2–44484.5 m	50.0	12	1	8.8 / 11.0 / 13.2
line-9 45473.0–45501.3 m	50.0	12	1	8.8 / 11.0 / 13.2
line-9 45549.6–45577.9 m	50.0	12	1	8.8 / 11.0 / 13.2
line-9 45654.4–45711.0 m	75.0	16	2	13.2 / 16.5 / 19.8
line-9 47664.6–47692.9 m	50.0	12	1	8.8 / 11.0 / 13.2
line-9 48195.7–48215.7 m	25.0	8	0	4.4 / 5.5 / 6.6
line-9 48372.3–48432.3 m	75.0	16	2	13.2 / 16.5 / 19.8
line-9 48810.3–48850.3 m	50.0	12	1	8.8 / 11.0 / 13.2
line-9 49023.4–49051.7 m	50.0	12	1	8.8 / 11.0 / 13.2
line-9 49148.3–49176.6 m	50.0	12	1	8.8 / 11.0 / 13.2
line-9 49261.4–49309.7 m	50.0	12	1	8.8 / 11.0 / 13.2
line-9 49567.7–49607.7 m	50.0	12	1	8.8 / 11.0 / 13.2
line-9 49692.6–49720.8 m	50.0	12	1	8.8 / 11.0 / 13.2
line-9 49749.1–49797.4 m	50.0	12	1	8.8 / 11.0 / 13.2
line-9 50015.4–51350.2 m	100.0	272	40	17.6 / 22.0 / 26.4
line-9 51378.5–51418.5 m	50.0	12	1	8.8 / 11.0 / 13.2
line-9 51816.5–51864.8 m	50.0	12	1	8.8 / 11.0 / 13.2
line-9 52057.9–52086.2 m	50.0	12	1	8.8 / 11.0 / 13.2
line-9 52426.2–52474.5 m	50.0	12	1	8.8 / 11.0 / 13.2
line-9 53197.3–53350.4 m	100.0	36	5	17.6 / 22.0 / 26.4
line-9 53398.7–53418.7 m	25.0	8	0	4.4 / 5.5 / 6.6
line-9 53438.7–53575.3 m	100.0	32	4	17.6 / 22.0 / 26.4
line-9 53623.6–53909.9 m	100.0	60	9	17.6 / 22.0 / 26.4
line-9 54450.7–54499.0 m	50.0	12	1	8.8 / 11.0 / 13.2
line-9 54828.7–54897.0 m	75.0	16	2	13.2 / 16.5 / 19.8
line-9 54953.5–54981.8 m	50.0	12	1	8.8 / 11.0 / 13.2
line-9 55602.6–55659.2 m	75.0	16	2	13.2 / 16.5 / 19.8

Elevated segment	Longest unit m	Bearings	Link slabs	Hot movement mm at 8 / 10 / 12 µstrain/K
line-9 55759.2–55799.2 m	50.0	12	1	8.8 / 11.0 / 13.2
line-9 55819.2–55919.2 m	100.0	20	3	17.6 / 22.0 / 26.4
line-9 56099.2–56119.2 m	25.0	8	0	4.4 / 5.5 / 6.6
line-9 57895.8–58915.8 m	100.0	208	30	17.6 / 22.0 / 26.4
line-9 60835.2–62219.7 m	100.0	280	42	17.6 / 22.0 / 26.4
line-9 63217.7–63246.0 m	50.0	12	1	8.8 / 11.0 / 13.2
line-9 63664.0–63692.3 m	50.0	12	1	8.8 / 11.0 / 13.2
line-9 64323.4–64403.4 m	100.0	20	3	17.6 / 22.0 / 26.4
line-9 64443.4–64483.4 m	50.0	12	1	8.8 / 11.0 / 13.2
line-9 64611.7–64668.2 m	75.0	16	2	13.2 / 16.5 / 19.8
line-9 67406.1–67426.1 m	25.0	8	0	4.4 / 5.5 / 6.6
line-9 67454.4–67494.4 m	50.0	12	1	8.8 / 11.0 / 13.2
line-9 68910.6–68967.2 m	75.0	16	2	13.2 / 16.5 / 19.8
line-9 68995.5–69023.8 m	50.0	12	1	8.8 / 11.0 / 13.2
line-9 69083.8–69123.8 m	50.0	12	1	8.8 / 11.0 / 13.2
line-9 69192.1–69248.6 m	75.0	16	2	13.2 / 16.5 / 19.8
line-9 69578.3–69606.6 m	50.0	12	1	8.8 / 11.0 / 13.2
line-9 69703.2–69763.2 m	75.0	16	2	13.2 / 16.5 / 19.8
line-9 70352.9–70392.9 m	50.0	12	1	8.8 / 11.0 / 13.2
line-9 70777.7–70834.3 m	75.0	16	2	13.2 / 16.5 / 19.8
line-9 71244.0–71272.3 m	50.0	12	1	8.8 / 11.0 / 13.2
line-9 71570.3–71618.6 m	50.0	12	1	8.8 / 11.0 / 13.2
line-9 71666.9–71695.1 m	50.0	12	1	8.8 / 11.0 / 13.2
line-9 71743.4–71791.7 m	50.0	12	1	8.8 / 11.0 / 13.2
line-9 71928.3–72064.8 m	100.0	32	4	17.6 / 22.0 / 26.4
line-9 72113.1–72153.1 m	50.0	12	1	8.8 / 11.0 / 13.2
line-9 72341.4–72381.4 m	50.0	12	1	8.8 / 11.0 / 13.2
line-9 72506.3–72534.5 m	50.0	12	1	8.8 / 11.0 / 13.2
line-9 73207.1–74401.1 m	100.0	240	36	17.6 / 22.0 / 26.4
line-9 74529.4–74589.4 m	75.0	16	2	13.2 / 16.5 / 19.8
line-9 74697.6–74717.6 m	25.0	8	0	4.4 / 5.5 / 6.6
line-9 74862.5–74919.1 m	75.0	16	2	13.2 / 16.5 / 19.8
line-9 75426.8–75446.8 m	25.0	8	0	4.4 / 5.5 / 6.6
line-9 75515.0–75543.3 m	50.0	12	1	8.8 / 11.0 / 13.2
line-9 75563.3–75631.6 m	75.0	16	2	13.2 / 16.5 / 19.8
line-9 75748.2–75776.5 m	50.0	12	1	8.8 / 11.0 / 13.2
line-9 75893.0–75921.3 m	50.0	12	1	8.8 / 11.0 / 13.2
line-9 76352.4–76400.7 m	50.0	12	1	8.8 / 11.0 / 13.2
line-9 76429.0–76485.6 m	75.0	16	2	13.2 / 16.5 / 19.8
line-9 76633.9–76662.1 m	50.0	12	1	8.8 / 11.0 / 13.2
line-9 76898.7–77038.7 m	100.0	32	4	17.6 / 22.0 / 26.4
line-9 77471.8–77500.1 m	50.0	12	1	8.8 / 11.0 / 13.2
line-9 77736.7–77805.0 m	75.0	16	2	13.2 / 16.5 / 19.8
line-9 78767.8–78856.1 m	100.0	20	3	17.6 / 22.0 / 26.4
line-9 79169.2–79197.5 m	50.0	12	1	8.8 / 11.0 / 13.2
line-9 79245.8–79314.1 m	75.0	16	2	13.2 / 16.5 / 19.8
line-9 79607.2–79627.2 m	25.0	8	0	4.4 / 5.5 / 6.6
line-9 79976.9–80016.9 m	50.0	12	1	8.8 / 11.0 / 13.2
line-9 80201.8–80230.1 m	50.0	12	1	8.8 / 11.0 / 13.2
line-9 80611.5–80671.5 m	75.0	16	2	13.2 / 16.5 / 19.8
line-9 80699.8–80728.1 m	50.0	12	1	8.8 / 11.0 / 13.2
line-9 80776.4–80832.9 m	75.0	16	2	13.2 / 16.5 / 19.8
line-9 81037.8–81057.8 m	25.0	8	0	4.4 / 5.5 / 6.6
line-9 81106.1–81126.1 m	25.0	8	0	4.4 / 5.5 / 6.6
line-9 81146.1–81166.1 m	25.0	8	0	4.4 / 5.5 / 6.6
line-9 81447.5–81467.5 m	25.0	8	0	4.4 / 5.5 / 6.6
line-9 81624.1–81652.3 m	50.0	12	1	8.8 / 11.0 / 13.2
line-9 82430.3–82478.6 m	50.0	12	1	8.8 / 11.0 / 13.2
line-9 82863.5–82911.7 m	50.0	12	1	8.8 / 11.0 / 13.2
line-9 83028.3–83048.3 m	25.0	8	0	4.4 / 5.5 / 6.6
line-9 83366.3–83394.6 m	50.0	12	1	8.8 / 11.0 / 13.2
line-9 83616.0–83644.3 m	50.0	12	1	8.8 / 11.0 / 13.2

Elevated segment	Longest unit m	Bearings	Link slabs	Hot movement mm at 8 / 10 / 12 µstrain/K
line-9 83769.1–83789.1 m	25.0	8	0	4.4 / 5.5 / 6.6
line-9 83990.6–84030.6 m	50.0	12	1	8.8 / 11.0 / 13.2
line-9 84087.1–85347.4 m	100.0	256	38	17.6 / 22.0 / 26.4
line-9 85432.2–85460.5 m	50.0	12	1	8.8 / 11.0 / 13.2
line-9 85528.8–85625.4 m	100.0	20	3	17.6 / 22.0 / 26.4
line-9 86028.2–86076.5 m	50.0	12	1	8.8 / 11.0 / 13.2
line-9 86096.5–86124.8 m	50.0	12	1	8.8 / 11.0 / 13.2
line-9 86173.1–86221.4 m	50.0	12	1	8.8 / 11.0 / 13.2
line-9 86446.2–87598.8 m	100.0	236	35	17.6 / 22.0 / 26.4
line-9 87783.6–87891.9 m	100.0	28	3	17.6 / 22.0 / 26.4
line-9 88028.5–88068.5 m	50.0	12	1	8.8 / 11.0 / 13.2
line-9 88136.7–88165.0 m	50.0	12	1	8.8 / 11.0 / 13.2
line-9 88185.0–88205.0 m	25.0	8	0	4.4 / 5.5 / 6.6
line-9 88293.3–88313.3 m	25.0	8	0	4.4 / 5.5 / 6.6
line-9 88438.2–88534.7 m	100.0	20	3	17.6 / 22.0 / 26.4
line-9 88872.7–88921.0 m	50.0	12	1	8.8 / 11.0 / 13.2
line-9 88981.0–89041.0 m	75.0	16	2	13.2 / 16.5 / 19.8
line-9 89405.9–89434.1 m	50.0	12	1	8.8 / 11.0 / 13.2
line-9 89534.1–89574.1 m	50.0	12	1	8.8 / 11.0 / 13.2
line-9 89879.0–89907.3 m	50.0	12	1	8.8 / 11.0 / 13.2
line-9 90115.6–91364.7 m	100.0	252	37	17.6 / 22.0 / 26.4
line-9 91473.0–91501.3 m	50.0	12	1	8.8 / 11.0 / 13.2
line-9 91521.3–91617.8 m	100.0	20	3	17.6 / 22.0 / 26.4
line-9 91637.8–91677.8 m	50.0	12	1	8.8 / 11.0 / 13.2
line-9 91826.1–91846.1 m	25.0	8	0	4.4 / 5.5 / 6.6
line-9 92119.2–92147.5 m	50.0	12	1	8.8 / 11.0 / 13.2
line-9 92275.8–92384.1 m	100.0	28	3	17.6 / 22.0 / 26.4
line-9 92484.1–92504.1 m	25.0	8	0	4.4 / 5.5 / 6.6
line-9 92564.1–92604.1 m	50.0	12	1	8.8 / 11.0 / 13.2
line-9 92672.4–92712.4 m	50.0	12	1	8.8 / 11.0 / 13.2
line-9 92960.7–93000.7 m	50.0	12	1	8.8 / 11.0 / 13.2
line-9 93028.9–93077.2 m	50.0	12	1	8.8 / 11.0 / 13.2
line-9 93362.1–93390.4 m	50.0	12	1	8.8 / 11.0 / 13.2
line-9 93446.9–93503.5 m	75.0	16	2	13.2 / 16.5 / 19.8
line-9 93560.1–93580.1 m	25.0	8	0	4.4 / 5.5 / 6.6
line-9 93600.1–93620.1 m	25.0	8	0	4.4 / 5.5 / 6.6
line-9 93640.1–93668.4 m	50.0	12	1	8.8 / 11.0 / 13.2
line-9 93756.6–93816.6 m	75.0	16	2	13.2 / 16.5 / 19.8
line-9 94646.3–94666.3 m	25.0	8	0	4.4 / 5.5 / 6.6
line-9 95179.5–95199.5 m	25.0	8	0	4.4 / 5.5 / 6.6
line-9 95307.8–95336.1 m	50.0	12	1	8.8 / 11.0 / 13.2
line-9 95857.5–95937.5 m	100.0	20	3	17.6 / 22.0 / 26.4
line-9 96502.3–96522.3 m	25.0	8	0	4.4 / 5.5 / 6.6
line-9 96622.3–96650.6 m	50.0	12	1	8.8 / 11.0 / 13.2
line-9 97072.0–97100.3 m	50.0	12	1	8.8 / 11.0 / 13.2
line-9 97446.6–98590.9 m	100.0	232	34	17.6 / 22.0 / 26.4
line-9 98619.1–98647.4 m	50.0	12	1	8.8 / 11.0 / 13.2
line-9 98800.6–98828.8 m	50.0	12	1	8.8 / 11.0 / 13.2
line-9 99151.7–99171.7 m	25.0	8	0	4.4 / 5.5 / 6.6
line-9 99191.7–99220.0 m	50.0	12	1	8.8 / 11.0 / 13.2
line-9 99248.3–99316.5 m	75.0	16	2	13.2 / 16.5 / 19.8

Mapped soil has 65 incomplete profiles. Prioritise investigation and settlement/drainage monitoring; mapped pH/texture do not determine bearing resistance, pile depth, sulfate/chloride exposure or a coating category.

[Location-specific soil investigation plan](#). Flag counts: {"coverage-gap": 65, "fine-soil-plasticity-and-shrink-swell-tests": 6821, "granular-density-and-groundwater-tests": 3972, "silt-moisture-frost-and-erosion-review": 4951}.

Joints, paints, finishes and cleaning

Steel, GFRP body and roof finish zones retain their shared qualified process. Protect electrical bond lands, seals, drains, PV glass and inspection areas. Radiative paint remains a trial option; no unverified cooling benefit enters the energy model.

Work package	Asset-level tasks generated
rs-finish-seal-joint	831
rs-controlled-wash	831
civil-joint-drain-finish	374
energy-soiling-cleaning	158

The operations bundle carries these inspection/cleaning triggers, owners and required work-order evidence. Cleaning intervals follow measured condition and access; material compatibility and wash-water handling must be qualified locally.

Release inputs still required:

- civil movement: component temperature extremes and erection temperature; measured concrete CTE/shrinkage/creep; bearing restraint, rail interaction, seismic and foundation movement; surveyed supports and abutment movement details.
- fasteners: joint loads and fatigue spectrum; fastener grade and proof load; actual finish/lubrication torque-tension trial; preload loss, slip/separation, bearing and thread checks.
- seals: supplier qualified compression range and ageing; actual thickness, land gap and tolerances; module thermal mismatch and vibration; water-ingress test after assembly and cleaning.
- finishes: local exposure and substrate preparation qualification; approved product system and batch TDS limits; DFT/adhesion/cure and fire/UV/wash qualification; masking, repair and inspection records.
- cleaning: measured washable area by material, not canopy area; trial person-minutes, water and safe access windows; detergent/nozzle/material compatibility; wash-water collection and local disposal route.

[Structured report](#) · [Workforce spreadsheet](#) · [Deployment gates](#)

Baghdad frozen-resource delivery and funding study

Source: cities/catalogue/west-asia/Iraq/Baghdad/engineering/delivery-risk/README.md

These are deterministic disturbances, not P80/P90 dates or measured failure rates. All 831 planned six-car trains must pass acceptance before their line opens. Selected stage cells, civil crew lanes and dispatch order remain fixed in every stress. No stress calls the capacity-sizing search.

The explicit two-path calendar books 16 exclusive running hours per train at the case availability, inside acceptance-bay occupation, with setup and clearance. This adds a conservative calendar audit to the published average-throughput check. Baseline lane order is preserved; a replanned optimal dispatch might recover some delay. Financial-close months include 30 pre-NTP working days and three commissioning months.

Case	First/full month	Peak IQD gap debt, tn	Interest/fees USD eq, bn	Debt cleared month
calendar_baseline	41/83	4.461	5.076	303
factory_delay_6months	47/88	4.669	5.128	306
supplier_shortage_6months	46/87	4.681	5.134	306
half_staff_late_12months	45/86	4.635	5.122	305
rework_10pct	41/86	4.603	5.122	305
one_test_path_outage_6months	41/84	4.561	5.107	304
civil_access_6months	42/89	4.453	5.067	303
availability_75pct	42/90	4.713	5.151	306
availability_65pct	45/99	4.978	5.215	310
combined	55/106	5.229	5.266	315
combined_second_test_shift	53/104	5.114	5.235	313
availability_75pct_second_test_shift	42/90	4.720	5.154	306
civil_cycles_20pct_faster	41/83	4.465	5.078	303
civil_earliest_unchanged_cycles	41/83	4.818	5.211	304
civil_earliest_20pct_faster	41/82	5.139	5.332	306
availability_75pct_costed	42/90	4.728	5.156	306
availability_75pct_test_shift_costed	42/90	4.735	5.159	307
hiring_ramp_costed	45/86	4.642	5.124	305
supplier_shortage_costed	46/87	4.690	5.137	306
availability_75pct_structural_shift	42/89	4.741	5.165	306
availability_75pct_electrical_shift	42/89	4.700	5.151	306
availability_75pct_composite_shift	42/89	4.704	5.151	306
availability_75pct_all_stage_shifts	40/83	4.542	5.114	304
recruitment_training_recovery	41/83	4.470	5.083	303
supplier_expedite_recovery	43/84	4.585	5.082	304
temporary_first_article	36/83	4.422	5.103	302
combined_delay_costs	55/106	5.308	5.291	316
combined_lower_demand	55/106	9.467	7.479	Unpaid
combined_escalation	55/106	7.810	7.060	354
combined_finance_downside	55/106	4.000	10.477	Unfunded
joint_downside	55/106	4.000	15.384	Unfunded
domestic_placement_interrupted_recovered	47/89	4.500	5.065	306
export_credit_delayed_recovered	47/89	4.342	5.012	304

Separate productivity from investment timing

The corrected civil_cycles_20pct_faster case retains every rephased start floor and changes only civil occupation durations. A 1.0 multiplier is tested as an exact schedule identity, including running-path reservations and line openings. Civil_earliest_unchanged_cycles removes the spending delays at original durations; civil_earliest_20pct_faster changes both. Their costs must not be attributed to productivity alone. Earlier completion may still advance completion/retention invoices even when mobilisation timing is retained. These are diagnostic cycle assumptions with no added crews or accepted acceleration price.

Recovery comparisons and priced assumptions

Factory cells and dispatch lane order remain fixed. Fixed curing, bonding, inspection and test holds are listed in `baghdad-delivery-risk.toml` as unqualified working-calendar equivalents. Shift compression applies only to the remaining staffed occupation; the additional 60-day first-article qualification is unchanged. At one shift the original schedule is preserved exactly. Cure elapsed hours and batch/test evidence must replace these assumed splits before adoption. The test-only option uses the same two segregated paths with a second eight-hour test shift: 52 incremental staff, USD 1.5m direct lighting/training plus 7% EPC, and indexed payroll before and after fares. It does not repair upstream stage throughput. The single structural, electrical and composite options add four staffed hours/day to the named stage at unchanged bay count; the coordinated option applies this to all seven stages and funds the second test shift. Incremental production FTE is half each selected stage crew, paid at a 25% premium, with nonlabour shift costs equal to 25% of added payroll. Each stage adds a USD 1m installation/training allowance and USD 10,000 per added FTE plus EPC. These assumed shift efficiencies, relief and wage premiums need qualification.

Use `availability_75pct_costed`, `hiring_ramp_costed` and `supplier_shortage_costed` as matching delay-cost baselines for their respective recovery options. Recruitment recovery funds USD 10,000 per delayed half of the 1044 production positions plus EPC and tests a six-month rather than twelve-month staffing ramp. Supplier recovery charges an assumed IQD local logistics fee of 2% of city imported invoice value, indexed to payment, and tests a shortage cut from 130 to 65 working days; this is a causal scenario assumption, not a guaranteed delivery improvement. Foreign-currency freight reimbursement and invoice eligibility need quotations. Neither purchases another train nor credits an unspecified subsidy.

The temporary first-article option adds USD 35m direct facility/tooling and 7% EPC, ready at day 260 (12 planning months from NTP), plus 60 incremental support FTE until the permanent plant is ready. Only the already-planned prototype assemblies there. Acceptance still waits for permanent bays and running paths at day 390, and series still waits for first-article qualification. All line fleets remain complete at opening. The temporary site and staff must have their own RFQs and acceptance; no shorter passenger section is folded into this option.

Delay costs and combined financial downside

Original delivery-only cases remain lower-bound comparisons. `Combined_delay_costs` additionally prices extended direct production staffing plus 100 support FTE at the existing Iraqi labour proxy; it compares staffed span lengths so a pure start delay does not buy the same crew-months twice. Plant storage/insurance/utilities carrying uses 1% of direct plant capital per extra/idle year excluding wages. Civil prolongation adds 150 supervision FTE and nonlabour site overhead at 0.5% of city capital per year beyond baseline civil completion. All these incremental operating costs are charged monthly at 5% annual indexing; baseline train labour/materials already inside procurement are not repeated. Rework prices 83 affected trains at USD 25,000 each. Monthly incremental-cost files reconcile each allowance to the case OPEX ledger. Contingency, tax and contractor claims remain unquoted; these are not a funded risk reserve.

The downside ladder isolates 30% fewer paid trips plus 25% lower existing retail/advertising receipts, 5% annual capital escalation applied to each invoice at actual payment month from financial close, and weaker financing. Weaker financing removes assumed green enhancement, climate grants, development rights and additional net income, increases core coupons two percentage points, and replaces the 2%/IQD 13tn gap sensitivity with 8% credit at a 1% draw fee and IQD 4tn maximum outstanding. Chinese credit and government import cash remain USD, all domestic credit/bonds/cash IQD; government remains 25% of escalated capital. Separately, `domestic_placement_interrupted_recovered` and `export_credit_delayed_recovered` stop procurement/construction/production for 130 working days and price local remobilisation plus prolongation; repayment is conditional on subsequently placing the refused debt. Permanent refusal has no opening or repayment date: see the [funding gates and evidence execution package](#), including denied amounts, escrow requirements and six-month placement shortfalls. No gap facility is used to conceal a refused core source.

Lower demand with combined delays leaves IQD 0.565tn terminal gap debt. `Joint_downside` applies all of those assumptions together and 7% rail OPEX inflation: IQD 26.862tn cumulative uncovered cash and IQD 4.000tn terminal gap debt. Uncovered support is a balancing requirement, not an extra government appropriation, loan or cash source. Its presence blocks any unconditional repayment claim even if the simulated debt eventually amortizes. Interest totals in such cases also assume the missing cash is supplied; they do not establish an executable financed programme.

Base fare and OPEX sensitivities remain 5% from financial close, with existing kiosks/advertising, separate receipts and each line revenue ramp included. No tickets are sold before opening. Capital escalation is absent from reference cases and explicit in the named escalation cases. Green/grant/rights and all financing availability remain uncommitted. No future national city cashflow supports Baghdad debt.

Evidence needed before adopting a recovery plan

The [qualification register](#) assigns owners, required measurements/RFQs and acceptance rules for production, suppliers, recruitment, civil quantities, depot/site fit, demand, funding and operation. Every row remains not demonstrated. No measurements or quotations were obtained by running this model. The baseline still has only 20 working days of fleet margin and 2.71% test-throughput margin. Shorter passenger sections require a separate route, turnback, charging, depot, fleet duty and safety/operating acceptance study; the present comparisons retain full fleets and existing operating scope.

Regenerate after the city controls, factory plan and Baghdad funding programme are current with `.venv/bin/python tools/automation/baghdad_delivery_stress.py`, then regenerate the proposal. Verify source/output bindings with `--check`.

[Scenario comparison](#) · [Machine-readable assumptions/results](#) · [Baseline half-year funding](#) · [Combined stress half-year funding](#) · [Joint downside half-year funding](#)

Baghdad deployment gaps

Source: [cities/catalogue/west-asia/Iraq/Baghdad/engineering/deployment/README.md](#)

3 closed checks; 14 open gates. Status follows the linked evidence; generating a receipt template does not count as receiving field data.

Soil inputs include a route/station investigation plan.

Gate	Status	Responsible function	Closure work
soil-desktop-inputs	closed	civil/geotechnical designer	Sample current station and civil geometry from the pinned soilDB source and retain means, uncertainty and nodata.
soil-coverage	open	geotechnical investigator	Fill mapped coverage gaps through local records and targeted sampling; keep estimated and measured records distinct.
control-processing	open	survey team	Receive survey control observations; process and check the project CRS, datum and residuals.
ground-model	open	survey team	Receive and inspect the terrain, point cloud and ground-model deliveries against independent control.
surveyed-alignment	open	alignment designer	Replace or confirm generated geometry with checked survey alignment and platform reconciliation.
route-station-fit	open	civil/station designer	Resolve utilities, land/access, flood levels, station access and construction staging against the current route.
drainage-ground	open	civil/geotechnical designer	Use the soil investigation plan to collect local geotechnics and groundwater, size foundations/treatment, and check drainage with local rainfall and surveyed levels.
structural-release	open	structural designer and checker	Complete route-specific support/span, foundation, movement and erection calculations and close checking comments.
model-timing-comparison	closed	simulation engineer	Reconcile current native reference journey times with SUMO using the same scenario dwells and route.
full-service-validation	closed	simulation engineer	Run current nominal and degraded full-service cases against the current scenario and simulator; retain failures and exact provenance.
operating-release	open	operator	Complete conflict-aware capacity and degraded-operation review and sign the bound operating evidence.
morning-fleet-allocation	open	service planner	Reconcile revenue fleet with required morning departures on each line; retain two station berths and same-line overflow storage.
continuous-stabling-replay (evidence missing)	open	simulation engineer	Run the selected line-local station/depot candidate across consecutive evenings and synchronised morning starts.
stabling-physical-fit	open	track/station designer	Locate usable station and line-local depot tracks, shared charging, isolation, inspection access and protected morning release slots.
solar-storage-endurance	open	energy designer	Bind declared station/ROW/dedicated PV to site storage and actual charging duty, reconcile conversion losses and prove replenishment across adverse weather; specify residual backup duty explicitly.
depot-placement-and-budget	open	depot and cost designer	Place the declared PV/storage inventory within the controlled site layout and reconcile itemised installed costs and renewal scope with existing allowances.
workforce-duty-and-workload	open	city operating lead	Reconcile the funded role plan with named shift/leave cover, measured inspection and cleaning workloads, simultaneous service-depot arrivals and actual local employment inputs.

city civil and operating deployment evidence; manufacturing and system certification remain in their own release registers.

baghdad depot scope reconciliation

Source: cities/catalogue/west-asia/Iraq/Baghdad/engineering/depot-scope/README.md

Depot energy quantities reconciled: yes. Physical/cost/stabling closure: open.

The policy assigns two revenue trains per selected powered station for coordinated morning starts and the remaining fleet to storage on its own line. Depot storage tracks are sized separately from maintenance bays; see the [station/depot allocation](#). The dispatch table below diagnoses the current simulator initialization; it is not a proposed overnight parking allocation or a requirement for more depots.

Depot station	PV kWp	Storage modules / kWh	Required / reference PV area m²	Additional equipment reference USD
line-6-0495-1669-s057516	5,000	80 / 40,000	33,333.3 / 4,000	5,930,000

The equipment reference is an unapproved sensitivity using existing repository rates. It is not added to CAPEX; allowance inclusion and installed scope remain unverified.

Initial dispatch station	Line	Trainsets	Train-body length m	Slot length with clearance m	Physical slots
line-1-2500-0775-s000000	line-1	48	5,328.0	5,808.0	unverified
line-1-0166-0866-s053822	line-1	47	5,217.0	5,687.0	unverified
line-2-1987-2192-s000000	line-2	52	5,772.0	6,292.0	unverified
line-2-0595-0296-s056994	line-2	51	5,661.0	6,171.0	unverified
line-3-2271-1120-s000000	line-3	54	5,994.0	6,534.0	unverified
line-3-0151-1637-s055819	line-3	54	5,994.0	6,534.0	unverified
line-4-0146-0199-s000000	line-4	43	4,773.0	5,203.0	unverified
line-4-1246-1628-s044443	line-4	42	4,662.0	5,082.0	unverified
line-5-1199-0260-s000000	line-5	49	5,439.0	5,929.0	unverified
line-5-0677-2304-s053737	line-5	49	5,439.0	5,929.0	unverified
line-6-2322-0251-s000000	line-6	56	6,216.0	6,776.0	unverified
line-6-0495-1669-s057516	line-6	55	6,105.0	6,655.0	unverified
line-7-1069-2206-s000000	line-7	42	4,662.0	5,082.0	unverified
line-7-1531-0418-s043502	line-7	42	4,662.0	5,082.0	unverified
line-8-2231-1783-s000000	line-8	50	5,550.0	6,050.0	unverified
line-8-0785-0290-s051366	line-8	50	5,550.0	6,050.0	unverified
line-9-0565-0620-s003116	line-9	24	2,664.0	2,904.0	unverified
line-9-0700-0568-s098647	line-9	23	2,553.0	2,783.0	unverified

- Operating PV/storage capacities are retained assumptions, not validated depot sizing. Reassess depot and station energy duties after distributed overnight placement and charging are represented.
- PV area is module area at the catalogue 0.15 kWp/m², not gross land area; setbacks, access and packing require layout.
- Battery footprint and separation require supplier/fire-engineering data; no area or approval is invented.
- Dispatch queues are initial simulator placements, not an overnight operating plan. Slot lengths use one train plus 5 m at each end from RFC 0014.
- Workshop bays and passenger platforms are not credited as verified overnight stabling slots.

Required closure records: a supplier equipment/footprint schedule; located PV and battery layout; itemised budget with explicit baseline inclusions; a station-by-station healthy-fleet allocation with track IDs, usable lengths, assigned trainsets, charger sharing, security/isolation and access; and coordinated morning departures with conflict-aware evening/morning repositioning evidence.

Baghdad detailed component register

Source: cities/catalogue/west-asia/Iraq/Baghdad/engineering/detail/README.md

Generated from the current city design, six-car profile, engineering parts and software/ERP contracts. Engineering release remains false. Cost and finance rates remain unchanged; unknown unit costs are open rather than zero.

Reference allocation: 831 trainsets, 4,986 cars, 9,972 bogies, 182 stations, 158 energy sites and one shared plant.

See [complete parts CSV](#), [source-bound register](#), [dimensioned ST6 reference](#) and [design and Iraqi manufacturing plan](#). The two-bogie-per-car allocation and repeated mechanical kit quantities are reference architecture, subject to Metro-6car family release. Nested wheelsets, clips and batteries are not extra charges on top of complete bought assemblies.

Track and parts quantities

The rail reference covers net mainline running length only. Actual procurement additionally needs alignment method zones, switches, stabling, depot track and cutting/weld/spares schedules. Slab panel totals remain unassigned until constrained zones are measured; rail-seat child quantities require a seat installation schedule.

ID	Part / assembly	Basis	Per basis	Network reference	Route
TR-01	60E1 reference running rail	route-km (m)	4000	2,066,070	import-or-qualified-local-source
TR-02	Resilient direct-fixation complete seat kit	rail-seat (each)	1	Open	import-first-local-assembly-later
TR-03	Elastic clips	rail-seat (each)	2	Open	import-qualified
TR-04	Rail pad / bonded resilient layer	rail-seat (each)	1	Open	import-qualified
TR-05	Lateral rail insulators	rail-seat (each)	2	Open	import-qualified
TR-06	Anchor assembly including insert, bolt and washer	rail-seat (set)	2	Open	local-inserts-after-qualification
TR-07	Vertical/lateral adjustment and levelling kit	rail-seat (set)	1	Open	local-machining-after-qualification
TR-08	OSR-ST6 precast slab panel	slab-panel (each)	1	Open	Iraqi-precast
TR-09	ST6 concrete, base and plinths only	slab-panel (m3)	5.0796	Open	Iraqi-batching
TR-10	ST6 reinforcement planning allowance	slab-panel (kg)	761.94	Open	Iraqi-cage-fabrication
TR-11	ST6 bedding / levelling grout	slab-panel (m3)	0.522	Open	local-qualified-material
TR-12	ST6 lifting sockets / engineered anchors	slab-panel (each)	4	Open	import-qualified-local-install
TR-13	Drainage channel and removable cover	route-km (m)	2000	1,033,035	Iraqi-precast
TR-14	Rail electrical continuity bond / monitored stray-current provision	project (set)	Open	Open	local-harness-import-terminals
TR-15	Turnout slab bearer, point drive, detection and locking kit	project (set)	Open	Open	mixed
TR-16	Rail/deck expansion interface and transition section	project (set)	Open	Open	mixed
TR-17	Utility ducts, draw pits, barriers and fencing	project (set)	Open	Open	Iraqi-manufacture
ME-01	Underframe and load-bearing car structure	car (each)	1	4,986	Iraqi-fabrication
ME-02	Modular FRP cladding, end frames and service rails	car (each)	1	4,986	Iraqi-composite-and-metalwork
ME-03	Powered bogie complete including frame, suspension and brakes	car (each)	1	4,986	import-first
ME-04	Trailer bogie complete including frame, suspension and brakes	car (each)	1	4,986	import-first
ME-05	Wheelset including wheels, axle, bearings and axleboxes	bogie (each)	2	19,944	included-in-bogie-import
ME-06	Motor and traction inverter/controller set	car (each)	2	9,972	import-qualified
ME-07	225 kWh nameplate traction battery allocation	car (each)	1	4,986	import-cells-local-pack-after-qualification
ME-08	Battery support, HV junction box, fuse, contactors and precharge	car (each)	1	4,986	mixed
ME-09	Battery/inverter liquid cooling loop	car (each)	1	4,986	local-assembly-import-pump-and-chiller

ID	Part / assembly	Basis	Per basis	Network reference	Route
ME-10	HV-to-LV DC/DC and battery-backed auxiliary distribution	car (each)	1	4,986	import-qualified-local-harness
ME-11	HVAC unit, ducts, filters, condensate and mounts	car (each)	1	4,986	mixed
ME-12	Passenger door package including leaves, drive, locks and detection	car (each)	4	19,944	import-qualified-local-fitting
ME-13	Side glazing, window frames, seals and adhesives	car (each)	Open	Open	import-glass-local-frame
ME-14	Intercar coupler, gangway, jumper and air/electrical interfaces	trainset (each)	5	4,155	mixed
ME-15	Train-end recovery coupler and sensor cowl	trainset (each)	2	1,662	mixed
ME-16	Brake supply/control and parking brake package	car (each)	1	4,986	mixed
ME-17	Roof PV, MPPT, mounts and protected DC wiring	car (each)	1	4,986	import-panels-local-mounts
ME-18	Interior, seats, grab rails, lighting and accessibility kit	car (each)	1	4,986	Iraqi-assembly
ME-19	Fire detection, alarm, segregation and suppression provisions	car (each)	1	4,986	mixed
ME-20	HV/LV/data harness, connector and clamp schedule	car (each)	1	4,986	Iraqi-harness-production
ME-21	Service consumables, fasteners, fluids and replacement spares	project (each)	Open	Open	mixed
EL-01	T-ECU/S end host assembly	trainset (each)	2	1,662	import-electronics-Iraqi-integration
EL-02	T-ECU/A application end host assembly	trainset (each)	2	1,662	import-electronics-Iraqi-integration
EL-03	T-OBS end host and external sensor assembly	trainset (each)	2	1,662	import-electronics-Iraqi-integration
EL-04	Safety evaluator MCU channel	t-obs (each)	2	3,324	import-electronics-Iraqi-integration
EL-05	Independent hardware heartbeat watchdog	t-obs (each)	2	3,324	import-electronics-Iraqi-integration
EL-06	Channel power conversion and voltage monitoring branch	t-obs (each)	2	3,324	import-electronics-Iraqi-integration
EL-07	Application compute module and carrier	t-obs (each)	1	1,662	import-electronics-Iraqi-integration
EL-08	Radar evaluation assembly and CAN interface adapter	t-obs (each)	1	1,662	import-electronics-Iraqi-integration
EL-09	Lidar HAP TX candidate and data/power harness	t-obs (each)	1	1,662	import-electronics-Iraqi-integration
EL-10	Stereo camera, carrier-compatible flex and rigid calibration bar	t-obs (each)	2	3,324	import-electronics-Iraqi-integration
EL-11	Ultrasonic sensing position	t-obs (each)	4	6,648	import-electronics-Iraqi-integration
EL-12	Isolated peer cross-check SPI link	t-obs (each)	1	1,662	import-electronics-Iraqi-integration
EL-13	Hardware series permission contacts and independent coil drivers	t-obs (each)	1	1,662	import-electronics-Iraqi-integration
EL-14	HV/LV input surge protection and branch fuse assemblies	t-obs (each)	1	1,662	import-electronics-Iraqi-integration
EL-15	Enclosure, heat spreader, glands, vents and grounding hardware	t-obs (each)	1	1,662	import-electronics-Iraqi-integration
EL-16	Train network switch, isolated CAN and external Ethernet controller set	trainset (each)	Open	Open	import-electronics-Iraqi-integration
EL-17	Train-to-ground radios, modem carrier, antennas and surge arresters	trainset (each)	Open	Open	import-electronics-Iraqi-integration
EL-18	W-SBC field cabinet and point/crossing I/O	project (each)	Open	Open	import-electronics-Iraqi-integration
EL-19	S-SBC station host and protected UPS/network enclosure	station (each)	1	182	import-electronics-Iraqi-integration
EL-20	Operator/server cluster, backup and disaster recovery	project (each)	Open	Open	import-electronics-Iraqi-integration

ID	Part / assembly	Basis	Per basis	Network reference	Route
CI-01	Pi20/Pi25 beam, pier cap, column and foundation package	project (each)	Open	Open	Iraqi-precast-and-site
CI-02	Walkway, barrier, cable tray and access cassette	project (each)	Open	Open	Iraqi-fabrication
CI-03	Station accessible platform, ramp/lift/stair and canopy package	station (each)	1	182	Iraqi-civil-and-metalwork
CI-04	Station fire, water, drainage, power and earthing services	station (each)	1	182	mixed
CI-05	Ticket validators, vending, gates and settlement terminals	station (each)	Open	Open	mixed
CI-06	Station PA/PIS, CCTV, help points and telecom cabinet	station (each)	1	182	mixed
CI-07	Charging contact, interlock, protection, DC/DC and site battery	site (each)	1	158	import-core-local-assembly
CI-08	Depot lifts, pits, wheel service, wash and recovery equipment	project (each)	Open	Open	mixed
CI-09	Precast plant moulds, cage jigs, batch/curing and survey tools	plant (each)	1	1	Iraqi-fabrication-import-special-tooling
CI-10	Rail weld/clip/anchor tools and track survey trolley	plant (each)	1	1	mixed
CI-11	Electrical bench, harness formboards and EMC/environmental test access	plant (each)	1	1	mixed

Corrected onboard power envelope

T-OBS simultaneous output allocation is 71.50 W; at 90% efficiency, input is 79.44 W. A 25% sizing margin requires 99.31 W, selecting a 100 W reference capacity. At the assumed 18 V minimum input, envelope current is 4.41 A, or 5.52 A including capacity margin. These are capacity assumptions, not energy-consumption measurements or qualified fuse/converter ratings.

Software host allocation

All 60 workspace packages have exactly one disposition record. This verifies inventory, not physical HAL, sensor driver, firmware timing or operational safety.

Component	Disposition	Hosts
osr-afc	composed-service	s-sbc
osr-afc-backoffice	deployable-endpoint	o-srv
osr-alignment	shared-library	design-tooling
osr-analytics	deployable-endpoint	o-srv
osr-ato	composed-service	t-ecu-a
osr-atp	composed-service	t-ecu-s
osr-aux-power	composed-service	t-ecu-a
osr-balise	composed-service	w-sbc
osr-bms	composed-service	t-ecu-s
osr-brake	composed-service	t-ecu-s
osr-cbm-backend	deployable-endpoint	o-srv
osr-cbm-onboard	composed-service	t-ecu-a
osr-city-studio	tooling	design-tooling
osr-consensus	composed-service	w-sbc, o-srv
osr-core	shared-library	all
osr-crypto	shared-library	all
osr-derailment	composed-service	t-ecu-s
osr-design	tooling	design-tooling
osr-door-control	composed-service	t-ecu-s
osr-energy-site	composed-service	w-sbc
osr-event-recorder	composed-service	t-ecu-a
osr-fire-safety	composed-service	t-ecu-s
osr-gui-shared	shared-library	o-srv, simulation
osr-historian	deployable-endpoint	o-srv
osr-hot-axle	composed-service	t-ecu-a

Component	Disposition	Hosts
osr-hot-axle-wayside	composed-service	w-sbc
osr-hvac	composed-service	t-ecu-a
osr-interlocking	composed-service	w-sbc
osr-intrusion-detect	composed-service	w-sbc
osr-level-crossing	composed-service	w-sbc
osr-lifecycle-identity	shared-library	o-srv, design-tooling
osr-lighting	composed-service	t-ecu-a
osr-obstacle-detect	composed-service	t-obs
osr-onboard-routing	composed-service	t-ecu-s
osr-passenger-assist	composed-service	t-ecu-a
osr-occ	deployable-endpoint	o-srv
osr-occ-gui	deployable-endpoint	o-srv
osr-odometry	composed-service	t-ecu-s
osr-pis-onboard	composed-service	t-ecu-a
osr-pis-station	composed-service	s-sbc
osr-proto	shared-library	all
osr-psd	composed-service	s-sbc
osr-ptp	shared-library	t-ecu-s, t-ecu-a, t-obs, w-sbc, s-sbc
osr-regen	composed-service	t-ecu-a
osr-routing	shared-library	o-srv, design-tooling
osr-safety-case	tooling	assurance-tooling
osr-secbus	shared-library	all
osr-selftest	deployable-endpoint	t-ecu-s, t-ecu-a, t-obs, w-sbc, s-sbc
osr-sim	simulation	simulation
osr-sim-gui	simulation	simulation
osr-station-scada	composed-service	s-sbc
osr-supervision-contract	shared-library	o-srv, simulation
osr-t2g	composed-service	t-ecu-a
osr-tcms	composed-service	t-ecu-a
osr-runtime	shared-library	design-tooling
osr-tcn	shared-library	t-ecu-s, t-ecu-a, t-obs
osr-traction	composed-service	t-ecu-s
osr-trainset-image	deployable-endpoint	t-ecu-s, t-ecu-a, t-obs
osr-tvm	composed-service	s-sbc
osr-wayside-points	composed-service	w-sbc

ERP coverage

Validated workflow contracts: assignment, budget, maintenance, manufacturing, quality, replenishment, service, training.

Enabled profile instances: training/city-induction. Other workflows exist in the app but require real deployment inputs before enablement. A valid reusable profile does not mean that production orders, maintenance assignments or capital budgets have been activated.

Profiles/templates only; legal company, physical bindings, approved schedules, suppliers, stock and production BOMs remain deployment-owned. Reference parts are not auto-created as live ERP Items.

Source hashes bind this report to its exact inputs. Rebuild with `.venv/bin/python engineering/baghdad_detail.py`; verify with `--check`.

Baghdad mixed joint-stock holding and ordinary-equity study

Source: cities/catalogue/west-asia/Iraq/Baghdad/engineering/equity/README.md

As of 2026-10-04. This is a proposed Iraqi holding company with 100%-owned rail, station-development, energy and manufacturing subsidiaries, not an incorporated company, offering, approved listing or committed funding. Baghdad alone supplies the order book. Future national projects remain uncontracted and supply zero revenue, collateral and terminal sale proceeds.

Ownership, new money and currencies

Government still provides USD 1.970123397bn equivalent: 25% of the original USD 7.880493587bn rail/energy/factory capital, rather than 25% of shares or of the enlarged property programme. Reclassifying that existing contribution as ordinary equity generates zero additional cash. At the same subscription price, ownership is government cash divided by government plus private subscribed cash. Borrowing earns no ordinary shares.

Transaction	Gross primary USD m eq	Company net primary USD m eq	Final government shares	Final Iraqi ownership	Government secondary sale USD m eq
primary_500m	500.0	490.0	79.76%	93.93%	0.0
primary_1000m	1000.0	980.0	66.33%	89.90%	0.0
primary_2000m	2000.0	1960.0	49.62%	84.89%	0.0
primary_1000m_premium	1000.0	980.0	67.28%	90.18%	0.0
secondary_500m	0.0	0.0	74.62%	92.39%	500.0

The share unit is an illustrative IQD 1 par and IQD 1 subscription price, not an accepted company valuation or legal share denomination. The premium sensitivity prices the later 25% issue at IQD 1.2, recording the difference in share premium and reducing dilution. Private subscriptions are 70% Iraqi and 30% foreign as an explicit scenario. Every paid-capital month observes the model's 25% state floor and 51% Iraqi floor; actual eligibility and beneficial ownership require counsel. Pending private subscriptions wait for sufficient paid government shares instead of treating promised government capital as paid. No shares or cash are credited for undelivered subscriptions. Land/concession/open designs are not priced into paid-up equity.

The government_grant_reference and government_equity_reference have identical debt, cash and physical costs before distributions. Government grants are a separately identified capital reserve without shares; cash-equity contribution is paid-in ordinary equity. Guarantees and public-service appropriations are uncommitted and never booked as subscription cash. A secondary sale transfers existing government shares: the seller receives the price, the company receives zero, and borrowing requirements remain unchanged. The secondary case omits sale fees as an explicit sensitivity; no government sale is authorised.

Original imported parts remain USD 1.793059933bn: government USD cash USD 896.529966m and Chinese USD credit USD 896.529966m. Only Chinese credit is USD debt. Remaining government cash, share subscriptions, bonds, bank/gap credit, fares, property proceeds and dividends are IQD. USD figures for those items are reference conversions at 1,300 IQD/USD, without a dollar-indexed obligation or promised conversion. Original USD capital intensity is 22.75%; local procurement prices still require actual imported-content and FX quotes. The additional USD 2.4bn-equivalent property construction is assumed IQD and makes physical group capital USD 10.280493587bn. It does not add government cash beyond the original 25% rule.

Conditional results and investor returns

Case	Actual gross primary USD m eq	Conditional first/full month	Peak IQD gap tn	Missing IQD cash tn	First dividend month	Private annual equity IRR
government_grant_reference	0.0	41/83	5.455	0.000	None	undefined
government_equity_reference	0.0	41/83	5.455	0.000	371	undefined
primary_500m	500.0	41/83	3.874	0.000	359	2.95%
primary_1000m	1000.0	41/83	2.631	0.000	335	3.00%
primary_2000m	2000.0	41/83	1.302	0.000	263	3.13%
primary_1000m_premium	1000.0	41/83	2.631	0.000	335	2.90%
secondary_500m	0.0	41/83	5.455	0.000	371	3.25%
undersubscribed_1000m	53.9	Blocked	0.000	0.000	None	undefined
failed_later_primary	750.0	Blocked	1.365	0.000	None	undefined
delayed_1000m	1000.0	47/89	2.479	0.000	323	3.28%
joint_downside_1000m	1000.0	55/106	4.000	39.132	None	undefined
joint_downside_undersubscribed	54.1	Blocked	0.000	0.000	None	undefined
aggregate_tax_proxy_1000m	1000.0	41/83	1.717	0.000	299	3.57%
coverage_dividends_1000m	1000.0	41/83	2.662	0.000	335	2.99%
rental_small_1000m	1000.0	41/83	2.676	0.000	335	3.02%
rental_medium_1000m	1000.0	41/83	2.646	0.000	323	3.22%
rental_medium_coverage_1000m	1000.0	41/83	2.669	0.000	323	3.21%
rental_medium_downside_1000m	1000.0	41/83	2.754	0.000	335	2.88%

The \$1bn primary case places IQD 1.300tn gross and IQD 1.274tn net after 2% issue fees. Peak IQD liquidity falls from 5.455tn in the matched holding-company equity reference to 2.631tn. These references pool four wholly owned businesses and include a conservative tax stress; they are different from the existing 4.461tn rail planning baseline and the independently financed 8.450tn SPV alternative. The original SPV private subscriptions of IQD 1.401tn are replaced, not added to holding subscriptions.

Private equity IRR for the \$1bn case is 3.00% nominal, with a 15% hurdle NPV of USD -507.474m for the 70% Iraqi private cohort. Its first dividend is month 335; ordinary shares have no guaranteed redemption or principal payment. With 5% general inflation, a roughly 3% nominal return does not establish an attractive real return. Actual ordinary dividends, retained profits, reserve balances, shareholder payments and ownership are exported monthly. IRR is undefined without positive shareholder receipts, or for ambiguous multi-sign cash flows; blocked-case NPV covers only funded investment to suspension and omits unresolved recovery/termination, so it is not a complete investor downside valuation.

Planned unlevered group resource NPV remains USD -2.945bn, or USD -3.412bn after the assumed unverified USD 600m land opportunity cost. It excludes debt/equity transfers, corporation tax and issuance fees and reconciles to the prior consolidated resource case. Debt replacement changes investor cash and finance costs, not this before-financing resource value. Blocked scenarios retain this planned-scope diagnostic while reporting only actual funded-prefix capital; no unbuilt assets or future earnings are presented as delivered. Financial resource NPV excludes wider social/economic benefits such as travel time and local employment, which need separate evidence.

Subscriptions, six-month placements and release gates

Founder/anchor subscriptions request 75% in six equal calls at months 0, 6, 12, 18, 24 and 30. The remaining 25% is a conditional primary issue at month 60, which may be private rather than listed; no admission is assumed. Actual settlements may extend beyond the requested dates to preserve the paid state floor. [primary_1000m-six-months.csv](#) reports actual subscriptions, net fees, restricted capital cash, government USD/IQD cash, native capital-credit draws, dividends and gap requirements every six months. Each monthly row reconciles physical invoices against capital sources; unspent subscription cash is restricted until construction invoices complete.

Net equity displaces the residual 25% bank/75% bond capital basket, bank first in each invoice. Domestic bank principal is amortised over 240 months and bond principal over 216 months, starting no earlier than full-network opening. This group-wide grace is more conservative than per-line cohorts and needs actual facility documents. Reference coupons stay 9%/8%; China retains 5%, 48-month draw grace and 180-month amortisation. Interest is paid throughout construction; arrangement fees, undrawn Chinese commitment charges, prepayment lock-ins/premiums, six months of debt service reserve, three months of OPEX buffer and factory warranty cash are included. The separate IQD 2% gap sensitivity has a 0.5% draw fee and 13tn cap; downside uses 8%, 1% and 4tn. None is committed.

Undersubscription and failed later issues hold the fully subscribed case's six-month capital-debt envelopes fixed. If the missing shares require capital credit beyond that envelope, the next physical invoice is withheld and the case reports no opening. The operating gap facility cannot cover refused capital. Already funded costs and debt remain; there is no forgiveness or assumed rescue. Resumption requires newly executable finance, rephasing, priced hold/restart and physical acceptance; resolution costs are unmodelled. The six-month delay sensitivity instead moves financial-close-to-NTP work and subscriptions together, retains fixed nominal property sales and indexes operations from close; it is a conditional whole-programme deferral, not a priced mid-construction suspension. Joint downside uses the existing 55/106-month physical stress, fare/nonfare shortfall, higher rates/inflation and weaker property sales, without uncontracted extra income.

Retained viaduct rentals, dividend policies and terminal cash

The [civil-linked rental portfolio](#) adds distinct, retained under-viaduct enclosures to the existing station-area sales. No sold parcel also supplies rent; actual title/overlap, height, fire/access and local tenant demand remain unaccepted. The 200,000 m² illustration exceeds the reference geometry screen and is omitted from group funding. Smaller cases include their own IQD fit-out capital, collected rents, rent-free/occupancy ramps, arrears/write-offs, vacant costs, insurance, refurbishment and cohort depreciation. Existing kiosk income is unchanged. Cash corporate tax is calculated once inside the group model; the rental ledger's separate tax diagnostic is not added again. Tenant deposits remain matched restricted cash and liabilities, supplying zero capital or dividends. Government original cash and Chinese USD imports remain unchanged; fit-outs borrow the domestic residual and use the existing parent subscription, with no additional partner cash in this 100%-owned alternative. Rental cases extend the domestic principal gate to the latest fit-out handover if it follows full-network opening; the conditional railway opening dates remain unchanged.

The medium rental case gives peak IQD gap debt 2.646tn, first dividend month 323, private nominal IRR 3.22% and planned before-tax resource NPV after existing station land USD -3.395bn. Additional corridor rights/opportunity cost is unknown, so this is not a complete new-land appraisal. The lower-rent/prolonged-vacancy case retains its costs rather than assuming free premises. Annual rental resource gains and investor gains are separately quantified.

Default ordinary dividends still require full debt repayment. coverage-and-reserves is an unapproved lender-policy sensitivity: annual distributions require 12 actual and 12 projected months of coverage at least 1.30, completed capital, cumulative distributable-profit proxy, no missing cash/current gap draw, restricted capital exhausted, intact OPEX/warranty reserves, six months of forward scheduled debt service and 12 months of upcoming rental refurbishment. CFADS excludes property-sale receipts and adds back their transaction costs; debt coverage includes scheduled interest/principal/fees and gap interest/fees, excluding voluntary principal sweeps. Forecast debt uses actual remaining vintages without future prepayment; forecast tax conservatively assumes no depreciation/interest/loss relief. No upcoming construction draw is permitted by the completion gate. The revolving gap sensitivity has no stated contractual amortisation, so its interest enters coverage and any distribution still needs an actual maturity/covenant agreement.

Up to 50% of cash surplus is held for possible dividends before optional debt sweeps; the balance repays the most expensive eligible credit. Final distributions also preserve cash for accrued tax and a liabilities/book-equity ceiling of 3.0. These are conservative group proxies, not subsidiary upstream permission, a legal reserve calculation or signed lender consent. Compared with the default first dividend at 335, the policy case first pays in month 335, private IRR 2.99%. With medium rentals, its first dividend is 323 and private IRR 3.21%. Monthly ledgers expose the actual coverage numerator/denominator, reserves, profit, dividends and debt balances. Earlier dividends reduce sweeps and can increase future finance costs; they are not new profit.

The original \$1bn default case ends with USD 3.767bn-equivalent unrestricted cash. A separately labelled `terminal_cash_sensitivity` allocates available final cash after nonsecurity debt/tax liabilities, at current proportional shares. It changes no actual dividend, company cash or share count and assumes zero transaction cost as an explicit illustration. Private IRR including this unapproved final cash return is 4.22%, versus 3.00% from dividends alone. Restricted deposits, reserves, unresolved funding, unvalued property/rail/factory assets and quoted share-price appreciation are excluded. Failed/unfunded cases cannot present a completed-horizon cash exit. No redemption, liquidation authority or proceeds are guaranteed; counsel must assess capital reduction, tax, transaction and continuing-service obligations before any distribution of capital. Each case exports both ordinary returns and the independent cash-only diagnostic, avoiding double counting.

Pro-forma accounts and consolidation

The group eliminates PPA, internal train invoices/capacity fees and station-rights charges using [the existing intercompany ledger](#). Factory input cash moves to group train assets at its actual purchase date; original manufacturing payroll/QA/overhead remains counted once. Subsidiary equity is replaced by parent subscriptions. Central treasury access, subsidiary distributions/guarantees and lender consent are assumptions requiring explicit agreements; 100% ownership alone does not permit unrestricted transfers. Business operating records remain separately attributable in [group-resource-monthly.csv](#). These are cash-based pro-forma accounts, not audited statutory statements or proof of admission.

Rail, train, solar and factory assets depreciate over 30, 20, 25 and 20 years respectively, phased into operation; depreciation is a noncash charge. Factory remaining book assets are conservatively impaired at Baghdad backlog/warranty close; there are no later orders or salvage receipts. Property construction is inventory, released once to cost of sales as sold; sold space generates no extra rental revenue. Existing maintenance/battery renewals remain in cash OPEX, plus the earlier explicit solar/caretaker allowance; reserves are not spent twice. Land opportunity cost remains a separate in-kind appraisal exposure, not a cash injection, paid-up shares or free balance-sheet asset.

The 15% corporate rate is the Iraqi GCT headline non-oil rate, not a tax ruling. Base scenarios tax each positive annual business operating-profit proxy after assigned depreciation/COGS, retaining internal business charges solely for a conservative tax stress. They assume no parent-interest deductions, subsidiary loss netting, exemptions, refunds or carryforward relief. This yields IQD 3.486tn lifetime cash tax in the normal case. The separately labelled `aggregate_tax_proxy_1000m` sensitivity taxes positive annual consolidated profit including finance costs; group eligibility is unestablished. Neither is statutory taxable income. Customs, property-transfer taxes, transaction withholding, relief and deductible rates require a priced Iraqi tax opinion before these budgets are investment-ready. [GCT corporate declaration guidance](#).

Annual tax is accrued at year end and paid next month, including final stub settlement. Net assets reconcile against nominal capital, share premium, issue-fee deductions, net retained profit and dividends; cash and book residuals must stay below USD 0.02/0.05. Missing cash is explicitly separate assumed unfunded support, never new government equity. Dividends require cumulative positive distributable-profit proxy, no missing funding, all debt repaid, no unused capital cash and intact OPEX/warranty/DSRA reserves. A further 10% positive-profit retention is contractual sensitivity, not the statutory reserve rule. Audited distributable profit, legal reserves, subsidiary upstreamability and board approval remain additional gates even when the model shows a possible dividend.

Iraqi legal, governance and admission gates

[Companies Law Articles 7 and 7 bis](#) support a mixed company with initial state capital at least 25%, holding subsidiaries and annual consolidated accounts. Mandate, concession, incorporation, capital approvals and enforceable public-service obligations still require the competent Iraqi bodies. Publish ordinary commercial contracts separately from affordability/service subsidies; optional future availability support requires an appropriation outside the 25% construction rule.

The [official 2019 amendment, Articles 12 and 28](#) contains the relevant 51% Iraqi participation condition and a liabilities limit of 300% of capital plus other owners' rights, with adjustment powers. Application, current amendments, foreign-investor exceptions, entity-level calculation and permitted treasury structure require counsel. A consolidated screening proxy flags 215 months above that limit or with insufficient book equity in the \$1bn scenario. It is not a legal compliance certificate; passing a share-floor check does not clear this debt constraint. The model is therefore not an executable capital structure until legal/capitalisation resolution is accepted.

[Regular-market Regulation 6](#) normally requires two years/history and audits, at least 100 shareholders, equity at least paid-up capital, actual business and two years' distributable profit; a discretionary shorter history provision does not constitute admission. [Secondary-market Regulation 15](#) provides a one-year activity/audit route and equity of at least 50% of paid-up capital, alongside approvals and disclosures. Apply the [ISC September 2026 amendments](#) and seek the regulator's current interpretation before fixing any offering date. Actual shareholder count, audits, eligible profits, applications and approvals are absent. All cases set `company_incorporated` and `listing_approved` false; month 60 is a funding assumption, not a listing prediction.

Use an independent board/audit committee, published related-party prices and conflicts policy, minority pre-emption/approval protections, consolidated and subsidiary disclosure, audited capital registers, procurement oversight and an explicit public-service contract. Government control and future retail ownership do not themselves demonstrate these controls. Open licences remain intact: investors own businesses, assets and enforceable contracts, without any valuation premium for exclusive ownership of openly licensed designs. New land/concession contributions require independent valuation, title and separate dilution treatment; none is included here.

ISC's [2025 market report](#) records IQD 531.5485bn share turnover and IQD 23.673365tn market capitalisation. The old IQD 4.461tn planning gap is approximately 8.4 times annual turnover; turnover is not a ceiling on primary subscriptions. The model therefore requires anchor placement evidence for each tranche rather than assumed retail absorption. The \$1bn primary gross IQD 1.3tn is about 2.45 times that turnover, which is a scale comparison, not proof of capacity or incapacity. Sources checked 3 October 2026.

Evidence programme

Six new native ERP work packages require Iraqi corporate/tax counsel, public mandate and capital register, anchor/debt placement proof, audit/consolidation and dividend policy, minority/open-licence controls, and actual admission evidence. Owners, quotes, signatures and approvals remain pending. Due days are relative to an approved programme start. The ERPNext Task import contains draft descriptions and no invented live IDs; local import receipts remain private. Existing ten physical/funding qualification tasks and six financing-redesign tasks continue to apply. Model outputs do not accept any evidence or permit operation. No external investor solicitation, issuance or expenditure is authorised.

Regenerate: `.venv/bin/python tools/automation/baghdad_equity.py`; validate source/output hashes with `--check`. Commitments and accepted physical evidence are required before any construction/opening/dividend decision.

Baghdad factory sized to the city programme

Source: cities/catalogue/west-asia/Iraq/Baghdad/engineering/factory/README.md

One factory is sized for 831 complete six-car trainsets / 4,986 cars, with physical cells, crews and test paths. Facility readiness is 18 months from notice to proceed (working day 390), replacing the old 24-month assumption. Financial close precedes NTP by 30 working days; these are conditional offsets, not dated construction commitments.

The unchanged infrastructure resource model finishes on working day 1682. The sized flow accepts the final train on day 1662, 20 working days earlier. Full-network commissioning therefore follows infrastructure completion, rather than waiting decades for four generic train-production slots. All 831 trains remain in the planned scope; there is no smaller opening-fleet substitution.

Flow and physical capacity

Cycles explicitly represent a whole six-car trainset, with structural, electrical and fit-out allowances doubled from the old generic cycle. One eight-hour shift and 85% productive availability are assumed. Occupation is $\text{ceil}(\text{productive cycle} / \text{availability})$; availability is not deducted again from that output. The first complete train belongs to the 831 and carries an additional 60-working-day qualification allowance. Every remaining material-kit task depends on that first-article acceptance; no series release is assumed before day 556.

Stage	Productive days	Occupied days	Cells/bays	Direct crew
Baghdad trainset kitting	3	4	4	12
Baghdad six-car structural assembly	24	29	24	432
Baghdad six-car composite kits	10	12	10	80
Baghdad six-car body installation	2	3	3	36
Baghdad six-car electrical integration	20	24	20	240
Baghdad six-car fitout and static test	16	19	16	192
Baghdad trainset acceptance bays	12	15	13	52

The limiting steady capacity is 215.2 trainsets / 1291 cars per working year. The resource scheduler simulates each train's linked stages, finite lanes and first-article gate. Cells are calculated from the city's fleet and the remaining infrastructure window, then increased to the first feasible balanced allocation; this is a reproducible capacity design, not a proof of minimum land or minimum cost.

Production bays use a 135 m by 6.5 m envelope for the 111 m consist, not a released building module. Composite and kitting cells have separate floor assumptions. Process and support space totals 107,582 m². The site screen is 36.9 hectares, including circulation/storage and two independently segregated 2 km test paths. These are site-reservation assumptions pending geometry, braking, fire, access, geotechnical and safe-operation design. Dynamic acceptance bays are not independent running tracks: the separate path calculation permits only 221.0 trainsets/year, based on 16 exclusive track-hours/train and 85% path availability. A shared route cannot be counted twice. The test-path margin above limiting production is only 2.71%; the final fleet has only 20 working days of schedule margin. See the [frozen-resource disruption, costed second-shift recovery and civil acceleration study](#). Those stresses preserve the selected cells rather than resizing them to hide delay.

Direct cell staffing totals 1,044 positions per staffed shift before management, stores, maintenance, relief and shift coverage. It is a proposed resource requirement, not measured job creation. Manufacturing labour and materials are already in train procurement CAPEX; they are not added again to plant CAPEX or railway OPEX.

Line handover and the early-line constraint

Line	Full fleet	Infrastructure day	Fleet day	Opening month
line-1	95	682	772	41
line-2	103	807	897	46
line-3	108	938	1028	52
line-4	85	1040	1130	57
line-5	98	1158	1248	62
line-6	111	1338	1383	69
line-7	84	1395	1485	73
line-8	100	1515	1605	79
line-9	47	1682	1662	83

The factory completes alongside the overall city civil programme. Full-fleet openings are retained. Noncritical infrastructure now moves within its existing crew lanes and dependency graph towards fleet handover, with a 90-working-day target buffer; critical infrastructure retains its original completion dates. Line 1's infrastructure completion moves from day 219 to day 682, reducing the idle interval before its fleet from 553 to 90 working days. No rolling-stock date, task duration, resource count or opening date changes. This is a conditional investment-timing proposal: surveys, land, utilities, permits and contract dates require approval before deferring site work. [Original and rephased line reconciliation](#) preserves both sets of dates. Line priority follows the original infrastructure requirements, with actual asset identifiers preserved.

The opening calculation retains the separate three-month integrated commissioning allowance after line infrastructure, the full fleet and shared depot/control work. The 60-day first-article allowance and three-month line allowance cover different activities. Required approvals, surveys, physical tests and independent acceptance remain open; neither allowance constitutes accepted safety evidence or an approved construction calendar.

Eighteen-month facility construction sequence

Activity	Working days	Accountable role
Land/access, survey, design and approvals	60	Sponsor and owner engineer
Mobilisation, earthworks, foundations and underground services	70	Local civil contractor
Factory shell, lifting structure and utility backbone	70	Local structural and MEP contractors
Production cells, moulds, cranes and test-track construction	80	Production engineer and equipment suppliers
Equipment installation, calibration, staffing and training	70	Factory operator and quality manager
Factory process proving, safe-work and facility acceptance	40	Owner engineer and independent inspectors

These six sequential planning packages total 390 working days. Long-lead tooling, cranes, battery handling equipment and imported systems must be ordered early enough to install and prove within that window. Land/access and financing availability are required at NTP. A delay to those gates changes the programme; the schedule is not a contractor-backed promise.

Factory capital and cashflow

Physical budget allocation	USD million
buildings	75.307
serviced site	12.913
stage tooling	97.700
test tracks	12.000
shared utilities stores labs logistics	35.000
design training qualification	20.000
contingency	50.584

The physical envelope is USD 303.505 million direct, compared with the old module allowance of USD 299.160 million. The programme budgets the larger amount, USD 303.505 million plus USD 21.245 million EPC, counted once. The resulting capital increase is USD 4.649 million; added bays are not treated as free capacity. Rates and quantities are engineering allocations requiring Iraqi contractor/vendor quotations, not market-verified prices. Actual equipment origin must also reconcile to the programme's imported/local allowances.

Factory payments are re-timed to the 390-day facility programme and city train payments follow the new production tasks. Baghdad's monthly debt, interest, 5% fare/OPEX indexing, liquidity gaps, surplus repayments and six-month bond/loan requirements are recalculated from those dates. Government remains 25% of capital; import cash and Chinese credit retain the 50:50 USD split, with remaining funding in IQD. The existing aggregate plant import allowance remains 20% of direct plant capital, allocated within tooling/test equipment rather than buildings; supplier origin and the actual equipment mix require RFQs and eligibility checks. EPC retains its separate procurement-origin allowance. Future national cities do not enter this plant's production load or Baghdad cashflow.

Procurement and qualification requirements

Before release, obtain measured six-car labour routes and prototype cycle times; trainset-level supplier delivery schedules; released mould/fixtures drawings and duplication plans; lifting/handling and HV/battery fire segregation; stores, quarantine and rework capacity; inspection/calibration capacity; crew recruitment and training; both independent test-path designs; utility availability; and a land/industrial permit package. Composite cells cannot be mistaken for the number of panel moulds: the final panel design and cure/release conditions determine mould duplication. The hypothetical whole-kit cycle must be validated against that tooling count. Existing civil precast/slab yards retain their own production resources and are not assigned to train bays.

The 85% factor is a deterministic allowance for non-productive time, not a quantified programme contingency. Supplier disruption, multi-month qualification, redesign and civil delays need risk scenarios before investment decisions. Expansion beyond Baghdad needs a new demand and capacity assessment; a national module budget is not a promise of parallel national production.

[Cell and tooling register](#) · [831 planned trainset acceptances](#) · [Line handover reconciliation](#) · [Machine-readable calculation](#) · [Factory capacity inputs](#)

Regenerate after operations with `.venv/bin/python tools/automation/generate-factory-plan.py`; validate the same command with `--check`. Status: reference planning engineering, no construction or manufacturing release.

Baghdad — Iraq funding appraisal

Source: <cities/catalogue/west-asia/Iraq/Baghdad/engineering/finance/FUNDING-MODEL.md>

Generated from current city CAPEX and procurement milestones. All facilities and appropriations remain uncommitted.

Government capital is 25% of total city CAPEX, before Chinese credit is deducted. The residual after that contribution and proposed Chinese proceeds is split 75% IQD bonds / 25% IQD bank credit. Imported purchases use 50% government USD cash and 50% proposed Chinese credit, assuming full-basket eligibility is later qualified; the remaining government contribution is IQD. Financing fees, construction interest, reserves and operating/debt support are additional funding requirements, not included in the 25% capital contribution. See the [Baghdad-only programme](#) for the uncovered cash requirements if no additional government cash is available.

Construction cash runs through month 84; full-network operations start in month 84 after retention. This conservative rollout follows the current resource-constrained CPM, not a five-year promise.

Capital sources and uses

USD is the comparison unit below. Chinese credit is USD debt; domestic bonds and bank credit are IQD debt. Baghdad imported purchases are funded 50% government USD cash and 50% proposed Chinese loan; the remaining government capital and local cash are IQD. The full Baghdad import basket is assumed eligible pending supplier-origin and lender qualification. Other city appraisals retain their own assumptions. Fares and local operating payments are budgeted in IQD.

Capital source	Contract / cash currency	Native amount at model FX	USD equivalent
chinese export credit	USD	864,586,115.98	864,586,115.98
domestic bonds	IQD	4,682,166,156,652.92	3,601,666,274.35
bank credit	IQD	1,560,722,052,217.64	1,200,555,424.78
government import cash	USD	864,586,115.98	864,586,115.98
government local cash	IQD	1,331,654,769,103.36	1,024,349,822.39
Total city capital uses	Mixed	—	7,555,743,753.49

Chinese buyer credit is proposed for eligible Chinese component invoices only. Government contributes its configured capital share, including the eligible-invoice downpayment; IQD bonds and bank credit finance the residual. The conditional ledger also calculates cash needed for fees, construction interest, reserves and operating/debt shortfalls. That additional support is uncommitted and is an unfunded requirement if Baghdad public cash is capped at its 25% capital contribution.

Proposed instruments

Instrument	Currency	Rate assumed	Grace from draw	Repayment
chinese export credit	USD	5.0%	48 months	180 months
domestic bonds	IQD	8.0%	24 months	180 months
bank credit	IQD	9.0%	0 months	84 months

Chinese component allocation

These allocations divide existing imported budgets; they are not additional costs, supplier quotes or confirmed origin.

Bucket	Component	Assumed eligible invoice USD
civil	other_imports_pending_lender_and_origin_qualification	584,171,601.60
stations	other_imports_pending_lender_and_origin_qualification	181,580,000.00
depots	other_imports_pending_lender_and_origin_qualification	2,000,000.00
rolling_stock	bogies	195,451,200.00
rolling_stock	traction_batteries	146,588,400.00
rolling_stock	windows	24,431,400.00
rolling_stock	doors	48,862,800.00
rolling_stock	other_imports_pending_lender_and_origin_qualification	73,294,200.00
solar_plant	solar_panels_and_inverters	366,690,920.62
signalling	other_imports_pending_lender_and_origin_qualification	12,912,936.00
charging_microgrid	solar_panels_and_inverters	9,464,000.00
charging_microgrid	station_batteries	10,816,000.00
charging_microgrid	other_imports_pending_lender_and_origin_qualification	6,760,000.00
epc_overhead	other_imports_pending_lender_and_origin_qualification	66,148,773.75

Annual cash requirements

Year 1 begins at assumed financial close; NTP follows 30 working days later. Amounts below are USD millions. DSCR excludes subsidy; public cash includes capital, fees, interest, reserves and support.

Year	CAPEX	Revenue	OPEX	Debt service	Public cash	DSCR before support
1	311.74	0.00	0.00	10.01	89.56	0.00
2	753.10	0.00	0.00	51.12	243.29	0.00
3	1313.82	0.00	0.00	127.62	463.19	0.00
4	1525.14	0.00	0.00	246.38	635.93	0.00
5	1362.01	0.00	0.00	370.64	718.56	0.00
6	1518.19	0.00	0.00	503.00	890.76	0.00
7	771.74	0.00	0.00	628.71	825.71	0.00
8	0.00	223.27	180.57	672.38	970.89	0.06
9	0.00	334.90	180.57	685.49	530.65	0.23
10	0.00	446.53	180.57	667.02	385.86	0.40
11	0.00	446.53	180.57	629.50	342.89	0.42
12	0.00	446.53	180.57	586.71	300.02	0.45
13	0.00	446.53	180.57	543.38	254.39	0.49
14	0.00	446.53	180.57	502.55	223.01	0.53
15	0.00	446.53	180.57	495.08	229.11	0.54
16	0.00	446.53	180.57	495.08	229.11	0.54
17	0.00	446.53	180.57	495.08	229.11	0.54
18	0.00	446.53	180.57	488.10	214.07	0.54
19	0.00	446.53	180.57	457.87	171.00	0.58
20	0.00	446.53	180.57	402.67	101.27	0.66
21	0.00	446.53	180.57	319.25	16.36	0.83
22	0.00	446.53	180.57	231.91	0.00	1.15
23	0.00	446.53	180.57	137.96	0.00	1.93
24	0.00	446.53	180.57	48.12	0.00	5.53
25	0.00	446.53	180.57	17.90	0.00	14.86
26	0.00	446.53	180.57	2.68	0.00	99.14
27	0.00	446.53	180.57	0.00	0.00	—
28	0.00	446.53	180.57	0.00	0.00	—
29	0.00	446.53	180.57	0.00	0.00	—
30	0.00	446.53	180.57	0.00	0.00	—
31	0.00	446.53	180.57	0.00	0.00	—
32	0.00	446.53	180.57	0.00	0.00	—
33	0.00	446.53	180.57	0.00	0.00	—
34	0.00	446.53	180.57	0.00	0.00	—
35	0.00	446.53	180.57	0.00	0.00	—
36	0.00	446.53	180.57	0.00	0.00	—
37	0.00	446.53	180.57	0.00	0.00	—

Sensitivities

Scenario	Peak annual public cash USD m	Minimum operating DSCR	Peak uncovered monthly capital USD m
base low demand	970.89	0.06	0.00
high capacity use	890.76	0.25	0.00
capex plus 25 percent	1224.29	0.05	0.00
demand minus 40 percent	1060.20	-0.07	0.00
iqd depreciation 35 percent	808.76	0.06	0.00
commissioning delay two years	949.83	0.06	0.00
china credit unavailable	882.75	0.07	16.90
four year bullet bonds	1718.28	0.04	0.00
government payment delay six months	991.72	0.06	12.33
interest plus three points	1162.74	0.05	0.00
combined downside	1266.07	-0.04	0.00

Assumptions and evidence

- Working-day milestones use 260 working days/year and 30 working days before NTP; holidays and an approved local calendar are pending.

- No signed Chinese loan, supplier-origin qualification, guarantee, bond mandate, bank facility, appropriation or tax/duty assessment.
- Chinese origin and lender eligibility remain unqualified proxies. Default advance is 85%; Baghdad assumes all imports can qualify for 50% USD loan / 50% government USD cash. Additional imported categories require lender and supplier approval; neither assumption is a verified lender rule.
- Domestic bonds assume a proposed MoF sovereign IQD programme; municipal borrowing powers and market demand are not assumed.
- Long amortising domestic bonds are an appraisal target; the four-year bullet sensitivity exposes redemption without automatic refinancing.
- Construction cohorts need separately approved facilities. A long rollout does not imply a lender offers decades of draw availability.
- Interest is paid on drawn native balances; mid-month draw convention, monthly repayment, fees paid by government; no interest capitalisation.
- All cash is nominal; no inflation/escalation in base. IQD revenue and OPEX are translated at scenario FX; imported energy/parts OPEX needs a separate currency survey.
- Unquoted CAPEX budgets remain nominal USD planning values. IQD tranches and appropriations convert at draw-date FX; no automatic local supplier price benefit from depreciation is assumed.
- Income used for labour and fares is the retained country-finance planning proxy, not a verified current Iraqi household median or an agreed wage/fare contract.
- Government payment delays and unavailable China credit produce explicit capital cash gaps; no committed bridge credit is assumed.
- Restricted DSRA targets six times current monthly service and uses available project cash before requesting support. Support cash is a required contribution, not a commitment; repayments and reserve balances are conditional on it. Baghdad programme caps government capital at 25% and separately discloses the additional unfunded requirement. Future-service covenant testing is pending.
- Battery renewal reserve remains inside existing rolling-stock maintenance OPEX; no second battery CAPEX is added.
- Demand is capacity-led, not a surveyed forecast. Passing reconciliation does not demonstrate affordability or bankability.

Long construction schedules can leave much of the debt already repaid by government before full-network opening. An operating DSCR above one therefore does not establish self-financing or remove the construction-period public cash burden. Uncovered capital gaps are conditional funding requirements, not actual cash receipts.

See [editable assumptions](#), [monthly cashflow](#), [annual cashflow](#), and [machine-readable model](#).

The [IMF Article IV](#) provides the historical FX anchor. [CBI](#) describes its role as fiscal agent for MoF bonds. [China Exim](#) describes export buyer credit; numeric project terms remain assumptions.

Conditional phased-opening sensitivity

The base above withholds all fares until every capital milestone. This separate sensitivity opens completed lines after the editable commissioning lag and gates every line on shared system/depot work. The city-sized Baghdad production plant is unavailable for its first 390 working days (18 months from NTP), followed by the scheduled first-article qualification gate. Opening also requires actual civil, train, depot, power and safety acceptance, which is not established by this calculation.

Revenue and variable OPEX use each line's share of controlled trainsets. Each line has its own 50% / 75% / 100% revenue ramp. Fixed OPEX is 25% of the full-network annual budget from first opening; remaining OPEX scales with opened fleet. These are explicit uncalibrated sensitivity assumptions, not a ridership survey or verified staffing plan.

First / last planned opening: month 41 / 83 from financial close (month zero). City additional funding requirement: USD 5,743.48 million. The plant remains separate. Future surplus is retained; gross additional support is not net lifetime loss or discounted cost.

Line	Opening month	Revenue / variable OPEX share
line-1	41	11.43%
line-2	46	12.39%
line-3	52	13.00%
line-4	57	10.23%
line-5	62	11.79%
line-6	69	13.36%

Line	Opening month	Revenue / variable OPEX share
line-7	73	10.11%
line-8	79	12.03%
line-9	83	5.66%

[Phased monthly cashflow](#) · [phased annual cashflow](#)

Baghdad financing reconciliation and six-month placement programme

Source: [cities/catalogue/west-asia/Iraq/Baghdad/engineering/finance/FUNDING-RECONCILIATION.md](https://cities.catalogue/west-asia/Iraq/Baghdad/engineering/finance/FUNDING-RECONCILIATION.md)

Scope: Baghdad city and one Baghdad manufacturing plant. Samawah, Mosul and every other city are excluded. Month zero is an assumed financial close, not an approved date. All financing and additional revenues remain uncommitted.

The numbers that must add up

The original capital sources and uses reconcile. The confusion was combining upfront capital, repayment of that same capital debt, gross liquidity injections and later retained cash. The revised schedule also spreads city EPC overhead across direct works instead of paying the entire programme allowance against the initial five-day baseline-freeze task. Capital cost is unchanged. An independent implementation rebuilds every debt vintage, interest charge, principal payment and financing fee from native draws; it does not copy the reported support total.

Capital source	Native currency	Native amount, billion	USD equivalent, million
chinese export credit	USD	0.896530	896.530
government import cash	USD	0.896530	896.530
government local cash	IQD	1,395.671459	1,073.593
domestic bonds	IQD	4,888.494218	3,760.380
bank credit	IQD	1,629.498073	1,253.460
Capital uses / sources	Mixed	—	7,880.494

Government cash remains 25% of capital, with government USD import cash inside that contribution. Imports are financed 50% government USD cash / 50% Chinese USD loan. Every other loan and bond is IQD. An international green lender must provide confirmed IQD on-lending or a priced currency hedge to preserve that strategy. A green label does not supply additional money.

Lifetime operating / finance reconciliation	USD equivalent, million
Fare and existing nonfare receipts	13,876.580
OPEX	5,822.337
Repayment of originally drawn capital principal	5,910.370
Interest on original capital financing	4,328.382
Original financing fees	42.309
Net nominal lifetime liquidity deficit	2,226.818
Gross extra cash required as bills fall due	6,168.154
Later unrestricted cash retained if those requirements are met	3,941.337

USD 6,168.154m gross injections minus USD 3,941.337m retained cash equals USD 2,226.818m net deficit. Capital purchases are already financed by their capital sources; principal repayments belong to the lifetime cash ledger and must not be added to the construction budget again. Net reserve deposits are zero at the end. This is a nominal identity, not discounted viability, and assumes city/plant cash can legally be pooled

A zero-interest, zero-fee revolving bridge is only a mathematical lower bound. Real gap funding must charge interest and fees on the borrowing needed to carry early deficits, repay itself from later surplus and show any terminal debt. Later surplus cannot be counted both as retained cash and as loan repayment.

Tickets, kiosk rents and advertising already included

Full-network steady annual receipts	IQD billion	USD equivalent, million
Passenger tickets	547.335	421.027
Station shops and kiosk leases	12.984	9.988
Advertising space	20.175	15.520
Total existing operating receipts	580.494	446.534

Tickets use 415.49m annual paid trips at IQD 1,317 average yield. This is assumed capacity use, not calibrated demand. Shops/kiosks use 31,112 m² at IQD 39,520/m²/month and 88% occupancy; advertising uses 5,720 boards at IQD 345,800/month and 85% occupancy. Rates derive from the retained income proxy, not Baghdad lease quotes. These are gross rental/advertising receipts; dedicated concession costs, tenant demand and collection losses require appraisal.

All three receipts enter the existing operating cashflows and therefore already reduce the financing gap. The phased sensitivity scales the combined receipts by opened-fleet share and the separate 50% / 75% / 100% revenue ramp for each line. Retail and advertising demand may follow different ramps in practice. Proposed naming rights, telecom leases, parking,

development proceeds or new commercial receipts must be genuinely additional and net of costs; existing rents cannot be booked again or sold upfront while also retained in future revenue.

What gap finance actually does

The illustrative maximum outstanding facility is IQD 13.000 trillion (USD 10.000bn equivalent). This is a size assumption, not proof that the Iraqi banking market can place it. All available surplus after core obligations and reserves is swept to its repayment. No refinancing or debt forgiveness is assumed at the final maturity.

Scenario	Gross new liquidity draws, IQD tn	Peak facility balance, IQD tn	Uncovered cash beyond facility, IQD tn	Terminal unpaid facility, IQD tn
unfunded reference	0.000	0.000	8.019	0.000
green label only	0.000	0.000	8.019	0.000
zero cost liquidity bound	8.019	8.019	0.000	2.895
commercial gap credit	13.000	13.000	18.472	13.000
concessional gap credit	10.320	10.320	0.000	8.083
blended candidate	8.259	8.074	0.000	5.153
green concessional only	9.386	9.386	0.000	7.455

Reference and label-only cases have no gap facility: their uncovered column is the required external cash contribution, not an approved government top-up. With a facility, uncovered cash excludes the separate terminal loan balance; both need a solution. Gross draws can exceed the maximum outstanding commitment when a revolving facility is repaid and redrawn. Conditional ledgers do not establish that an unfunded payment occurred.

Commercial gap credit assumes 8% interest and 1.0% draw fees. Concessional gap credit assumes 2% interest and 0.5% draw fees. Neither is a term sheet or an available GCF line. Funding old debt-service deficits is not automatically eligible green use of proceeds.

Explicit blended sensitivity

Candidate climate expenditure is USD 2,278.55m of electric rolling stock, solar and charging provision, including USD 1,396.19m assigned locally. Climate benefit, supplier qualification and green eligibility require independent review. Candidate green bonds replace only domestic bonds attributable to those expenditures; civil works and unrelated overhead are not automatically relabelled.

The blended sensitivity replaces that eligible bond portion with proposed IQD green bonds/sukuk at 4%, 48 months' grace and 240 months' amortisation, with 0.75% arrangement fees and 0.5%/year enhancement fees on outstanding principal. The lower rate depends on a concessional or guarantee structure; a green label alone retains the original 8% terms and changes no financial outcome.

Additional targets are a USD 25m climate capital grant translated into IQD receipts, replacing domestic borrowing against eligible local capital after month 24; USD 300m net development-rights proceeds in 6 six-month instalments from month 24; and USD 25m/year genuinely new net local receipts at full opening, scaled with the opened fleet. These are editable targets, not appraisals, GCF approvals, developer contracts or statutory levies. Existing rents, advertising and fares cannot be counted again; transaction/development costs and displaced future rents must be deducted. Public land contribution also needs economic valuation.

Solving the model rather than declaring the targets sufficient requires approximately USD 113.78m/year additional NET receipts (IQD 147.92bn/year at full opening) to eliminate both uncovered cash and the terminal facility under the other blended assumptions. This is a break-even funding requirement, not a forecast or a recommended passenger fare increase. Any covenant, inflation, FX or demand change can raise it.

Variable tickets, annual price changes and OPEX inflation

The following cases retain the same conditional blended capital/grant/rights/2% IQD facility assumptions. They change fares, OPEX and income separately. A 5% tariff increase with flat costs is an optimistic sensitivity; the paired 5% OPEX case is the requested inflation comparison. None is an adopted fare, a lender commitment or an Iraqi inflation forecast.

Indices compound in annual steps from financial close, including the years before first opening. No tickets are sold before service starts; the nominal launch price reflects elapsed inflation. Existing retail/advertising are held flat unless the case explicitly indexes those rents. New net income and rights targets remain fixed nominal amounts. CAPEX escalation and future FX changes still require a separate full nominal appraisal.

Demand responds to fare relative to the indexed income proxy with an illustrative elasticity of -0.30, except the explicitly fixed-demand cases. Income growth is a separate assumption, not evidence that wages will keep pace. Discount-led demand growth is capped at 2.0 times the baseline, within the original practical capacity. There is no calibrated Baghdad

peak-switching or household demand model.

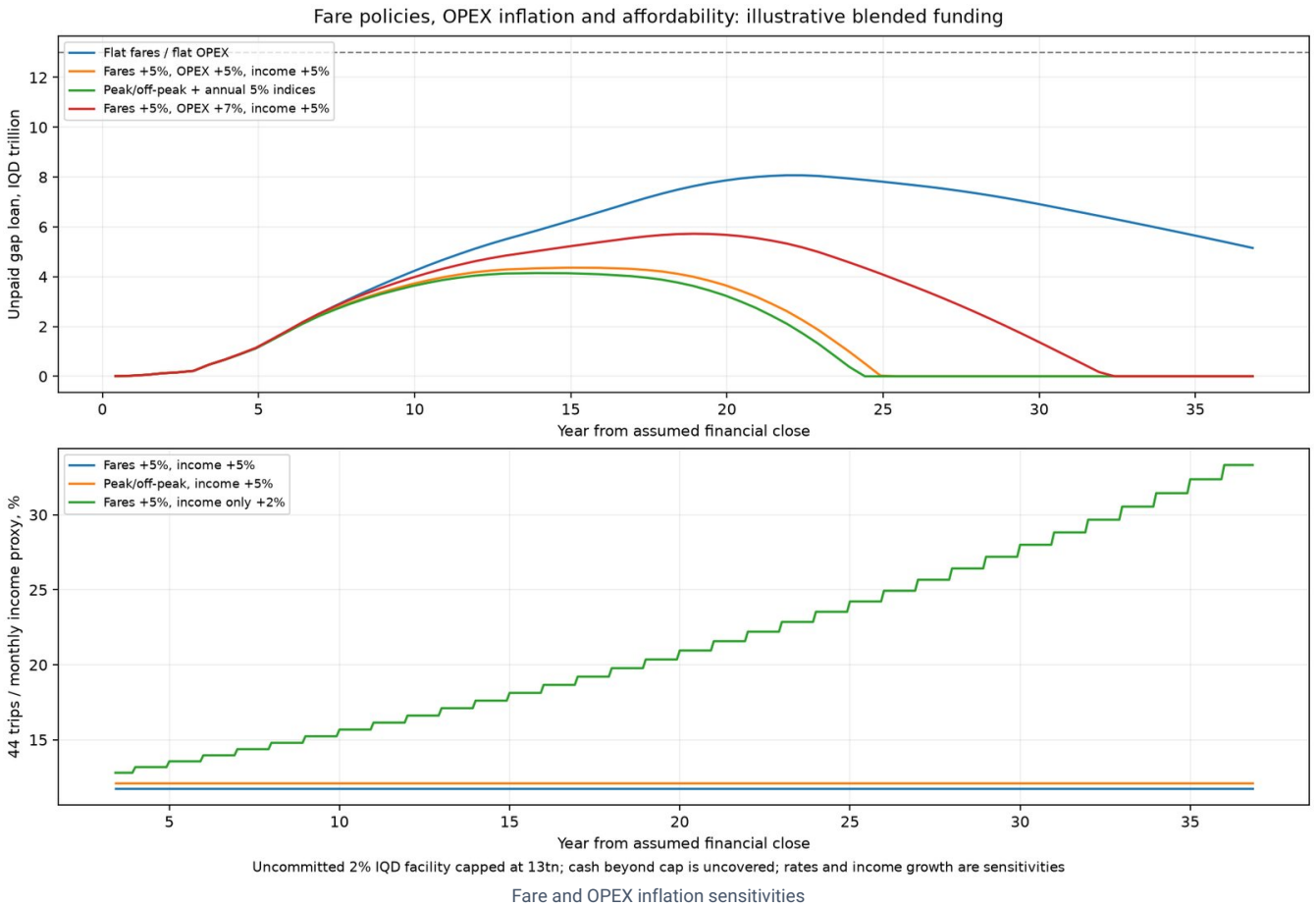
Variable tickets assume 40% of baseline trips in a peak tier at 1.25 times the standard fare, and remaining trips at 0.90 times it. Each tier gets its own demand response before revenues are combined. Concessions, commuter caps, integrated transfers and off-peak eligibility require an actual tariff design and funding contract; their costs are not silently waived.

Pricing sensitivity	Fare / OPEX / income growth	Full-opening average fare, IQD	44 trips / income at full opening	Peak facility, IQD tn	Uncovered cash, IQD tn	Terminal debt, IQD tn
fare 5pct flat costs fixed demand	5% / 0% / 0%	1,765	15.7%	3.321	0.000	0.000
fare 5pct flat costs elastic	5% / 0% / 0%	1,765	15.7%	4.033	0.000	0.000
fixed fare 5pct opex	0% / 5% / 5%	1,317	8.8%	13.000	9.457	13.000
fare 5pct opex 5pct	5% / 5% / 5%	1,765	11.7%	4.362	0.000	0.000
fare 5pct opex 7pct	5% / 7% / 5%	1,765	11.7%	5.723	0.000	0.000
fare 5pct opex 5pct income 2pct	5% / 5% / 2%	1,765	14.0%	5.501	0.000	0.000
variable fare 5pct opex 5pct	5% / 5% / 5%	1,822	12.1%	4.144	0.000	0.000
fare 5pct opex 5pct rents indexed	5% / 5% / 5%	1,765	11.7%	4.166	0.000	0.000

In the paired 5% case, the first line's average nominal fare is IQD 1,525 in month 41; at full opening in month 83 it is IQD 1,765. Its maximum supplemental facility is IQD 4.362tn, with IQD 0.000tn uncovered cash and IQD 0.000tn unpaid at the end. Later surplus can repay a priced facility under these assumptions; it does not remove the early borrowing requirement.

At an 5% general-price assumption, the 8% real discount assumption becomes 13.4% nominal. The paired-case unlevered NPV is USD -3.299bn equivalent before grants/new rights/net-income targets and with capital still un-escalated. Large distant nominal balances are not present-value wealth or proof of project viability.

If income rises with fares, the modelled commuting share stays broadly constant. With income rising only 2%, the same ticket policy becomes progressively less affordable and reduces paid trips. If OPEX rises 7% against 5% fares, the tested financing again leaves uncovered requirements and terminal debt. Maintaining an affordable real tariff, collecting revenue and placing the required early IQD facility matter as much as the nominal annual increase.



[Paired 5% six-month funding requirements](#) · [paired monthly prices and affordability](#) · [variable-ticket funding requirements](#) · [variable monthly tickets](#)

The initial real-fare multiplier required to remove both terminal debt and uncovered cash under paired 5% indices and the other blended assumptions is approximately 1.000 times the existing baseline. This is a conditional break-even sensitivity, not a tariff recommendation; it does not guarantee the facility can be placed or that households' incomes will grow.

Surplus cash and early repayment

These cases use identical paired 5% fare/OPEX/income assumptions and the same conditional blended capital sources. Each pays OPEX, scheduled principal, interest and fees, then funds the debt-service reserve and a three-month current-OPEX buffer before voluntary repayment. Borrowed gap proceeds and uncovered external cash cannot fund early payments. The extra buffer remains cash held at the terminal horizon and earns no interest.

Compare strategies against buffered gap-only, rather than attributing the buffer change to repayment savings. Loans-first pays bank credit, Chinese credit and gap credit before bonds. Cost priority pays bank (9%), ordinary bonds (8%), Chinese credit (5%), green bonds (4% plus 0.5% annual guarantee), then gap credit (2%). It is an interest-rate heuristic, not a proof of the globally best strategy.

Surplus strategy	All debt cleared, month from close	Finance cost saving vs buffered gap-only, USD equivalent m	Early-payment premiums, USD equivalent m	Peak gap debt, IQD tn	Terminal unrestricted cash, IQD tn
gap only buffered	371	0.000	0.000	4.505	16.797
gap then core	305	13.320	1.896	4.505	16.815
loans then bonds	304	92.811	7.121	4.461	16.918
cost priority	303	195.029	23.681	4.461	17.051
cost priority zero premium	302	225.425	0.000	4.457	17.090
cost priority noncallable bonds	371	78.731	5.175	4.461	16.900

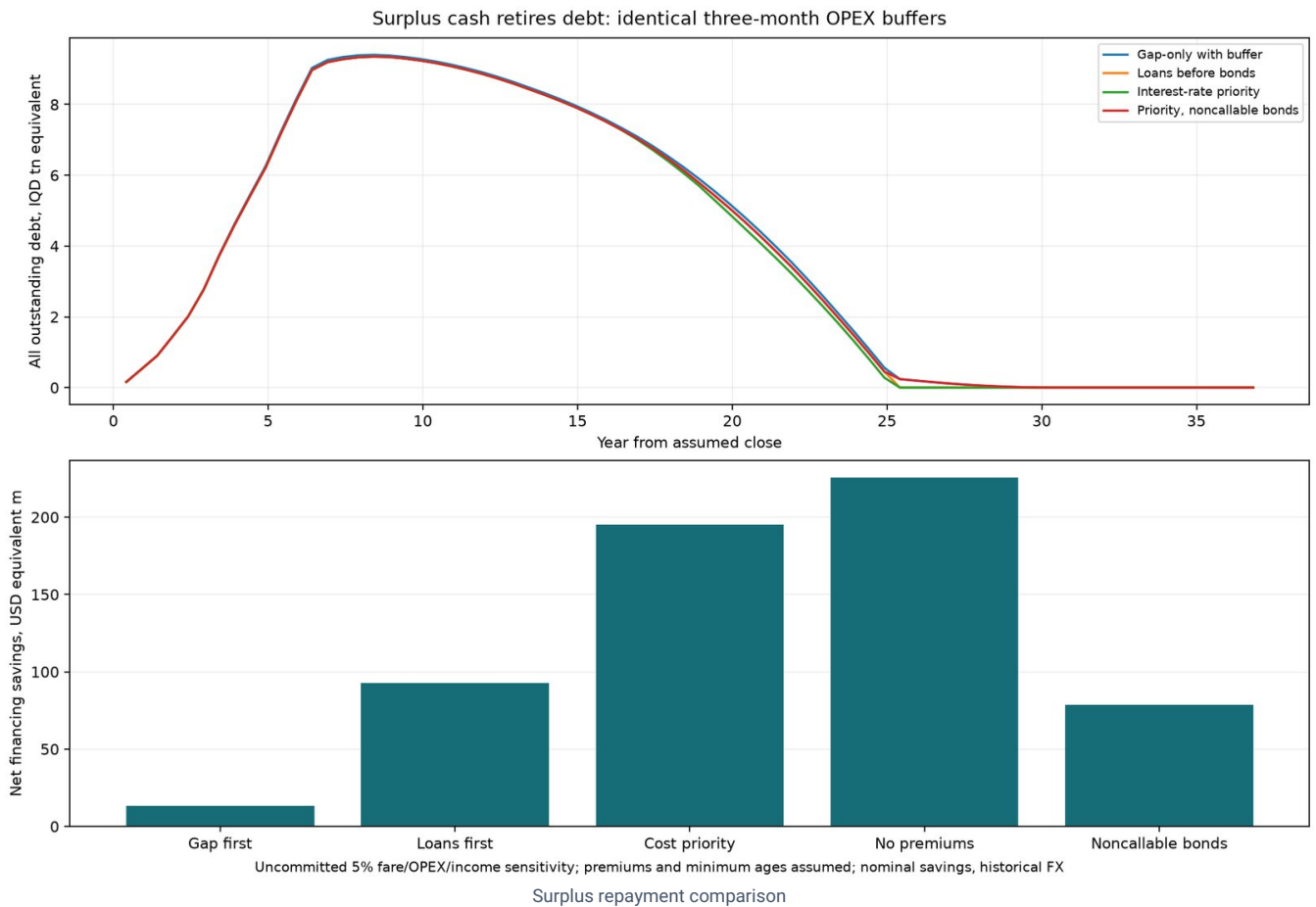
Cost priority clears all debt in month 303, compared with 371 for buffered gap-only. Net nominal financing savings are USD 195.029m equivalent, after USD 23.681m assumed early-payment premiums. Savings include core interest, annual green guarantee charges and supplemental interest/draw fees; they exclude principal, which is returned once. All cases retain IQD 0.340tn operating buffer separately from unrestricted cash. There is no additional government contribution above 25% of CAPEX in these cases.

Assumed premiums are 1% of bank/Chinese/green principal and 2% of ordinary bond principal. A minimum draw age of 6 months (bank), 12 (Chinese) and 24 (both bonds) prevents immediate issue-and-redemption. Eligible vintages are repaid oldest first; contractual instalments are kept and maturity shortens. Notice, issuer call rights, investor consent, buyback price, remaining-maturity compensation, tax and FX require actual agreements. Noncallable-bond sensitivity makes no voluntary bond payments; zero-premium sensitivity removes only the assumed premium, retaining minimum ages.

Facility	Buffered gap-only final principal payment month	Cost-priority final principal payment month	Early principal, native currency	Premium, native currency
bank credit	167	167	IQD 189.902bn	IQD 1.899bn
domestic bonds	287	231	IQD 963.845bn	IQD 19.277bn
chinese export credit	311	236	USD 266.436m	USD 2.664m
green bonds	371	246	IQD 614.570bn	IQD 6.146bn

The gap facility's final principal payment is in month 303. Later surplus remains unrestricted cash after debt retirement. Each month's native debt balance equals prior balance plus draws minus scheduled and early principal. Each six-month closing balance is its final month's balance; payments, premiums, interest and reserve movements are period sums. Contractual repayment windows in tranche files describe the original draw terms; the actual final-payment months above incorporate early repayment.

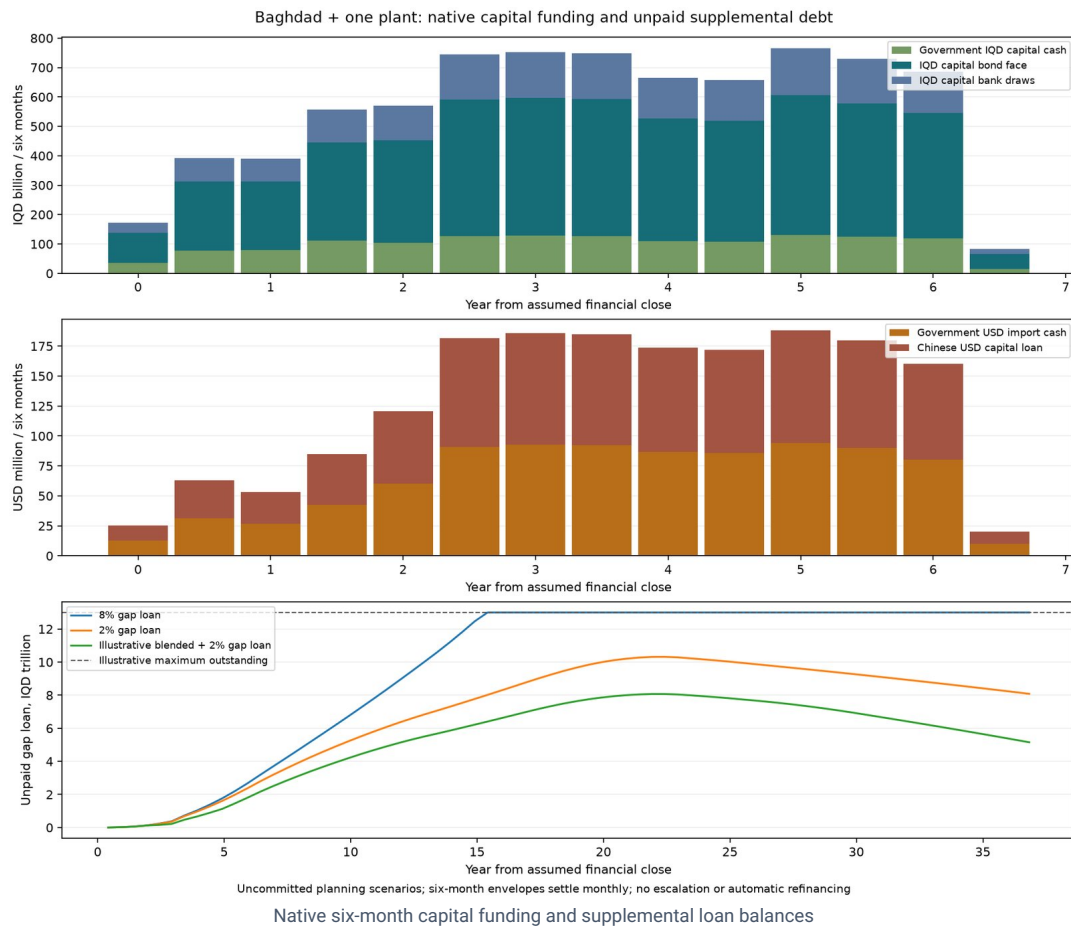
[Monthly cost-priority repayments](#) · [six-month repayment tranches](#) · [loans-first tranches](#) · [all six complete calculations](#)



Loan prepayment can carry redeployment/unwind charges: the [World Bank Treasury FAQ](#) illustrates the concept, without establishing Chinese or Iraqi loan terms. The [US Treasury buyback FAQ](#) and [published purchase results](#) illustrate issuer authority and priced buybacks rather than automatic par redemption; they do not supply an Iraqi legal mandate.

This repayment allocation does not change the unlevered project NPV or validate demand. Nominal savings over several decades are not present-value gains. CAPEX escalation, FX, lifecycle replacement, floating-rate risk and the uncommitted early funding still require appraisal.

Six-month bond sales and capital loan requirements



Each half-year is a placement/draw envelope, with monthly settlement against capital milestones. Bonds assume par proceeds and an illustrative IQD 1,000,000 denomination. Face-value requirements below use unrounded values; unit counts in CSV round upwards and separately disclose the resulting indicative face amount. Sale discounts, issuance fees and investor capacity need actual bookbuilding. Selling every half-year amount upfront changes interest/carry and grace clocks and is not simulated here.

The government USD cash and Chinese USD loan are separate; government local cash, bonds, bank draws and uncovered liquidity are native IQD. Chinese facility commitment fees retain the existing annual procurement-cohort assumption; six-month envelopes do not assert a new signing schedule. Grace starts on each actual monthly settlement. First/last repayments for a half-year series therefore differ by draw date.

Tranche	Months from close	Government USD m	Government IQD bn	Chinese loan USD m	Bond face IQD bn	Bank loan IQD bn	Additional cash need IQD bn
BGD-H001	0–5	12.63	35.03	12.63	103.44	34.48	4.82
BGD-H002	6–11	31.51	77.51	31.51	235.84	78.61	16.82
BGD-H003	12–17	26.67	80.20	26.67	232.45	77.48	36.65
BGD-H004	18–23	42.52	111.55	42.52	333.91	111.30	59.87
BGD-H005	24–29	60.40	103.18	60.40	349.94	116.65	102.39
BGD-H006	30–35	90.90	127.12	90.90	463.27	154.42	123.87
BGD-H007	36–41	92.85	127.88	92.85	468.79	156.26	323.85
BGD-H008	42–47	92.43	126.92	92.43	465.82	155.27	267.47
BGD-H009	48–53	86.76	109.79	86.76	416.21	138.74	298.17
BGD-H010	54–59	85.84	108.48	85.84	411.47	137.16	319.31
BGD-H011	60–65	93.99	130.42	93.99	476.71	158.90	352.08
BGD-H012	66–71	89.94	123.89	89.94	454.13	151.38	370.42
BGD-H013	72–77	80.04	119.69	80.04	425.38	141.79	385.89
BGD-H014	78–83	10.04	14.02	10.04	51.13	17.04	358.21
Total capital envelopes	—	896.53	1395.67	896.53	4888.49	1629.50	—

Displayed two-decimal values can differ from full-precision totals by rounding. The six-month CSVs continue through the entire operating/repayment horizon, including periods with no new capital but interest, principal, liquidity draws, reserve movements and cash sweeps. Every tranche independently reconciles its capital sources and uses.

Financing routes to investigate in Iraq

Route	What it can finance	Cashflow treatment and next requirement
IQD green bonds / green sukuk	Qualified electric transport, solar and charging capital	Replace conventional debt. Define eligible use, segregate proceeds, obtain independent review and report allocation/impact. Sukuk requires an approved asset/usufruct, legal and Sharia structure; the model does not assume a qualified sukuk issue.
GCF / GEF preparation and climate capital support	Climate studies, proven incremental mitigation/adaptation expenditure and eligible capital	Grants reduce debt only when approved; concessional debt still repays. Seek an accredited entity and Iraq NDA support with an emissions/resilience baseline. No grant is booked in the reference.
CBI renewable-energy initiative	Qualifying solar installations under the latest participating-bank rules	Obtain bank confirmation of borrower/project eligibility, current caps and terms. This existing Iraqi route is not assumed available for the whole metro or booked as a new capital source.
MDB / IFC credit enhancement and IQD syndication	Mobilising Iraqi banks and institutional bond investors	Guarantees improve placement or terms, supply no standalone cash, cost fees and may create sovereign contingent liabilities. Confirm IQD availability, hedge pricing and lender limits.
Station-area joint development and development rights	A contracted contribution towards stations/access or unrestricted net proceeds	Independently value parcels and rights, confirm title and revenue powers, tender transparently, deduct costs and future rent foregone. Early rights proceeds can reduce peak liquidity more than late rents.
Parking/congestion proceeds, employer transport subscriptions and business improvement charges	Incremental local recurring cash after authorised net collection costs	Separate each revenue stream from existing fare/nonfare receipts; validate legal powers, demand, affordability, exemptions and enforcement. No unverified tax power is assumed.
Solar/BESS project company, energy-service or availability contracts	Private financing of an accepted energy package	Remove transferred CAPEX and related city debt once; add the PPA/lease/service payments and guarantees. Compare their NPV with existing electricity/OPEX to avoid a free-CAPEX saving. No assumed saving is booked here.
Manufacturer / tooling leases and supplier finance	Qualified equipment and assembly capacity	Deferred invoices or leases are debt-like obligations with priced future payments. Do not duplicate Chinese invoice credit. Factory customer receipts and actual margins need a consolidated business case.
Diaspora and institutional IQD issues	Local capital subscription within the bond programme	These are purchaser channels for existing bond face, not another capital source. Disclose investor FX risk, maturity and redemption; no automatic rollover or compulsory pension purchase.
Carbon / results-based climate receipts	Verified additional emission reductions after delivery	Treat as contingent upside until methodology, monitoring, ownership and an offtake are contracted; no carbon revenue in the reference or blended case.

Priority: first align civil spending and production with accepted openings, then seek eligible cheaper/longer capital financing and contractual early rights proceeds. Obtain a funded recurring-revenue plan before borrowing a multi-trillion-dinar gap facility. Green finance and creative structures do not cure an unfunded timing mismatch or unsupported demand on their own.

Primary evidence checked 3 October 2026

The [GCF Iraq country profile](#) lists two projects with USD 49.5m total GCF finance and USD 5.2m approved readiness support when checked; these are existing country programmes, not Baghdad rail allocations. The [Iraq country programme](#) provides national priorities. An additional USD 25m application target is therefore a sensitivity requiring a new proposal, not money already available.

[GCF investment framework](#) and [financial instruments](#) support the grant/concessional/guarantee concepts and require viable revenue-generating activities for loans. [ICMA Green Bond Principles](#) require a transparent use-of-proceeds process, allocation controls and impact reporting; they do not promise a reduced coupon.

The [CBI renewable-energy amendment of April 2025](#) establishes a local financing route to investigate for eligible energy packages. Its limits and bank eligibility cannot be extrapolated into a multi-trillion-dinar metro facility. [IsDB green/sustainability sukuk guidance](#) supplies a structuring reference, not a Baghdad issuance mandate.

[World Bank guarantee programme](#) supports foreign/local-currency obligations subject to project and sovereign approvals. [IFC local-currency syndications](#) explains domestic parallel loans and hedge structures; it does not establish an IQD loan offer for OSR. [IFC Iraq investment activity](#) establishes an existing country presence, not project eligibility or earmarked money.

[World Bank land-value-capture guidance](#) and [innovative revenues](#) provide structuring examples, not Iraq borrowing/tax authority or a Baghdad parcel valuation.

[World Bank integrated-fare guidance](#) supports examining peak/off-peak, transfer and integrated ticket structures. Its [Bogotá affordability study](#) illustrates the need for household-specific demand evidence; neither calibrates the -0.30 sensitivity for Baghdad or endorses a 5% annual increase.

Remaining modelling and term-sheet limits

- Six-month bond placement envelopes settle monthly at par; they are not advance-funded bond sales. Advance issuance changes carry, reserve and interest costs and needs a separate cash forecast.
- Full unrounded placement values reconcile; illustrative IQD 1 million unit counts round upwards and are not executable orders. No auction price, investor demand or mandate is established.
- A green label alone uses unchanged domestic pricing. The lower coupon/longer tenor and 0.5% annual guarantee fee are explicit sensitivities, not donor or lender terms.
- Climate capital grants replace domestic capital borrowing only, against candidate local expenditure. They do not pay old debt interest, imported invoices already funded 50:50, or increase government cash above 25%.
- Incremental land/rights/local receipts are zero in the reference and must be genuinely additional to existing fare, retail, rent and advertising forecasts. Net proceeds after costs, title, statutory powers and independent valuation require qualification.
- The supplemental IQD facility pays its own interest and origination fees. Surplus is swept to repayment; terminal balances are exposed rather than assumed refinanced or forgiven.
- The illustrative IQD facility has a 13 trillion maximum outstanding commitment; residual cash requirements beyond that cap remain explicitly uncovered. No committed lender or future market capacity is assumed.
- Supplemental facilities have no extra DSRA or undrawn commitment fee and no fixed amortisation before the terminal maturity; term sheets can materially increase costs. Long draw availability and the terminal balloon are uncommitted assumptions.
- City/plant cash is pooled subject to legally permitted transfers and creditor consent. Factory income/OPEX remain outside the capital-only plant appraisal; no free factory profit is credited.
- All non-Chinese debt is modelled in IQD. Foreign-funded concessional programmes need confirmed IQD on-lending or a priced hedge; an unhedged USD development loan would change the currency strategy.
- The reference has constant nominal fares, OPEX and FX. Dedicated pricing cases explicitly index fares, OPEX and income independently from financial close. Capital escalation, FX inflation pass-through and separate lifecycle replacement inflation remain outside this sensitivity; full nominal appraisal must include them.
- Pricing cases retain fixed nominal core and gap debt terms; supplier terms, floating rates and imported OPEX exposure need qualification. New net revenue targets are held nominal rather than assumed inflation-protected.
- Price elasticity is an uncalibrated response to fare relative to indexed income; no transfer, peak switching or low-income household demand survey is modelled. Off-peak trip increases are bounded by the same practical capacity, not new physical capacity.
- Early repayment is a contractual sensitivity, not an exercised call right. Bank/Chinese loan eligibility, bond call or voluntary buyback rights, notice, price and premiums require actual terms. The noncallable case forbids all early bond principal. Illustrative minimum draw ages are 6 months for bank credit, 12 for Chinese credit and 24 for bonds; these are not actual covenants.
- Cost priority is a coupon/guarantee-fee heuristic, not a globally optimal treasury strategy. Premiums, tax, remaining maturities, future draws and FX can change the best order; no refinancing or capital-issue cancellation is modelled.
- A three-month OPEX buffer is funded before discretionary repayment in every early-policy comparison. It is retained at the terminal horizon and earns no income. Original baseline cases have no such additional buffer; savings are compared with a buffered gap-only case as well as the original.

[Editable options](#) · [existing capital assumptions](#) · [original city appraisal](#) · [Baghdad-only programme](#)

Baghdad financing redesign – 2026-10-04

Source: cities/catalogue/west-asia/Iraq/Baghdad/engineering/financing-redesign/README.md

Twelve executable alternatives retain 9 lines, 516.5175 km, 182 stations and 831 six-car trains. Financial close, title, investor equity, insurance eligibility and physical acceptance remain unestablished. The [existing baseline](#) remains the planning reference. Conditional full-line openings are 41, 46, 52, 57, 62, 69, 73, 79, 83 months; the [independently operable first section](#) is a separately costed sensitivity, not silently added to these full-network cases.

Results and what they mean

Case	Peak aggregate IQD liquidity tn	Cumulative missing IQD funding tn	Terminal IQD gap debt tn	Lifetime interest/fees USD bn eq	Consolidated resource NPV USD bn eq
reference	4.461	0.000	0.000	5.076	n/a
opening_linked_debt	3.850	0.000	0.000	6.713	n/a
development_rights_1bn	2.395	0.000	0.000	5.170	n/a
insured_15year_debt	4.870	0.000	0.000	7.009	n/a
integrated	8.450	0.070	2.898	7.797	-2.945
integrated_low_ppa	5.243	0.000	0.000	6.427	-2.945
integrated_high_ppa	8.450	3.512	7.588	8.401	-2.945
integrated_public_availability	3.684	0.000	0.000	5.775	-2.945
integrated_joint_downside	3.400	48.418	3.400	18.501	-6.207
integrated_rental_small	8.450	0.143	2.898	7.816	-2.950
integrated_rental_medium	8.450	0.111	2.898	7.824	-2.928
integrated_rental_medium_downside	8.450	0.517	2.898	7.863	-2.981

Next financing priority: obtain opening-linked IQD terms and independently valued station proceeds, then compare quoted PPA prices against continued ownership. The integrated high-PPA structure shifts substantial cash to the energy company while rail remains short; private equity needs an acceptable return as well as debt repayment. The tested property margin improves resource NPV before land cost, but the assumed in-kind land cost more than offsets that improvement. None of these scenarios establishes financial feasibility.

[Summary with exact figures and appraisal definitions](#); every case has a complete JSON model, monthly native-currency CSVs, six-month placements and contractual loan-vintage dates. Missing funding must be filled before any reported conditional repayment date is credible. Independently funded entities cannot use another company's cash without a separately approved agreement. Liquidity caps are allocated once: 65% rail, 10% energy, 5% factory and 20% development. The sum is IQD 13.0tn reference; the downside divides the existing IQD 4tn cap in the same proportions.

Integrated borrower	Capital uses USD m eq	Private equity IQD bn	Peak gap IQD tn	Missing funding IQD tn	Terminal gap IQD tn
rail	6740.875	0.000	8.450	0.070	2.898
energy	814.869	211.866	0.018	0.000	0.000
factory	324.750	84.435	0.294	0.000	0.000
development	3400.000	1105.000	0.368	0.000	0.000

These are pre-tax sensitivities. Corporate tax, duties, VAT, sponsor equity hurdle rates, dividend restrictions, lender DSCR and termination liabilities need quoted/legal terms. Surplus is swept after monthly OPEX, scheduled debt service, reserve funding and the existing lock-ins/premiums. A positive cash balance or an annuity model does not establish bankability. The factory has no assumed national orders or residual asset sale.

Opening-linked credit and insured green instruments

Bank amortisation becomes 240 months and ordinary bonds 216 months at the unchanged 9%/8% reference rates. Monthly draw cohorts use the latest opening of the assets included in that cohort; shared contracts use full opening. This conservative aggregation needs actual line-linked facility documents. Interest is paid throughout grace, funded by the separate IQD liquidity account if cash is insufficient; no interest is silently forgiven or capitalised. Arrangement/commitment charges, buffer and surplus rights remain priced. Chinese USD loan terms remain unchanged.

Loan-vintage and six-month CSVs show actual draw, first principal and final contractual principal dates. Cheaper/longer debt changes liquidity and finance cost, while core unlevered NPV stays USD -3.299bn. Core excludes grant, development rights and new net receipt targets.

[ICIEC's official product](#) describes sovereign/sub-sovereign sukuk with coverage up to 15 years. The tested insured instrument has grace plus amortisation capped at 180 months from each draw, replacing the previous 48+240-month assumption in that sensitivity. A 6% coupon, 0.75% annual insurance and 1% annual IQD conversion/on-lending cost are assumptions, not insurer

pricing or proof that an IQD issue is eligible. No climate grant is booked in insured/integrated cases. Obtain written eligibility, binding terms and a priced local-currency route before using an institution's name in funding commitments.

Station-area partnership and public land

15 station candidates are selected only by network archetypes, with actual station IDs/coordinates. Parcel title, legal ownership, survey area, encumbrances, independent values, displacement and developer interest are null/unverified. No selection is a claimed commercial valuation. First investigate 10–20 sites through independent market/title/safeguards work and competitive long leases or voluntary pooling.

USD 1bn total net development-rights replaces USD 300m; the increase is USD 700m. Six equal tranches begin month 24, every six months. Receipts are IQD using the model's 1,300 IQD/USD reference; they are not USD debt or dollar-indexed obligations. In the integrated sensitivity developers finance both rights and buildings. The hypothesis is 1.2m m² land, FAR 2.5, 3m m² buildings at USD 800/m², USD 4.2bn sales in months 60–96 and 5% sales costs. There is no rent revenue additionally booked against sold floor space. Public land's assumed USD 600m opportunity cost is an in-kind exposure, never spendable cash. All prices are unverified and require an Iraqi valuation/tax review. The downside halves rights, reduces sales 30%, delays both 24 months and escalates construction.

Rights payments are a developer use and a rail receipt. They cancel in consolidated appraisal; they cannot also be counted as programme profit. Developer equity is 25% of its rights/build uses, and its borrowing is IQD bank credit. Property debt/transaction costs and missing funding are included in its own ledger. [World Bank land-value guidance](#) and [MTR's model](#) provide precedents, not Baghdad valuations.

Energy and manufacturing contracts

Dedicated solar capital is transferred once, without inventing an additional EPC removal. Charging/microgrid equipment and its existing EPC remain rail assets. The railway pays a PPA of IQD 104/kWh, indexed 5% annually; the annual planning purchase is 1.940bn kWh, USD 155.179m equivalent at reference prices. Dispatch/load follows open-line weights, not fare demand. Energy receives exactly what rail pays. Existing solar maintenance transfers once; a further 0.5% of solar capital/year is an explicit renewal-spending sensitivity. Timetable-derived annual generation is not 8760-hour dispatch proof: grid backup, curtailment, storage adequacy and metered purchases require qualification. Export sales are zero, with no signed buyer. Low/high PPA cases redistribute company cash and leave the consolidated resource NPV unchanged. [World Bank PPA guidance](#) explains the contractual financing route.

Energy and factory each test 20% private IQD equity. Government remains 25% of original physical rail/energy/factory capital. The factory receives Baghdad's existing train invoices and pays the same resource budgets one month earlier, exposing working capital. Original train prices already include manufacturing labour, QA, ordinary overhead and warranty; 1,044 production staff wages are not added again. An additional USD 250,000-equivalent capacity fee per train, annually indexed and paid in IQD, is a separately visible investor-return sensitivity. It cancels on consolidation but increases rail's operating obligations. Warranty cash is locked at 5% of cumulative train payments until 12 months after the final invoice, then released; it is cash restriction, not a duplicate warranty expense. Fifteen caretaker staff and 0.25% plant carrying cost are added during that tail. Plant costs, working-capital funding, interest and outstanding debt at backlog close remain visible in [integrated.json](#). New orders and salvage are zero until contractual.

At the illustrative capacity fee, factory cash remaining at contract/warranty close is USD 6.750m against USD 64.950m invested equity. The 15% equity-hurdle NPV is USD -54.159m. Debt repayment therefore does not establish an investable factory partnership. Developer equity-hurdle NPV is USD -194.940m. These close-date distribution diagnostics deduct any remaining debt and disclose missing funding; no dividend or sponsor return is assumed to be committed. Competitive pricing, equity terms and a bankable capacity contract require further work.

Appraisal and public exposure

Three retained under-viaduct rental variants add a fifth independently financed entity to the integrated case: small, medium and lower-rent/prolonged-vacancy medium. The [civil-linked register, pilot unit, costs, leases and hazards](#) have no accepted site/demand evidence. External rent adds revenue; actual collection follows fit-outs, physical line availability, initial rent-free periods, occupancy ramp, arrears/recovery, tenant turnover and lease-end refunds. Fit-out capital, occupied/vacant maintenance/insurance, 12-year refurbishment and a standalone 15% cash-tax proxy are included. No station sale parcel also earns rent; no existing kiosk income is duplicated. Tenant deposits are restricted cash and matching liabilities, never construction finance or revenue.

The rental partner supplies 25% fit-out capital as private IQD equity and the residual is IQD bank capital, with opening-linked principal and 240-month amortisation. It has zero new operating-gap capacity: missing construction interest, fees or reserves remains an explicit funding requirement. The original four borrowers keep their existing allocated caps; combined cap remains 13tn. Rental transfers to rail are zero until an approved agreement exists, so rail cannot silently use the new company's cash.

Each borrower exports monthly native ledgers, six-month placements and loan vintages. These ring-fenced structures are alternative to the wholly owned holding rental cases, whose subscriptions replace rather than add partner equity.

Consolidated resource NPV adds collected rent less its OPEX and physical fit-out capital, before financing/tax, to the original resource case. A separately labelled after-rental-tax diagnostic deducts rental tax once; the original four businesses retain their existing before-tax basis. Additional rental-site rights/land opportunity cost is unknown, so the existing station-land subtraction is not a complete new-site appraisal. The 200,000 m² area illustration is blocked as unmapped and supplies no integrated cash.

Core rail/energy/factory unlevered NPV: USD -3.299bn. Rail entity NPV including its PPA, capacity fees and internal rights: USD -3.134bn. Consolidated resource NPV including property sales/building costs and additional renewals/caretaking: USD -2.945bn; including land opportunity cost: USD -3.412bn. Debt/equity/grants are financing transfers and excluded from resource NPV; intercompany transfers cancel to the cent. Nominal discount is 13.4%, combining 8% real and 5% general inflation.

Government original-capital cash is USD 1.970bn equivalent (25%). Its USD import cash is USD 896.530m and Chinese USD credit is USD 896.530m, splitting imports 50:50. All other debt, private equity, property payments, fares and subsidy payments are IQD. Original programme USD capital intensity therefore remains 22.75%; SPV transfers do not remove those USD import requirements. Property construction is assumed contracted in IQD; actual imported content/FX exposure needs quotes.

Public cash capital PV is USD 1.262bn; in-kind land PV USD 0.467bn. The optional availability case pays USD 100m equivalent/year at full opening, indexed 5% and phased by available lines; its PV is USD 0.790bn. It sits outside the 25% capital limit and needs appropriation, service performance terms and affordability review. Recipient NPV adds public cash explicitly; the public-plus-private resource appraisal cancels that transfer. Contingent bond/guarantee/termination exposures remain unquantified rather than assumed zero. [World Bank public-support guidance](#) describes these obligations.

Existing fares and kiosk/advertising receipts are retained with their original demand/price/OPEX indices. Integrated cases omit the old USD 25m grant and USD 25m/year new-net-income targets. Fibre, naming rights, parking and employer contracts require incremental net quotes; prepaid fares change timing of existing receipts and do not create revenue twice. At 13.4%, USD 50m/year for 20 end-of-year payments has PV USD 342.962m at the beginning of the payment period, lower when discounted back to close.

Iraqi financing routes and 90-day evidence programme

The [World Bank's USD 930m national railway project](#) is a separate Umm Qasr–Mosul programme through Baghdad, not metro money. IFC's [September 2025 Iraqi announcement](#) combines investments and engagements across industrial/property/financial partnerships, not an available USD 1bn fund for this proposal. [ICIEC's 2025 impact report](#) records reinsurance for Iraqi substations; it does not commit rail credit. Sources checked 3 October 2026. [GCF's published Iraq country programme](#) and [transport programme](#) establish potential application routes, not an award. Climate funding requires emissions baselines, eligibility, accredited-entity engagement and actual terms; no grant is presumed.

[Six workstreams](#) cover lender terms, land/valuation, PPA, standalone factory, existing independently operable phase, and independent model/public-affordability review. Due days are relative to an actual approved start. Named people, legal opinions, quotes and accepted evidence remain pending. [ERPNext Task JSON](#) is a draft native-import package with stable evidence subjects, not live IDs or accepted evidence. Local deployments record actual IDs privately and preserve assignments/status when refreshing descriptions. No external lender/developer contact, issuance or expenditure is authorised by these model files.

Later construction commitments require executable native-currency funding, separately accepted physical evidence and a conservative demand case. Integrated joint downside retains the existing 55/106-month physical stress and adds weaker property conditions: missing funding is IQD 48.418tn, terminal gap debt IQD 3.400tn. The proposed structure is a market-test programme, not a claim that the financing gap is solved.

Regenerate with `.venv/bin/python tools/automation/baghdad_financing_redesign.py`; validate exact inputs/outputs with `--check`. No new financial commitments or physical release are implied.

Baghdad qualification, funding gates and first operating section

Source: cities/catalogue/west-asia/Iraq/Baghdad/engineering/qualification/README.md

This package turns the review findings into executable draft work. It supplies no physical acceptance, supplier quotation, named owner assignment or financing commitment. The full 831-train Baghdad baseline and its month 41/83 openings remain unchanged.

Temporary first-article industrial package

[Site block layout](#), [quantities and reconciliation](#), and [RFQ schedule](#) specify six prototype cells, 4,960.0 m² process floor, 6,696.0 m² total covered floor and 16,740.0 m² fenced-site allowance. The metric 180 x 93 m envelope is a reference arrangement, not a selected or surveyed site. Access/fire widths, hazardous materials, drainage, foundations, heavy-haul route, crane coverage and utilities require detailed vendor/civil drawings. No production acceptance paths are counted at this temporary site.

The quantity-based direct subtotal is USD 19.873m; contingency USD 3.975m; allocated total USD 23.848m. USD 11.152m remains unallocated within the existing USD 35m allowance. EPC remains USD 2.45m. Nothing in this package is added a second time to the USD 37.45m recovery case. Rates are editable assumptions, not received quotations or realised savings.

Vendor interfaces: bogie/body datum and lifting reactions; composite mould temperature/cure traceability and ventilation; door/window fit and tolerance stack; battery transport SOC/fire segregation and HV isolation; electrical test connectors and calibrated instruments; ERP lot/serial traveller and revision interfaces. RFQ returns must identify imported tooling versus Iraqi buildings/labour, actual origin, tax, delivery slots and financed invoice eligibility; the recovery model's 20% tooling/import fraction remains unqualified until these are reconciled.

Transfer sequence: permanently identify TFA-C01 through C06 tooling in the asset register; release the prototype traveller and first-article configuration; move the complete six-car train on certified frames/haulage with clearance and escort approval to permanent acceptance at day 390; retain all temporary tools until prototype acceptance and production continuity are secured; lock out, pack, transfer, reinstall and recalibrate each cell into its named permanent work centre; validate first-off dimensions/HV tests before release. Permanent plant tooling is already fully budgeted: temporary equipment is incremental and any reuse credit needs an approved asset/cost change, never an automatic reduction.

Readiness gates: survey/land and supplier interfaces before procurement; civil/fire/utility/lifting acceptance and calibrated equipment before day-260 assembly; permanent segregated test paths and bays before day-390 dynamic tests; the additional 60 working-day first-article qualification remains fixed; no series release before signed acceptance. Day 260/390 and month-36 first service are conditional dates, with independent ERP evidence tasks below.

Executable evidence work and ERPNext

[Ten source-bound work packages](#) map every delivery qualification row to an accountable role, measurement fields, units, numeric criteria, source content revision, ERPNext Task subject and result template. A named accountable person is deliberately pending rather than fabricated. [Task import JSON](#) and [CSV](#) are drafts; they are not live ERP records or accepted evidence. Reconcile them with the ERP app's `qualification.import_tasks` method; preview is the default. Assign actual users/owners through ERP and preserve assignment when refreshing descriptions.

Populate each BAG-EVID result template using real measurements and raw evidence file checksums. Use `.venv/bin/python tools/automation/baghdad_qualification.py --verify-result RESULT --envelope ENVELOPE --signature SIGNATURE --policy TRUSTED_POLICY`. The verifier rejects null/wrong-unit/out-of-threshold measurements, missing raw files, source changes, absent or unauthorized review and self-review. It reuses the subsystem qualification's externally controlled Ed25519 reviewer policy and envelope (`osr-subsystem-review/1`), with evidence fingerprint = SHA256 of the exact result bytes, configuration fingerprint = package source revision, required role = assessor, acceptance reference and expiry. Trusted reviewer identities/keys must be supplied independently; the generator provides no signing key and no acceptance. An accepted evidence package still does not authorize public operation.

Coverage ratios require item/lot/phase-level underlying files and independent scope review, not an unsupported declaration of 1. Production records must separate elapsed cure clocks and inspection holds from staffed hours. Baseline throughput and the shifted recoveries remain unmeasured. Funding evidence needs the appropriation, actual native-currency term sheets, investor/underwriter placement mandates, collateral/guarantees, maturity/fees, and every six-month sources-and-uses gate, including reserves and interest. Demand work must include a surveyed OD matrix, affordable fare/elasticity tests and net kiosk/advertising concessions after vacancy, collection and management costs.

Denied and interrupted funding

[Funding gate cases](#) and each placement-monthly/six-month CSV distinguish requested source face, available fraction, refused amount, invoices held and undrawn capital capacity reserved for deferred invoices. Recovered domestic placement assumes half the ordinary IQD bond placement fails for months 29–34; all unfinished procurement, civil work, production and running tests pause at working day 600 for 130 working days. Recovered export credit pauses the programme at NTP for 130 working days and requires the refused initial USD credit to be placed before imported procurement restarts. The costed schedules and financing are in the delivery-risk cases `domestic_placement_interrupted_recovered` and `export_credit_delayed_recovered`. A USD 10m equivalent IQD local remobilisation assumption plus 7% EPC is added once, with staff/site carrying costs charged in the six suspension months for the already-mobilised domestic case. An export refusal at proposed NTP shifts the entire unstarted programme without buying another construction/production staffed span; only the quoted-later remobilisation/reservation allowance is added. Cure/inspection restart requirements could cost more and need vendor evidence. Longer delay, storage damage, cancellation penalties, FX changes and lender acceleration/default are outside these recovery prices.

The placement files diagnose the original invoice request. Recovered cashflow files retime actual invoice milestones using the held schedule; they must not be added together. The available-capital column is an undrawn capacity requirement, not received or free operating cash. Recovered cases retain uncommitted liquidity for interest and hold costs; it never replaces the refused core capital. Investor commitment/carry fees on undrawn domestic placement are unquoted and excluded. Partial grants/green placements already in the baseline remain assumed; refusing ordinary bonds does not prove green placements will succeed. Bank refusal can be run with the same gate mechanism. Recovered dates and repayment require renewed placement and restart appropriations, which remain uncommitted. Permanent domestic/export refusal has no operating opening or debt-clearance date. Existing debt is not forgiven; cancellation, default/workout, liquidation proceeds and prior-creditor claims require a separate resolution model and are not represented by zero balances. Government is not automatically increased beyond 25%; the original import split stays 50% government USD / 50% Chinese USD loan; only actual import invoices can qualify. Domestic cash, bonds, bank/green/gap credit remain IQD, while foreign donor receipts would have separately identified FX conversion requirements.

Gate	Currency	Refused amount	Recovery assumed
<code>domestic_half_placement_recovered</code>	IQD	162,454,126,156.67	True
<code>domestic_half_placement_permanent</code>	IQD	1,258,199,231,929.58	False
<code>export_unavailable_recovered</code>	USD	12,630,671.95	True
<code>export_denied_permanent</code>	USD	896,529,966.45	False

Separate independently operable first-section option

[First-section duty, assets and finance](#) takes the actual first five line-1 stations, from the Mahmudiya terminal to line-1-1848-0797-s014034, spanning 14.034 km. This outer corridor is a demonstrator candidate, not a proven best catchment. It needs its own OD/accessibility study and comparison against better central corridors. Existing station/track/wayside tasks alone are pulled forward within the frozen lane/predecessor graph; their already-budgeted capital is retimed, not purchased again. Other route quantities and the 831-city fleet remain unchanged.

At the reference six-minute headway and 40 km/h average running speed, the round trip is 74.10 minutes including intermediate dwells and ten-minute charging/turnaround at each end. It requires 13 peak + 2 spare + 1 cold reserve = 16 existing six-car trains. Allocated train IDs are in the data file and must all pass acceptance before section operation. A six-minute section service is an explicit early-phase option rather than the ultimate line's three-minute design demand.

Each end needs 2 independent 121 m clear berths for the 111 m train, No.9 crossover/points, a safeguarded 185 m terminal arrangement, 2 x 2,000 kW chargers and 4,000 kW feeder capacity. Dwell plus clearance occupies each berth for 11 minutes against a 12-minute two-berth arrival cycle. These are geometry/utility requirements pending surveyed access, swept path, braking, interlocking and electrical studies; a generic standard station cannot silently serve as the new terminal.

Round-trip traction plus the conservative auxiliary proxy is 587.9 kWh, compared with 600.0 kWh delivered terminal charge at 90% efficiency. Battery definitions are 1,350 kWh nameplate / 1,080 kWh usable from the controlled six-car profile. The conservative section-specific 40% floor is applied to usable energy, leaving 648 kWh; one-leg duty fits that window. Normal charging margin is only 2.05%. A 10% reduction in delivered charging gives -8.15% balance and is not qualified. Roof PV and intermediate charging are excluded. The battery acceptance power, HVAC electrical demand, gradients, ambient heat, degraded charging and rescue case require measured qualification; the small energy margin is not operational robustness.

The [section RFQs](#) provide four incremental packages excluding already-budgeted route, station and fleet equipment. Reuse one original charger per end and add two berth chargers in total, upgrading each end to 4 MW. Independent maintenance/stabling, battery quarantine, lifting, rescue and road access avoid reliance on the distant line-6 depot. The provisional 3.5 ha maintenance site has eight stabling roads of 250 m usable length (two 111 m trains per road plus 28 m clearance) and two 135 x 6.5 m maintenance bays; site rights, turnout ladders, fire/access geometry and measured depot duty

remain open. Turnback/platform adaptation USD 5m, charging/grid USD 8m, maintenance/access USD 12m and controls/acceptance USD 3m give USD 29.96m including EPC, assuming 20% imported direct scope pending RFQs. The section has its own reference 200 operating FTE and USD 9.382m/year OPEX before indexation, replacing the whole-network fixed OPEX before month 41. It covers 16 hours/day over 330 days/year, USD 0.10/kWh grid electricity with no PV generation credit, Iraqi wage proxies, proportional existing civil/solar/signalling maintenance and train maintenance including the existing battery reserve, plus 2% maintenance on additional support assets. These cost/demand/utility assumptions need quotations and surveys. Separately, sixty temporary construction/qualification support FTE are indexed and charged from support readiness until the full line opens, then integrated into the baseline allowance; they are not counted as section operating staff. Civil/support acceptance at day 560 is an additional unqualified support-package assumption, not an existing accepted site.

The first 16 accepted trains finish at day 677; selected civil at day 666; independent support at day 560. The resulting conditional section opening is month 36, including the same three-month commissioning allowance; complete line opening stays month 41. No passengers or fares occur before that opening. Section paid demand is only half the smaller of fleet-capacity and route-length allocation proxies, weight 0.009627, deducted from later full-line demand to avoid double counting. This is a scenario, not population coverage evidence. Opening acceptance must cover section-specific turnbacks/chargers/depot, full allocated section fleet, isolation from works, evacuation/rescue, software configuration, hazards and independently signed operator/authority release; the full-line BAG-EVID-009 remains separate.

[Monthly finance](#), [six-month tranches](#) and [incremental cost ledger](#) include early baseline invoice timing, new support capital, indexed support payroll, fares/nonfare ramp, fixed/variable OPEX, reserves, native debt and prepayments. Peak gap debt is IQD 4.544tn; clearance month 304. Its unlevered NPV is USD -3.341bn. Compare against the baseline IQD 4.461tn peak / month 303 clearance / USD -3.299bn NPV. At the conservative demand proxy, mature section receipts are only USD 4.299m/year versus its USD 9.382m OPEX before indexing/ramp: it is not independently self-financing. Earlier service requires an explicit operating bridge; the extra facilities and advanced invoices must be justified by measured access benefits, demand or lower costs.

Financial feasibility and funding evidence

The baseline remains conditionally financeable under uncommitted cheap IQD gap credit and green/rights/receipt assumptions. Its unlevered financial NPV is USD -3.299bn at 13.4% nominal (8% real plus 5% inflation), before additional grant/rights/receipt targets. Clearing debt through later nominal fare growth is not the same as positive discounted investment value. [Feasibility data](#) supplies an illustrative annual external-benefit threshold of USD 417.4m in reference prices, phased with baseline openings and indexed 5%, whose discounted value would offset that financial deficit. This is not a completed economic appraisal: economic CBA must remove fares/taxes/financing transfers, use shadow resource costs, count time/safety/emissions benefits without double counting, and assess displacement/opportunity cost of labour rather than call every wage a net benefit.

Additional funding routes to investigate are competitively leased station land/development rights, employer travel contracts, advertising/kiosk concessions, utility/telecom corridor leases, climate grant/guarantee applications, and Iraqi institutional or retail IQD placements. Their net contract proceeds, authority, collection costs, timing and senior-creditor covenants are required before cash enters the ledger; a loan/guarantee is financing, not new operating revenue. The existing USD 300m net rights and USD 25m/year additional net receipts are targets, not a valuation or signed concession, and must not be added again as these same mechanisms. Replicable land value capture depends on land rights, planning powers and market demand ([World Bank financing guidance](#)). Iraq's [GCF country programme](#) establishes a project-development framework, not a Baghdad rail award or a 2% IQD facility. No fund application, lender term sheet or placement was obtained in this work.

Compare liquidity, repayment and discounted value separately

[Financial comparison data](#) uses each case's actual capital and indexed operating ledger, excluding debt and new grant/rights/receipt targets from unlevered NPV. All dates require physical acceptance and financing availability.

Case	First/full month	Capital USD bn	NPV USD bn	Peak IQD gap tn	Debt cleared month
calendar_baseline	41/83	7.880	-3.299	4.461	303
temporary_first_article	36/83	7.918	-3.329	4.422	302
availability_75pct_costed	42/90	7.880	-3.336	4.728	306
availability_75pct_all_stage_shifts	40/83	7.895	-3.332	4.542	304
domestic_placement_interrupted_recovered	47/89	7.891	-3.179	4.500	306
export_credit_delayed_recovered	47/89	7.891	-3.069	4.342	304
independent_first_section	36/83	7.910	-3.341	4.544	304

Temporary first-article production opens the complete first line earlier and lowers peak gap/repayment time, yet worsens discounted reference investment value after its additional capital/early operating costs. The independent shorter section also

adds capital and runs a conservative standalone operating deficit. Neither is adopted merely because fares arrive sooner. Coordinated shifts should be compared with the matching 75% availability case, not assumed to save money against the stronger baseline. Funding-delay NPV can look less negative because nominal un-escalated purchases are deferred; this discounting effect is not a benefit of denied credit. Real delay appraisal needs contract escalation, FX, cancellation/default costs and lost passenger benefits. Existing escalation/downside scenarios demonstrate their materiality.

Rebuild after delivery stresses with `.venv/bin/python tools/automation/baghdad_qualification.py`; validate with `--check`. Physical evidence and finance gates remain open.

Baghdad operations cross-check

Source: <cities/catalogue/west-asia/Iraq/Baghdad/engineering/simulation/operations-crosscheck.md>

- Status: running-time-screen-passed-awaiting-junction-evidence
- Automatic running-time cross-check: passed
- Retained full-service replay matches current inputs/tools: yes
- Junction occupancy evidence: pending
- Authority accepted: no

Line	OSR reference	SUMO mean	Difference	Tolerance	Result
line-1	4846.1 s	4480.0 s	-366.1 s	726.9 s	pass
line-2	5210.3 s	4860.0 s	-350.3 s	781.5 s	pass
line-3	5516.8 s	5150.0 s	-366.8 s	827.5 s	pass
line-4	4239.8 s	3935.5 s	-304.3 s	636.0 s	pass
line-5	4912.2 s	4538.0 s	-374.2 s	736.8 s	pass
line-6	5554.0 s	5153.0 s	-401.0 s	833.1 s	pass
line-7	4197.1 s	3901.0 s	-296.1 s	629.6 s	pass
line-8	4954.4 s	4602.0 s	-352.4 s	743.2 s	pass
line-9	9642.1 s	9142.0 s	-500.1 s	1446.3 s	pass

The automatic result is a deterministic planning-model timing comparison, not proof of safe headways, signalling performance or junction capacity.

Junction occupancy must be checked in an independently reviewed conflict-capable model and the operator or authority must sign the bound evidence before operational release.

Remaining junction gate

- independently reviewed junction-occupancy evidence not received

Remaining acceptance gate

- signed operational acceptance record not received

Baghdad civil soil screening

Source: cities/catalogue/west-asia/Iraq/Baghdad/engineering/soil/README.md

7,287 route/station sample locations; 7,222 complete profiles; 65 profiles with missing data.

Published soilDB 2020–2022 predictions, 120 m pixels, 0–30 / 30–60 / 60–100 cm depths. Mean and p16–p84 are retained for clay, sand, silt, bulk density, organic carbon and pH.

The [samples](#), [source receipt](#), [map](#) and [civil investigation plan](#) are reproducible planning inputs. Each station and civil segment has an investigation scope. These records do not close the surveyed ground-model or foundation-design gates.

Investigation trigger	Sample locations
coverage-gap	65
fine-soil-plasticity-and-shrink-swell-tests	6821
granular-density-and-groundwater-tests	3972
silt-moisture-frost-and-erosion-review	4951

- No bearing capacity, CBR, friction angle, cohesion, groundwater, contamination, sulfate/chloride or deep stratigraphy is inferred from these maps.
- Desert, water, urban fill and other nodata remain unknown; no nearest-pixel or climate-based substitution.
- Texture fractions are independent predictions; they are not renormalised or converted to a geotechnical soil class.
- Prioritisation thresholds are editable OSR screening rules, not statutory limits or evidence of a hazard.
- No excavation depths, foundation dimensions, treatment quantities or civil costs are changed from pedological predictions.

Source: [OpenLandMap soilDB](#), Hengl et al., DOI 10.5194/essd-2025-336, CC BY 4.0. Exact source revision, URLs and raster metadata are retained in the receipt.

Sampled property ranges

Ranges below span the sampled locations; they are not a city-wide characteristic soil value. The uncertainty envelope spans the lowest p16 to highest p84.

Depth	Property	Unit	Mean range	Uncertainty envelope	Available locations
0..30cm	clay	%	15–32	2–42	7222
0..30cm	sand	%	24–58	5–86	7222
0..30cm	silt	%	27–45	8–61	7222
0..30cm	bd.core	kg/m3	1410–1510	1200–1720	7222
0..30cm	soc	g/kg	2.1–6	0.5–14.8	7222
0..30cm	ph.h2o	pH	7.5–8.4	6.6–9.2	7222
30..60cm	clay	%	17–34	1–42	7222
30..60cm	sand	%	25–57	5–91	7222
30..60cm	silt	%	25–42	4–60	7222
30..60cm	bd.core	kg/m3	1410–1560	1170–1780	7222
30..60cm	soc	g/kg	1.6–4	0–11.2	7222
30..60cm	ph.h2o	pH	7.6–8.7	6.3–10.1	7222
60..100cm	clay	%	17–34	2–43	7222
60..100cm	sand	%	27–58	5–93	7222
60..100cm	silt	%	23–41	1–59	7222
60..100cm	bd.core	kg/m3	1400–1630	1200–1940	7222
60..100cm	soc	g/kg	1.5–3.2	0–7.6	7222
60..100cm	ph.h2o	pH	7.6–8.7	5.7–10.3	7222

Station and depot overnight allocation

Source: [cities/catalogue/west-asia/Iraq/Baghdad/engineering/stabling/README.md](#)

Plan: 298 trainsets at stations + 533 at depots = 831 total. Two revenue trainsets per selected station support coordinated morning starts; the remaining revenue trains and reserves stay at storage on their own line.

Allocation check: FAIL. Depot stabling positions are planning requirements, separate from workshop bays. Physical release remains open.

Depot/storage station	Line	Site basis	Stabling positions required	Usable slot length m	Workshop bays
line-1-0166-0866-s053822	line-1	storage-at-existing-powered -service-point	57	6,897.0	0
line-2-0595-0296-s056994	line-2	storage-at-existing-powered -service-point	63	7,623.0	0
line-3-0151-1637-s055819	line-3	storage-at-existing-powered -service-point	72	8,712.0	0
line-4-1246-1628-s044443	line-4	storage-at-existing-powered -service-point	57	6,897.0	0
line-5-0677-2304-s053737	line-5	storage-at-existing-powered -service-point	68	8,228.0	0
line-6-0495-1669-s057516	line-6	declared-depot	81	9,801.0	125
line-7-1531-0418-s043502	line-7	storage-at-existing-powered -service-point	56	6,776.0	0
line-8-0785-0290-s051366	line-8	storage-at-existing-powered -service-point	74	8,954.0	0
line-9-0700-0568-s098647	line-9	storage-at-existing-powered -service-point	5	605.0	0

Line	Location	Type	Direction	Role	Trainsets
line-1	line-1-0166-0866-s053822	station	reverse	revenue	2
line-1	line-1-0319-0836-s050133	station	forward	revenue	1
line-1	line-1-0319-0836-s050133	station	reverse	revenue	1
line-1	line-1-0463-0845-s047129	station	forward	revenue	1
line-1	line-1-0463-0845-s047129	station	reverse	revenue	1
line-1	line-1-0557-0867-s044984	station	forward	revenue	1
line-1	line-1-0557-0867-s044984	station	reverse	revenue	1
line-1	line-1-0643-0866-s043055	station	forward	revenue	1
line-1	line-1-0643-0866-s043055	station	reverse	revenue	1
line-1	line-1-0731-0859-s041121	station	forward	revenue	1
line-1	line-1-0731-0859-s041121	station	reverse	revenue	1
line-1	line-1-0821-0812-s038113	station	forward	revenue	1
line-1	line-1-0821-0812-s038113	station	reverse	revenue	1
line-1	line-1-0940-0787-s035101	station	forward	revenue	1
line-1	line-1-0940-0787-s035101	station	reverse	revenue	1
line-1	line-1-1063-0773-s032095	station	forward	revenue	1
line-1	line-1-1063-0773-s032095	station	reverse	revenue	1
line-1	line-1-1182-0792-s029092	station	forward	revenue	1
line-1	line-1-1182-0792-s029092	station	reverse	revenue	1
line-1	line-1-1306-0780-s026078	station	forward	revenue	1
line-1	line-1-1306-0780-s026078	station	reverse	revenue	1
line-1	line-1-1429-0829-s023054	station	forward	revenue	1
line-1	line-1-1429-0829-s023054	station	reverse	revenue	1
line-1	line-1-1554-0820-s020402	station	forward	revenue	1
line-1	line-1-1554-0820-s020402	station	reverse	revenue	1
line-1	line-1-1702-0808-s017045	station	forward	revenue	1
line-1	line-1-1702-0808-s017045	station	reverse	revenue	1
line-1	line-1-1848-0797-s014034	station	forward	revenue	1
line-1	line-1-1848-0797-s014034	station	reverse	revenue	1
line-1	line-1-2017-0784-s010529	station	forward	revenue	1
line-1	line-1-2017-0784-s010529	station	reverse	revenue	1
line-1	line-1-2180-0774-s007014	station	forward	revenue	1
line-1	line-1-2180-0774-s007014	station	reverse	revenue	1
line-1	line-1-2343-0771-s003515	station	forward	revenue	1
line-1	line-1-2343-0771-s003515	station	reverse	revenue	1
line-1	line-1-2500-0775-s000000	station	forward	revenue	2
line-2	line-2-0595-0296-s056994	station	reverse	revenue	2
line-2	line-2-0654-0439-s052456	station	forward	revenue	1

Line	Location	Type	Direction	Role	Trainsets
line-2	line-2-0654-0439-s052456	station	reverse	revenue	1
line-2	line-2-0679-0342-s054719	station	forward	revenue	1
line-2	line-2-0679-0342-s054719	station	reverse	revenue	1
line-2	line-2-0728-0584-s047170	station	forward	revenue	1
line-2	line-2-0728-0584-s047170	station	reverse	revenue	1
line-2	line-2-0753-0460-s050189	station	forward	revenue	1
line-2	line-2-0753-0460-s050189	station	reverse	revenue	1
line-2	line-2-0828-0672-s044160	station	forward	revenue	1
line-2	line-2-0828-0672-s044160	station	reverse	revenue	1
line-2	line-2-0923-0772-s041136	station	forward	revenue	1
line-2	line-2-0923-0772-s041136	station	reverse	revenue	1
line-2	line-2-0947-0911-s038129	station	forward	revenue	1
line-2	line-2-0947-0911-s038129	station	reverse	revenue	1
line-2	line-2-1022-1022-s035123	station	forward	revenue	1
line-2	line-2-1022-1022-s035123	station	reverse	revenue	1
line-2	line-2-1095-1097-s031056	station	forward	revenue	1
line-2	line-2-1095-1097-s031056	station	reverse	revenue	1
line-2	line-2-1103-1047-s033035	station	forward	revenue	1
line-2	line-2-1103-1047-s033035	station	reverse	revenue	1
line-2	line-2-1141-1176-s029095	station	forward	revenue	1
line-2	line-2-1141-1176-s029095	station	reverse	revenue	1
line-2	line-2-1249-1269-s026078	station	forward	revenue	1
line-2	line-2-1249-1269-s026078	station	reverse	revenue	1
line-2	line-2-1361-1359-s023069	station	forward	revenue	1
line-2	line-2-1361-1359-s023069	station	reverse	revenue	1
line-2	line-2-1461-1452-s020064	station	forward	revenue	1
line-2	line-2-1461-1452-s020064	station	reverse	revenue	1
line-2	line-2-1509-1487-s018636	station	forward	revenue	1
line-2	line-2-1509-1487-s018636	station	reverse	revenue	1
line-2	line-2-1550-1543-s017049	station	forward	revenue	1
line-2	line-2-1550-1543-s017049	station	reverse	revenue	1
line-2	line-2-1648-1703-s012612	station	forward	revenue	1
line-2	line-2-1648-1703-s012612	station	reverse	revenue	1
line-2	line-2-1895-2056-s003494	station	forward	revenue	1
line-2	line-2-1895-2056-s003494	station	reverse	revenue	1
line-2	line-2-1987-2192-s000000	station	forward	revenue	2
line-3	line-3-0151-1637-s055819	station	reverse	revenue	2
line-3	line-3-0242-1604-s053550	station	forward	revenue	1
line-3	line-3-0242-1604-s053550	station	reverse	revenue	1
line-3	line-3-0350-1616-s051291	station	forward	revenue	1
line-3	line-3-0350-1616-s051291	station	reverse	revenue	1
line-3	line-3-0433-1532-s048276	station	forward	revenue	1
line-3	line-3-0433-1532-s048276	station	reverse	revenue	1
line-3	line-3-0528-1521-s046268	station	forward	revenue	1
line-3	line-3-0528-1521-s046268	station	reverse	revenue	1
line-3	line-3-0620-1509-s044279	station	forward	revenue	1
line-3	line-3-0620-1509-s044279	station	reverse	revenue	1
line-3	line-3-0711-1492-s042273	station	forward	revenue	1
line-3	line-3-0711-1492-s042273	station	reverse	revenue	1
line-3	line-3-0830-1419-s039265	station	forward	revenue	1
line-3	line-3-0830-1419-s039265	station	reverse	revenue	1
line-3	line-3-0943-1486-s036258	station	forward	revenue	1
line-3	line-3-0943-1486-s036258	station	reverse	revenue	1
line-3	line-3-1046-1445-s033249	station	forward	revenue	1
line-3	line-3-1046-1445-s033249	station	reverse	revenue	1
line-3	line-3-1111-1426-s030244	station	forward	revenue	1
line-3	line-3-1111-1426-s030244	station	reverse	revenue	1
line-3	line-3-1162-1305-s027239	station	forward	revenue	1
line-3	line-3-1162-1305-s027239	station	reverse	revenue	1
line-3	line-3-1265-1319-s024226	station	forward	revenue	1
line-3	line-3-1265-1319-s024226	station	reverse	revenue	1
line-3	line-3-1391-1293-s021226	station	forward	revenue	1
line-3	line-3-1391-1293-s021226	station	reverse	revenue	1

Line	Location	Type	Direction	Role	Trainsets
line-3	line-3-1502-1256-s018214	station	forward	revenue	1
line-3	line-3-1502-1256-s018214	station	reverse	revenue	1
line-3	line-3-1613-1236-s015796	station	forward	revenue	1
line-3	line-3-1613-1236-s015796	station	reverse	revenue	1
line-3	line-3-1702-1235-s014007	station	forward	revenue	1
line-3	line-3-1702-1235-s014007	station	reverse	revenue	1
line-3	line-3-2271-1120-s000000	station	forward	revenue	2
line-4	line-4-0146-0199-s000000	station	forward	revenue	2
line-4	line-4-0298-0305-s004261	station	forward	revenue	1
line-4	line-4-0298-0305-s004261	station	reverse	revenue	1
line-4	line-4-0515-0750-s016457	station	forward	revenue	1
line-4	line-4-0515-0750-s016457	station	reverse	revenue	1
line-4	line-4-0594-0786-s018376	station	forward	revenue	1
line-4	line-4-0594-0786-s018376	station	reverse	revenue	1
line-4	line-4-0666-0827-s020297	station	forward	revenue	1
line-4	line-4-0666-0827-s020297	station	reverse	revenue	1
line-4	line-4-0708-0930-s023314	station	forward	revenue	1
line-4	line-4-0708-0930-s023314	station	reverse	revenue	1
line-4	line-4-0789-1036-s026330	station	forward	revenue	1
line-4	line-4-0789-1036-s026330	station	reverse	revenue	1
line-4	line-4-0834-1141-s029342	station	forward	revenue	1
line-4	line-4-0834-1141-s029342	station	reverse	revenue	1
line-4	line-4-0936-1241-s032367	station	forward	revenue	1
line-4	line-4-0936-1241-s032367	station	reverse	revenue	1
line-4	line-4-1020-1350-s035373	station	forward	revenue	1
line-4	line-4-1020-1350-s035373	station	reverse	revenue	1
line-4	line-4-1080-1447-s038373	station	forward	revenue	1
line-4	line-4-1080-1447-s038373	station	reverse	revenue	1
line-4	line-4-1154-1476-s040395	station	forward	revenue	1
line-4	line-4-1154-1476-s040395	station	reverse	revenue	1
line-4	line-4-1217-1551-s042429	station	forward	revenue	1
line-4	line-4-1217-1551-s042429	station	reverse	revenue	1
line-4	line-4-1246-1628-s044443	station	reverse	revenue	2
line-5	line-5-0677-2304-s053737	station	reverse	revenue	2
line-5	line-5-0853-1593-s036330	station	forward	revenue	1
line-5	line-5-0853-1593-s036330	station	reverse	revenue	1
line-5	line-5-0895-1451-s032907	station	forward	revenue	1
line-5	line-5-0895-1451-s032907	station	reverse	revenue	1
line-5	line-5-0904-1340-s029889	station	forward	revenue	1
line-5	line-5-0904-1340-s029889	station	reverse	revenue	1
line-5	line-5-0974-0947-s018085	station	forward	revenue	1
line-5	line-5-0974-0947-s018085	station	reverse	revenue	1
line-5	line-5-0993-1151-s023847	station	forward	revenue	1
line-5	line-5-0993-1151-s023847	station	reverse	revenue	1
line-5	line-5-1000-1254-s026861	station	forward	revenue	1
line-5	line-5-1000-1254-s026861	station	reverse	revenue	1
line-5	line-5-1004-0809-s015076	station	forward	revenue	1
line-5	line-5-1004-0809-s015076	station	reverse	revenue	1
line-5	line-5-1008-1046-s020829	station	forward	revenue	1
line-5	line-5-1008-1046-s020829	station	reverse	revenue	1
line-5	line-5-1062-0705-s012053	station	forward	revenue	1
line-5	line-5-1062-0705-s012053	station	reverse	revenue	1
line-5	line-5-1077-0591-s009034	station	forward	revenue	1
line-5	line-5-1077-0591-s009034	station	reverse	revenue	1
line-5	line-5-1116-0417-s003985	station	forward	revenue	1
line-5	line-5-1116-0417-s003985	station	reverse	revenue	1
line-5	line-5-1122-0491-s006033	station	forward	revenue	1
line-5	line-5-1122-0491-s006033	station	reverse	revenue	1
line-5	line-5-1170-0351-s002077	station	forward	revenue	1
line-5	line-5-1170-0351-s002077	station	reverse	revenue	1
line-5	line-5-1199-0260-s000000	station	forward	revenue	2
line-6	line-6-0495-1669-s057516	station	reverse	revenue	2
line-6	line-6-0647-1607-s052578	station	forward	revenue	1

Line	Location	Type	Direction	Role	Trainsets
line-6	line-6-0647-1607-s052578	station	reverse	revenue	1
line-6	line-6-0691-1529-s050169	station	forward	revenue	1
line-6	line-6-0691-1529-s050169	station	reverse	revenue	1
line-6	line-6-0734-1414-s047145	station	forward	revenue	1
line-6	line-6-0734-1414-s047145	station	reverse	revenue	1
line-6	line-6-0837-1343-s044142	station	forward	revenue	1
line-6	line-6-0837-1343-s044142	station	reverse	revenue	1
line-6	line-6-0973-1337-s041141	station	forward	revenue	1
line-6	line-6-0973-1337-s041141	station	reverse	revenue	1
line-6	line-6-1081-1255-s038128	station	forward	revenue	1
line-6	line-6-1081-1255-s038128	station	reverse	revenue	1
line-6	line-6-1156-1145-s035101	station	forward	revenue	1
line-6	line-6-1156-1145-s035101	station	reverse	revenue	1
line-6	line-6-1219-1042-s032087	station	forward	revenue	1
line-6	line-6-1219-1042-s032087	station	reverse	revenue	1
line-6	line-6-1332-0995-s029084	station	forward	revenue	1
line-6	line-6-1332-0995-s029084	station	reverse	revenue	1
line-6	line-6-1417-0951-s026078	station	forward	revenue	1
line-6	line-6-1417-0951-s026078	station	reverse	revenue	1
line-6	line-6-1515-0927-s023073	station	forward	revenue	1
line-6	line-6-1515-0927-s023073	station	reverse	revenue	1
line-6	line-6-1580-0848-s020342	station	forward	revenue	1
line-6	line-6-1580-0848-s020342	station	reverse	revenue	1
line-6	line-6-1694-0774-s017029	station	forward	revenue	1
line-6	line-6-1694-0774-s017029	station	reverse	revenue	1
line-6	line-6-2322-0251-s000000	station	forward	revenue	2
line-7	line-7-1069-2206-s000000	station	forward	revenue	2
line-7	line-7-1185-1622-s012693	station	forward	revenue	1
line-7	line-7-1185-1622-s012693	station	reverse	revenue	1
line-7	line-7-1206-1546-s014599	station	forward	revenue	1
line-7	line-7-1206-1546-s014599	station	reverse	revenue	1
line-7	line-7-1277-1491-s016522	station	forward	revenue	1
line-7	line-7-1277-1491-s016522	station	reverse	revenue	1
line-7	line-7-1314-1261-s022537	station	forward	revenue	1
line-7	line-7-1314-1261-s022537	station	reverse	revenue	1
line-7	line-7-1328-1386-s019526	station	forward	revenue	1
line-7	line-7-1328-1386-s019526	station	reverse	revenue	1
line-7	line-7-1396-0881-s031947	station	forward	revenue	1
line-7	line-7-1396-0881-s031947	station	reverse	revenue	1
line-7	line-7-1398-1020-s028935	station	forward	revenue	1
line-7	line-7-1398-1020-s028935	station	reverse	revenue	1
line-7	line-7-1401-1143-s025930	station	forward	revenue	1
line-7	line-7-1401-1143-s025930	station	reverse	revenue	1
line-7	line-7-1412-0741-s034958	station	forward	revenue	1
line-7	line-7-1412-0741-s034958	station	reverse	revenue	1
line-7	line-7-1420-0682-s036232	station	forward	revenue	1
line-7	line-7-1420-0682-s036232	station	reverse	revenue	1
line-7	line-7-1434-0612-s037982	station	forward	revenue	1
line-7	line-7-1434-0612-s037982	station	reverse	revenue	1
line-7	line-7-1487-0523-s040748	station	forward	revenue	1
line-7	line-7-1487-0523-s040748	station	reverse	revenue	1
line-7	line-7-1531-0418-s043502	station	reverse	revenue	2
line-8	line-8-0785-0290-s051366	station	reverse	revenue	2
line-8	line-8-0835-0366-s048955	station	forward	revenue	1
line-8	line-8-0835-0366-s048955	station	reverse	revenue	1
line-8	line-8-0957-0411-s045951	station	forward	revenue	1
line-8	line-8-0957-0411-s045951	station	reverse	revenue	1
line-8	line-8-0976-0467-s044485	station	forward	revenue	1
line-8	line-8-0976-0467-s044485	station	reverse	revenue	1
line-8	line-8-1004-0531-s042945	station	forward	revenue	1
line-8	line-8-1004-0531-s042945	station	reverse	revenue	1
line-8	line-8-1085-0596-s039937	station	forward	revenue	1
line-8	line-8-1085-0596-s039937	station	reverse	revenue	1

Line	Location	Type	Direction	Role	Trainsets
line-8	line-8-1168-0688-s036929	station	forward	revenue	1
line-8	line-8-1168-0688-s036929	station	reverse	revenue	1
line-8	line-8-1281-0738-s033926	station	forward	revenue	1
line-8	line-8-1281-0738-s033926	station	reverse	revenue	1
line-8	line-8-1349-0838-s030923	station	forward	revenue	1
line-8	line-8-1349-0838-s030923	station	reverse	revenue	1
line-8	line-8-1422-0924-s027911	station	forward	revenue	1
line-8	line-8-1422-0924-s027911	station	reverse	revenue	1
line-8	line-8-1518-0993-s024893	station	forward	revenue	1
line-8	line-8-1518-0993-s024893	station	reverse	revenue	1
line-8	line-8-1645-1094-s021347	station	forward	revenue	1
line-8	line-8-1645-1094-s021347	station	reverse	revenue	1
line-8	line-8-2231-1783-s000000	station	forward	revenue	2
line-9	line-9-0487-0760-s006980	station	reverse	revenue	1
line-9	line-9-0502-0673-s005049	station	forward	revenue	1
line-9	line-9-0502-0673-s005049	station	reverse	revenue	1
line-9	line-9-0538-1065-s015166	station	forward	revenue	1
line-9	line-9-0538-1065-s015166	station	reverse	revenue	1
line-9	line-9-0549-0960-s012159	station	forward	revenue	1
line-9	line-9-0557-0867-s010067	station	forward	revenue	1
line-9	line-9-0557-0867-s010067	station	reverse	revenue	1
line-9	line-9-0565-0620-s003116	station	forward	revenue	1
line-9	line-9-0565-0620-s003116	station	reverse	revenue	1
line-9	line-9-0619-1569-s028209	station	forward	revenue	1
line-9	line-9-0619-1569-s028209	station	reverse	revenue	1
line-9	line-9-0620-1509-s026894	station	forward	revenue	1
line-9	line-9-0620-1509-s026894	station	reverse	revenue	1
line-9	line-9-0637-1439-s025194	station	forward	revenue	1
line-9	line-9-0647-1607-s029412	station	reverse	revenue	1
line-9	line-9-0853-1593-s033809	station	forward	revenue	1
line-9	line-9-0853-1593-s033809	station	reverse	revenue	1
line-9	line-9-0911-1579-s035217	station	forward	revenue	1
line-9	line-9-0976-0467-s090748	station	forward	revenue	1
line-9	line-9-1116-0417-s087021	station	reverse	revenue	1
line-9	line-9-1185-1622-s041363	station	forward	revenue	1
line-9	line-9-1185-1622-s041363	station	reverse	revenue	1
line-9	line-9-1190-0372-s084695	station	forward	revenue	1
line-9	line-9-1190-0372-s084695	station	reverse	revenue	1
line-9	line-9-1200-0443-s082619	station	reverse	revenue	1
line-9	line-9-1246-0493-s080552	station	forward	revenue	1
line-9	line-9-1246-0493-s080552	station	reverse	revenue	1
line-9	line-9-1246-1628-s042633	station	forward	revenue	1
line-9	line-9-1335-0562-s077540	station	forward	revenue	1
line-9	line-9-1335-0562-s077540	station	reverse	revenue	1
line-9	line-9-1389-0675-s074529	station	forward	revenue	1
line-9	line-9-1430-1529-s048432	station	forward	revenue	1
line-9	line-9-1430-1529-s048432	station	reverse	revenue	1
line-9	line-9-1477-0761-s071522	station	forward	revenue	1
line-9	line-9-1477-0761-s071522	station	reverse	revenue	1
line-9	line-9-1509-1487-s050681	station	forward	revenue	1
line-9	line-9-1509-1487-s050681	station	reverse	revenue	1
line-9	line-9-1580-0848-s068248	station	reverse	revenue	1
line-9	line-9-1613-1236-s058396	station	reverse	revenue	1
line-9	line-9-1645-1094-s061501	station	forward	revenue	1
line-9	line-9-1645-1094-s061501	station	reverse	revenue	1
line-1	line-1-0166-0866-s053822	depot	—	revenue	48
line-1	line-1-0166-0866-s053822	depot	—	spare	8
line-1	line-1-0166-0866-s053822	depot	—	cold_reserve	1
line-2	line-2-0595-0296-s056994	depot	—	revenue	53
line-2	line-2-0595-0296-s056994	depot	—	spare	9
line-2	line-2-0595-0296-s056994	depot	—	cold_reserve	1
line-3	line-3-0151-1637-s055819	depot	—	revenue	62
line-3	line-3-0151-1637-s055819	depot	—	spare	9

Line	Location	Type	Direction	Role	Trainsets
line-3	line-3-0151-1637-s055819	depot	—	cold_reserve	1
line-4	line-4-1246-1628-s044443	depot	—	revenue	49
line-4	line-4-1246-1628-s044443	depot	—	spare	7
line-4	line-4-1246-1628-s044443	depot	—	cold_reserve	1
line-5	line-5-0677-2304-s053737	depot	—	revenue	59
line-5	line-5-0677-2304-s053737	depot	—	spare	8
line-5	line-5-0677-2304-s053737	depot	—	cold_reserve	1
line-6	line-6-0495-1669-s057516	depot	—	revenue	70
line-6	line-6-0495-1669-s057516	depot	—	spare	10
line-6	line-6-0495-1669-s057516	depot	—	cold_reserve	1
line-7	line-7-1531-0418-s043502	depot	—	revenue	48
line-7	line-7-1531-0418-s043502	depot	—	spare	7
line-7	line-7-1531-0418-s043502	depot	—	cold_reserve	1
line-8	line-8-0785-0290-s051366	depot	—	revenue	64
line-8	line-8-0785-0290-s051366	depot	—	spare	9
line-8	line-8-0785-0290-s051366	depot	—	cold_reserve	1
line-9	line-9-0700-0568-s098647	depot	—	spare	4
line-9	line-9-0700-0568-s098647	depot	—	cold_reserve	1

Native hybrid candidate unavailable: morning station allocation is incomplete.

Station-only native benchmark

The runnable scenario below tests station holding and restart behaviour. It does not yet execute the station/depot allocation above or depot yard movements. Its station overflow is a diagnostic result, not the overnight design allocation.

Operating allocation: 831 trainsets at 158 stations; largest initial station queue 10. Physical release: open.

This candidate preserves all non-fleet scenario inputs and the existing fleet counts/service windows. It enables station holding and 150 kW top-up to 95% SoC, subject to shared site limits. Existing canonical simulation evidence still describes the retained endpoint-dispatch scenario.

Fleet roles: 751 revenue, 71 spare, 9 cold reserve. Reserves are held out of routine dispatch.

Two-train station-capacity check: FAIL. Selected stations provide 316 positions; 515 fleet positions exceed station-only provision. The initial allocation exceeds the limit at 126 stations. Four-berth reference platforms do not override the two-train provision.

Line	Station	Direction	Role	Initial trainsets	Verified track slots
line-1	line-1-2500-0775-s000000	forward	revenue	3	pending
line-1	line-1-2343-0771-s003515	forward	revenue	3	pending
line-1	line-1-2343-0771-s003515	reverse	revenue	3	pending
line-1	line-1-2180-0774-s007014	forward	revenue	3	pending
line-1	line-1-2180-0774-s007014	reverse	revenue	3	pending
line-1	line-1-2017-0784-s010529	forward	revenue	3	pending
line-1	line-1-2017-0784-s010529	reverse	revenue	3	pending
line-1	line-1-1848-0797-s014034	forward	revenue	3	pending
line-1	line-1-1848-0797-s014034	reverse	revenue	3	pending
line-1	line-1-1702-0808-s017045	forward	revenue	3	pending
line-1	line-1-1702-0808-s017045	reverse	revenue	3	pending
line-1	line-1-1554-0820-s020402	forward	revenue	3	pending
line-1	line-1-1554-0820-s020402	reverse	revenue	3	pending
line-1	line-1-1429-0829-s023054	forward	revenue	3	pending
line-1	line-1-1429-0829-s023054	reverse	revenue	2	pending
line-1	line-1-1306-0780-s026078	forward	revenue	2	pending
line-1	line-1-1306-0780-s026078	reverse	revenue	2	pending
line-1	line-1-1182-0792-s029092	forward	revenue	2	pending
line-1	line-1-1182-0792-s029092	reverse	revenue	2	pending
line-1	line-1-1063-0773-s032095	forward	revenue	2	pending
line-1	line-1-1063-0773-s032095	reverse	revenue	2	pending
line-1	line-1-0940-0787-s035101	forward	revenue	2	pending
line-1	line-1-0940-0787-s035101	reverse	revenue	2	pending
line-1	line-1-0821-0812-s038113	forward	revenue	2	pending
line-1	line-1-0821-0812-s038113	reverse	revenue	2	pending
line-1	line-1-0731-0859-s041121	forward	revenue	2	pending

Line	Station	Direction	Role	Initial trainsets	Verified track slots
line-1	line-1-0731-0859-s041121	reverse	revenue	2	pending
line-1	line-1-0643-0866-s043055	forward	revenue	2	pending
line-1	line-1-0643-0866-s043055	reverse	revenue	2	pending
line-1	line-1-0557-0867-s044984	forward	revenue	2	pending
line-1	line-1-0557-0867-s044984	reverse	revenue	2	pending
line-1	line-1-0463-0845-s047129	forward	revenue	2	pending
line-1	line-1-0463-0845-s047129	reverse	revenue	2	pending
line-1	line-1-0319-0836-s050133	forward	revenue	2	pending
line-1	line-1-0319-0836-s050133	reverse	revenue	2	pending
line-1	line-1-0166-0866-s053822	reverse	revenue	2	pending
line-1	line-1-1429-0829-s023054	reverse	spare	1	pending
line-1	line-1-1306-0780-s026078	forward	spare	1	pending
line-1	line-1-1306-0780-s026078	reverse	spare	1	pending
line-1	line-1-1182-0792-s029092	forward	spare	1	pending
line-1	line-1-1182-0792-s029092	reverse	spare	1	pending
line-1	line-1-1063-0773-s032095	forward	spare	1	pending
line-1	line-1-1063-0773-s032095	reverse	spare	1	pending
line-1	line-1-0940-0787-s035101	forward	spare	1	pending
line-1	line-1-0940-0787-s035101	reverse	cold_reserve	1	pending
line-2	line-2-1987-2192-s000000	forward	revenue	3	pending
line-2	line-2-1895-2056-s003494	forward	revenue	3	pending
line-2	line-2-1895-2056-s003494	reverse	revenue	3	pending
line-2	line-2-1648-1703-s012612	forward	revenue	3	pending
line-2	line-2-1648-1703-s012612	reverse	revenue	3	pending
line-2	line-2-1550-1543-s017049	forward	revenue	3	pending
line-2	line-2-1550-1543-s017049	reverse	revenue	3	pending
line-2	line-2-1509-1487-s018636	forward	revenue	3	pending
line-2	line-2-1509-1487-s018636	reverse	revenue	3	pending
line-2	line-2-1461-1452-s020064	forward	revenue	3	pending
line-2	line-2-1461-1452-s020064	reverse	revenue	3	pending
line-2	line-2-1361-1359-s023069	forward	revenue	3	pending
line-2	line-2-1361-1359-s023069	reverse	revenue	3	pending
line-2	line-2-1249-1269-s026078	forward	revenue	3	pending
line-2	line-2-1249-1269-s026078	reverse	revenue	3	pending
line-2	line-2-1141-1176-s029095	forward	revenue	3	pending
line-2	line-2-1141-1176-s029095	reverse	revenue	3	pending
line-2	line-2-1095-1097-s031056	forward	revenue	2	pending
line-2	line-2-1095-1097-s031056	reverse	revenue	2	pending
line-2	line-2-1103-1047-s033035	forward	revenue	2	pending
line-2	line-2-1103-1047-s033035	reverse	revenue	2	pending
line-2	line-2-1022-1022-s035123	forward	revenue	2	pending
line-2	line-2-1022-1022-s035123	reverse	revenue	2	pending
line-2	line-2-0947-0911-s038129	forward	revenue	2	pending
line-2	line-2-0947-0911-s038129	reverse	revenue	2	pending
line-2	line-2-0923-0772-s041136	forward	revenue	2	pending
line-2	line-2-0923-0772-s041136	reverse	revenue	2	pending
line-2	line-2-0828-0672-s044160	forward	revenue	2	pending
line-2	line-2-0828-0672-s044160	reverse	revenue	2	pending
line-2	line-2-0728-0584-s047170	forward	revenue	2	pending
line-2	line-2-0728-0584-s047170	reverse	revenue	2	pending
line-2	line-2-0753-0460-s050189	forward	revenue	2	pending
line-2	line-2-0753-0460-s050189	reverse	revenue	2	pending
line-2	line-2-0654-0439-s052456	forward	revenue	2	pending
line-2	line-2-0654-0439-s052456	reverse	revenue	2	pending
line-2	line-2-0679-0342-s054719	forward	revenue	2	pending
line-2	line-2-0679-0342-s054719	reverse	revenue	2	pending
line-2	line-2-0595-0296-s056994	reverse	revenue	2	pending
line-2	line-2-1095-1097-s031056	forward	spare	1	pending
line-2	line-2-1095-1097-s031056	reverse	spare	1	pending
line-2	line-2-1103-1047-s033035	forward	spare	1	pending
line-2	line-2-1103-1047-s033035	reverse	spare	1	pending
line-2	line-2-1022-1022-s035123	forward	spare	1	pending
line-2	line-2-1022-1022-s035123	reverse	spare	1	pending

Line	Station	Direction	Role	Initial trainsets	Verified track slots
line-2	line-2-0947-0911-s038129	forward	spare	1	pending
line-2	line-2-0947-0911-s038129	reverse	spare	1	pending
line-2	line-2-0923-0772-s041136	forward	spare	1	pending
line-2	line-2-0923-0772-s041136	reverse	cold_reserve	1	pending
line-3	line-3-2271-1120-s000000	forward	revenue	3	pending
line-3	line-3-1702-1235-s014007	forward	revenue	3	pending
line-3	line-3-1702-1235-s014007	reverse	revenue	3	pending
line-3	line-3-1613-1236-s015796	forward	revenue	3	pending
line-3	line-3-1613-1236-s015796	reverse	revenue	3	pending
line-3	line-3-1502-1256-s018214	forward	revenue	3	pending
line-3	line-3-1502-1256-s018214	reverse	revenue	3	pending
line-3	line-3-1391-1293-s021226	forward	revenue	3	pending
line-3	line-3-1391-1293-s021226	reverse	revenue	3	pending
line-3	line-3-1265-1319-s024226	forward	revenue	3	pending
line-3	line-3-1265-1319-s024226	reverse	revenue	3	pending
line-3	line-3-1162-1305-s027239	forward	revenue	3	pending
line-3	line-3-1162-1305-s027239	reverse	revenue	3	pending
line-3	line-3-1111-1426-s030244	forward	revenue	3	pending
line-3	line-3-1111-1426-s030244	reverse	revenue	3	pending
line-3	line-3-1046-1445-s033249	forward	revenue	3	pending
line-3	line-3-1046-1445-s033249	reverse	revenue	3	pending
line-3	line-3-0943-1486-s036258	forward	revenue	3	pending
line-3	line-3-0943-1486-s036258	reverse	revenue	3	pending
line-3	line-3-0830-1419-s039265	forward	revenue	3	pending
line-3	line-3-0830-1419-s039265	reverse	revenue	3	pending
line-3	line-3-0711-1492-s042273	forward	revenue	3	pending
line-3	line-3-0711-1492-s042273	reverse	revenue	3	pending
line-3	line-3-0620-1509-s044279	forward	revenue	3	pending
line-3	line-3-0620-1509-s044279	reverse	revenue	3	pending
line-3	line-3-0528-1521-s046268	forward	revenue	3	pending
line-3	line-3-0528-1521-s046268	reverse	revenue	3	pending
line-3	line-3-0433-1532-s048276	forward	revenue	3	pending
line-3	line-3-0433-1532-s048276	reverse	revenue	3	pending
line-3	line-3-0350-1616-s051291	forward	revenue	3	pending
line-3	line-3-0350-1616-s051291	reverse	revenue	2	pending
line-3	line-3-0242-1604-s053550	forward	revenue	2	pending
line-3	line-3-0242-1604-s053550	reverse	revenue	2	pending
line-3	line-3-0151-1637-s055819	reverse	revenue	2	pending
line-3	line-3-0350-1616-s051291	reverse	spare	1	pending
line-3	line-3-0242-1604-s053550	forward	spare	1	pending
line-3	line-3-0242-1604-s053550	reverse	spare	1	pending
line-3	line-3-0151-1637-s055819	reverse	spare	1	pending
line-3	line-3-2271-1120-s000000	forward	spare	1	pending
line-3	line-3-1702-1235-s014007	forward	spare	1	pending
line-3	line-3-1702-1235-s014007	reverse	spare	1	pending
line-3	line-3-1613-1236-s015796	forward	spare	1	pending
line-3	line-3-1613-1236-s015796	reverse	spare	1	pending
line-3	line-3-1502-1256-s018214	forward	cold_reserve	1	pending
line-4	line-4-0146-0199-s000000	forward	revenue	3	pending
line-4	line-4-0298-0305-s004261	forward	revenue	3	pending
line-4	line-4-0298-0305-s004261	reverse	revenue	3	pending
line-4	line-4-0515-0750-s016457	forward	revenue	3	pending
line-4	line-4-0515-0750-s016457	reverse	revenue	3	pending
line-4	line-4-0594-0786-s018376	forward	revenue	3	pending
line-4	line-4-0594-0786-s018376	reverse	revenue	3	pending
line-4	line-4-0666-0827-s020297	forward	revenue	3	pending
line-4	line-4-0666-0827-s020297	reverse	revenue	3	pending
line-4	line-4-0708-0930-s023314	forward	revenue	3	pending
line-4	line-4-0708-0930-s023314	reverse	revenue	3	pending
line-4	line-4-0789-1036-s026330	forward	revenue	3	pending
line-4	line-4-0789-1036-s026330	reverse	revenue	3	pending
line-4	line-4-0834-1141-s029342	forward	revenue	3	pending
line-4	line-4-0834-1141-s029342	reverse	revenue	3	pending

Line	Station	Direction	Role	Initial trainsets	Verified track slots
line-4	line-4-0936-1241-s032367	forward	revenue	3	pending
line-4	line-4-0936-1241-s032367	reverse	revenue	3	pending
line-4	line-4-1020-1350-s035373	forward	revenue	3	pending
line-4	line-4-1020-1350-s035373	reverse	revenue	3	pending
line-4	line-4-1080-1447-s038373	forward	revenue	3	pending
line-4	line-4-1080-1447-s038373	reverse	revenue	3	pending
line-4	line-4-1154-1476-s040395	forward	revenue	3	pending
line-4	line-4-1154-1476-s040395	reverse	revenue	3	pending
line-4	line-4-1217-1551-s042429	forward	revenue	3	pending
line-4	line-4-1217-1551-s042429	reverse	revenue	3	pending
line-4	line-4-1246-1628-s044443	reverse	revenue	2	pending
line-4	line-4-1246-1628-s044443	reverse	spare	1	pending
line-4	line-4-0146-0199-s000000	forward	spare	1	pending
line-4	line-4-0298-0305-s004261	forward	spare	1	pending
line-4	line-4-0298-0305-s004261	reverse	spare	1	pending
line-4	line-4-0515-0750-s016457	forward	spare	1	pending
line-4	line-4-0515-0750-s016457	reverse	spare	1	pending
line-4	line-4-0594-0786-s018376	forward	spare	1	pending
line-4	line-4-0594-0786-s018376	reverse	cold_reserve	1	pending
line-5	line-5-1199-0260-s000000	forward	revenue	4	pending
line-5	line-5-1170-0351-s002077	forward	revenue	4	pending
line-5	line-5-1170-0351-s002077	reverse	revenue	4	pending
line-5	line-5-1116-0417-s003985	forward	revenue	4	pending
line-5	line-5-1116-0417-s003985	reverse	revenue	4	pending
line-5	line-5-1122-0491-s006033	forward	revenue	3	pending
line-5	line-5-1122-0491-s006033	reverse	revenue	3	pending
line-5	line-5-1077-0591-s009034	forward	revenue	3	pending
line-5	line-5-1077-0591-s009034	reverse	revenue	3	pending
line-5	line-5-1062-0705-s012053	forward	revenue	3	pending
line-5	line-5-1062-0705-s012053	reverse	revenue	3	pending
line-5	line-5-1004-0809-s015076	forward	revenue	3	pending
line-5	line-5-1004-0809-s015076	reverse	revenue	3	pending
line-5	line-5-0974-0947-s018085	forward	revenue	3	pending
line-5	line-5-0974-0947-s018085	reverse	revenue	3	pending
line-5	line-5-1008-1046-s020829	forward	revenue	3	pending
line-5	line-5-1008-1046-s020829	reverse	revenue	3	pending
line-5	line-5-0993-1151-s023847	forward	revenue	3	pending
line-5	line-5-0993-1151-s023847	reverse	revenue	3	pending
line-5	line-5-1000-1254-s026861	forward	revenue	3	pending
line-5	line-5-1000-1254-s026861	reverse	revenue	3	pending
line-5	line-5-0904-1340-s029889	forward	revenue	3	pending
line-5	line-5-0904-1340-s029889	reverse	revenue	3	pending
line-5	line-5-0895-1451-s032907	forward	revenue	3	pending
line-5	line-5-0895-1451-s032907	reverse	revenue	3	pending
line-5	line-5-0853-1593-s036330	forward	revenue	3	pending
line-5	line-5-0853-1593-s036330	reverse	revenue	3	pending
line-5	line-5-0677-2304-s053737	reverse	revenue	3	pending
line-5	line-5-1122-0491-s006033	forward	spare	1	pending
line-5	line-5-1122-0491-s006033	reverse	spare	1	pending
line-5	line-5-1077-0591-s009034	forward	spare	1	pending
line-5	line-5-1077-0591-s009034	reverse	spare	1	pending
line-5	line-5-1062-0705-s012053	forward	spare	1	pending
line-5	line-5-1062-0705-s012053	reverse	spare	1	pending
line-5	line-5-1004-0809-s015076	forward	spare	1	pending
line-5	line-5-1004-0809-s015076	reverse	spare	1	pending
line-5	line-5-0974-0947-s018085	forward	cold_reserve	1	pending
line-6	line-6-2322-0251-s000000	forward	revenue	4	pending
line-6	line-6-1694-0774-s017029	forward	revenue	4	pending
line-6	line-6-1694-0774-s017029	reverse	revenue	4	pending
line-6	line-6-1580-0848-s020342	forward	revenue	4	pending
line-6	line-6-1580-0848-s020342	reverse	revenue	4	pending
line-6	line-6-1515-0927-s023073	forward	revenue	4	pending
line-6	line-6-1515-0927-s023073	reverse	revenue	4	pending

Line	Station	Direction	Role	Initial trainsets	Verified track slots
line-6	line-6-1417-0951-s026078	forward	revenue	4	pending
line-6	line-6-1417-0951-s026078	reverse	revenue	4	pending
line-6	line-6-1332-0995-s029084	forward	revenue	4	pending
line-6	line-6-1332-0995-s029084	reverse	revenue	4	pending
line-6	line-6-1219-1042-s032087	forward	revenue	4	pending
line-6	line-6-1219-1042-s032087	reverse	revenue	4	pending
line-6	line-6-1156-1145-s035101	forward	revenue	4	pending
line-6	line-6-1156-1145-s035101	reverse	revenue	4	pending
line-6	line-6-1081-1255-s038128	forward	revenue	4	pending
line-6	line-6-1081-1255-s038128	reverse	revenue	3	pending
line-6	line-6-0973-1337-s041141	forward	revenue	3	pending
line-6	line-6-0973-1337-s041141	reverse	revenue	3	pending
line-6	line-6-0837-1343-s044142	forward	revenue	3	pending
line-6	line-6-0837-1343-s044142	reverse	revenue	3	pending
line-6	line-6-0734-1414-s047145	forward	revenue	3	pending
line-6	line-6-0734-1414-s047145	reverse	revenue	3	pending
line-6	line-6-0691-1529-s050169	forward	revenue	3	pending
line-6	line-6-0691-1529-s050169	reverse	revenue	3	pending
line-6	line-6-0647-1607-s052578	forward	revenue	3	pending
line-6	line-6-0647-1607-s052578	reverse	revenue	3	pending
line-6	line-6-0495-1669-s057516	reverse	revenue	3	pending
line-6	line-6-1081-1255-s038128	reverse	spare	1	pending
line-6	line-6-0973-1337-s041141	forward	spare	1	pending
line-6	line-6-0973-1337-s041141	reverse	spare	1	pending
line-6	line-6-0837-1343-s044142	forward	spare	1	pending
line-6	line-6-0837-1343-s044142	reverse	spare	1	pending
line-6	line-6-0734-1414-s047145	forward	spare	1	pending
line-6	line-6-0734-1414-s047145	reverse	spare	1	pending
line-6	line-6-0691-1529-s050169	forward	spare	1	pending
line-6	line-6-0691-1529-s050169	reverse	spare	1	pending
line-6	line-6-0647-1607-s052578	forward	spare	1	pending
line-6	line-6-0647-1607-s052578	reverse	cold_reserve	1	pending
line-7	line-7-1069-2206-s000000	forward	revenue	3	pending
line-7	line-7-1185-1622-s012693	forward	revenue	3	pending
line-7	line-7-1185-1622-s012693	reverse	revenue	3	pending
line-7	line-7-1206-1546-s014599	forward	revenue	3	pending
line-7	line-7-1206-1546-s014599	reverse	revenue	3	pending
line-7	line-7-1277-1491-s016522	forward	revenue	3	pending
line-7	line-7-1277-1491-s016522	reverse	revenue	3	pending
line-7	line-7-1328-1386-s019526	forward	revenue	3	pending
line-7	line-7-1328-1386-s019526	reverse	revenue	3	pending
line-7	line-7-1314-1261-s022537	forward	revenue	3	pending
line-7	line-7-1314-1261-s022537	reverse	revenue	3	pending
line-7	line-7-1401-1143-s025930	forward	revenue	3	pending
line-7	line-7-1401-1143-s025930	reverse	revenue	3	pending
line-7	line-7-1398-1020-s028935	forward	revenue	3	pending
line-7	line-7-1398-1020-s028935	reverse	revenue	3	pending
line-7	line-7-1396-0881-s031947	forward	revenue	3	pending
line-7	line-7-1396-0881-s031947	reverse	revenue	3	pending
line-7	line-7-1412-0741-s034958	forward	revenue	3	pending
line-7	line-7-1412-0741-s034958	reverse	revenue	3	pending
line-7	line-7-1420-0682-s036232	forward	revenue	3	pending
line-7	line-7-1420-0682-s036232	reverse	revenue	3	pending
line-7	line-7-1434-0612-s037982	forward	revenue	3	pending
line-7	line-7-1434-0612-s037982	reverse	revenue	3	pending
line-7	line-7-1487-0523-s040748	forward	revenue	3	pending
line-7	line-7-1487-0523-s040748	reverse	revenue	2	pending
line-7	line-7-1531-0418-s043502	reverse	revenue	2	pending
line-7	line-7-1487-0523-s040748	reverse	spare	1	pending
line-7	line-7-1531-0418-s043502	reverse	spare	1	pending
line-7	line-7-1069-2206-s000000	forward	spare	1	pending
line-7	line-7-1185-1622-s012693	forward	spare	1	pending
line-7	line-7-1185-1622-s012693	reverse	spare	1	pending

Line	Station	Direction	Role	Initial trainsets	Verified track slots
line-7	line-7-1206-1546-s014599	forward	spare	1	pending
line-7	line-7-1206-1546-s014599	reverse	spare	1	pending
line-7	line-7-1277-1491-s016522	forward	cold_reserve	1	pending
line-8	line-8-2231-1783-s000000	forward	revenue	4	pending
line-8	line-8-1645-1094-s021347	forward	revenue	4	pending
line-8	line-8-1645-1094-s021347	reverse	revenue	4	pending
line-8	line-8-1518-0993-s024893	forward	revenue	4	pending
line-8	line-8-1518-0993-s024893	reverse	revenue	4	pending
line-8	line-8-1422-0924-s027911	forward	revenue	4	pending
line-8	line-8-1422-0924-s027911	reverse	revenue	4	pending
line-8	line-8-1349-0838-s030923	forward	revenue	4	pending
line-8	line-8-1349-0838-s030923	reverse	revenue	4	pending
line-8	line-8-1281-0738-s033926	forward	revenue	4	pending
line-8	line-8-1281-0738-s033926	reverse	revenue	4	pending
line-8	line-8-1168-0688-s036929	forward	revenue	4	pending
line-8	line-8-1168-0688-s036929	reverse	revenue	4	pending
line-8	line-8-1085-0596-s039937	forward	revenue	4	pending
line-8	line-8-1085-0596-s039937	reverse	revenue	4	pending
line-8	line-8-1004-0531-s042945	forward	revenue	4	pending
line-8	line-8-1004-0531-s042945	reverse	revenue	4	pending
line-8	line-8-0976-0467-s044485	forward	revenue	4	pending
line-8	line-8-0976-0467-s044485	reverse	revenue	3	pending
line-8	line-8-0957-0411-s045951	forward	revenue	3	pending
line-8	line-8-0957-0411-s045951	reverse	revenue	3	pending
line-8	line-8-0835-0366-s048955	forward	revenue	3	pending
line-8	line-8-0835-0366-s048955	reverse	revenue	3	pending
line-8	line-8-0785-0290-s051366	reverse	revenue	3	pending
line-8	line-8-0976-0467-s044485	reverse	spare	1	pending
line-8	line-8-0957-0411-s045951	forward	spare	1	pending
line-8	line-8-0957-0411-s045951	reverse	spare	1	pending
line-8	line-8-0835-0366-s048955	forward	spare	1	pending
line-8	line-8-0835-0366-s048955	reverse	spare	1	pending
line-8	line-8-0785-0290-s051366	reverse	spare	1	pending
line-8	line-8-2231-1783-s000000	forward	spare	1	pending
line-8	line-8-1645-1094-s021347	forward	spare	1	pending
line-8	line-8-1645-1094-s021347	reverse	spare	1	pending
line-8	line-8-1518-0993-s024893	forward	cold_reserve	1	pending
line-9	line-9-0565-0620-s003116	forward	revenue	1	pending
line-9	line-9-0565-0620-s003116	reverse	revenue	1	pending
line-9	line-9-0502-0673-s005049	forward	revenue	1	pending
line-9	line-9-0502-0673-s005049	reverse	revenue	1	pending
line-9	line-9-0487-0760-s006980	reverse	revenue	1	pending
line-9	line-9-0557-0867-s010067	forward	revenue	1	pending
line-9	line-9-0557-0867-s010067	reverse	revenue	1	pending
line-9	line-9-0549-0960-s012159	forward	revenue	1	pending
line-9	line-9-0538-1065-s015166	forward	revenue	1	pending
line-9	line-9-0538-1065-s015166	reverse	revenue	1	pending
line-9	line-9-0637-1439-s025194	forward	revenue	1	pending
line-9	line-9-0620-1509-s026894	forward	revenue	1	pending
line-9	line-9-0620-1509-s026894	reverse	revenue	1	pending
line-9	line-9-0619-1569-s028209	forward	revenue	1	pending
line-9	line-9-0619-1569-s028209	reverse	revenue	1	pending
line-9	line-9-0647-1607-s029412	reverse	revenue	1	pending
line-9	line-9-0853-1593-s033809	forward	revenue	1	pending
line-9	line-9-0853-1593-s033809	reverse	revenue	1	pending
line-9	line-9-0911-1579-s035217	forward	revenue	1	pending
line-9	line-9-1185-1622-s041363	forward	revenue	1	pending
line-9	line-9-1185-1622-s041363	reverse	revenue	1	pending
line-9	line-9-1246-1628-s042633	forward	revenue	1	pending
line-9	line-9-1430-1529-s048432	forward	revenue	1	pending
line-9	line-9-1430-1529-s048432	reverse	revenue	1	pending
line-9	line-9-1509-1487-s050681	forward	revenue	1	pending
line-9	line-9-1509-1487-s050681	reverse	revenue	1	pending

Line	Station	Direction	Role	Initial trainsets	Verified track slots
line-9	line-9-1613-1236-s058396	reverse	revenue	1	pending
line-9	line-9-1645-1094-s061501	forward	revenue	1	pending
line-9	line-9-1645-1094-s061501	reverse	revenue	1	pending
line-9	line-9-1580-0848-s068248	reverse	revenue	1	pending
line-9	line-9-1477-0761-s071522	forward	revenue	1	pending
line-9	line-9-1477-0761-s071522	reverse	revenue	1	pending
line-9	line-9-1389-0675-s074529	forward	revenue	1	pending
line-9	line-9-1335-0562-s077540	forward	revenue	1	pending
line-9	line-9-1335-0562-s077540	reverse	revenue	1	pending
line-9	line-9-1246-0493-s080552	forward	revenue	1	pending
line-9	line-9-1246-0493-s080552	reverse	revenue	1	pending
line-9	line-9-1200-0443-s082619	reverse	revenue	1	pending
line-9	line-9-1190-0372-s084695	forward	revenue	1	pending
line-9	line-9-1190-0372-s084695	reverse	revenue	1	pending
line-9	line-9-1116-0417-s087021	reverse	revenue	1	pending
line-9	line-9-0976-0467-s090748	forward	revenue	1	pending
line-9	line-9-0976-0467-s090748	reverse	spare	1	pending
line-9	line-9-0889-0471-s092604	forward	spare	1	pending
line-9	line-9-0832-0577-s095621	forward	spare	1	pending
line-9	line-9-0832-0577-s095621	reverse	spare	1	pending
line-9	line-9-0700-0568-s098647	forward	cold_reserve	1	pending

Reference platform capacity comparison

464 trainsets exceed the reference platform envelope, requiring 56,144.0 m of additional usable slots under this initial allocation. This does not establish the location or need for new infrastructure; existing sidings require evidence before crediting them.

Station	Trainsets	Reference platform berths	Beyond envelope	Additional usable slot length m
line-1-0166-0866-s053822	2	2	0	0.0
line-1-0319-0836-s050133	4	2	2	242.0
line-1-0463-0845-s047129	4	2	2	242.0
line-1-0557-0867-s044984	4	4	0	0.0
line-1-0643-0866-s043055	4	2	2	242.0
line-1-0731-0859-s041121	4	2	2	242.0
line-1-0821-0812-s038113	4	2	2	242.0
line-1-0940-0787-s035101	6	4	2	242.0
line-1-1063-0773-s032095	6	2	4	484.0
line-1-1182-0792-s029092	6	2	4	484.0
line-1-1306-0780-s026078	6	2	4	484.0
line-1-1429-0829-s023054	6	2	4	484.0
line-1-1554-0820-s020402	6	4	2	242.0
line-1-1702-0808-s017045	6	4	2	242.0
line-1-1848-0797-s014034	6	2	4	484.0
line-1-2017-0784-s010529	6	2	4	484.0
line-1-2180-0774-s007014	6	2	4	484.0
line-1-2343-0771-s003515	6	2	4	484.0
line-1-2500-0775-s000000	3	2	1	121.0
line-2-0595-0296-s056994	2	2	0	0.0
line-2-0654-0439-s052456	4	2	2	242.0
line-2-0679-0342-s054719	4	2	2	242.0
line-2-0728-0584-s047170	4	4	0	0.0
line-2-0753-0460-s050189	4	2	2	242.0
line-2-0828-0672-s044160	4	2	2	242.0
line-2-0923-0772-s041136	6	4	2	242.0
line-2-0947-0911-s038129	6	2	4	484.0
line-2-1022-1022-s035123	6	4	2	242.0
line-2-1095-1097-s031056	6	2	4	484.0
line-2-1103-1047-s033035	6	2	4	484.0
line-2-1141-1176-s029095	6	4	2	242.0
line-2-1249-1269-s026078	6	2	4	484.0
line-2-1361-1359-s023069	6	2	4	484.0
line-2-1461-1452-s020064	6	2	4	484.0

Station	Trainsets	Reference platform berths	Beyond envelope	Additional usable slot length m
line-2-1509-1487-s018636	6	4	2	242.0
line-2-1550-1543-s017049	6	2	4	484.0
line-2-1648-1703-s012612	6	2	4	484.0
line-2-1895-2056-s003494	6	2	4	484.0
line-2-1987-2192-s000000	3	2	1	121.0
line-3-0151-1637-s055819	3	2	1	121.0
line-3-0242-1604-s053550	6	2	4	484.0
line-3-0350-1616-s051291	6	2	4	484.0
line-3-0433-1532-s048276	6	2	4	484.0
line-3-0528-1521-s046268	6	2	4	484.0
line-3-0620-1509-s044279	6	4	2	242.0
line-3-0711-1492-s042273	6	2	4	484.0
line-3-0830-1419-s039265	6	2	4	484.0
line-3-0943-1486-s036258	6	2	4	484.0
line-3-1046-1445-s033249	6	4	2	242.0
line-3-1111-1426-s030244	6	2	4	484.0
line-3-1162-1305-s027239	6	2	4	484.0
line-3-1265-1319-s024226	6	2	4	484.0
line-3-1391-1293-s021226	6	2	4	484.0
line-3-1502-1256-s018214	7	2	5	605.0
line-3-1613-1236-s015796	8	4	4	484.0
line-3-1702-1235-s014007	8	2	6	726.0
line-3-2271-1120-s000000	4	2	2	242.0
line-4-0146-0199-s000000	4	2	2	242.0
line-4-0298-0305-s004261	8	2	6	726.0
line-4-0515-0750-s016457	8	4	4	484.0
line-4-0594-0786-s018376	8	2	6	726.0
line-4-0666-0827-s020297	6	2	4	484.0
line-4-0708-0930-s023314	6	2	4	484.0
line-4-0789-1036-s026330	6	2	4	484.0
line-4-0834-1141-s029342	6	2	4	484.0
line-4-0936-1241-s032367	6	2	4	484.0
line-4-1020-1350-s035373	6	2	4	484.0
line-4-1080-1447-s038373	6	4	2	242.0
line-4-1154-1476-s040395	6	2	4	484.0
line-4-1217-1551-s042429	6	4	2	242.0
line-4-1246-1628-s044443	3	2	1	121.0
line-5-0677-2304-s053737	3	2	1	121.0
line-5-0853-1593-s036330	6	4	2	242.0
line-5-0895-1451-s032907	6	2	4	484.0
line-5-0904-1340-s029889	6	2	4	484.0
line-5-0974-0947-s018085	7	2	5	605.0
line-5-0993-1151-s023847	6	2	4	484.0
line-5-1000-1254-s026861	6	2	4	484.0
line-5-1004-0809-s015076	8	2	6	726.0
line-5-1008-1046-s020829	6	4	2	242.0
line-5-1062-0705-s012053	8	2	6	726.0
line-5-1077-0591-s009034	8	4	4	484.0
line-5-1116-0417-s003985	8	4	4	484.0
line-5-1122-0491-s006033	8	2	6	726.0
line-5-1170-0351-s002077	8	4	4	484.0
line-5-1199-0260-s000000	4	2	2	242.0
line-6-0495-1669-s057516	3	2	1	121.0
line-6-0647-1607-s052578	8	4	4	484.0
line-6-0691-1529-s050169	8	2	6	726.0
line-6-0734-1414-s047145	8	2	6	726.0
line-6-0837-1343-s044142	8	2	6	726.0
line-6-0973-1337-s041141	8	2	6	726.0
line-6-1081-1255-s038128	8	2	6	726.0
line-6-1156-1145-s035101	8	4	4	484.0
line-6-1219-1042-s032087	8	2	6	726.0
line-6-1332-0995-s029084	8	2	6	726.0
line-6-1417-0951-s026078	8	4	4	484.0

Station	Trainsets	Reference platform berths	Beyond envelope	Additional usable slot length m
line-6-1515-0927-s023073	8	2	6	726.0
line-6-1580-0848-s020342	8	4	4	484.0
line-6-1694-0774-s017029	8	4	4	484.0
line-6-2322-0251-s000000	4	2	2	242.0
line-7-1069-2206-s000000	4	2	2	242.0
line-7-1185-1622-s012693	8	4	4	484.0
line-7-1206-1546-s014599	8	4	4	484.0
line-7-1277-1491-s016522	7	2	5	605.0
line-7-1314-1261-s022537	6	2	4	484.0
line-7-1328-1386-s019526	6	2	4	484.0
line-7-1396-0881-s031947	6	2	4	484.0
line-7-1398-1020-s028935	6	2	4	484.0
line-7-1401-1143-s025930	6	2	4	484.0
line-7-1412-0741-s034958	6	2	4	484.0
line-7-1420-0682-s036232	6	4	2	242.0
line-7-1434-0612-s037982	6	2	4	484.0
line-7-1487-0523-s040748	6	2	4	484.0
line-7-1531-0418-s043502	3	2	1	121.0
line-8-0785-0290-s051366	4	2	2	242.0
line-8-0835-0366-s048955	8	2	6	726.0
line-8-0957-0411-s045951	8	2	6	726.0
line-8-0976-0467-s044485	8	4	4	484.0
line-8-1004-0531-s042945	8	2	6	726.0
line-8-1085-0596-s039937	8	4	4	484.0
line-8-1168-0688-s036929	8	2	6	726.0
line-8-1281-0738-s033926	8	2	6	726.0
line-8-1349-0838-s030923	8	2	6	726.0
line-8-1422-0924-s027911	8	4	4	484.0
line-8-1518-0993-s024893	9	2	7	847.0
line-8-1645-1094-s021347	10	4	6	726.0
line-8-2231-1783-s000000	5	2	3	363.0
line-9-0487-0760-s006980	1	4	0	0.0
line-9-0502-0673-s005049	2	2	0	0.0
line-9-0538-1065-s015166	2	2	0	0.0
line-9-0549-0960-s012159	1	2	0	0.0
line-9-0557-0867-s010067	2	4	0	0.0
line-9-0565-0620-s003116	2	2	0	0.0
line-9-0619-1569-s028209	2	2	0	0.0
line-9-0620-1509-s026894	2	4	0	0.0
line-9-0637-1439-s025194	1	2	0	0.0
line-9-0647-1607-s029412	1	4	0	0.0
line-9-0700-0568-s098647	1	4	0	0.0
line-9-0832-0577-s095621	2	2	0	0.0
line-9-0853-1593-s033809	2	4	0	0.0
line-9-0889-0471-s092604	1	2	0	0.0
line-9-0911-1579-s035217	1	2	0	0.0
line-9-0976-0467-s090748	2	4	0	0.0
line-9-1116-0417-s087021	1	4	0	0.0
line-9-1185-1622-s041363	2	4	0	0.0
line-9-1190-0372-s084695	2	4	0	0.0
line-9-1200-0443-s082619	1	2	0	0.0
line-9-1246-0493-s080552	2	2	0	0.0
line-9-1246-1628-s042633	1	4	0	0.0
line-9-1335-0562-s077540	2	2	0	0.0
line-9-1389-0675-s074529	1	4	0	0.0
line-9-1430-1529-s048432	2	2	0	0.0
line-9-1477-0761-s071522	2	2	0	0.0
line-9-1509-1487-s050681	2	4	0	0.0
line-9-1580-0848-s068248	1	4	0	0.0
line-9-1613-1236-s058396	1	4	0	0.0
line-9-1645-1094-s061501	2	4	0	0.0

- Initial queues are balanced across powered station/direction choices; this is not a surveyed parking layout.

- Declared spares and cold reserves remain parked and charge; automatic substitution, defect routing and maintenance release are not modelled.
- Returning trains hold at selected powered stations after service ends; no train is teleported back to its initial placement.
- Reference platform berths are an optimistic length/count comparison, not verified parking capacity; dedicated sidings and workshop bays receive no automatic credit.
- Station berths, crossovers and shared junction conflicts are outside the simplified interstation movement-authority graph.
- Existing depot/station energy quantities and service schedules are preserved, not accepted as correctly sized.

Regenerate this report and its local runnable scenario with:

```
.venv/bin/python tools/automation/generate-stabling-plan.py --design 'cities/catalogue/west-asia/Iraq/Baghdad/design.toml'
```

The scenario is generated under `build/engineering/stabling/`; its hash is recorded in `summary.json`. Source-linked operating replay evidence, where available, is in `operating-screen.json` beside this report.

Baghdad survey-control processing

Source: cities/catalogue/west-asia/Iraq/Baghdad/engineering/survey/control-processing-readiness.md

- Status: awaiting-field-data
- RTKLIB runtime: not probed
- Processing completed: no
- Technical screen passed: no
- Survey authority accepted: no

The automated result is a reproducibility and solution-quality screen. It is not a network adjustment, survey certification or permission to use the control for design or construction.

Only the appointed deployment survey authority can accept the control network after redundant occupations, independent checks, network adjustment, datum/geoid review and monument documentation.

Current gates

- Missing processing roles: rover_observation, base_observation, navigation, rtklib_configuration
- Duplicate processing roles: none
- Processing-configuration findings: 0
- Missing authority-review roles: rover_observation, base_observation, navigation, rtklib_configuration, field_log, monument_schedule, network_adjustment_report, authority_acceptance_record

Controlled-storage workflow

RINEX observations and navigation data remain in access-controlled project storage. Git records only the receipt manifest, file hashes, requirements and non-sensitive processing/acceptance summaries.

- 1 Put received files in access-controlled project storage.
- 2 Add one SUR-CTRL manifest row per file, including `file_role`, metadata and SHA-256.
- 3 Freeze the required RTKLIB settings, then run this processor; the four processing roles must occur exactly once.
- 4 Review RTKLIB outputs, complete independent checks and network adjustment, then add the controlled acceptance record.
- 5 Re-run; only the appointed survey authority's explicit record can close the authority gate.

Baghdad drainage and ground-design gate

Source: cities/catalogue/west-asia/iraq/baghdad/engineering/survey/drainage-ground-readiness.md

- Status: awaiting-drainage-ground-evidence
- Lines/stations: 9 / 182
- SWMM/ground technical screen passed: no
- OpenGeoSys required by received decision: no
- Authority accepted: no

The automated gate reruns the received SWMM model, checks continuity and hydraulic performance for normal/blockage/backwater scenarios, reconciles exact reviewed assets/supports, and checks numerical site comparisons for catalogue foundation/ground-treatment selections. It does not validate rainfall, flood levels, soil parameters, hydraulic boundaries, structural capacity, settlement predictions or construction suitability.

The drainage and geotechnical engineers of record, asset owner, relevant authority and information manager must accept the project inputs, calculations, selected variants, residual risks and immutable evidence record.

Current gates

- Missing technical roles: ground_model_readiness, route_station_fit_readiness, accepted_hydrology_basis, swmm_model, swmm_processing_report, geotechnical_ground_model, foundation_ground_schedule, ground_design_verification_report, groundwater_coupling_decision, civil_asset_register
- Duplicate roles: none
- Authority findings:
 - drainage/ground acceptance record not received

[Desktop soil inputs and route/station investigation priorities](#) are available. These do not replace the received geotechnical ground model.

Controlled workflow

Keep borehole logs, laboratory results, sensitive asset data, native calculations and large model results in controlled project storage. Git carries requirements, receipts, hashes, reproducible exchange inputs and non-sensitive summaries.

- 1 Accept the ground model, route fit, hydrology basis and geotechnical model.
- 2 Run the project SWMM model; retain its input, report, source hashes and continuity results.
- 3 Size a checked catalogue foundation or ground-treatment system for every line and station scope.
- 4 Record whether groundwater/coupled analysis is warranted; require OpenGeoSys evidence only when the reviewed triggers say yes.
- 5 Close residual risks and obtain the signed drainage/geotechnical acceptance record.

Baghdad field-evidence brief

Source: <cities/catalogue/west-asia/iraq/baghdad/engineering/survey/field-evidence-brief.md>

Deterministic pre-mobilisation requirements generated from the shared field-evidence template. It issues the information request; it does not claim that field data or approvals exist.

- Brief status: brief-issued-awaiting-signatures
- Mobilisation authorized: no
- Field evidence accepted: no
- Horizontal CRS: EPSG:32638 candidate UTM zone 38N; survey authority to confirm or replace
- Vertical datum: authority-to-confirm-before-mobilisation
- Canonical requirements: lib/templates/field-evidence.toml

The appointed survey authority, architect/engineer of record, operator and information manager approve this brief, responsibility matrix, CRS, vertical datum, accuracy schedule, access plan and data rights before field capture.

Required packages

ID	Dataset	Owner role	Delivery
SUR-CTRL	Primary and secondary survey control	appointed survey authority	RINEX plus field logs, adjustment report, coordinate schedule and signed control report
SUR-TOPO	Corridor and site topographic survey	survey delivery lead	GeoPackage 3D points/lines/polygons plus LandXML terrain and accepted PDF plans
SUR-UTIL	Above- and below-ground utilities	utility survey lead	GeoPackage 3D utility features, investigation records, photographs and confidence/quality attributes
SUR-LAND	Property, right-of-way and access constraints	client land and legal lead	GeoPackage boundaries/rights plus controlled title, consent and access register
SUR-FLOOD	Hydrology, flood and drainage evidence	drainage and flood lead	GeoPackage water/flood/outfall layers plus rainfall, level, model and authority records
GEO-GI	Ground investigation and laboratory testing	geotechnical engineer of record	AGS or equivalent structured logs, coordinates, samples/tests, factual report and interpretive design report
OPS-WORKSHOP	Existing workshop and depot capability audit	operator engineering lead	asset/capability register, measured layouts, utilities, photographs and gap assessment
OPS-FLEET	Existing fleet and operating interface audit	operator rolling-stock lead	fleet register, measured interface sheets, duty/condition records and migration constraints
SUR-IMAGE	Georeferenced photography and photogrammetry	reality-capture lead	source-image manifest and hashes, camera logs, GCP/check points, orthophoto/DSM and processing report
SUR-SCAN	Point-cloud and dimensional interface capture	reality-capture lead	E57/LAS/LAZ plus station setup, targets/control, registration and CloudCompare QA report
INF-GOV	Information ownership, security and acceptance	client information manager	responsibility matrix, naming/revision rules, common-data-environment plan, licence/consent register and acceptance record

Scope and acceptance

SUR-CTRL — Primary and secondary survey control

Stable primary/secondary monuments covering every corridor, station, depot, bridge, energy site and construction compound.

Provisional accuracy/quality requirement: Primary control ≤ 10 mm horizontal and ≤ 15 mm vertical expanded uncertainty; interface control ≤ 5 mm relative to accepted project control. Survey authority must approve or replace these project targets.

Acceptance evidence:

- approved projected CRS, epoch, units, geoid model and vertical datum
- instrument/antenna serials and calibration status
- redundant occupations and independent checks

- least-squares residuals, covariance/uncertainty and rejected-observation record
- signed monument descriptions and protection/inspection plan

SUR - TOP0 — Corridor and site topographic survey

Full corridor, station, depot, access, watercourse, road, structure, compound and lifting-route coverage with breaklines and feature coding.

Provisional accuracy/quality requirement: Hard-surface/detail points ≤ 25 mm horizontal and ≤ 20 mm vertical; general ground ≤ 50 mm. Survey authority must approve class-by-feature targets and check density.

Acceptance evidence:

- feature code and layer dictionary
- check-point residual report
- coverage/void map and inaccessible-area register
- capture date/epoch and change notes
- signed completeness review against route limits

SUR - UTIL — Above- and below-ground utilities

Electricity, telecoms, water, sewer, drainage, fuel, irrigation and unknown services within the works influence zone.

Provisional accuracy/quality requirement: Every feature carries horizontal/vertical uncertainty and detection/verification method; trial holes or records alone must not be silently treated as surveyed truth.

Acceptance evidence:

- records search and authority contacts
- detection method, equipment and calibration
- surface-feature correlation
- verification exposure/trial-hole schedule where design risk warrants
- unresolved and abandoned-service issue register

SUR - LAND — Property, right-of-way and access constraints

Ownership, occupation, easements, road reserve, temporary access, compounds, lifting routes and acquisition constraints.

Provisional accuracy/quality requirement: Boundary geometry remains indicative until reconciled to authoritative records and verified by the competent land authority.

Acceptance evidence:

- authoritative source and retrieval date
- boundary/status confidence
- owner/occupier and consent state
- temporary/permanent requirement
- dispute and encroachment register

SUR - FLOOD — Hydrology, flood and drainage evidence

Catchments, channels, historical levels, culverts, outfalls, tailwater, groundwater indicators and exceedance routes affecting the works.

Provisional accuracy/quality requirement: Levels use the accepted vertical datum; rainfall return periods and climate allowances require local authority confirmation.

Acceptance evidence:

- source dates and return periods
- surveyed channel/outfall/culvert levels
- historical event and community evidence

- blockage/tailwater assumptions
- authority-approved design storm and discharge constraints

GEO - GI — Ground investigation and laboratory testing

Boreholes, CPTu, test pits, groundwater monitoring and laboratory programme zoned for at-grade, elevated, station, depot and energy works.

Provisional accuracy/quality requirement: Investigation spacing/depth and tests are risk-based and approved by the geotechnical engineer after desk study and walkover.

Acceptance evidence:

- desk study, walkover and conceptual ground model
- location/elevation tied to accepted control
- factual logs and chain of custody
- strength, settlement, corrosivity and groundwater parameters with uncertainty
- signed geotechnical design recommendations and residual risks

OPS - WORKSHOP — Existing workshop and depot capability audit

Buildings, pits, cranes, lifting, machine tools, power, compressed air, fire systems, stores, access and workforce capability.

Provisional accuracy/quality requirement: Critical lifting, clearance and interface dimensions are measured to the tolerance required by the selected equipment supplier.

Acceptance evidence:

- asset identity, condition and certification
- measured capacity/clearance and utility availability
- repair/reuse/replace decision
- production and maintenance throughput
- operator-approved upgrade work packages

OPS - FLEET — Existing fleet and operating interface audit

Existing vehicles, gauges, couplers, wheel/rail, depot interfaces, rescue, operating rules and reusable components relevant to the pilot.

Provisional accuracy/quality requirement: Safety and physical interfaces use independently checked measurements and controlled vehicle/configuration identity.

Acceptance evidence:

- vehicle/configuration and maintenance history
- measured gauge/coupler/wheel/depot interfaces
- operating duty and failure records
- rescue/recovery constraints
- signed compatibility and migration decisions

SUR - IMAGE — Georeferenced photography and photogrammetry

Continuous corridor and detailed station/depot/structure/utility-context coverage, subject to permissions and privacy controls.

Provisional accuracy/quality requirement: Ground sampling distance and independent check-point limits are agreed by use case; neither EXIF position nor self-reported processing precision is acceptance evidence.

Acceptance evidence:

- capture plan, permissions and privacy controls
- camera/lens identity and calibration

- GCP versus independent check-point separation
- processing software/version/settings
- check-point residuals, voids, artefacts and obscuration map

SUR - SCAN — Point-cloud and dimensional interface capture

Complex structures, station/depot interfaces, constrained utilities and clearance-critical areas selected by the design team.

Provisional accuracy/quality requirement: Registration RMS is reported but not used alone; independent control/check residuals and local interface comparisons govern acceptance.

Acceptance evidence:

- scanner identity/calibration and setup log
- target/control schedule
- registration method and residuals
- independent check points and cloud-to-cloud sections
- density, range, noise, void and movement/change register

INF - GOV — Information ownership, security and acceptance

All raw and derived field information, personal/sensitive content, third-party records and authoritative project derivatives.

Provisional accuracy/quality requirement: Not applicable; this package governs provenance, access, revision, retention and authorized use.

Acceptance evidence:

- owner, producer, checker and accepting authority per dataset
- licence, reuse, publication and personal-data constraints
- file naming, metadata, revision and immutable hash rules
- backup, retention and access-control plan
- signed receipt/rejection/reissue workflow

Data handling and handoff

Keep raw survey, imagery, scan, utility and personal data in access-controlled project storage; Git contains requirements, manifests, hashes, accepted summaries and non-sensitive derivatives only.

Complete [survey-input-manifest.csv](#) for every delivery. Each accepted row must name the producer/checker, CRS and vertical datum, capture date, controlled project-storage path, SHA-256 digest and acceptance state. RTKLIB processing configurations/status/residual outputs, QGIS/CloudCompare/ODM settings and signed review records accompany their respective source data.

Baghdad surveyed-ground-model gate

Source: cities/catalogue/west-asia/Iraq/Baghdad/engineering/survey/ground-model-readiness.md

- Status: awaiting-ground-model-data
- Technical inspection completed: no
- Technical screen passed: no
- Survey authority accepted: no

Automated inspection checks file integrity, GeoPackage structure, independent-check residuals and processing-report completeness. It does not prove feature completeness, datum correctness or fitness for design.

Only the appointed survey authority and information manager can accept the master ground model after control, coverage, void, epoch, uncertainty, CRS and vertical-datum review.

Current gates

- Missing technical roles: control_acceptance_report, topographic_features, terrain_dtm, checkpoint_residuals, odm_processing_report, orthophoto, registered_point_cloud, cloudcompare_qa_report, void_register, master_ground_model
- Duplicate roles: none
- Missing authority-review roles: control_acceptance_report, topographic_features, terrain_dtm, checkpoint_residuals, odm_processing_report, orthophoto, registered_point_cloud, cloudcompare_qa_report, void_register, master_ground_model, ground_model_acceptance_record
- Authority-record findings:
 - ground-model acceptance record not received

Controlled workflow

Keep source imagery, raw scans and intermediate dense clouds in access-controlled project storage. Git records requirements, file hashes, non-sensitive QA summaries and accepted derivatives only.

- 1 Complete accepted survey control before registering terrain or clouds.
- 2 Process imagery in OpenDroneMap and registration/comparison in CloudCompare; preserve settings, source/output hashes and QA reports.
- 3 Build the federated master GeoPackage in the approved horizontal CRS and vertical datum.
- 4 Add every derivative to the shared receipt with its immutable hash and independent checker.
- 5 Run inspection, resolve residual/coverage findings, and obtain the controlled authority acceptance record.

Baghdad route and station fit gate

Source: cities/catalogue/west-asia/iraq/baghdad/engineering/survey/route-station-fit-readiness.md

- Status: awaiting-route-fit-evidence
- Lines: 9
- Stations: 182
- Technical screen passed: no
- Coordinated authority acceptance: no

The automated gate proves that every current line and station is covered by hash-locked, internally consistent route-fit evidence and that no unresolved critical/high issue is hidden. It does not determine land rights, utility position, flood risk, accessibility, traffic safety, lifting feasibility or possession sufficiency.

Only the relevant utility, land, drainage, highway, operations and design authorities can accept their evidence, followed by the engineer of record, operator and information manager accepting the coordinated route-fit record.

Current gates

- Missing technical roles: ground_model_readiness, surveyed_alignment_readiness, utility_coordination, land_access, flood_drainage, station_access, intercity_yard_integration, road_traffic_management, construction_logistics, possession_staging, route_fit_verification_report, route_fit_issue_register
- Duplicate roles: none
- Authority findings:
 - route-fit acceptance record not received

Controlled workflow

Store titles, personal data, utility-sensitive records, permits and native studies in the controlled project environment. Git carries requirements, empty receipts, immutable hashes and non-sensitive acceptance summaries.

- 1 Accept survey control, the ground model and every surveyed alignment.
- 2 Register checked utility, land, flood, station-access, yard/intercity, road, logistics and possession evidence.
- 3 Reconcile exactly one resolved result for every current line and station, with source hashes.
- 4 Close or formally transfer all high/critical issues in the controlled issue register.
- 5 Obtain the coordinated authority acceptance record against the immutable evidence hashes.

Baghdad structural-release gate

Source: cities/catalogue/west-asia/Iraq/Baghdad/engineering/survey/structural-release-readiness.md

- Status: awaiting-structural-evidence
- Lines: 9
- Technical screen passed: no
- Authority accepted: no

The automated gate checks actual solver output files and numerical limits, reviewed asset/support coverage, load-case/result reconciliation and independent-check completeness. It does not validate modelling assumptions, codes, loads, mesh quality, reinforcement, prestress, soil springs, fatigue detail, seismic response or structural safety.

Only the structural and geotechnical engineers of record, independent checker, asset owner and approving authority can release the project structures in a signed controlled record.

Current gates

- Missing technical roles: drainage_ground_readiness, structural_design_basis, structural_asset_schedule, load_case_register, opensees_model, opensees_report, calculix_input, calculix_report, structural_verification_report, independent_check_record, civil_asset_register
- Duplicate roles: none
- Authority findings:
 - structural acceptance record not received

Controlled workflow

- 1 Accept drainage/ground design and freeze the project structural basis and load combinations.
- 2 Schedule every span, pier, abutment, foundation and special structure against line chainage.
- 3 Preserve OpenSees global/seismic/soil-spring and CalculiX component input/output hashes and convergence evidence.
- 4 Reconcile foundation, wind, seismic, fatigue and bearing/movement results per scheduled asset.
- 5 Close independent-check comments and obtain the signed structural release.

Baghdad surveyed-alignment gate

Source: cities/catalogue/west-asia/iraq/baghdad/engineering/survey/surveyed-alignment-readiness.md

- Status: awaiting-surveyed-alignments
- Lines expected: line-1, line-2, line-3, line-4, line-5, line-6, line-7, line-8, line-9
- Technical screen passed: no
- Authority accepted: no

The automated gate proves receipt integrity, OSR-ALN hard-gate compliance, explicit horizontal/vertical/cant content, line and station reconciliation, deterministic LandXML re-import, and recorded interface tolerances. It does not establish that the survey, fit, clearances, possessions or construction design are correct.

Only the appointed alignment designer, survey authority, track engineer and information manager can accept an alignment for design use in a controlled signed record.

Current gates

- Missing technical roles: *:ground_model_readiness, *:interface_verification_report, line-1:surveyed_osr_aln, line-1:landxml_export, line-1:landxml_roundtrip_report, line-2:surveyed_osr_aln, line-2:landxml_export, line-2:landxml_roundtrip_report, line-3:surveyed_osr_aln, line-3:landxml_export, line-3:landxml_roundtrip_report, line-4:surveyed_osr_aln, line-4:landxml_export, line-4:landxml_roundtrip_report, line-5:surveyed_osr_aln, line-5:landxml_export, line-5:landxml_roundtrip_report, line-6:surveyed_osr_aln, line-6:landxml_export, line-6:landxml_roundtrip_report, line-7:surveyed_osr_aln, line-7:landxml_export, line-7:landxml_roundtrip_report, line-8:surveyed_osr_aln, line-8:landxml_export, line-8:landxml_roundtrip_report, line-9:surveyed_osr_aln, line-9:landxml_export, line-9:landxml_roundtrip_report
- Duplicate roles: none
- Authority findings:
 - alignment acceptance record not received

Controlled workflow

Keep signed source surveys, native authoring files and large point clouds in controlled project storage. Git carries the empty receipt, requirements, reproducible checks and non-sensitive accepted exchange derivatives.

- 1 Accept survey control and the surveyed ground model.
- 2 Fit each line in the confirmed CRS/datum; explicitly issue horizontal, vertical and cant schedules.
- 3 Export one OSR-ALN and LandXML file per line and record the deterministic re-import hash and comparison tolerances.
- 4 Reconcile every platform and record yard, turnout and clearance dispositions per line.
- 5 Run this inspection and obtain the controlled multi-discipline acceptance record.

Baghdad retained under-viaduct rental portfolio — 2026-10-04

Source: [cities/catalogue/west-asia/Iraq/Baghdad/engineering/viaduct-rentals/README.md](https://cities.catalogue/west-asia/Iraq/Baghdad/engineering/viaduct-rentals/README.md)

Rental income is a conditional addition to the existing station-development sale case, with different assets and external tenants. No site is commercially accepted, surveyed, leased or funded. [Network Rail's arch portfolio](#) provides a precedent for railway SME premises. [Places for London's guidance](#) emphasises access, services, insulation, fit-out responsibilities and permitted uses. These UK precedents do not establish Iraqi demand, prices or building permissions. Sources checked 4 October 2026.

Geometry, eligibility and pilot

The controlled design has 1,143 elevated civil segments totalling 75.8105 km, with median length 40 m. Each [register entry](#) carries its civil index, line/chainage, planning coordinates, existing parent track assets and nearest station. 14 segments lack a materialised parent track link and need asset-register reconciliation; no ID is invented. Coordinates interpolate the corridor and are not surveyed footprints. Clear height, frontage, ownership, title, utilities, permitted use and street/rail access remain null or unaccepted.

The reference screen excludes segments shorter than 75 m, reserves 10 m at each approach and counts complete 25 m planning bays. Each bay tests three independent 30 m² internal units, with 2 m protected support-end inspection zones and 1.2 m independent access. It suggests 128,070 m², before unknown road crossings, utility space, pier positions, access, ownership, flooding, station-sale overlap and fire exclusions. Confirmed eligible area is zero. Elevation alone never grants eligibility. Pier height is not usable clear height beneath a cap/deck.

The 200,000 m² illustration exceeds this screen by 71,930 m² and is unmapped-area-blocked, with no lease receipts in executable integrated cases. It requires a different surveyed footprint/reference layout; there is no automatic widening of a viaduct. Small/medium models allocate assumed area against identified civil segments and phase construction by their infrastructure dates. These are conditional surveys-to-test, not valuations. [Thirty draft pilot units](#) use one long candidate selected by station-chainage proximity and bay count, without a claim of commercial desirability. Market-test 20–50 units through comparable IQD rents, tenant interest, permissions and costed fit-outs before expansion. No external contact has been made.

Open reference unit and costs

[Unit plan](#) gives a 6 × 5 m internal module in a 6.4 × 5.4 m independent shell, requiring 3.2 m measured clear height. [Parts and allowances](#) total USD 21,000 equivalent per 30 m² unit: USD 18,000 direct, USD 2,000 design/shared services and USD 1,000 contingency, contracted in IQD. These are planning allowances, not quotes. Walls/roof, thermal/acoustic insulation, drainage, slab, independent foundations, doors/windows, separate metered/isolation services, fire/egress equipment, vehicle protection and removable inspection panels are included. Actual soil, materials, fire protection and service capacity may change cost materially. No structure is dimensioned from rental value alone; compare net value against costs before modifying supports or spans. Open licences remain intact.

The shell is independent of railway piers/caps/deck: no drilling, welding or load attachment. Preserve bearing/jacking, drainage and electrical access, surveyed support clearances and maintenance easements. Independent utility meters are paid directly by tenants in the model; no utility reimbursement is booked as extra profit. Retail, services and studios are candidate uses only; workshops, cooking, hot works, fuel/solvent storage and battery activities require specific assessments. There is no generic permission for a use because it fits the floor area.

Illustrations versus priced cash-flow scenarios

Area illustration	Lettable m²	Monthly rent USD eq/m²	Occupancy	Annual landlord NOI USD m eq
small	40,000	10	70%	2.52
medium	100,000	15	80%	10.80
large	200,000	20	85%	30.60

These illustrations deduct 25% of occupied receipts and exclude capital, financing and tax. For the medium illustration, 30 annual receipts, first at month 48, rising 5% each year, discounted at 13.4%, have PV USD 79.405m. Explicit payment dates matter: this is receipts-only PV, not project NPV or the phased portfolio forecast.

Executable rental scenario	Status	Fit-out capital USD m eq	First conditional cash month	Resource NPV before tax USD m eq	Standalone after-tax NPV USD m eq
small	conditional-unverified-retained-portfolio	38.908	44	-4.493	-5.591

Executable rental scenario	Status	Fit-out capital USD m eq	First conditional cash month	Resource NPV before tax USD m eq	Standalone after-tax NPV USD m eq
medium	conditional-unverified-retained-portfolio	93.532	44	17.244	10.745
large	unmapped-area-blocked	Unmodelled	Blocked	Unmodelled	Unmodelled
medium_downside	conditional-unverified-retained-portfolio	93.532	56	-35.427	-35.797

Monthly ledgers phase fit-outs after the corresponding infrastructure-completion month and delay tenant handover until railway opening or fit-out completion, whichever is later. Six-month enclosures, three initial rent-free months and an 18-month occupancy ramp precede steady receipts. The last fractional module is constructed in full; spare floor area receives no rent. Rent and OPEX escalate 5% from financial close, and unquoted capital escalates 5% to each invoice. The downside uses rent USD 9/m², 55% target occupancy, 12 extra months of vacancy, a 36-month ramp, 8% arrears and 7% OPEX inflation.

Market vacancy is separate from modeled tenant turnover (10%/year, three-month re-letting vacancy and one-month free concession). Three-percent arrears have 50% recovery after six months; unrecovered debt is written off, and final unresolved receivables have zero recoverable value. Revenue is cash collected in these pro-forma sensitivity accounts; statutory accrual/tax treatment needs advice. Occupied maintenance/insurance uses 25% of reference occupied face rent with its own OPEX index, including during rent-free concessions, plus USD 10/m²/year for vacant-unit maintenance/insurance. Costs persist through concessions or defaults. Major refurbishment is additional cash expense at 20% of reference fit-out every 12 years, with OPEX inflation; it is not also capitalised. Existing station kiosk receipts are unchanged.

Security deposits of two months' indexed face rent are assumed topped up/refunded with occupancy. They are separately restricted cash and tenant liabilities, with zero revenue, zero construction funding and no shareholder distribution. Ending leases refund all deposits. No upfront lease premium or sold rental parcel is booked. Parcel overlap with the existing 15 station sale candidates is unknown and must be accepted as disjoint before either case can become executable. Current site-right cash is explicitly assumed zero, with additional land opportunity cost unknown: full site economics cannot be concluded from the reported before-land NPV. No USD credit or additional government cash is assumed for fit-outs.

Assets depreciate over 20 years from each cohort handover. Standalone tax applies the illustrative 15% positive annual cash-profit-after-depreciation proxy, with no loss relief or interest deductions, paid the next month; it is not an Iraqi tax ruling. The holding model instead includes the same portfolio inside its separate-business tax stress, so standalone tax is not deducted twice. The project horizon caps late cohorts' rental years and leaves no unsupported terminal sale/rent beyond the railway forecast. These cases improve only to the extent that collected rent exceeds all additional resources and costs.

Financing, dividends and terminal cash

[Financing redesign](#) adds a separately financed retained-property entity with 25% private partner fit-out equity and IQD construction bank debt. Its operating-gap capacity is zero; missing carry/fees/reserves are exposed rather than adding an uncommitted facility to the existing 13tn cap. Internal distributions are zero pending lender consent. External tenant cash adds group revenue; future property-to-rail transfers would cancel exactly. The [holding-equity study](#) uses the same resources with a 100%-owned business, replacing partner equity with the parent subscription case and borrowing the residual in IQD. These are alternative ownership/funding structures, not cumulative sources.

The holding study separately compares all-debt-repaid dividends with an unapproved coverage/reserve/profit-based policy, and reports a diagnostic of proportional final unrestricted cash net of liabilities. Cash at the horizon is not a guaranteed redemption, quoted share price or assumed sale of railway/property assets. Do not confuse that diagnostic with actual company distributions or count cash twice. Company accounts remain separately reconciled.

Digital twin, common hazards and evidence

Every draft pilot unit links its civil segment/parent track, proposed enclosure design, unknown lease and tenant IDs, meters, isolation, inspection rights, permitted use and [hazard records](#). The [native ERP mapping](#) uses existing Asset/Contract/Customer/Sales Invoice/Payment Entry/Asset Maintenance/Issue types when actual records and permissions exist; null IDs are not live leases, meters or safety records. No fake tenant, permit, contract or asset is created from an unverified footprint.

Tenant fire, flooding/drainage, delivery impact, blocked inspection access and utility/attachment faults propagate from enclosure to pier/foundation, bearings and shared spans and can affect railway operations. Redundant onboard controllers provide no mitigation for these common physical hazards. Proposed inspections include pre-lease engineering acceptance, monthly access/drain/isolation checks, post-incident checks, annual structure-linked inspection and controlled jacking/maintenance closures; durations and legal/fire requirements need actual assessment. Native maintenance/Issue records must trace the landlord obligation to the railway asset and escalation owner. Occupancy and any railway restriction require the authorised engineer's decision, not automatic evidence acceptance by this model.

Six 90-day evidence packages require surveyed eligibility and rights/no-overlap, independent civil/fire/utility access acceptance, market-tested pilot leases, supplier fit-out/refurbishment quotes, native lease/meter/maintenance controls, and independent cash/tax/funding/distribution review. Named people, surveys, quotes and leases remain pending. Native ERP Task imports are open drafts; no evidence or operating release is granted.

Regenerate `.venv/bin/python tools/automation/baghdad_viaduct_rentals.py`, then financing redesign, equity and proposal; validate source/output hashes with `--check`.

OpenSourceRail Baghdad — concept and FEED offer

Source: [cities/catalogue/west-asia/Iraq/Baghdad/offer/README.md](https://github.com/OpenSourceRail/cities/catalogue/west-asia/Iraq/Baghdad/offer/README.md)

This package presents a city-specific, evidence-linked proposal for a Baghdad urban railway. It is intended to start an owner-led feasibility, requirements and front-end engineering design (FEED) engagement with the relevant Iraqi and Baghdad authorities. It is not a construction release, supplier quotation, regulatory approval or safety certificate.

Offer documents

- [Baghdad system offer \(PDF\)](#)
- [Network design and reproducible engineering evidence](#)
- [Offer evidence manifest](#)
- [Baghdad operations dashboard](#)
- [Baghdad project twin](#)
- [Baghdad QA gates](#)
- [Engineering GIS view](#)
- [Native simulation dashboard](#)
- [SUMO timetable validation](#)

Proposed system

The current planning baseline contains 9 lines, 516.5 km of double-track route, 182 unique stations, 23 interchange complexes and 831 six-car trainsets. The service plan targets a three-minute peak headway and a 05:30–02:00 operating day. The civil screen identifies 428.3 km at grade, 75.8 km elevated and 12.4 km of bridge works. Open geospatial screening must be confirmed by survey, property, utilities, ground, hydraulic and alignment evidence during FEED.

The energy concept includes 52.1 MW of station/depot PV, 354.0 MWh of storage, 316.0 MW of connected charging and 1018.6 MW of dedicated solar. Operating energy, islanding, connections, protection, land and duty remain unaccepted.

City planning CAPEX is USD 7.56 billion before owner-confirmed land, utilities, tax/duty and escalation. The \$8 M depot allowance is not reconciled to surveyed stabling, workshops, energy, fire and security. The Baghdad manufacturing plant is outside city CAPEX and counted once in the Baghdad-only programme.

Iraq financing proposal

The [city funding model](#) and [Baghdad-only programme](#) divide eligible Chinese component invoices, government capital, IQD bonds and IQD bank credit. Government capital is 25% of total CAPEX, including USD 896.53 million for half the imported-parts budget. Proposed Chinese USD credit covers the other half. Remaining government capital, bonds and bank credit are in IQD. Full import basket lender and supplier-origin qualification remains pending. Fees, interest, reserves and cash support beyond that contribution are separately disclosed funding needs. They include staged draws, native-currency principal/interest, fees, reserves, cash support and downside funding gaps. The rates, maturities and 50% invoice advance are uncommitted appraisal assumptions. The resource-constrained construction cash schedule is not a five-year funding promise. Monthly and annual ledgers and charts are generated from the same controlled model.

The programme comparison covers the historical 148 km / USD 18 billion proposal against this 516.5 km / 182 station planning network. Total USD capital funding (government USD cash plus Chinese loan) is USD 1.793 billion; the rest is IQD. The older all-USD financing basis is the requested comparison scenario, not verified final contract terms. The modelled fare is IQD 1,317 per paid trip. Population access uses a 46.4% anchor-weighted planning score, not a surveyed resident catchment. Local procurement is USD 6.087 billion equivalent; indicative operating employment is 2,350 FTE. Construction job counts require validated hours and productivity. The model leaves USD 6.601 billion in additional cash requirements in the full-network-only case if public funding is capped at the 25% capital contribution. The conditional phased case reduces this to USD 6.168 billion, with first / last line openings in months 41 / 83. These are gross nominal liquidity needs, not net lifetime loss. Opening dates require actual plant, depot, line and safety acceptance; fleet-based phase demand and the 25% fixed / 75% variable OPEX split remain planning assumptions.

The [independent reconciliation](#) and [six-month bond and loan requirements](#) separate capital from debt repayment and price additional liquidity. Pooling city/plant cash yields USD 6.168 billion gross early cash needs and USD 3.941 billion later retained cash: a USD 2226.82 million net nominal deficit before gap-finance interest and fees. The illustrative green/concessional/grant/development mix still leaves USD 3.964 billion equivalent unpaid gap debt with constant nominal

fares and OPEX. It is not a funded programme. Steady annual tickets, shop/kiosk leases and advertising already contribute IQD 547.3, 13.0 and 20.2 billion respectively; new revenue targets cannot count those receipts again.

The separate paired sensitivity increases fares and OPEX 5% annually from financial close. With 5% income growth and the other blended assumptions, peak supplemental debt is IQD 4.362 trillion, with no terminal unpaid facility. Average nominal tickets reach IQD 1,525 at first opening and IQD 1,765 at full opening; commuting uses 11.7% of the indexed income proxy. This conditional sensitivity does not demonstrate household income growth or placed lending. Separate cases test peak/off-peak tickets, slower income growth and higher OPEX inflation. Capital escalation and future FX changes remain open. The [paired 5% six-month schedule](#) and [monthly prices](#) show those cashflows and affordability assumptions.

Surplus cash now also tests early repayment after scheduled obligations, debt-service reserves and a three-month current-OPEX buffer. Under assumed call/prepayment rights, cost priority clears debt in month 303 against month 371 for buffered gap-only repayment, saving USD 195.029 million equivalent in nominal interest and fees after USD 23.681 million assumed premiums. The order is bank credit, ordinary bonds, Chinese credit, green bonds, then the cheaper gap facility; loans-first and noncallable bonds are separate comparisons. Native principal is deducted once; no gap draw funds voluntary repayment. Premiums, minimum draw ages, calls, buyback prices, notice and permissions need actual contracts. Government capital remains 25%. [Six-month early-repayment tranches](#) and [all six strategies](#) retain the repayment, fees, outstanding debt, buffer and cash reconciliation.

Rolling stock and CRRC component strategy

The proposed fleet uses the OpenSourceRail six-car battery-electric platform with a candidate CRRC component package. Candidate scope includes traction motors, converters/inverters, auxiliary conversion and train-control interfaces, bogies, air springs, dampers, couplers/draft gear, brake equipment and passenger-information interfaces. These product families are supported by CRRC's public component catalogue:

- [CRRC components overview](#)
- [CRRC vehicle components](#)
- [CRRC electric-control equipment](#)
- [CRRC Dalian traction and control](#)

This repository has no CRRC partnership, endorsement, selected supplier, quotation or configuration approval. Any CRRC parts remain subject to open and competitive procurement, supplier offers, interface control, RAMS/FMEA, cybersecurity, EMC, fire/environmental qualification, hardware-in-the-loop and first-article testing, factory/site acceptance, spares and obsolescence terms, licensing and the competent authorities' acceptance.

Digital delivery and management system

The Baghdad package already instantiates 1,862 assets, 8,012 manufacturing and verification tasks, 15,804 material/procurement rows, 10,785 maintenance tasks and 6,689 QA actions. These feed the project twin, operations portal, ERPNext/Frappe adapters, supervision/SCADA boundary, maintenance planning, controlled evidence, QR identity, and multi-model AI advisory council.

The AI council may prepare and independently cross-check reversible management drafts. It cannot grant movement authority, bypass train protection, command SCADA, hire or dismiss staff, commit funds, sign contracts, certify work or exercise legal or safety authority. Named humans retain approval and accountability, and model identity, inputs, dissent and decision evidence are recorded.

Assurance and present release boundary

The deterministic package uses five fail-closed assurance gates:

- 1 G0 – identity and source baseline: configuration, provenance and hashes.
- 2 G1 – design assurance: requirements, hazards, interfaces and applicable standards.
- 3 G2 – implementation qualification: actual supplier/product evidence and tests.
- 4 G3 – installed integration: site, vehicle, system and operational validation.
- 5 G4 – independent/legal acceptance: competent authority and independent assessor decisions.

The GIS, finance generation, native simulation and SUMO timetable checks pass their planning gates. The city package is deliberately marked incomplete because physical depot and distributed stabling positions have not been surveyed or verified. Construction release also requires a surveyed alignment/property/utility baseline; calibrated demand and an operator-owned timetable; geotechnical, drainage, structural and fire acceptance; supplier-frozen battery, charger, traction and mechanical

equipment; and independent safety assessment and legal authorization.

Recommended engagement

- 1 Establish the Iraqi owner, governance, data room and acceptance authorities.
- 2 Run a 12–16 week data-validation and corridor-selection phase.
- 3 Complete survey, demand calibration, depot/stabling option selection and system requirements for a priority corridor.
- 4 Obtain competing supplier packages—including, but not limited to, CRRC—and freeze validated component interfaces and whole-life support.
- 5 Deliver a pilot corridor through G1–G4 before authorizing network rollout.

The PDF is generated from the current city design, engineering summaries and screenshots by
tools/automation/build-baghdad-offer.py; manifest.json records the input and output hashes.

Baghdad Acceptance And Accreditation Evidence Basis

Source: <cities/catalogue/west-asia/Iraq/Baghdad/operations/acceptance-evidence-report.md>

This report is the generated evidence basis for acceptance, trial-running readiness, and accreditation review. It is not itself an approval or certificate; it is the traceability index that shows which assets, manufacturing packages, BOM/material refs, QA hold points, evidence records, release authorities, and predecessor controls must be closed.

Summary

Item	Count / Status
Assets in register	1,862
Manufacturing schedule rows	8,012
Manufacturing material/BOM rows	15,804
Manufacturing QA verification rows	8,012
Construction QA action rows	6,689
Maintenance handover schedule rows	10,785
Manufacturing rows with material refs	8,012 / 8,012
Manufacturing rows with verification refs	8,012 / 8,012
Manufacturing rows linked to QA actions	8,012 / 8,012
Unresolved external predecessors	0

Material / BOM Basis

Source	Rows
project_kit	15,804

Rolling-stock rows link to the generated rolling-stock BOM and COTS fit-out BOM. Infrastructure rows use controlled `project_kit : *` refs until detailed civil/station/energy BOMs are added.

QA Gate Coverage

QA gate	Verification rows
qa-00-design-freeze	1
qa-10-carbody-structure	2,493
qa-11-bogie-wheelset	831
qa-12-traction-brake-battery	831
qa-13-passenger-systems	831
qa-15-first-article-trainset	831
qa-20-survey-geotech	173
qa-21-earthworks-drainage	173
qa-22-trackform-rail	455
qa-24-stations-depots-plant	366
qa-25-power-energy	317
qa-26-wayside-comms-safety	710

Acceptance Control Logic

- Every manufacturing package has a controlled material/BOM row set.
- Every manufacturing package has a QA verification row.
- Every verification row links to a generated QA action by `qa_uid`.
- Resolved predecessor ids are generated for schedule blocking.
- The portal blocks successor manufacturing work until predecessor work orders are closed with pass evidence.
- The portal blocks manufacturing closeout until the selected work order has pass evidence.
- Failed evidence creates a defect/NCR and puts the work order on hold.
- SQLite-backed Ops Core stores work orders, inspections, defects/NCR, and audit records for handover.

Remaining External Release Gates

This generated evidence basis organizes acceptance work, but it does not close release gates that require independent review, field data, hardware bring-up, supplier freeze, first-article tests, or authority acceptance. The open gates remain:

Gate family	Still required before accreditation / revenue release
Independent safety assessment and residual risk	Named ISA/assessor review, action log, residual-risk acceptance, and deployment-specific ALARP/tolerability decision.
Formal safety integration	Consensus refinement proof from TLA+ to implementation behavior, plus signed safety-log integration tests for forged, replayed, stale, and unknown-issuer entries.
Pilot hardware evidence	Exact COTS BOM freeze, wiring/harness maps, enclosure/mounting, power/thermal margins, SD image checksums, self-test logs, bench/safety evidence, and commissioning records for T-ECU/S, T-ECU/A, T-OBS, W-SBC, and S-SBC.
Rolling-stock production release	Supplier envelope freeze, EN structural/FEA reports, weld/WPS/NDT packages, manufacturing drawings, flat patterns/NC output, harness routing, weight/balance, first-car build hold point, and first-article inspection.
Field validation	Obstacle/intrusion sensor calibration and representative hot-weather, dust/soiling, night, and rain datasets with false-positive/false-negative analysis.
Charging and site energy	Site-specific solar yield, grid interconnect, charger thermal study, protection settings, utility approval, and train charging interface tests.
Operations validation	Operator workshops, translated/deployment rulebook where needed, competence records, emergency exercises, maintenance access trials, and trial-running records.

Those items are tracked in the [certification release gap register](#), [control-electronics release checklist](#), and [rolling-stock v2 release checklist](#). The acceptance matrix should be treated as the evidence index that collects those closures, not as the closure itself.

Review Artifacts

- Tracked compact asset register: [baghdad-assets.csv](#)
- Tracked operations manifest: [baghdad-operations-manifest.json](#)

The following high-volume files are regenerated in the local city package and intentionally excluded from Git. Their names are recorded for handover without presenting unavailable GitHub links:

- [baghdad-operations.json.gz](#)
- [baghdad-acceptance-evidence-matrix.csv](#)
- [baghdad-manufacturing-schedule.csv](#)
- [baghdad-manufacturing-materials.csv](#)
- [baghdad-manufacturing-verification.csv](#)
- [baghdad-qa-register.csv](#)
- [baghdad-maintenance-schedule.csv](#)

Accreditation Use

The evidence matrix can be filtered by asset, package, QA gate, release authority, or material source. During acceptance, each row should be matched to completed Ops Core evidence, defects/NCR status, and the release authority signoff before the related asset is accepted into trial running or passenger operation.

Baghdad-only funding programme

Source: cities/catalogue/west-asia/Iraq/IRAQ-FUNDING-PROGRAMME.md

Scope: [Baghdad](#) and one manufacturing plant sized for Baghdad. Samawah, Mosul and every other city are excluded. All funding is proposed and uncommitted.

Independent reconciliation, priced gap finance and six-month tranches

Capital sources still total USD 7,880.494m. The independent pooled-cash reconstruction requires USD 6,168.154m of gross extra cash and retains USD 3,941.337m later, leaving a USD 2,226.818m nominal net deficit before pricing additional gap finance. These three amounts answer different questions. The original capital principal is repaid in the lifetime cash ledger, not added to CAPEX a second time. City EPC is now spread over direct works; the initial baseline-freeze task no longer receives the full programme overhead allowance.

[Detailed arithmetic and Iraqi financing routes](#) · [six-month reference bond/loan requirements](#) · [six-month blended candidate independent calculation](#).

Green debt replaces qualified conventional borrowing; grants replace domestic capital debt; guarantees supply credit enhancement rather than cash. The new sensitivity charges supplemental IQD funding for interest and fees, sweeps later surplus to repayment, and exposes cash beyond its illustrative IQD 13tn cap and any unpaid terminal loan. The earlier gross cash-support figures below assume external contributions; they are not a priced bridge-loan requirement.

Priced gap-finance sensitivity	Gross gap draws, IQD tn	Uncovered cash, IQD tn	Unpaid terminal loan, IQD tn
unfunded reference	0.000	8.019	0.000
green label only	0.000	8.019	0.000
zero cost liquidity bound	8.019	0.000	2.895
commercial gap credit	13.000	18.472	13.000
concessional gap credit	10.320	0.000	8.083
blended candidate	8.259	0.000	5.153
green concessional only	9.386	0.000	7.455
fare 5pct flat costs fixed demand	3.509	0.000	0.000
fare 5pct flat costs elastic	4.220	0.000	0.000
fixed fare 5pct opex	13.179	9.457	13.000
fare 5pct opex 5pct	4.545	0.000	0.000
fare 5pct opex 7pct	5.904	0.000	0.000
fare 5pct opex 5pct income 2pct	5.683	0.000	0.000
variable fare 5pct opex 5pct	4.328	0.000	0.000
fare 5pct opex 5pct rents indexed	4.349	0.000	0.000

The candidate mix is uncommitted: eligible IQD green bonds at an assumed 4% plus enhancement fees; USD 25m equivalent climate capital grant; USD 300m equivalent net development-rights proceeds; and USD 25m equivalent annual new net local receipts at full opening. The grant, valuation, legal powers and IQD concessional facility need evidence. Existing rents/fare receipts cannot be counted again. With constant nominal fares and OPEX, a terminal unpaid loan means that sensitivity has not achieved self-financing.

Additional pricing and OPEX inflation sensitivities

The requested paired sensitivity increases fares and OPEX 5% annually from financial close, while testing income growth separately. At 5% income growth and assumed -0.30 real-price elasticity, the illustrative blended case peaks at IQD 4.362tn supplemental debt and ends with IQD 0.000tn unpaid. This can repay the priced facility under the assumptions, but still requires placed early financing, the candidate grant/rights receipts, fixed nominal debt terms and income growth. It is not a committed funding outcome.

Average nominal tickets move from IQD 1,525 at first opening to IQD 1,765 at full opening; 44 trips remain 11.7% of the indexed income proxy when incomes grow 5%. Separate cases test 2% income growth, 7% OPEX inflation, peak/off-peak tiers, fixed demand, and rental indexation. Capital escalation and future FX changes remain outside these sensitivities.

[Paired 5% six-month financing](#) · [monthly tickets and affordability](#) · [variable-ticket six-month financing](#). The detailed report contains the NPV, all assumptions and downside cases.

Surplus cash and early repayment

These cases use identical paired 5% fare/OPEX/income assumptions and the same conditional blended capital sources. Each pays OPEX, scheduled principal, interest and fees, then funds the debt-service reserve and a three-month current-OPEX buffer before voluntary repayment. Borrowed gap proceeds and uncovered external cash cannot fund early payments. The extra buffer remains cash held at the terminal horizon and earns no interest.

Compare strategies against buffered gap-only, rather than attributing the buffer change to repayment savings. Loans-first pays bank credit, Chinese credit and gap credit before bonds. Cost priority pays bank (9%), ordinary bonds (8%), Chinese credit (5%), green bonds (4% plus 0.5% annual guarantee), then gap credit (2%). It is an interest-rate heuristic, not a proof of the globally best strategy.

Surplus strategy	All debt cleared, month from close	Finance cost saving vs buffered gap-only, USD equivalent m	Early-payment premiums, USD equivalent m	Peak gap debt, IQD tn	Terminal unrestricted cash, IQD tn
gap only buffered	371	0.000	0.000	4.505	16.797
gap then core	305	13.320	1.896	4.505	16.815
loans then bonds	304	92.811	7.121	4.461	16.918
cost priority	303	195.029	23.681	4.461	17.051
cost priority zero premium	302	225.425	0.000	4.457	17.090
cost priority noncallable bonds	371	78.731	5.175	4.461	16.900

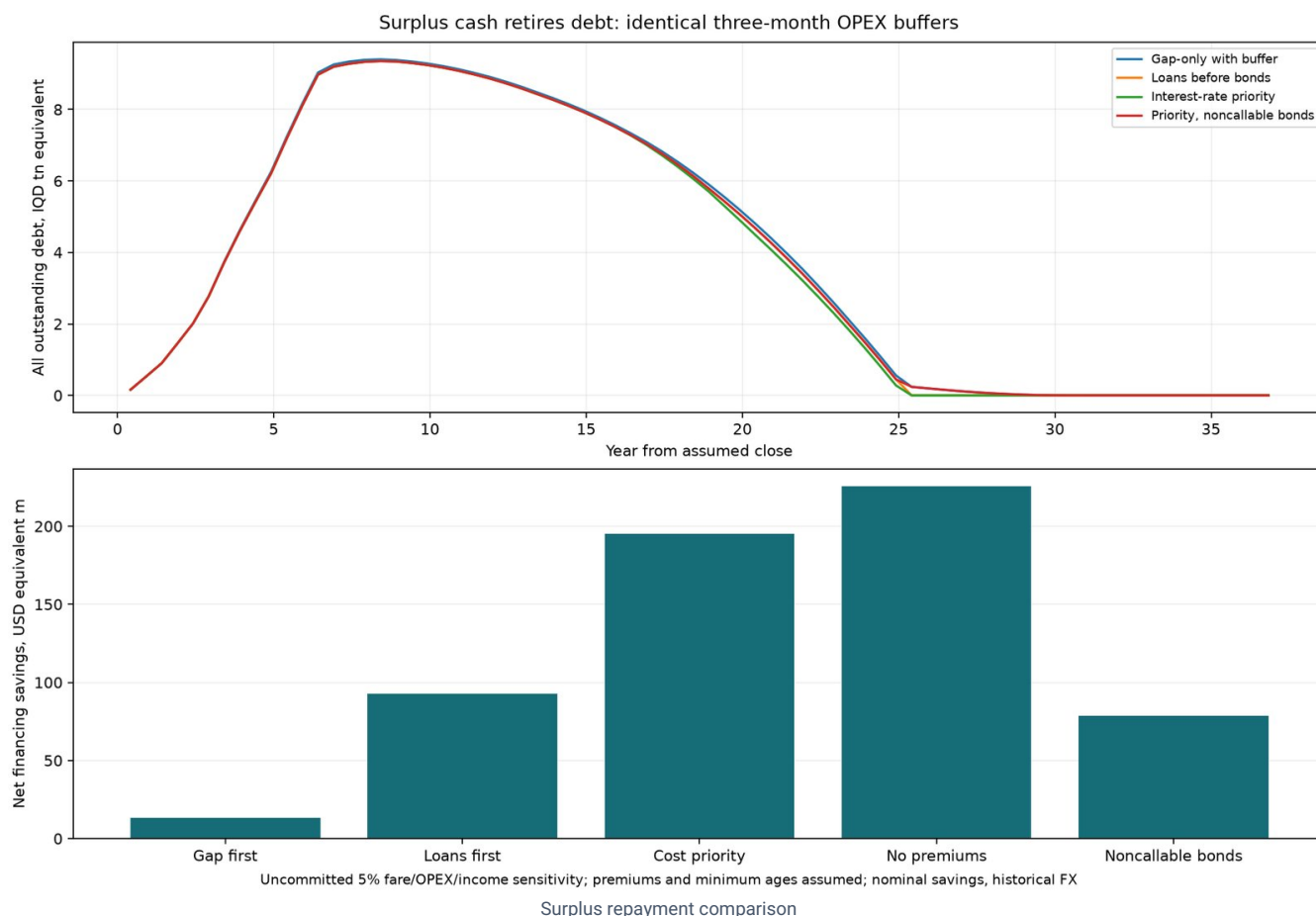
Cost priority clears all debt in month 303, compared with 371 for buffered gap-only. Net nominal financing savings are USD 195.029m equivalent, after USD 23.681m assumed early-payment premiums. Savings include core interest, annual green guarantee charges and supplemental interest/draw fees; they exclude principal, which is returned once. All cases retain IQD 0.340tn operating buffer separately from unrestricted cash. There is no additional government contribution above 25% of CAPEX in these cases.

Assumed premiums are 1% of bank/Chinese/green principal and 2% of ordinary bond principal. A minimum draw age of 6 months (bank), 12 (Chinese) and 24 (both bonds) prevents immediate issue-and-redemption. Eligible vintages are repaid oldest first; contractual instalments are kept and maturity shortens. Notice, issuer call rights, investor consent, buyback price, remaining-maturity compensation, tax and FX require actual agreements. Noncallable-bond sensitivity makes no voluntary bond payments; zero-premium sensitivity removes only the assumed premium, retaining minimum ages.

Facility	Buffered gap-only final principal payment month	Cost-priority final principal payment month	Early principal, native currency	Premium, native currency
bank credit	167	167	IQD 189.902bn	IQD 1.899bn
domestic bonds	287	231	IQD 963.845bn	IQD 19.277bn
chinese export credit	311	236	USD 266.436m	USD 2.664m
green bonds	371	246	IQD 614.570bn	IQD 6.146bn

The gap facility's final principal payment is in month 303. Later surplus remains unrestricted cash after debt retirement. Each month's native debt balance equals prior balance plus draws minus scheduled and early principal. Each six-month closing balance is its final month's balance; payments, premiums, interest and reserve movements are period sums. Contractual repayment windows in tranche files describe the original draw terms; the actual final-payment months above incorporate early repayment.

[Monthly cost-priority repayments](#) · [six-month repayment tranches](#) · [loans-first tranches](#) · [all six complete calculations](#)



Loan prepayment can carry redeployment/unwind charges: the [World Bank Treasury FAQ](#) illustrates the concept, without establishing Chinese or Iraqi loan terms. The [US Treasury buyback FAQ](#) and [published purchase results](#) illustrate issuer authority and priced buybacks rather than automatic par redemption; they do not supply an Iraqi legal mandate.

This repayment allocation does not change the unlevered project NPV or validate demand. Nominal savings over several decades are not present-value gains. CAPEX escalation, FX, lifecycle replacement, floating-rate risk and the uncommitted early funding still require appraisal.

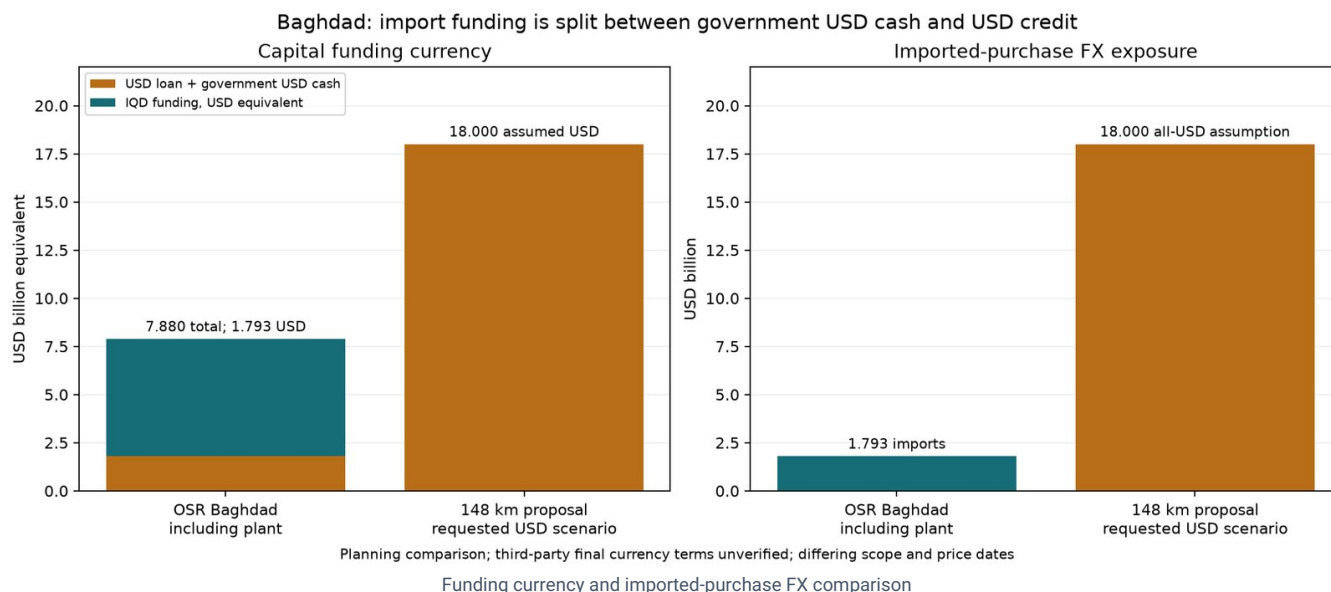
Consolidated sources and uses

Government contributes 25% of total capital uses, including the plant and EPC. Imported purchases are split 50% government USD cash and 50% proposed Chinese USD credit, assuming the full imported basket can qualify. That USD government cash is inside the 25% total contribution; the rest of the government contribution is IQD. The remaining balance after those two sources is split 75% domestic IQD bonds and 25% IQD bank term credit.

Proposed capital source	Currency	Native amount, billions	USD equivalent, million	Share of total capital
chinese export credit	USD	0.897	896.53	11.38%
government import cash	USD	0.897	896.53	11.38%
government local cash	IQD	1,395.671	1,073.59	13.62%
domestic bonds	IQD	4,888.494	3,760.38	47.72%
bank credit	IQD	1,629.498	1,253.46	15.91%
Total capital uses	Mixed	—	7,880.49	100%

Baghdad city CAPEX is USD 7,555.74 million. The plant is counted once at USD 303.50 million plus USD 21.25 million EPC. Its sizing basis is 4,986 vehicle/car modules from Baghdad's controlled fleet and car count. Imported tooling receives proposed Chinese credit within that plant budget. The physical cell/floor/tooling envelope replaces the smaller module allowance where necessary; both remain unquoted engineering assumptions.

Currency and USD capital intensity



Only Chinese credit is USD-denominated debt: USD 896.53 million. Government additionally provides USD 896.53 million cash for imports. Combined USD capital funding is 22.75% of uses; the remaining 77.25% is IQD government cash, IQD bonds and IQD bank credit. Revenue and local OPEX are also budgeted in IQD. At the historical planning conversion of 1,300 IQD/USD, the local government cash allowance is IQD 1.396 trillion, alongside its separate USD import cash. Together they equal 25% of total capital. USD columns are comparison equivalents, not a requirement to borrow or appropriate those domestic amounts in dollars.

Funding currency and procurement currency differ. Estimated imported purchases total USD 1,793.06 million (22.75% of capital), including plant imports. The government supplies half directly in USD cash; proposed Chinese credit supplies half as USD debt. The full import pool, including categories beyond the initially selected solar, bogies, batteries, windows, doors and tooling, is assumed lender-eligible pending origin qualification. If that expanded basket cannot qualify, its loan funding is uncovered; it is not automatically replaced by IQD bank credit or another grant. IQD borrowing reduces the revenue/debt currency mismatch; it does not remove Chinese debt-service FX risk, imported maintenance costs, domestic interest, inflation or placement constraints.

Eligibility sensitivity: the original named-component categories cover USD 863.01 million of invoices. At a 50% advance, they alone support USD 431.50 million of proposed credit. If the additional imported categories cannot qualify, USD 465.03 million of planned capital loan proceeds remains uncovered, in addition to the base support requirement below. Even the original categories remain unqualified; this screen is not a lender approval. No extra government contribution is assumed to replace that missing loan.

What the 25% government limit leaves unfunded

The 25% capital contribution is USD 1,970.12 million. If that is also the limit on all public cash, the model leaves USD 6,600.60 million of additional funding requirements across the full construction, operating and debt horizon. These are interest, fees, reserve and operating/debt cash needs after modelled revenue; they are not additional approved government contributions.

Peak annual capital contribution is USD 381.28 million. Peak annual additional funding requirement is USD 998.36 million. If all additional support were provided publicly, conditional lifetime public cash would be USD 8,570.72 million, with a combined annual peak of USD 998.36 million. This exceeds the requested contribution and is shown only to expose the funding gap.

Across the 37.0-year nominal model horizon, additional requirements divide into USD 1,978.06 million before Baghdad's full-network opening, USD 4,197.74 million during Baghdad operations/debt tail, and USD 424.80 million for plant capital financing. Principal repayment before opening is part of the pre-opening requirement. This base withholds all fares until full-network completion; the separate phased sensitivity below tests earlier revenue. Gross additional support is a liquidity requirement over time, not net lifetime loss: later retained cash cannot repay earlier obligations without an approved bridge.

The programme CSVs distinguish the capped capital contribution, additional funding requirement and conditional payment requirements. Underlying city and factory ledgers calculate the support needed to pay scheduled obligations; their positive reserve balances and completed repayments are conditional on that support being raised. They are not a cash-solvent forecast under the public cash cap. No bridge, equity investor, rollover or extra appropriation is invented to close the gap.

Cash and contractual structure

- 1 An Iraqi public sponsor seeks MoF authority for proposed sovereign IQD bonds through the established MoF/CBI issuance route, subject to legal review and placement. Municipal borrowing powers are not assumed. The 25% limit is the direct capital contribution, not a limit on sovereign bond liabilities, guarantees or lifetime public exposure.
- 2 Chinese export buyer credit is allocated to qualified Chinese invoices for solar equipment, bogies, batteries, windows, doors and Baghdad plant tooling. The requested 50% loan advance is matched by 50% government USD cash. All other imported categories also require Chinese supplier-origin and lender qualification before the full-basket financing assumption is valid. Noneligible imports still need foreign currency. Eligibility and origin require supplier evidence.
- 3 Government's 25% capital contribution covers the planned eligible downpayment; bonds and bank credit cover the remaining capital balance. Fees, construction interest and reserve funding are separate cash requirements, not silently capitalised or counted as capital-source proceeds.
- 4 Each draw starts its own grace and repayment clock. Planned receipts match procurement milestones; no actual payment or lender commitment is asserted. Chinese credit uses USD, bonds and bank credit IQD. Terms remain unconfirmed appraisal assumptions: 5%/48-month grace/180-month repayment, 8%/24/180, and 9%/0/84 respectively.
- 5 Revenue pays OPEX first and debt service next; subsidy is excluded from DSCR. Additional support shown by the city and plant models is an unfunded requirement under the cap. The factory model covers capital financing only; manufacturing income and factory OPEX need a separate business case.
- 6 Declined Chinese finance, delayed contributions, FX changes and bullet redemption are separately tested in the Baghdad appraisal. None creates an automatic facility or refinancing source.

Rollout and financial close

Baghdad's current capital milestones span 84 calendar months. Full-network fare revenue begins after all budget milestones. Lowering the capital grant increases borrowing and cash-service needs; it does not shorten this schedule. The city-sized Baghdad plant becomes available after 390 working days (18 months from NTP), followed by the separately scheduled first-article series-release gate. The separate phased sensitivity uses this integrated planning schedule; actual plant and line acceptance and a commissioned operating baseline remain pending.

Required before financial close: approved sponsor and borrowing powers; appropriation limited to the agreed government contribution; a funded solution for the additional cash gap; supplier quotations and origin evidence; signed term sheets, guarantees and insurance; IQD placement/redemption plan; lender draw windows and FX access; surveyed demand/fare policy; tax, duty, land and utility pricing; accepted resource calendar and reserve covenants.

Model and evidence

[Programme monthly cashflow](#) · [annual cashflow](#) · [machine-readable programme](#) · [Baghdad city appraisal](#) · [editable assumptions](#). Other Iraqi city appraisals remain standalone examples outside this programme.

[IMF 2025 Article IV](#) supplies the historical 2024 FX anchor, not a current dealing rate. [CBI fiscal-agent description](#) and [Injaz issue example](#) support sovereign-bond context. [China Exim export buyer credit](#) supports the instrument concept; numeric terms and eligibility are unconfirmed assumptions.

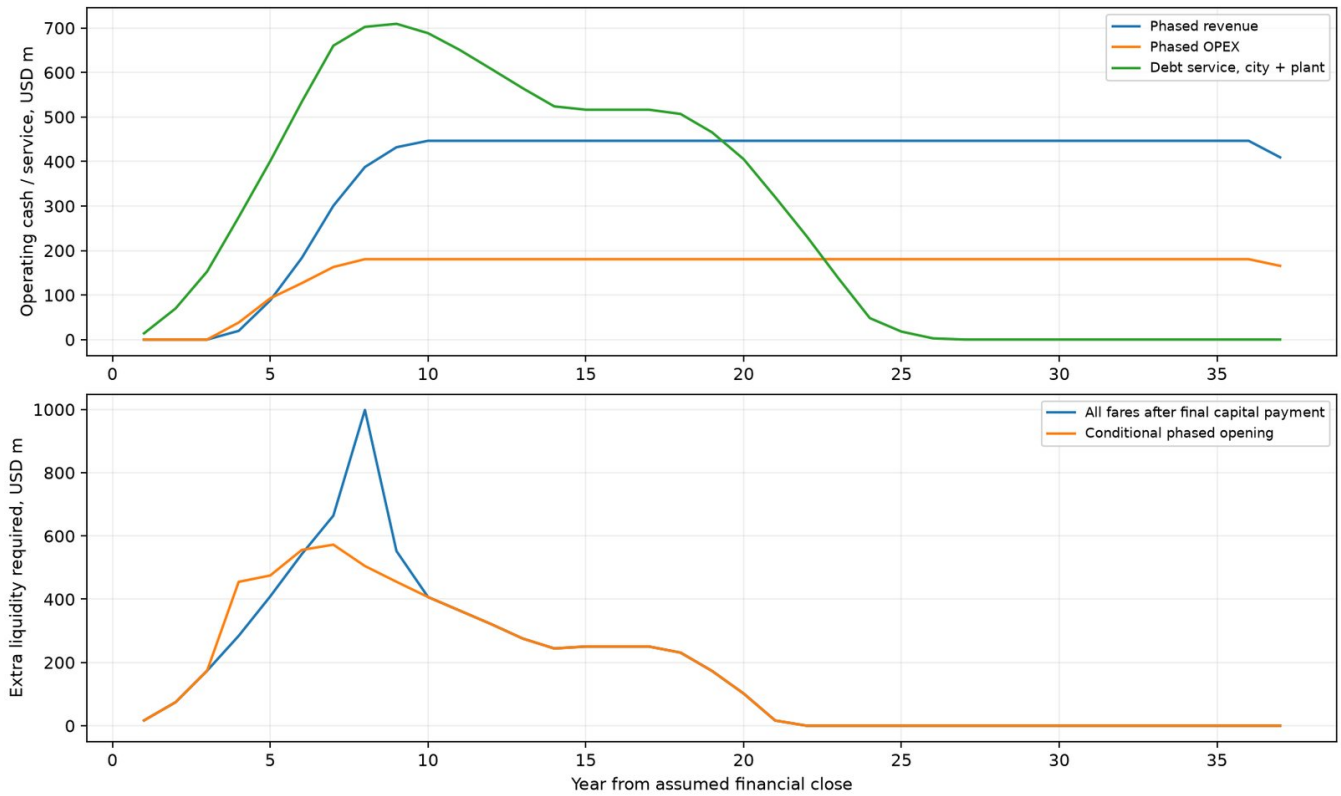
Recalculation: conditional phased opening

The earlier USD 9.890 billion result combined whole-fleet stage batching, no fares before the final capital payment and reserve deposits always funded with additional public cash. Ready-task dispatch now pipelines work within the same resource limits, gates train production on the new plant, and funds reserves from available project cash before seeking support. The full-network-only result above remains a conservative comparator.

With line openings tied to completed line and shared/depot work plus a 3-month commissioning allowance, first revenue begins in month 41, and all nine lines operate from month 83 after financial close. These dates are conditional planning milestones, not authorisation to run trains.

Each line's controlled trainset share allocates revenue and variable OPEX; its own 50% / 75% / 100% ramp applies. From first opening, 25% of full-network OPEX is fixed and 75% scales with opened fleet. This proxy can misstate early demand, central staffing and transfer benefits; a surveyed phase-specific operating plan is required. The same nominal fares, capital total, debt terms and 25% grant apply in every case.

Baghdad + one plant: phased receipts and additional annual liquidity



Conditional nominal planning flows; 25% capital grant excluded from extra liquidity; opening and funding uncommitted

Phased operating cash and additional liquidity

Opening / demand scenario	Additional funding beyond 25% capital, USD bn	Conditional lifetime public cash, USD bn
All fares after final capital payment	6.601	8.571
Phased: commissioning delay two years	6.700	8.670
Phased: demand minus 40 percent	8.983	10.953
Phased: low demand	6.168	8.138

Line	Planned opening month	Fleet / variable-cost share
line-1	41	11.43%
line-2	46	12.39%
line-3	52	13.00%
line-4	57	10.23%
line-5	62	11.79%
line-6	69	13.36%
line-7	73	10.11%
line-8	79	12.03%
line-9	83	5.66%

All these are gross nominal liquidity contributions, conditional on funding. Later operating surplus is retained and is not netted against earlier required injections. The phased city ledger ends with USD 3.941 billion of unrestricted cash; no distribution or return to the sponsor is assumed. The plant ledger has no manufacturing income/OPEX: it remains a capital-financing allowance. A 25% direct grant also does not cap sovereign IQD bond liabilities or guarantees.

[Phased programme monthly cashflow](#) · [phased programme annual cashflow](#)

Comparison with the historical 148 km third-party proposal

The [July 2024 report](#), citing the prime minister's adviser, gives USD 18 billion, 148 km, seven lines and 64 stations. The [NIC procurement notice](#) describes DBOMFT and requires bidders to submit a funding plan. Neither establishes a final all-USD financing contract. The 100% USD foreign-loan/government-cash comparator below is the user's requested scenario, not a verified split of debt and grants. The older USD estimate has not been restated to a common price date.

Measure	Current OpenSourceRail Baghdad plan	Historical proposal / requested USD scenario
Route km (not track km)	516.5	148
Lines / stations	9 / 182	7 / 64
Capital uses, USD equivalent bn	7.880, including one plant	18.000, reported estimate
Capital intensity, USD equivalent m / route km	15.26	121.62

Measure	Current OpenSourceRail Baghdad plan	Historical proposal / requested USD scenario
USD-denominated capital funding, USD bn	1.793, half loan / half government cash	18.000, assumed USD cash + loans
IQD-denominated capital funding, IQD tn	7.914	Zero in requested scenario
Imported-purchase FX exposure, USD bn	1.793	18.000 only under all-USD purchase assumption
USD purchase intensity, USD m / route km	3.47	121.62 under that assumption
Contracted fares / debt terms	Planning fare and uncommitted facilities	Not established by these sources

The OSR plan has 3.49 times the route length and 2.84 times the stations. Its planning capital estimate is 56.2% below USD 18 billion. Under the requested all-USD funding scenario, USD-denominated capital funding is 90.0% lower; under an all-USD purchase scenario, imported-purchase exposure is 90.0% lower. These measure different exposures and are not interchangeable debt-service savings.

The planning comparison is not a like-for-like qualified bid: route geometry, tunnelling/structures, land, utilities, taxes/duties, contingency, escalation, supplier qualification and acceptance maturity differ or remain unresolved. The older report's four-year completion expectation also differs substantially from OSR's current 84-month capital schedule. A lower capital estimate does not establish an earlier or more sustainable delivered service.

Fares, population access and financial sustainability

The modelled average paid-trip fare is IQD 1,317 (USD 1.01 comparison equivalent), denominated in IQD rather than automatically indexed to the USD loan. It derives from the retained USD 380/month income proxy, equivalent to IQD 494,000/month at the planning anchor. Thirty paid trips consume 8.0% of that proxy income; 44 commuter trips cost IQD 57,963, or 11.7%. This is an average-trip yield assumption, not an adopted tariff or unlimited monthly pass; concessions, transfers and family affordability need an explicit tariff and household survey.

Low/high capacity-use cases assume 1,138,320 / 1,821,312 paid trips/day, not unique people or surveyed demand. Low-case steady annual revenue is IQD 580.5 billion; OPEX is IQD 234.7 billion. At that trip volume, the OPEX-only neutral fare is IQD 485, with nonfare receipts held constant. It excludes capital, debt and reserve funding.

In the first complete operating year, the 50% revenue ramp produces 207.7 million paid trips. Holding those trips and nonfare receipts fixed, covering OPEX, scheduled debt service and fees requires approximately IQD 5,258 per paid trip, excluding reserve deposits and factory debt. This is a cash threshold, not a recommended tariff; higher fares can reduce demand. Minimum annual operating DSCR before support is 0.06. Higher utilisation improves later cashflow but cannot fund the long pre-opening debt-service period. Neither this fare nor IQD denomination closes the additional funding gap shown above.

The design retains a planning population of 9,780,429. Its 46.4% anchor-weighted coverage score gives an indicative proxy of 4,538,119 residents. That proxy is not measured population within a real 800 m walking network: survey, population grid, station access, river crossings and feeder coverage remain unvalidated. The older report's 80% city-coverage ambition uses no published comparable denominator or access method; it cannot establish that either plan reaches more residents. The 182 stations and nine routes offer a larger network to test, not proof of superior population access.

Direct Iraqi labour and industrial benefits

Approximately USD 6.087 billion equivalent (77.2%) of programme procurement is assigned to local suppliers under the editable origin assumptions. This is potential local expenditure, not wages, GDP added or guaranteed Iraqi content. City civil works alone allocate USD 3.310 billion locally; rolling stock allocates USD 907.45 million locally, with plant local expenditure separately counted.

Local train assembly and fabrication create work in body modules, fit-out, wiring, coatings, systems integration, inspection and maintenance while bogies, batteries, windows and doors remain imported inputs requiring supplier and process qualification. Infrastructure work supports Iraqi concrete/precast production, civil erection, stations, utilities, solar installation and supervision. Tooling, training, process qualification and supplier access can leave reusable industrial capacity, shorter repair chains and retained skills after construction. Local employment and supplier income circulate in IQD and can generate Iraqi tax receipts; no multiplier or tax recovery is booked without evidence.

The operating allowance supports 2,350 indicative FTE with annual labour cost of IQD 19.50 billion. These are operating positions, not construction or manufacturing job counts. Construction employment requires validated work hours, productivity, wage rates, shift cover and local-content contracts; no fabricated job total is assigned. The third-party proposal may also use Iraqi civil labour, so its local share cannot be assumed zero.

A sustainable appraisal must demonstrate phased service before full-network completion, realistic paid demand and affordable tariffs, funded debt/interest/reserves within the public limit, placed IQD facilities, supplier and labour qualification, lifecycle replacement funding and an accepted environmental/physical design. Current software and ledger checks establish planning consistency, not those outcomes.

Iraq National OpenSourceRail Strategy

Source: <cities/catalogue/west-asia/Iraq/NATIONAL-BRIEF.md>

This page contains only Iraq-specific aggregation. Shared network, service, energy, civil, cost, finance, QA and validation methods are defined once in the [deployment planning reference](#).

Important

Foreign-capital advantage: against the default equivalent foreign-turnkey sensitivity, this national programme avoids \$36.95 B (86.9%) of external capital and \$45.43 B of external interest. Capital plus saved interest totals \$82.39 B.

The [Baghdad physical factory plan](#) sizes production cells and test paths for its 831 six-car trainsets to finish alongside the overall city infrastructure programme, with facility readiness at 18 months from NTP. The national aggregation uses the larger of that physical capital envelope and the original module allowance, counted once. Future city loads are not concurrent factory commitments or part of Baghdad finance.

Iraq financing

The catalogue-wide figures below are generic capital/benchmark aggregations. They do not establish a five-year rollout or an Iraq lender commitment. The scheduled proposal covers Baghdad only, including one manufacturing plant. It uses government capital at 25% of total CAPEX, imports split 50% government USD cash / 50% proposed Chinese USD credit, with the remaining government capital, bonds and bank credit in IQD. Full-basket Chinese eligibility remains unqualified. Additional cash requirements beyond that public contribution remain visible in the [Baghdad funding programme](#). Samawah, Mosul and every other Iraqi city are excluded from its cashflows.

National Programme

Local measure	Planning value
Catalogue cities	18
Represented population	29,491,199
Trainsets / vehicle modules	3,650 / 15,996
City infrastructure and fleet CAPEX	\$23.30 B
Shared national factory	\$303.5 M
Factory sizing basis	4,986 modules for Baghdad, then reused nationally
Total national programme	\$23.62 B

Capital And Funding

Local funding measure	Planning value
Imported / external capital	\$5.56 B (23.5%)
Domestic / local capital	\$18.06 B (76.5%)
Default foreign-turnkey external capital	\$42.52 B
External capital saved	\$36.95 B
Capital + lifetime external interest saved	\$82.39 B

Procurement-Origin Composition

CAPEX bucket	Total	Imported	Local value
Civil works	\$10.32 B	\$1.55 B	\$8.77 B
Stations	\$3.98 B	\$796.1 M	\$3.18 B
Depots	\$144.0 M	\$36.0 M	\$108.0 M
Rolling stock	\$4.55 B	\$1.59 B	\$2.96 B
Dedicated solar plants	\$2.62 B	\$1.18 B	\$1.44 B
Residual train control	\$114.4 M	\$57.2 M	\$57.2 M
Charging microgrids	\$218.7 M	\$87.5 M	\$131.2 M
EPC / project services	\$1.37 B	\$206.1 M	\$1.17 B
Shared national trainset factory	\$303.5 M	\$60.7 M	\$242.8 M
Total	\$23.62 B	\$5.56 B	\$18.06 B

City Programme

City	Population	Fleet	City CAPEX	External capital	Local capital
Baghdad	9,780,429	831	\$7.56 B	\$1.73 B	\$5.83 B
Basra	3,955,000	450	\$3.11 B	\$793.5 M	\$2.32 B
Sulaymaniyah	2,150,000	129	\$1.03 B	\$243.5 M	\$783.0 M
Erbil	1,952,000	212	\$1.02 B	\$262.0 M	\$758.5 M
Mosul	1,940,000	256	\$1.93 B	\$431.6 M	\$1.50 B
Kirkuk	1,780,000	179	\$1.17 B	\$273.5 M	\$893.6 M
Najaf	1,540,000	229	\$1.52 B	\$354.4 M	\$1.17 B
Karbala	1,390,000	198	\$1.37 B	\$319.5 M	\$1.05 B
Nasiriyah	705,000	147	\$538.7 M	\$125.7 M	\$413.0 M
Hillah	700,000	125	\$461.2 M	\$111.8 M	\$349.4 M
Amarah	660,000	101	\$433.0 M	\$99.4 M	\$333.6 M
Ramadi	525,000	104	\$430.3 M	\$101.4 M	\$328.9 M
Baqubah	470,000	130	\$479.2 M	\$116.4 M	\$362.8 M
Diwaniyah	440,000	106	\$434.5 M	\$101.5 M	\$333.1 M
Kut	410,000	101	\$426.6 M	\$97.9 M	\$328.7 M
Samawah	373,770	108	\$415.3 M	\$100.0 M	\$315.4 M
Duhok	360,000	122	\$518.5 M	\$127.7 M	\$390.8 M
Fallujah	360,000	122	\$452.4 M	\$109.5 M	\$342.9 M

Local Basis And Regeneration

Country finance parameters use IQ in `lib/templates/country-finance.toml`. The factory is counted once nationally and excluded from city CAPEX. City values come from each local `design.toml` and expanded scenario; common limitations and interpretation are not repeated here.

```
python3 tools/automation/generate-national-briefs.py
```

RFC 0033 – TACS runtime and committed resource control

Source: docs/rfcs/0033-tacs-runtime-and-resource-control.md

Status: proposed development target; software evidence generated/unreviewed. Supersedes: RFC 0032's separate authority/protection implementation. Deployment: synthetic reference only; sectional-authority pilot retained.

Responsibility and interface map

Function	Decision owner	Physical or deployment boundary
Shared railway identities/configuration	osr-core::deployment	Exact model-byte hash; frozen startup parameters; designed revision distinct from installed and approved records
Authoritative resource lifecycle	osr-interlocking::resources, resource_log, DerivedState	Only verified committed ResourceControl entries mutate ownership
Ordered authoritative decisions	Existing osr-consensus::step	Three static voters; independent power/network placement must be designed and assessed
Physical proving	Identified point/station I/O hosts	Asset-scoped occupancy, integrity, points, restrictions and qualified no-reentry proof; synthetic ports in the reference
Onboard movement authority	Existing compute_self_ma_from_state	Validated committed view, exact frozen model, bounded full train footprint and evidence age
Speed and emergency protection	Existing osr-atp::atp_evaluate	Grade-adjusted illustrative braking profile; measured braking/adhesion/latency remains open
Mission and station requests	Existing osr-ato::ato_evaluate, station_step	Doors, charger isolation, connector clearance and energy independently gate departure
Final brake request	Existing osr-brake::brake_evaluate	Application request to separately qualified safety-output channel; software host is not qualified output hardware
Authentication and message contracts	osr-proto, osr-crypto, osr-secbus, consensus proposal verifier	Registered issuer, asset permissions, configuration/session/sequence/time checks before state effects
Process execution and persistence	Thin osr-runtime hosts	Framed local ports, event scheduling, disk writes before outbound effects, conservative cold start
OCC	Service intentions and observations	No authority, ownership release or latch-clear bypass
ERPNext/FUXA	Existing observation/maintenance integrations	Observation only; no new railway command interface

There is one MA implementation and one ATP implementation, executed in each logical protection channel. Resource contracts from the earlier prototype move into existing crates; osr-tacs and its parallel MA/protection algorithms are retired. osr-runtime contains orchestration and adapters. Route selection remains intent, per osr-onboard-routing; agreement cannot supersede missing permission or physical proving. Logical peer observations may traverse common infrastructure; no direct radio or osr-t2t crate is added.

Configuration and communications

railway-model.json schema osr-tacs-railway/2 supplies resource IDs, existing section IDs, conflicts, extents, asset identities, train/session/formation parameters, station locations, static voters and role-scoped public keys. deployment.json is an exact-byte-hash-bound generated startup export. FrozenRailway rejects unsupported versions, invalid topology/identities, changed startup parameters and operational activation. This source is a designed synthetic configuration; installed_state=not-installed and approved_operating_profile=prototype-unapproved are separate explicit facts. Changes require a new bundle and restart, never live digital-twin mutation.

The reference transport uses signed osr-runtime/1 packets, postcard encoding, existing signed consensus proposals and explicit internal interlocking entries. Legacy protobuf/postcard osr-proto::Entry remains a separate versioned format; it must not be deserialised directly into internal entries. Float-based legacy position inputs require checked unit conversion at an explicitly selected adapter. The runtime path publishes integer measured speed and bounded, non-zero uncertainty. Unsupported/malformed packets fail closed.

The envelope binds configuration hash, registered session, monotonic sequence, original observation time and message kind. Limits: 512 KiB application payload, 768 KiB signed payload, 4 MiB framed command, 16 publications per batch, 12,000 log entries and a 256-message reference queue. One-second input freshness and 100 ms maximum configured clock error are prototype assumptions; PTP and hardware timing must establish their operational applicability. Future-dated packets are rejected. Send/receive counters persist across restart. Asset permissions are separate from signature validity. All load-bearing events, including positions, use Safety. The legacy simulator consensus publisher also classifies occupancy-changing position reports as Safety and retains its last committed view when quorum confirmation is unavailable. No advisory route/position can unlock track.

The published deterministic reference keys are test fixtures. Operational provisioning, protected key storage, rotation, revocation distribution, offline replacement and recommissioning require a controlled maintenance pack. Changing keys or sessions changes the configuration hash; interoperability stops until the complete bundle is reconciled. Revoked participants

are rejected. Crash-fault Raft is assumed: a signed member's committed-prefix assertion is not a Byzantine quorum certificate. Authenticated erroneous sensors and common software, configuration, power and clock causes remain FMEA concerns.

Ownership and evidence freshness

Unknown qualified clear available reserve/prove/lock occupy explicit withdrawal qualified no-reentry/clearance available. Identified occupied startup admission can bind the proved train/session when no conflicting owner exists; it cannot transfer another owner's lock. Leases time out to ReleasePending, retaining their owner. Restart increments the committed controller epoch, retains owners as unknown and quarantines empty clearance for the maximum prior permission plus clock error. Identified retained owners can be reconciled without transferring protection. Blocked possessions cannot be cleared by these recovery APIs.

An empty detector does not prove cancellation safe. Qualified release requires bounded full rear clearance with integrity, or committed withdrawal and a proved stop/protected approach that prevents old-permission re-entry. These are physical interface assumptions to validate. Platform resources remain owned at arrival. Depot/work resource is explicitly protected; rescue, coupled formations, turnbacks, arbitrary topology and large fleets remain separate packages.

MA expiry is the minimum of the new MA window and the original position, local proof and held permission deadlines. Recalculation cannot refresh evidence. Every traversed section must have the correct active owner/configuration/epoch, current point proof and integrity. Local full-footprint uncertainty checking adds a restrictive gate before ATP. Recovery requires stopped, valid local facts and a privileged local procedure, never a dashboard acknowledgement. Civil conditions enter through asset-scoped signed restrictions/closures; urgent protection must have an engineered path independent of alarm viewing. Removal requires qualified inspection/handback, not an ERP status change.

Persistence and process reference

osr-consensus::disk::DiskJournal has one OS-locked writer. Frames carry node, configuration, term, vote, complete log deltas, commit index, application state and replay fences, with length and CRC bounds. Every host persists and fsyncs file and directory before exposing messages or actuator requests. Torn/corrupt journals inhibit startup; repair never silently discards an accepted lock. Storage compaction atomically replaces a full-log checkpoint after fsync. Logical Raft snapshotting/log truncation is not implemented. Capacity exhaustion inhibits further publication; offline servicing requires retained-protection and configuration reconciliation. A power interruption during rename/fsync requires storage/hardware qualification; SIGKILL tests are software evidence.

The default Raft batch remains one entry. The reference host uses bounded contiguous batches of twelve; Rust handles each entry under the same checked prefix semantics. This is outside the one-entry formal refinement and needs independent refinement review. Quorum confirmation records fresh per-peer responses in the current term, including empty-log heartbeats; stale historical replication indices cannot refresh a quorum. Durable reboot restores no leader role, quorum confirmation or volatile acknowledgements.

Run `python3 tools/automation/tacs_reference.py --output /tmp/reference.json`. The coordinator spawns four osr-train-agent processes (A/B for each train) and six osr-wayside-agent processes: three voters, point I/O and west/east station I/O. Charger requests and local proving pass through the station process ports; their asset-scoped interlocks can inhibit power and cannot grant movement. The depot/work resource is committed blocked and remains blocked across expiry and controller restart. Two further osr-safety-output processes each host the paired osr-brake::dual::DualGuard comparator. Both channels must agree to permit; either emergency, missing/stale/replayed request, disagreement or unhealthy feedback latches an emergency output with zero torque. The reference feeds at 50 ms, before the 100 ms output deadline. A late feed cannot replace an overdue request to hide a missed interval; only stopped, privileged, fresh recovery can clear the latch. These ports block on coordinator commands and use virtual time. They do not establish autonomous real-clock execution, independent physical timers or de-energised hardware outputs. Processes exchange signatures through asynchronous bounded queues; publication does not wait for commitment. Simulation truth enters only own-train sensors and local infrastructure proving ports, never the MA state. Real journals and process kill/restart are used. The same frozen bundle feeds every process. The physics fixture checks its supported extents, conflict identities, formation/braking and berth/energy parameters before startup; unsupported bundles require a new calibrated fixture. osr-sim's legacy synchronous cluster publisher remains isolated from this reference; it is not claimed as the deployed asynchronous path. Full integration of its scenarios/physics, shared clocks, radio/T2G/TCN failover and direct-peer performance is future work.

Station sequence is stop/secure exchange/charge isolate physical departure proof ready. The reference measures synthetic energy and closed-loop speed; per-car BMS/thermal/door/PSD/traction integrations remain pending HIL. Failed charging can hold service without erasing track ownership. Emergency departure cannot bypass a connected charger or unlocked door. Missing application output must trip a separately qualified deadline/output channel; the isolated reference output ports model deadlines and latches but do not qualify that hardware function.

Default redundant protection and platform review follow-up

The [executable policy](#) sets two logical channels, two-out-of-two permission and either-channel trip as the development default for train protection, point proving/locking, crossings, doors/platforms, charging isolation and obstacle protection. Loss of either channel stops the affected function. Automatic single-channel operation or takeover is excluded. Availability after a channel failure requires a separately assessed architecture. Three Raft voters replicate authoritative state; they are not protection outputs. The conservative deployable pilot profile and its release gates remain in force.

Each train A/B host uses a separate OS process, journal and restored committed view. The binary requires `--safety-channel A|B`; a persisted journal cannot change channel on restart. A publishes the train's network proposals; B evaluates the same committed domain inputs without publishing duplicate proposals. Loss of A does not authorise B takeover. Pair comparison uses exact synthetic cycle sequence/time, brake and torque agreement. It has no end-to-end source-issued output identity yet. Shared software, sensor fixture, host, test keys, coordinator clock and comparator are explicit common causes. Physical power/output separation, channel-local clocks and independent feedback must be engineered and qualified. Only train protection's logical pair is executed; the policy explicitly keeps all five other local pairs open. Two processes are not installed board quantities.

The 2 October 2026 review of 17b31f055c9e16b5439c18b4a4d83f004890bcad identified a stale FMEA applicability check and a concealed deadline gap. This change rejects stale failure subjects as well as stale configuration baselines, retains the late-feed regression and adds missing-A, missing-B, disagreement, replay and feedback-failure process cases. Each new case must first permit motion, actually inject its fault, trip within one 50 ms reference step, stop with retained protection and remain latched. These are virtual-time software criteria, not measured hardware response times. The existing TLC abstraction covers resource ownership only; it does not prove the new paired output function.

Next work closes named integration gaps in this order. Owners and reviewers are roles to appoint; acceptance is still open.

Work	Owner / independent reviewer	Required exit evidence
Autonomous watchdog and output paths	Embedded/electronics / safety-I/O reviewer	Coordinator/application/comparator freeze and kill, delayed scheduling, clock rollback/skew, power loss and stuck feedback; measured traction inhibition and effective braking without a new application command
Sensors and emergency union	Train integration / protection reviewer	Typed adapters to existing localisation, fire, derailment, obstacle and BMS/thermal evaluators; wheel slip/bias and simultaneous resource faults independently reach the final output; shared bad sensors do not establish independence
Repeated station missions	ATO/station integration / operations reviewer	Explicit mission/stop identity and arrival/departure transitions; at least three stops, short charge, delayed doors, turnback/resume, per-car charging and no moving door release
Remaining local protection pairs	Wayside/station / safety-I/O reviewer	Point/crossing proving, door/PSD, charger isolation and obstacle channels; independent release and inspected handback
Calibrated plant and city conversion	Simulation/configuration / braking and site reviewers	Sealed city revision converted to supported resources and shared osr - sim plant with mass/load, adhesion, gradients, build-up/delay, thermal/energy limits; full LM3 stopping envelope under uncertainty sweeps
Duty, recovery and storage	Simulation/runtime / operations and storage reviewers	Representative city's station capacity, overnight storage, morning dispatch, charging outage/recovery; separate rescue/tow/split/reverse/handover missions and sustained duty without capacity exhaustion or unbounded restart; safe snapshots/replay fences retain locks
Formal and scenario evidence	Assurance / independent formal reviewer	Property-to-code/test mapping, overlapping multi-resource routes, sequences/epochs/crash points, counterexample replay and specific hazard activation/deadline/excursion/queue/CPU/storage/recovery metrics
Integration and lifecycle quantities	Platform/hardware / configuration and cost reviewers	Observation-only A/B health/discrepancy/feedback/age/degraded/revision data through supervision gateway to FUXA/ERP; no acknowledgement resets; actual serial/power/firmware/calibration/replacement records and matched-service redundancy/spares/training/assessment costs
Physical release	Hardware/operations / independent assessor	HIL, measured braking/integrity/radio, power interruption, closed-track recovery and maintenance drills before profile approval

`station_step(Ready, fresh berthed/secured arrival)` currently remains Ready; the host targets the final stop. Nominal arrival therefore does not demonstrate another passenger exchange or a multi-station mission. The bounded two-train fixture retains geometry checks; expanding it cannot substitute for calibrated city conversion. Civil closures must bind precise operational resources and inspected handback separately from ERP repair completion. No duplicated ERP/GUI or comfort software is implied by protection redundancy. Commodity hosts, the existing small Rust evaluators, reusable local controllers and open replacement/ calibration procedures remain the implementation approach. Hardware quantities and hot-standby/two-MCU mappings must be reconciled at the hardware freeze.

Evidence and staged delivery

The shared connected-engineering compiler carries part subassembly controller function train/infrastructure hazard control evidence operating response. Source changes stale execution hashes and propagate through configuration-bound claims and blocked release decisions. TLC checks a bounded resource abstraction with qualified physical proving assumptions; it is neither a Rust refinement proof nor an unbounded liveness result. G0–G4 decisions remain blocked pending review and measurements.

Owners/reviewers below are accountable roles to appoint, not fabricated named approvals. No package is accepted merely because a generated checkbox is true.

Package	Technical owner	Independent reviewer	Exit evidence / dependency
1 Architecture	Control architect	Independent railway safety engineer	Reviewed responsibility map; current baseline
2 Shared configuration	Configuration/lifecycle engineer	Independent configuration auditor	Identity/schema/mixed-version tests; 1
3 Resource/durability	Interlocking/consensus engineer	Independent formal/storage reviewer	Lifecycle, crash and restart evidence; 2
4 Authority/protection	ATP/localisation engineer	Independent braking engineer	Freshness/uncertainty/integrity properties; 2–3
5 Communications	Communications/security engineer	Independent security/timing reviewer	Replay/loss/restart/vendor-neutral transport tests; 2
6 Runtime	Embedded runtime engineer	Independent safety-I/O reviewer	Multi-process reference and physical power-loss tests; 3–5
7 Station	Station/energy integration engineer	Independent doors/charging reviewer	Complete station-cycle and per-car tests; 4–6
8 Twin/campaign	Simulation engineer	Independent test engineer	Reproducible raw results; starts after 2
9 FMEA/GSN	Assurance engineer	Independent hazard reviewer	Connected change impact and applicability; continuous from 1
10 HIL/hardware	Electronics integration engineer	Independent hardware assessor	Measured output/timing/fault records; 6–9
11 Closed track	Test/operations engineer	Independent field safety reviewer	Braking, integrity, recovery drills; 10
12 Profile release	Railway release authority	Independent assessor	Named gates, rollback and approved installed bundle; 9–11

Hardware preparation proceeds with software and assurance using existing t-ecu-s, t-ecu-a, t-obs, w-sbc, s-sbc families. Each pack needs exact module/SKU/substitutes, wiring/connectors/enclosure, power/thermal/environmental budgets, fail-safe outputs/watchdogs, calibration, controlled firmware, fixtures, bench records, and complete-unit replacement/recommissioning instructions. Existing COTS and custom-board release tracks both remain valid; safety-channel selection follows control-electronics/safety-controller-selection.md. This change supplies no invented SKUs, installed serials or physical test records.

Cost comparison uses equivalent service and availability, separately counting train controllers, static voters, non-voting sites, radios/antennas/power, retained detection, installation, spares/calibration/maintenance, integration, assessment, recovery/disruption and local/import content. No equipment removal or city-model savings is authorised without replacement function and evidence. The reference's ten processes are software roles, not a bill of materials. Cost quantities/prices remain deployment-specific inputs; an unknown input must keep the comparison incomplete. Track duplicate safety implementations, configuration variants, core dependencies and physical interfaces alongside money.

Distributed Onboard Control Profile

Source: <docs/certification/distributed-onboard-control-profile.md>

Status: development target; not the deployable pilot baseline.

This profile reduces dependence on centralised wayside computation and permits continued route selection during a communications outage. It does not remove the safety functions commonly implemented by interlocking, ATS or ATP. Those functions may move between onboard and distributed equipment only after a hazard analysis, independent assessment and deployment approval demonstrate equivalent or better safety.

The deployable reference remains the conservative sectional-authority profile in [pilot-signalling-profile.md](#).

Executable train-centred milestone

[RFC 0033](#) and the [TACS assurance package](#) use the existing committed interlocking MA and ATP/ATO/brake components through thin, authenticated process hosts. The reference includes A/B agents for each of two trains, three static voters, point I/O and two station/charging interfaces. The [development default](#) requires both protection channels to permit and either to trip; loss stops the affected function. Train logical pairs are executed; point/crossing/door/charger/obstacle pairs and physical independence remain open. Raft replication cannot substitute for safety outputs. Generated software evidence remains unreviewed. HIL, measured braking/integrity, field recovery, independent assessment and a named approved installed bundle are required to change the deployable profile. The conservative pilot remains the default.

Three independent route inputs

The onboard selector compares:

- 1 a signed, versioned plan stored onboard;
- 2 route and position inferred from onboard sensors against immutable topology;
- 3 an authenticated network command from operations control.

An input is eligible only while trusted, fresh and internally valid. Exactly matching route, next-section and plan-epoch values from two eligible sources are required. If all three disagree, only one source is eligible, or the two available sources disagree, the result is Ho l d. A network outage can produce OnboardAutonomous only when the onboard plan and sensor-derived input agree.

Route selection is not permission to move. Every accepted selection carries an explicit `movement_authority_required` condition and is passed through the independent authority, localisation, speed-envelope and emergency-brake path.

Functions that must remain

Any implementation must retain, even if equipment placement changes:

- movement-authority generation and expiry;
- train separation and opposing-move exclusion;
- route locking and detected/locked point position;
- occupancy or safety-integrity localisation;
- overspeed supervision and emergency braking;
- intrusion/obstacle gates required by the operating envelope;
- degraded-mode supervision, event recording and accountable override.

“Eliminating interlocking/ATS/ATP” is therefore interpreted as reducing duplicated central equipment, not deleting their safety outcomes. A model, sensor fusion algorithm or radio link cannot independently claim equivalence.

Narrow single-track operation

Short single-track sections are permitted only on exclusive guideway and with passing loops at passenger stops or protected portals. Entry requires locked route exclusivity, detected points, clear occupancy, a valid authority and a proved timetable/degraded-mode strategy. Loss of required state prevents entry; it does not rely on two trains negotiating informally over the radio.

Recovery sites and shunt robots

The reference planner reserves a recovery siding at or immediately beyond every third passenger station. Each site has a protected mainline interface, independent stop detection and remote isolation. Its small shunt robot is remote-controlled, has local emergency stop and fail-held braking, and cannot autonomously enter the main line. Radio loss inhibits traction. Mainline entry, towing and return to service remain controlled recovery operations.

Evidence gates

Before this profile can replace any pilot equipment, the project must freeze the safety allocation and demonstrate source independence/common-cause analysis, target-hardware timing, localisation accuracy, radio-loss behaviour, point and occupancy interfaces, cybersecurity, HIL fault injection, shadow running, representative field trials, recovery drills and independent safety assessment. The deterministic report records design coherence; it does not close these physical or regulatory gates.

Rolling-stock top-down / bottom-up design system

Source: [docs/rolling-stock/design-system.md](https://docs.rolling-stock/design-system.md)

This is the design authority layer above the parametric CAD. It gives the project a systemic way to move from whole-train intent down to bought-in parts, fabricated parts, subassemblies, final assemblies, and then back up through scoring and evidence.

The rule of thumb is simple:

- 1 Start top-down with requirements: platform fit, axle load, range, traction margin, thermal margin, cost, mass, manufacturability, and safety evidence.
- 2 Decompose into the owned design hierarchy: external components, fabricated parts, subassemblies, final assemblies.
- 3 Iterate candidate definitions automatically inside an explicit design space.
- 4 Promote only feasible candidates into FreeCAD/FEM regeneration.
- 5 Freeze only when the requirements scorecard, supplier evidence, CAD, FEM, BOM, and manufacturing plan agree.

Hierarchy

Layer	Meaning	Examples	Source of truth
Top-level requirements	What the train must achieve	platform margin, range, axle load, cost, mass, traction/HVAC margin	<code>osr_mech.design_definition.REQUIREMENTS</code>
External components	Bought-in or BID/SOURCE modules	traction motors, HVAC, batteries, doors, glazing, sensors, couplers	ExternalComponent records plus supplier evidence
Fabricated parts	Parts OSR owns and regional workshops make	shell, bogie frame, motor cradle, cowl rings, articulation adapters	FabricatedPart records plus CAD templates
Subassemblies	Integration units that can be inspected	car module, powered bogie, trailer bogie, cowl, articulation, roof services	Subassembly records plus FreeCAD review groups
Final assemblies	Released trainset configurations	promoted light-metro 3-car trainset	FinalAssembly records plus generated CAD/FEM/BOM

Automated iteration

The first implementation is deterministic and dependency-free:

```
tools/automation/design-iterate.sh
```

It exhausts the declared discrete design space for the default light-metro-3car family, evaluates every candidate, and writes:

- `design/component-catalogue/catalog/design-system/design-iteration.json`
- `design/component-catalogue/catalog/design-system/design-iteration-summary.md`

To turn the winning candidate into a buildable product tree and review the current CAD/BOM baseline against it:

```
tools/automation/buildable-trainset.sh
```

That writes the buildable manifest and buildability review under `design/component-catalogue/catalog/buildable-trainset/`.

To run a bounded exploratory pass:

```
tools/automation/design-iterate.sh --family metro-4car --max-iterations 250
```

The current scorecard is intentionally planning-grade. It is good enough to rank candidate architecture choices before spending FreeCAD/FEM time, but it is not a certification result. As the project matures, replace or augment the simple metrics with:

- FreeCAD-derived mass properties and package clearances;
- CalculiX margins from `tools/automation/freecad-generate.sh --fem`;
- supplier quotations and evidence status;
- lifecycle energy and maintenance models;
- risk penalties from the hazard log and compliance matrix.

Promotion loop

Use the generated best candidate as a design review input, then run:

```
tools/automation/freecad-generate.sh --models --assemblies --fem  
tools/automation/freecad-generate.sh --screenshots --station-scenes
```

If CAD/FEM/supplier evidence rejects the candidate, update the component catalogue or constraints in `osr_mech.design_definition`, rerun `tools/automation/design-iterate.sh`, and promote the next best feasible candidate. That is the closed loop: requirements drive definitions, definitions drive CAD/FEM, evidence updates definitions.

RFC 0011 — Civil Infrastructure Design Standard

Source: <docs/rfcs/0011-civil-infrastructure-design-standard.md>

Status: Draft — planning only, no structural drawings ship with this RFC Date: 2026-04-22 Updated: 2026-08-17 — OSR Rapid Viaduct Kit correction Depends on: [RFC 0008 Rolling-Stock Reference Design](#), [RFC 0009 Track Design Standard](#), [RFC 0010 Station Design Standard](#)

1. Summary

OpenSourceRail commits to two civil classes only — at-grade and elevated — with bridges for water crossings as a narrow third class. No tunnels. Every corridor the auto-gen pipeline produces, and every hand-authored corridor the operator commits, will sit on one of:

Class	Share (target)	Description
at-grade	$\geq 70\%$	Track sits on prepared subgrade at natural ground level (possibly with minor cut/fill ≤ 3 m). The cheapest and fastest class by 1–2 orders of magnitude.
elevated	$\leq 25\%$	Track sits on a viaduct structure 5–12 m above ground (precast OSR-Pi20/Pi25 decked beams, standardised span).
bridge	$\leq 5\%$	Over a water crossing or deep gap. Separately engineered special span with site-specific substructure.

The fourth class the first-pass auto-gen inference used — tunnel (cut-and-cover and bored) — is removed from the catalogue. Where an alignment previously would have gone underground, it now goes elevated over the same footprint, or deviates around the constrained zone on at-grade. Operators who deliberately want a tunnel are forking the project; OpenSourceRail upstream does not carry tunnel designs, tunnel cost models, or tunnel-specific signalling / ventilation / evacuation engineering.

2. Non-goals

- Not a tunnel standard. By construction. The project's cost model, the simulator's energy budget, the sensor-tuning for GNSS + balise, and the operations rulebook (RFC 0013) are all calibrated against above-ground operation. Going underground reopens every one of those calibrations.
- Not a geotechnical rulebook. Soil survey, pile type, foundation depth, seismic detailing — owned by the deploying operator's civil team. This RFC fixes the functional envelope (span range, clearance, load class) not the method.
- Not a bridge design manual. Spans above 30 m, water crossings, turnouts, and exceptional crossings use a separately engineered segmental, I-girder, or steel-composite product. They are not a stretched version of the standard full-span trough.
- Not a land-acquisition policy. How a deployment acquires right-of-way (ROW) — expropriation, purchase, existing-rail corridor reuse, new-build — is municipal work.
- Not a standards body. We reference EN 1990 (Eurocode basis), EN 1991-2 (traffic loads on bridges / viaducts), EN 1998 (seismic), UIC 777-2 (infrastructure gauge), local building codes where deployed. We do not publish new ones.

3. Why no tunnels

A planning-grade cost comparison for a 1 km segment in typical dense-urban conditions across the target regions:

Class	OSR target / retained benchmark (USD/km)	Build time	Post-build maintenance
At-grade dedicated ROW	2.584 M / 3.0 M	6–12 months	Low — ballastless slab/embedded trackform
Elevated viaduct	9.748 M / 12.0 M	12–18 months	Medium — bearing + expansion-joint inspection
Bridge over water	18.0 M / 18.0 M	18–24 months	Medium — same as viaduct + scour inspection
Cut-and-cover tunnel	60–120 M conventional	30–48 months	High — ventilation + pumping + egress drills
Bored tunnel	90–200 M conventional	48–72 months	High — same as above, plus tunnel-boring-machine commissioning

Numbers are planning-grade, calibrated against generated deployment instances including Samawah context ([RFC 0003 §2.3](#)).

A tunnel segment is 10–40× the CAPEX and 4–8× the build-time of an equivalent at-grade segment. It also brings a permanent operational overhead: mechanical ventilation, flood pumping, fire suppression, emergency evacuation plans, and radio-coverage retrofit. Every one of those is an additional subsystem the deploying operator must staff and maintain.

The mission is "urban rail affordable enough that a developing nation can finance it domestically." A corridor with any tunnelling in it is outside that envelope. The design decision is: if the alignment can't go at-grade or elevated, the alignment should deviate around the constraint. Even at a 20 % longer corridor, the detour is cheaper than the tunnel it avoids.

4. At-grade – the 70 % case

4.1 Envelope

- Subgrade: compacted fill or in-situ soil graded to +/-25 mm of design. Capping layer of 150 mm crushed stone, then ballastless reinforced slab / embedded direct-fixation plinths for the running rails. The default urban trackform avoids ballast migration, tamping cycles, and loose aggregate at crossings.
- Earthworks: cut / fill up to 3 m depth only. Beyond 3 m the segment is either:
 - Re-classified to elevated (bridge over the gap / viaduct over the hill), or
 - Deviated around the constraint.
- Drainage: transverse slope $\geq 2\%$ either side; longitudinal channels every 50 m; culverts at every flow-path crossing. Drainage is not optional: a flooded slab undermines fastener integrity, corrodes embedded hardware, and damages the subgrade.
- Fencing: dedicated ROW is fenced the full length on both sides. No uncontrolled pedestrian crossings. Level crossings only where a city street intersects and an underpass / overpass is prohibitively expensive. See [RFC 0012](#) for the level-crossing equipment standard.
- ROW width: minimum 10.5 m for double-track at-grade (two tracks $\times$ 3.5 m wide plus 1 m each side clearance plus fencing allowance); 7 m for single-track.

4.1.1 Narrow old-town single-track option

Some historic cores and market streets cannot fit a 10.5 m dedicated double-track reservation without demolition, but also do not justify viaduct intrusion. OpenSourceRail therefore permits a constrained single-track at-grade segment as an at-grade variant, not a fourth civil class.

Rules:

- Use only for short constrained sections where double track would require demolition or permanent removal of the street's essential pedestrian/service function.
- Fit the running way inside the 7 m single-track envelope, with protected pedestrian edges and no mixed running with road traffic.
- Provide double-track passing loops only at stations, portals, or immediately outside the constrained street.
- Interlocking treats the whole single-track section as one opposing-move exclusion zone unless a deployment-specific signalling design divides it into shorter protected blocks.
- Timetables must prove the desired headway through the single-track section before the option is accepted. At 3-minute peak headway, a single-track section longer than roughly 800-1,000 m will usually force either lower frequency, directional tidal operation, or a bypass.
- Emergency walkways, evacuation gates, drainage, charging isolation, and intrusion detection remain mandatory; the narrow option saves width, not safety systems.

The auto-generation pipeline should prefer this option before elevated only when a tagged old-town / constrained-street corridor is short, station-loop spacing is feasible, and the resulting timetable still meets the line's published service class. Otherwise it should stay double-track or go elevated/deviate.

4.2 When at-grade is inappropriate

Even with the default bias to at-grade, certain conditions push toward elevated:

- The alignment crosses a river wider than 30 m bridge.
- The alignment crosses a highway or mainline rail at grade that cannot be broken by a level crossing (too many crossings per hour would make the crossing barrier close more than open) elevated.

- The alignment runs through a dense built-up core where ROW acquisition would require demolition of more than 3 buildings per 100 m of corridor elevated or deviate.
- The alignment runs along a street where at-grade light-rail running would conflict with road vehicles at every junction elevated or route to a parallel street.

The auto-gen pipeline's civil inference (§6) now maps these conditions to Elevated where it previously would have emitted BoredTunnel.

5. Elevated – the 25 % case

5.1 Envelope

- Structure: paired single-track prestressed decked pi-beams. OSR-Pi25 is the primary product and OSR-Pi20 is the closure/utility- avoidance product. Each beam is 2.9 m wide with stems beneath the rail lines; independent outer walkway/barrier cassettes produce an 8.5–9.0 m controlled twin-track envelope. The catalogue contains no 30 m Pi beam.
- Piers: single-column reinforced concrete, 1.5 m × 2.0 m rectangular, heights 5–12 m, with a 6.5–7.5 m hollow/precast-shell cap. Foundation type comes from the soil/access catalogue and actual deep-element lengths.
- Deck clearance: 5.0 m vertical under the girder soffit over any road; 5.5 m over a truck route; 7.5 m over a rail corridor (UIC 777-2 minimum).
- Pier spacing: 25 m nominal, using 20 m closure bays where constrained. Spans above 30 m use OSR-SP or another separately engineered structure.
- Bearings: scheduled fixed/guided/free bearings. A semi-continuous internal pier has four bearings in one longitudinal line; an expansion-unit boundary has eight in two independent lines; an end support has four. Permanent checked jacking interfaces permit replacement.
- Movement: four normal spans form a short semi-continuous unit using a released reinforced diaphragm or link slab. Expansion gaps occur every 80–125 m, never as one kilometre-long continuous structure. The project issues a bearing/movement schedule and continuous-welded-rail interaction, braking, temperature, seismic and foundation-flexibility analyses.
- Parapets: 1.4 m solid concrete panel, continuous, with a handrail on the inspection walkway.
- Walkway: separate outer inspection/escape cassette, 1.0 m clear, escape ramps every 250 m down to at-grade.
- Acoustic screens: fit only where the project noise model demonstrates a requirement; use replaceable secondary panels rather than widening the primary member by default.
- PV integration: concentrate PV at stations, depots and canopies. Normal viaduct spans do not carry a default PV attachment.

5.2 Rationale for one reference girder

- One long-line mould family across the whole project. The precast yard casts one Pi-beam section with movable Pi20/Pi25 stop ends; every normal span of an OpenSourceRail deployment uses that shape. This is the biggest single CAPEX / schedule lever in elevated construction and matches the "one spares pool, one CAD reuse" principle from RFC 0008 / 0009 / 0010.
- Double-track substructure, repeated single-track beams. One shared hollow/precast-shell cap carries two identical Pi-beams at 3.5 m centres. A single-track initial line still builds the shared substructure and may defer the second girder until capacity demands it.
- Short semi-continuity, not kilometre-long continuity. Transport and erect ordinary individual beams, then connect four spans with a released link slab or diaphragm. This removes most deck gaps and one bearing line at ordinary internal supports while retaining manageable thermal units.

6. Bridge – the 5 % case

- Reference class: separately engineered OSR-SP segmental, I-girder, or steel-composite span. The standard full-span U catalogue ends at 30 m.
- Pier: when a pier has to sit in the water, foundation is a reinforced-concrete caisson or a pile group with a protection cutwater. Scour assessment per EN 1991-2 §13 with a 100-year design flood.

- Navigation clearance: per the navigable waterway's published class. The operator commissions the bridge design and the standard OSR-U catalogue ends at the approach where the site demands a non-catalogue structure.
- Thermal: project-specific bearing, joint, and rail/bridge interaction design for the complete temperature, creep, shrinkage, and seismic range.

7. Stations-as-structures

Station archetypes from [RFC 0010](#) map to civil class as follows:

Archetype	Typical civil class	Notes
halt	at-grade	No canopy PV. Single side platform on direct-ground footing.
standard	at-grade	Solar canopy + side platforms.
major	at-grade	Island platform, larger canopy.
interchange	elevated + at-grade	One line is on the viaduct, the other at-grade. The two platforms stack vertically at the junction. Never two elevated lines stacked – the second level's wind and seismic loads rule that out of the upstream catalogue.
terminal	at-grade	End-of-line; turnback tracks sit at grade.
depot-terminal	at-grade	Yard footprint is large; elevating it is prohibitively expensive.

Underground stations are not in the catalogue. The previous draft of RFC 0010 used the word "stacked" to describe interchange layouts; this RFC clarifies that stacked means vertical separation between at-grade and elevated levels at the junction, not between two underground levels.

8. Auto-gen pipeline – the civil inference, revised

The civil-class inference in [crates/osr-routing/src/civil.rs](#) currently classifies a cell based on the underlying cost surface. Under the no-tunnel invariant:

- Cells with cost ≤ 40 at-grade (unchanged).
- Cells with cost 40..100 at-grade (was already; unchanged).
- Cells with cost 100..400 (water) bridge (unchanged).
- Cells with cost ≥ 400 (building / dense built-up) elevated (was BoredTunnel now elevated).

This aligns with the rule: "if the corridor can't be at-grade, it goes above, not below."

Two downstream consequences, surfaced in `design-quality.yaml`:

- 1 Elevated fraction as a soft gate. If the share of elevated structure for a line exceeds 30 %, the quality file raises a warning. Designs with > 50 % elevated either need deviation or are inherently expensive and should prompt review.
- 2 No tunnel class. Quality output reports only at-grade, elevated, and bridge fractions.

9. Cost-anchor lookup

Total CAPEX drives the national-treasury conversation. The auto-gen pipeline emits a full `[costs]` block per city in `design.toml`, broken down by bucket below.

The source currency is USD because the marketplace and country-finance templates quote in USD. Generated `*_eur` fields are converted reporting views at 0.92 USD->EUR. Non-civil units and retained civil benchmarks live in [capex-costs.toml](#). The active civil rates are generated from [civil-cost-calibration.toml](#) and CAD quantities into [civil-cost-model.toml](#); this RFC keeps the narrative summary.

9.1 Civil works (USD/km × civil mix)

Class	Active USD planning target	Retained benchmark
at-grade	2 584 000 / route-km	3 000 000 / route-km
elevated	9 748 000 / route-km	12 000 000 / route-km
bridge	18 000 000 / route-km	18 000 000 / route-km
elevated-interchange premium	4 500 000 / site	—

9.2 Stations (RFC 0010 archetype catalogue)

Archetype	USD planning-grade
halt	600 000
standard	2 500 000
major	4 500 000
terminal	4 500 000
depot-terminal	5 000 000
interchange	8 000 000
interchange-elevated	12 000 000

The elevated-junction premium in §9.1 covers the viaduct + upper deck at interchanges; the table above covers the at-grade lower platform, ground-level pedestrian access, and protected crossing equipment where needed. Overbridges, lifts, and stairs are not the standard station assumption; they are carried only by elevated/stacked sites or local road-barrier overrides.

9.3 Depots (RFC 0014 archetype catalogue)

Archetype	USD planning-grade
main-heavy	8 000 000
secondary-medium	3 000 000
layup-minimal	400 000

These rates assume RFC 0014 distributed overnight stabling: healthy trainsets may sleep at powered passenger stations and layups, so the main-heavy no longer carries parking capacity for every train every night and layup-minimal is a powered-station stabling/isolation kit rather than a small standalone yard.

9.4 Rolling stock (RFC 0008 local-owner production × fleet count)

Rolling-stock city CAPEX uses a local-owner trainset-family planning unit, not an inflated per-car multiplier and not the raw marketplace BOM floor. The unit includes direct material, local assembly/labour, nominal per-train QA/acceptance evidence, and modest local handover logistics. Fixtures/tooling live in the railway production plant line; warranty, spares, and routine commissioning support are OPEX.

Family	USD / trainset local-owner unit
urban-shuttle-1car	280 000
tram-2car	560 000
light-metro-3car	900 000
metro-4car	1 120 000
metro-6car	1 680 000

Fleet count = peak-revenue + spare + cold-reserve per RFC 0014 §4.

9.5 Railway production plant

Each country carries one lean shared railway production-plant setup allowance for all city fleets it will build or assemble locally. Cities do not duplicate this asset in city CAPEX. The national factory is sized to the largest single-city fleet programme, costed per supported vehicle/car module, reused through a phased rollout, and kept separate from rolling-stock procurement.

Item	Planning-grade basis
Base local plant setup	60 000 USD / vehicle-car module
High sensitivity check	120 000 USD / vehicle-car module

The allowance covers reusable 1 m fiberglass panel moulds, clip/drill gauges, basic steel fixtures, plant services, commissioning bay setup, material handling, and production-readiness work. It excludes a full-length body mould and production adhesive cure hall. Six parallel two-person crews can install and release a three-car exterior body in one eight-hour shift after painted-frame dimensional release; doors, glazing, equipment, bogies, commissioning, homologation, and first-article testing remain separate work. Distributed overnight stabling reduces the number of depot-centred commissioning and parking bays needed in the production/depot complex. A three-module increment of supported factory capacity therefore carries 180 000 USD of base plant allowance, while the high sensitivity is 360 000 USD.

9.6 Systems

Item	Planning-grade basis
Residual train-control wayside (RFC 0015 GoA 4)	50 000 USD / route-km
Station/depot charging microgrid interfaces	120 000–1 000 000 USD / stop, by station archetype

RFC 0015 mandates battery-electric operation with station charging: there is no overhead catenary, third rail, feeder substation, or route traction-power budget. The energy infrastructure bucket covers the conductive chargers, local switchgear, inverter interface, and station/depot microgrid tie-in. PV/storage quantities are sized in the energy-site catalogue rather than as a continuous per-km rail-power system.

9.7 EPC overhead

Integration + project management is 7 % of the subtotal across §9.1 – §9.6.

Rates are deployment-localised via the [country-costs.toml](#) template (labour, materials, finance cost adjustments). The base rates above are what the upstream catalogue quotes; actual numbers in any given deployment can be $\pm 40\%$.

`total_usd` is the procurement-basis headline per city. `total_eur` is also emitted for compatibility with earlier generated schemas.

10. Constructability constraints

- Concrete supply. Precast concrete yards exist in every target region; the Pi section needs a supplier-designed railway prototype and a reusable long-line mould fabricated from established formwork systems (PERI, Meva, Doka). Operators without a nearby precast yard bootstrap one — a 5 km² yard with 20 formworks produces enough girders for 50 km of viaduct in 18 months.
- Steel reinforcement. All target regions have domestic rebar mills in the 12–40 mm range we use.
- Post-tensioning tendons. Commodity; local supply exists in every target region.
- Lifting and transport. Pi25 is 2.9 m wide and targets 65–75 t. The project must release the route, transporter, lifting points, portal/launcher or actual crane load chart, synchronized lift controls, ground pressure, wind limits and contingency landing method.
- Launching. Where full-span access is constrained, OSR-US uses match-cast 2.5–3.0 m segments, post-tensioning, and an overhead launcher with specialist geometry, epoxy-joint, tendon-grouting, and staged-load QA.

No restricted-export equipment. No bespoke machinery. The construction envelope matches what a mid-size regional civil contractor can mobilise.

11. Pitfalls and decisions

- No tunnels is the non-negotiable decision. An operator who believes their corridor absolutely requires a 3 km tunnel under a city centre has exactly two options: (a) deviate the alignment; (b) fork the project. Upstream stays on the at-grade / elevated / bridge envelope.
- At-grade dominates even where it's unfashionable. Modern metro vendors push elevated + tunnel because their business model prefers CAPEX depth. OSR's mission is the opposite: every km pushed to at-grade saves 4–10× cost. A Samawah-scale deployment that lands 70 % at-grade is the successful design; any design above 50 % elevated is a red flag.
- One reference viaduct. Operators who want a different girder geometry (ornamental, signature architecture) are free to substitute; the upstream catalogue fixes one shape for CAPEX / spares reasons, not for aesthetic ones.
- Special crossings are separate products. OSR-Pi25 is not stretched to 40 m. OSR hands off to OSR-SP or a local engineering tender at the approach.
- Noise mitigation is structural, not operational. Acoustic screens on viaducts > operational speed restrictions. A 2 m concrete panel costs almost nothing and avoids a permanent speed-limit loss. Operators sometimes skip this; upstream builds it in.
- Seismic detailing is per deployment. The reference viaduct envelope handles $PGA \leq 0.3$ g; above that, the deployment's structural team ups the rebar + bearing spec per EN 1998. The shape stays the same.

12. Rollout

Phase	Deliverable	Dependencies
v0	This RFC ratified	—
v1	<code>osr-routing::civil</code> updated to drop tunnel classes (in this RFC's commit)	v0
v2	Emit <code>[costs]</code> block (<code>at_grade / elevated / bridge / total_eur</code>) per city in the auto-gen output, using the §9 rate table (done 2026-04-22)	v1
v2.1	Extend <code>[costs]</code> to the full CAPEX stack – stations (RFC 0010 archetypes), depots (RFC 0014 archetypes), rolling stock (RFC 0008 families), GoA 4 onboard autonomy/residual wayside train control, station/depot charging microgrids, and EPC overhead. USD is the procurement basis and EUR values are explicit converted views.	v2
v2.2	Move the source cost basis to USD direct-procurement planning rates, add <code>*_usd</code> fields, keep <code>*_eur</code> mirrors, and drop EPC overhead to 7 % (done 2026-06-04).	v2.1
v2.3	Add the railway production-plant setup bucket, now 60 000 USD per vehicle/car module with 120 000 USD retained as a high sensitivity check after 1 m clip-on fiberglass body and distributed-stabling scope reductions (updated 2026-08-15).	v2.2
v2.4	Replace duplicated city production plants with one shared national factory sized to the largest city fleet programme; city CAPEX is zero for this bucket and national briefs carry the asset once. Add imported-value and local-capital financing boundaries.	v2.3
v3	Parametric OSR-Pi20/Pi25 decked beams, transportable OSR-ST6 panels, 3.5 m elevated track centres, 6.5–7.5 m hollow caps, soil/access foundation catalogue, geometry-derived quantities and resource-driven production equations are controlled under design/component-catalogue/src/osr_mech/civil/ . Legacy U-troughs remain special products; OSR-US and OSR-SP cover exceptional geometry. The controlled design-basis, load, egress, movement, erection, first-article, foundation, production and quantity packages live under docs/civil/ .	v0
v4	Surveyed civil design package for the first pilot line, generated from the current city model	v3, RFC 0003
v5	First-article viaduct erected for a deployment instance	v3

13. Relationship to existing work

- [lib/templates/structures.toml](#) – the Lego-block schema. v1 of this RFC removes the `cut-and-cover-tunnel` and `bored-tunnel` entries.
- [crates/osr-routing/src/civil.rs](#) – the inference function. v1 of this RFC collapses the top threshold to `Elevated` (was `BoredTunnel`) and removes the `CutAndCoverTunnel` and `BoredTunnel` variants from `CivilClass`.
- [crates/osr-design/src/emit.rs](#) – the quality YAML emitter. v1 stops counting tunnel length (always zero now) and adds the `elevated-fraction` soft gate.
- RFC 0009 §5 and RFC 0010 §13 reference tunnel alignment / stacked-underground interchange respectively – both amended in this commit to match the no-tunnel invariant.
- [lib/templates/platform-doors.toml](#) references "tunnel stations"; that line is updated to describe full-height PSDs as an optional upgrade for climate-sealed stations regardless of civil class.

14. Open questions

- 1 Grade-separated level crossings – a road underpass below an at-grade rail corridor counts as a structure; does it get its own civil class? v1 folds it into the road authority's scope, not rail's.
- 2 Cut-and-fill extent limits. 3 m is arbitrary. Real deployments in hilly terrain may need 5 m cuts. Revisit with a hilly-pilot case study.
- 3 Acoustic screen effectiveness in hot climates. Concrete screens expand/contract; joints need seasonal inspection. How does that feed the CBM schedule? Candidate `osr-cbm-*` extension.
- 4 Wooden / bamboo viaduct for ultra-low-cost deployments? Real regional-rail projects in parts of South and Southeast Asia have used timber structures historically. Out of scope upstream; a local fork can layer that on.
- 5 Green-roof viaducts – planting integrated into the deck for heat-island mitigation. Candidate v4 addendum.

15. Done criteria

- [x] Two classes only — at-grade + elevated — with bridges narrow (§1)
- [x] No-tunnel rationale explicit with cost comparison (§3)
- [x] At-grade envelope specified (§4)
- [x] Elevated envelope specified, one reference girder (§5)
- [x] Bridge class as a viaduct variant (§6)
- [x] Stations-as-structures mapping (§7)
- [x] Civil inference updated (§8)
- [x] Cost anchors named (§9)
- [x] Constructability constraints stated (§10)
- [x] Pitfalls + alternatives (§11)
- [x] Rollout ordered (§12)
- [x] Relationship to existing code + templates (§13)

The next session picks up at v2 — cost estimates in auto-gen output (a short follow-on to the v1 tunnel-removal already landed in this commit).

RFC 0014 — Depot Design Standard

Source: docs/rfcs/0014-depot-design-standard.md

Status: Draft — planning only, no site drawings ship with this RFC Date: 2026-04-22 Depends on: [RFC 0008 Rolling-Stock Reference Design](#), [RFC 0010 Station Design Standard](#), [RFC 0011 Civil Infrastructure Design Standard](#), [RFC 0013 Operations Rulebook](#)

1. Summary

OpenSourceRail commits to three depot archetypes covering the full range of deployment sizes. Every deployment has exactly one main depot plus optional secondaries and layups. The main depot co-locates with a depot-terminal station ([RFC 0010 §1](#)).

The default operating pattern is distributed overnight stabling: clean, telemetry-healthy trainsets may finish service at powered passenger stations or layups near their first morning trip instead of all running empty to a terminal depot. This reduces depot stall requirements, avoids wasteful dead mileage, and lets first departures begin across the line at the same time. The main-heavy remains the maintenance authority. Defective trains are held and recovered to a service point on their own line; maintenance planning must not assume an interline rail connection. Two revenue trains per selected station form the morning launch stock; remaining revenue trains and reserves stable at a powered service point on their own line. Storage tracks are sized separately from workshop bays, using reusable depot/layup blocks.

Archetype	Catalogue max stalls	Heavy maintenance	Overhaul throughput	Notes
main-heavy	20	Yes — overhaul bay, wheelset lathe	≤ 40 trainsets	Larger workshops require a controlled site-specific extension
secondary-medium	12	Light only — brake disc, door actuator, HVAC modules	—	Second depot on a long line
layup-minimal	6	None — overnight stabling only	—	Remote-terminal layup

The catalogue max is the reference layout envelope. The per-deployment workshop requirement comes from §4 ($\max(4, \text{ceil}(\text{fleet} \times 0.15))$). It is not an overnight parking allowance. Requirements beyond that envelope need an explicit site design; a template ceiling must not silently truncate them.

Plus one optional type for edge cases:

Archetype	Purpose
training-wing	Co-located operations-training facility — OCC dispatcher-console simulators, maintenance-technician workshop mock-ups, and recovery-mode-crew briefing rooms. OSR is GoA 4 driverless (RFC 0015), so there are no revenue-service drivers to train — the wing exists for dispatchers, maintainers, station staff, and the recovery-mode crew who operate the manual-fallback joystick (RFC 0015 §8.2). Opt-in on main-heavy; not a standalone depot.

All depots are at-grade. Per [RFC 0011 §1](#) depots are never elevated — the footprint is too large to economically put on viaduct. A deployment that can't site its depot at-grade has an alignment problem, not a depot problem.

2. Non-goals

- Not a workshop management system. How the depot schedules its work — CMMS, parts management, labour planning — is owned by the operator and overlays with the osr-cbm-backend output.
- Not an off-line overhaul facility. Major overhauls (wheel reprofiling at 150 000 km, bogie overhaul at 600 000 km, body overhaul at 10 years) happen in the main-heavy depot's overhaul bay. Deployments without a main-heavy either contract overhaul out or depot-trip their fleet to a sister network.
- Not a yard-design manual. Track yard geometry within the depot boundary uses RFC 0009's standard-urban or (for large shunting yards) a depot-specific geometry variant. Detailed yard layout per facility.
- Not an HR / workforce document. Shift patterns and headcount from the ops rulebook (RFC 0013) and the workforce template.
- Not a brownfield-rehabilitation scope. Where a deployment inherits an existing rail workshop (Samawah, Khartoum-Atbara, Karachi Cantt, Maputo Machava, etc.), [RFC 0027](#) governs the conversion of the existing site into an OSR depot. This RFC's archetypes are the greenfield envelope; the brownfield path swaps greenfield depot CAPEX (use the current [cost template](#) and reconcile energy inclusions) for workshop rehabilitation (~\$0.5–2 M building + \$1–3 M new OSR-specific tooling) — an order-of-magnitude saving that the per-deployment plan should commit to whenever assets are real.

3. Why three archetypes

Depots are where the simplicity bet pays off most. The three- archetype set covers:

- **main-heavy** – the one non-negotiable facility per deployment, where heavy maintenance happens.
- **secondary-medium** – for deployments whose geography makes defect and light-maintenance access to the main depot impractical.
- **layup-minimal** – storage at a line's powered service location for stock beyond the two-train station provision; no separate heavy workshop or interline connection is implied.

The catalogue provides reference envelopes. A workshop exceeding the reference bay count needs a controlled extension at the deployment main depot, including access, equipment and staffing capacity.

4. Fleet-sizing formula

Workshop bays are sized for concurrent maintenance and inspection. Two revenue sets stable at each selected powered passenger station where fleet permits; remaining revenue trains and reserves use separate depot stabling tracks:

```
main_depot_workshop_bays = max(4, ceil(total_fleet × 0.15))
secondary_depot_bays = site-specific exception
service_rotation = 0 # depot service is prohibited in peak windows
total_fleet = peak_revenue_trainsets + service_rotation + spare + cold_reserve
station_overnight_positions = 2 × unique_selected_station_count
depot_overnight_trainsets = total_fleet - station_overnight_trainsets
```

The two-train station provision is an overnight allocation constraint. Allocate remaining revenue trains and all reserves to same-line storage, preserving total fleet inventory. Depot storage positions and usable track lengths are sized separately from workshop bays. Samawah allocates 40 trains to stations and 68 to storage on their own lines: 37 on Line 1, 16 on Line 2 and 15 on Line 3. These comprise 57 revenue, eight spare and three cold reserve trains. The storage requirement totals 4,046 m of usable slots at 59.5 m per train, separately from the existing 17 main-depot workshop bays. Lines 2 and 3 reuse their powered service terminals. Local storage layouts need detailed design; interline depot connections are not part of the operating requirement.

- **peak_revenue_trainsets** = what the schedule requires at peak. Sized from the physical round-trip cycle vs. the peak headway:

```
traversal_energy_kwh = line_length_km × cars × 2.4 × 1.25 × 1.25
charging_dwell_s = ceil_30s(3600 × traversal_energy_kwh / sum(charger_kw))
charging_dwell_s = clamp(charging_dwell_s, 120, 600)
added_dwell_min = charging_stops × max(charging_dwell_s / 60 - 1, 0)
stop_count = station_count - 2 # radial; station_count for a ring
one_way_min = (line_length_km / cruise_speed_kmh) × 60
               + stop_count × stop_penalty_min
radial_cycle_min = 2 × one_way_min + 2 × 3 + 2 × added_dwell_min
ring_cycle_min = one_way_min + added_dwell_min
peak_revenue_trainsets = max(2, ceil(cycle_min × recovery_factor / headway_min))
```

Current **native emitter** parameters:

Family	Cruise km/h	Stop penalty min	Recovery factor	500 kW cabinets per powered station	--- --- --- --- ---														
tram-2car	48	1.45	1.10	1	light-metro-3car	62	1.35	1.62	1	metro-4car	72	1.30	1.30	3	metro-6car	72	1.30	1.45	4

Radials use a three-minute peak headway and a three-minute turnback at each end; rings use a six-minute peak headway. The stop penalty includes the one-minute reference dwell, and only charging dwell above that reference is added. The energy balance uses a 25% climate uplift and a separate 25% charging margin. The larger recovery factors include the emitter's retained onboard-energy resilience allowance; they are planning assumptions requiring city-specific service and energy evidence, not a uniform 10% timetable margin.

These parameters replace the former written 35 km/h proxy and uniform 1.10 recovery factor. Existing city fleet inventories remain declared inputs; reconcile regeneration changes with timetable, energy, cost and physical stabling evidence before adopting a different count. This formula does not enforce two trainsets per station. Station capacity must be checked separately.

- **spare** = max(1, floor(peak_revenue_trainsets / 10)), for planned maintenance rotation.
- **cold_reserve** = 1 per line, for unplanned incidents per [RFC 0013 §7](#).
- **service_rotation** = zero dedicated sets. Depot service is prohibited in the 3-minute peak windows and uses surplus peak-fleet capacity when the timetable relaxes to 6 or 12 minutes.

The 12-minute service happens at one designated service depot per line only in lower-frequency windows. During 07:00–09:00 and 15:00–17:00, trains make the normal quick terminal turnback, can use the platform connector, and may deplete the

battery further while the 20% section-entry reserve remains mandatory. Interior cleaning, exterior/running-gear walk-around, door, coupler and emergency-equipment release checks, fault-log download, and a 150 kW low-C recharge run concurrently. A red defect removes the set from rotation and uses a planned spare; a clear inspection releases it straight back to revenue service. The opposite terminal may still top up or stable a train, but does not repeat the full inspection on the same round trip.

The 15% workshop concurrency allowance covers overhaul, defect repair, incoming inspection, and planned maintenance. Growth beyond the 20-bay catalogue envelope requires a controlled workshop extension; it does not restore fleet-wide depot parking.

Distributed overnight stabling does not increase the service-rotation fleet. End-of-service dispatch assigns healthy trainsets to powered stations with CCTV monitoring, remote isolation, and at least low-C charging access capped by installed equipment. The 150 kW rate is a top-up target, not a minimum station rating. Morning dispatch releases those sets from their stabled stations so outer stops do not wait for trains to run out from the terminal depot. The main-heavy stall formula remains sized for maintenance, inspection, overhaul, spares, and growth rather than for parking every healthy set every night.

Example (current generated Samawah output, 3-min peak headway):

- The authoritative generated summary is [cities/catalogue/west-asia/Iraq/Samawah/README.md](#).
- The September 2026 design has fleets of 53, 28 and 27 trainsets: 108 total. The [depot reconciliation](#) derives initial placements and slot-length requirements from the stored scenario. Physical overnight allocation remains unverified.
- `design.toml` emits the depot archetype, terminal assignment, and `fleet_stalls` as workshop/inspection bays from the same run. Older 1.25x fleet-parking examples and hand-calculated 6/4/8 trainset examples were removed from the docs and must not be copied into current deployment documents.

The emitter (RFC 0014 v2) computes this from the `design.toml`'s `fleet_sizing` block.

5. main-heavy envelope

5.1 Functions

- Workshop holding for inspection, scheduled work and defective trainsets (bay count per §4); overnight fleet allocation requires a separate plan.
- Daily inspection of every incoming trainset.
- Weekly inspection (7-day service).
- 30-day inspection (A-class service per EN 50126).
- Light repair (door actuator, HVAC module, brake-disc replacement, battery module swap).
- Heavy overhaul: bogie swap, wheel reprofiling on depot lathe, body-work repaint.
- Wash: automated two-pass wash track.
- Battery-module inspection, coolant service, and sealed-module exchange.

5.2 Physical layout

Area	Target footprint (per consist bay)
Stabling track	1x consist length + 5 m clearance at each end
Daily-inspection bay	Roofed, pit-equipped, overhead power, compressed air, data sync drop (TCN-E monitoring)
Weekly bay	Pit + roof walkway for upper-body access
Overhaul bay	Pit + overhead crane (40 t) + wheel-lathe trench
Wash track	Two-pass automatic with water recycling
PV canopy	4 000 m² reference allowance; full operating PV inventory needs additional located area (§5.3)
Classroom + office	≥ 200 m² for operators + trainers
Stores	≥ 600 m² with rack storage for spare modules

Total main-heavy footprint: ~5 000 m² gross for a 10-stall facility, ~8 000 m² for a 20-stall. These historical workshop envelopes do not include the complete operating PV/storage inventory and cannot be used as total deployment land requirements.

5.3 Energy

- `depots.toml` selects the canonical [energy-sites.toml](#) tier. main-heavy uses depot-main: 5,000 kWp PV and 40,000 kWh storage, comprising 80 × 500 kWh stationary modules. Scenario, BOM and layout requirements resolve the

same inventory; explicit site overrides must remain traceable. These are retained operating assumptions, not validated depot sizing. Reassess energy duties with distributed overnight station charging before freezing procurement or depot land requirements.

- The old 4,000 m² / 600 kWp / 2,000 kWh scope is a reference allowance. At the catalogue density of 0.15 kWp/m², 5,000 kWp requires 33,333 m² of modules, before access, setbacks and packing. Additional land/canopy and the battery compound are not yet located by the reference CAD.
- Annual PV generation must come from the city energy study with declared weather and loss assumptions. A nameplate rating alone is not annual yield.
- Stationary storage feeds the RFC 0021 DC charging system. Supplier dimensions, conversion/protection equipment, fire separation and installed cost require explicit reconciliation with the fixed depot allowance.
- Grid import/export must satisfy the declared connection limits and utility study. Per RFC 0002, every depot is one physical osr-energy-site, including when co-located with a passenger station.

5.4 Signalling

- A single W-SBC plus an interlocking zone boundary at the depot entry.
- Inside the depot, speed limit 15 km/h, manual operation under mode M3 (RFC 0013 §5).
- Shunting signals at each stall entry – mechanical dwarf signals driven by osr-wayside-points extensions.
- No ATP envelope inside the depot; osr-atp disables below 15 km/h in depot-tagged geofence, per a safety-bounded configuration. The operational safety envelope inside the depot is line-of-sight + hand-signal coordination.

5.5 Access

- Personnel access: turnstile-gated with ID-card entry, separate from the fleet access.
- Fleet access: one track connection to the line, TPS-style merge per RFC 0012 §4.3 (trains go in – it's a trailing move).
- Road access: for parts delivery, emergency services, depot staff parking. Minimum one 4 m-wide road connection.
- Rail access for deliveries of new trainsets is via the same line connection (new rolling stock rolls in on its own wheels).

6. secondary-medium envelope

- Functions: overnight stabling for ≤ 16 trainsets, daily inspection, weekly inspection, light module swap.
- No overhaul bay. No wheel lathe. No bogie-lift. Anything heavier is trips back to the main-heavy.
- Footprint: ~2 500 m² for 8 stalls.
- Reference PV canopy: 1,500 m² / 225 kWp, with 500 kWh baseline storage.
- Operating depot - secondary tier: 1,500 kWp PV and 5,000 kWh storage (10 × 500 kWh modules); additional placement and installed cost remain open.
- Signalling: same as main-heavy but smaller zone.
- Classroom: none. Reuse the nearest major station's community room if needed.

7. layup-minimal envelope

- Functions: overnight stabling, low-C top-up, cleaning, and a ground-level safety-release walk-around. Trainsets drive in at end of service, can pass through the same 12-minute turnaround service if the site is designated for its line, sit overnight, and drive out at morning service start. There is no pit inspection, repair work, or heavy maintenance.
- Footprint: ~1 000 m² for 4 stalls.
- PV canopy: 600 m² 90 kWp.
- Operating depot - layup tier: one 150 kWh stationary package for opportunity charging during dwell; the main-heavy is primary.
- Train top-up: 150 kW low-C per occupied stabling road, used while a set is held between duties and overnight; passenger-platform fast charging remains a separate terminal function.
- Signalling: single W-SBC at the layup entry; rest is manual line-of-sight.
- No personnel on-site overnight; the OCC monitors via CCTV.

8. Integration with the auto-gen pipeline

Today's emitter picks one depot - terminal archetype ([crates/osr-design/src/emit.rs](#), `compute_archetypes`, priority rule). That choice is where the main-heavy sits. v2 of the emitter extends this:

- 1 Main depot: co-located with the longest radial line's far terminal. The station archetype is `depot-terminal`; the depot archetype is `main-heavy`.
- 2 Powered station stabling: every other terminal and selected intermediate powered stations receive healthy sets overnight; they do not emit depots.
- 3 Secondary/layup exception: emit only after a site-specific security, utility, access, or defect-recovery study shows station stabling is not viable.

Every depot writes a record into `design.toml`:

```
[[depots]]
station      = "line-a-east"
archetype    = "main-heavy"
fleet_stalls = 4 # illustrative workshop bays; calculate per $4
energy_site_tier = "depot-main" # optional: selected by the archetype
```

The scenario generator resolves depot energy tiers into `[[sites]]` and operating service settings into the simulator input. The simulator ([osr-sim](#)) models service countdowns and energy flows. A future physical distributed-stabling model must satisfy these requirements:

- Charges each trainset at depot pad power while held and overnight.
- Treat powered passenger stations and layups as valid overnight stabling points when the distributed-stabling policy is enabled in [lib/templates/depots.toml](#).
- Holds a returning train for the configured clean/inspect/recharge slot at one designated depot per line and emits service-start/service-complete evidence.
- Preserves the zero dedicated service-rotation policy and distinguishes planned spares and cold reserve from revenue fleet requirements.
- Feeds the depot battery bank into the network-wide energy balance.

8.1 Executable station-stabling candidate

The simulator supports `station_stabling = true` in scenario `[[fleets]]`, with selected powered-station `dispatch_points`. Stations may use solar/storage without grid import. Waiting trains use up to 150 kW, capped by installed station and shared cabinet/source limits, for low-C top-up to 95% SoC; after closing, returning trains hold at selected stations and resume under schedule, energy and movement-authority gates. This does not require returning healthy trains to the main depot. Before departure, a distributed train must retain the 20% reserve over the route to the next selected charging station, including unpowered stops and terminal reversals. A failed pad or unavailable energy source excludes a destination. A grid outage alone does not disable a battery-backed charger. The gate checks current source availability; it does not reserve future energy against other arrivals. Actual charging debits shared storage, and each next departure rechecks the onboard reserve. Continuous replay verifies the duty. Optional `spare_count` and `cold_reserve_count` are included in each fleet's total inventory and default to zero. Candidates use the city's declared counts; these trains remain parked and can charge without routine dispatch.

The [distributed-stabling workflow](#) generates separate candidates for all catalogue cities and records the Samawah overnight comparison, checking each planned line/station/direction at opening. Plans compare allocated train lengths with the reference platform envelope. The planning/replay acceptance gate also limits overnight allocation to two trainsets per selected station, including reserves. Shared station IDs count once; a four-berth reference platform does not override this provision. The simulator exposes over-capacity queues for diagnosis, and the combined replay fails even when holding, charging and morning departures otherwise pass. Verified physical slots, security, inspection release, reserve activation, defective-train routing, full-day energy duty and deployment acceptance remain open.

9. Pitfalls and decisions

- At-grade only, per RFC 0011. Depots are large; elevating the footprint is prohibitively expensive. The auto-gen emitter rejects depot siting on an elevated section.
- One reference `main-heavy` everywhere. Operators with unique geography (coastal, mountainside) will push for bespoke depots. The upstream catalogue stays at one shape; the deployment's civil team handles the site-specific adaptation.
- Wheel lathe at the main-heavy only. Decentralising lathes is tempting (saves depot-trip time) but the operator depth for wheel-reprofiling is thin in target regions; centralising at one site lets that thin team do all the work.

- No pit at layup-minimal. A pit adds cost, safety maintenance, and lighting — just for overnight parking. Skip.
- Depot simulation in `osr-sim`. The sim models depot top-up while held plus an explicit service countdown at the designated depot. Cleaning, release inspection, diagnostics, and recharge are concurrent; deeper maintenance remains a separate availability/CMMS model.
- Distributed stabling is not distributed maintenance. Overnight passenger-station stabling is allowed only for healthy sets with remote isolation and CCTV. Any trainset needing inspection beyond the daily release walk-around, fault rectification, wheel work, or battery-module exchange goes to the main-heavy.

10. Rollout

Phase	Deliverable	Dependencies
v0	This RFC ratified	—
v1	Shared energy tier, BOM quantities and layout requirements reconciled (2026-09-09); full equipment placement and installed cost remain open	v0
v2	Current Rust emitter uses one main depot and powered-station stabling; legacy Python planner exception policy still needs alignment	v0, RFC 0010 v2
v3	<code>osr-sim</code> turnaround-service state (clean / inspect / diagnostics / recharge), service countdown and event evidence (done 2026-08-12)	v2
v4	Generated site plan for the current Samawah depot and distributed stabling locations from <code>cities/catalogue/west-asia/Iraq/Samawah/design.toml</code>	RFC 0003 §5, v3
v5	Reference depot CAD under CERN-OHL-S v2	v4

11. Relationship to existing work

- [lib/templates/depots.toml](#) — schema this RFC ratifies.
- [crates/osr-design/src/emit.rs](#) — auto-gen emitter. `depot-terminal` detection today; the `[[depots]]` block emission is v2.
- [crates/osr-energy-site](#) — every depot is one `osr-energy-site` instance; existing integration holds.
- [crates/osr-sim](#) — modest extension for v3; charging semantics already work.
- [RFC 0010](#) `depot-terminal` archetype pairs 1:1 with this RFC's `main-heavy` or a mix of `main-heavy` + `layup`.

12. Open questions

- 1 Co-located vs detached training wing. Some operators prefer the training simulator in a quiet city building (closer to drivers' homes). The catalogue default is co-located with the main depot; per-deployment override.
- 2 Decentralised battery swapping. A future opt-in where trainsets exchange their battery pack at depot instead of recharging — halves the charging-pad occupancy. Out of scope upstream until battery-swap mechanical designs mature.
- 3 Night-shift head count. Ops rulebook §4.5 (RFC 0013) fixes the shift structure; what's the minimum overnight staff headcount at a `layup-minimal`? Currently zero (CCTV-monitored); confirm when the first `layup-minimal` deploys.
- 4 Depot expansion path. A `secondary-medium` that outgrows its 8 stalls — does it upgrade to `main-heavy` in place, or is a `second-depot-nearby` preferred? Likely the latter given construction-on-live-facility complexity.

13. Done criteria

- [x] Three archetypes committed (§1)
- [x] Fleet-sizing formula (§4)
- [x] Main-heavy envelope (§5) + secondary-medium (§6) + layup-minimal (§7)
- [x] Auto-gen pipeline integration (§8)
- [x] Pitfalls (§9)
- [x] Rollout ordered (§10)
- [x] Relationship to existing software + templates (§11)

Next: locate the full PV/storage inventory, reconcile installed cost, and provide track-by-track overnight allocation and conflict-aware run-in/run-out evidence. The quantity contract alone does not satisfy those release gates.

RFC 0013 – Operations Rulebook

Source: <docs/rfcs/0013-operations-rulebook.md>

Status: Draft — planning only Date: 2026-04-22 Depends on: [RFC 0008 Rolling-Stock Reference Design](#), [RFC 0009 Track Design Standard](#), [RFC 0010 Station Design Standard](#), [RFC 0011 Civil Infrastructure Design Standard](#), [RFC 0012 Switches & Crossings](#)

1. Summary

Every OpenSourceRail deployment inherits one baseline operations rulebook — the standing rules for drivers, dispatchers, station staff, and maintenance workers. The rulebook is deliberately short (target ≤ 60 pages in the rendered PDF) and shared across deployments. Operators adapt it for national regulatory overlays but do not re-invent it.

This RFC fixes the rulebook's table of contents and the governing rule-granularity principle. The rule text itself lives in markdown under `docs/operations/` and is owned by a standing committee once the project has one; until then, the project maintainer ratifies additions.

2. Non-goals

- Not a national regulatory document. Every country that adopts OSR has a national rail safety authority (NRSA or equivalent) that must accept the rulebook. This RFC produces a baseline that a deployment's local legal counsel adapts.
- Not a driver training manual. Training content comes from the rulebook, but the pedagogy (lesson plans, sim scenarios, exam questions) is a per-deployment deliverable handled by workforce materials ([lib/templates/workforce.toml](#)).
- Not a safety-case evidence file. The rulebook is an operational artefact; the safety case ([docs/safety-case/](#)) references rules as supporting evidence, not the other way around.
- Not an incident-response playbook. Playbooks are scenario- specific and owned per-deployment (fires, passenger illness, derailment, signal failure); this RFC specifies the categories each playbook must exist for, not the specific response.

3. The simplicity principle

Every rule in the rulebook must pass this test:

A new driver who has never seen this system before can read the whole rulebook in one morning and have an accurate enough mental model to operate safely under supervision by the end of the day.

This means:

- Every rule is one sentence, one decision. No compound rules.
- Every rule has exactly one author (the project maintainer until the standing committee exists).
- Every rule is numbered; operators cite rules by number in incident reports.
- Every rule has a why: a short justification paragraph. No rule is an unexplained ritual.

The target is a ≤ 60 -page PDF. Rail industry rulebooks that hit 200 + pages are artefacts of 150 years of accretion — we're not accruing that.

4. Table of contents — one section per role

The rulebook is organised by role, not by scenario. A driver reads one section; a dispatcher reads another; the sections overlap only where the roles interact.

4.1 Dispatcher rulebook (target ≤ 15 pages)

Section	Scope
S1	Shift start — OCC login, fleet inventory check, schedule ready
S2	Routine dispatching — route grants, timetable adherence, headway maintenance
S3	Incident handling — single-train fault, section unavailable, mass-casualty
S4	Weather and environmental — dust storms, heatwave, flood warning
S5	Maintenance interlock — work-on-track grants, blocking a section from revenue
S6	Shift end — handover, outstanding-event log, system state summary
S7	Intrusion response (RFC 0016) — track-patrol dispatch, return-to-service gating, sustained-Unknown escalation

4.2 Station staff rulebook (target ≤ 10 pages)

Section	Scope
T1	Station opening — power up, PIS check, fare-gate test
T2	Passenger boarding — platform supervision, PSD status, wheelchair assist
T3	Passenger incidents — medical, fare dispute, unattended item
T4	Fare enforcement — inspector on-board, non-payment, appeal
T5	Station closure — evacuation, power down

4.3 Maintenance worker rulebook (target ≤ 10 pages)

Section	Scope
M1	Depot safety — hazard awareness, PPE, lone-worker protocol
M2	Work-on-track authorisation — formal blocks, protection signals, access to active ROW
M3	Routine track inspection — walking the line, visual checks, GPM/RECD recording-car cadence (per RFC 0009 §9)
M4	Switch maintenance — blade lubrication, detection-sensor cleaning, 30-day check
M5	Fleet maintenance at depot — inspection intervals (daily / 7-day / 30-day / overhaul)
M6	Incident site access — derailment response, fire site, crane operations
M7	Wayside sensor maintenance (RFC 0016) — 30-day walkthrough, post-weather inspection, sensor replacement + calibration

4.4 Control-centre rulebook (target ≤ 5 pages)

Section	Scope
C1	OCC watch roles — dispatcher, supervisor, engineer-on-call
C2	Communications — radio discipline, PA authority, external-party notifications
C3	Shift handover — written log, outstanding items, emergency-on-call list

Total target: 40 pages across four sections.

5. Degraded-mode operations — the important part

Modern CBTC metros routinely run in "normal mode" (full ATO under a fresh MA) and have short, mission-critical manual-restoration procedures when the signalling degrades. OpenSourceRail keeps exactly three degraded modes:

Mode	Trigger	Operator action
M1 — Automated restricted MA	ATO is healthy and ATP can enforce a shortened MA	OCC authorises reduced-speed automatic movement; ATP remains fully active and speed is capped at 80 % of normal.
M2 — Controlled stop	Consensus cannot issue a trustworthy MA	Every affected train remains stopped while OCC establishes a work block and restores the signalling quorum. No written order may bypass ATP.
M3 — Protected depot shunt	Movement is required inside an isolated depot worksite	An authenticated local maintenance pendant requests automatic movement at ≤ 15 km/h under ATP, direct CCTV observation, and a locked work zone.

No intermediate modes. No "sort-of-ATP" states. Either the safety envelope holds (modes M1 / M3) or the train remains stopped (mode M2). This is a forcing decision: ambiguous degraded modes are the dominant cause of metro-world incidents.

6. Weather-dependent rules

Target regions routinely experience weather conditions that challenge legacy rolling stock:

- Dust storm (haboob): visibility < 100 m automatic speed cap at 50 % (osr - ato enforces). Dispatcher issues a network-wide speed restriction at the OCC. Passenger announcements via osr - pis - onboard. See also [lib/templates/climate-adapters.toml](#).
- Heatwave (> 45 °C sustained): rail-expansion risk triggers a pre-emptive 15 % speed cap on mainline-mixed and standard-metro presets with continuous-welded-rail. Switch detection sensors recalibrate at dawn.
- Flooding: any rainfall > 50 mm / h triggers immediate inspection of at-grade segments; the dispatcher can preventively block sections via osr - interlocking's maintenance-override. Elevated segments per RFC 0011 §5 are unaffected by surface flooding — deliberate design bet.
- Lightning: nearest-strike distance < 2 km Cat.22 5G backhaul and LoRa backhaul are both watched for outage; if both fail for > 10 s the operator notifies the OCC and cooperates with osr - atp's default fail-restrictive MA.

7. Incident categorisation

Every incident fits one of six categories. This drives the notification chain and the regulatory report:

Category	Example	Dispatcher action
I1 – fleet self-recovery	Door interlock fault on a single train, resolved at next terminal	Log in event recorder; notify depot; no regulatory report.
I2 – service disruption	Single-train immobile; next train coupled to rescue it	Passenger PA, timetable recovery plan; 24h incident report.
I3 – short safety event	Emergency brake triggered by spurious trip; no injury	Log, investigate root cause, 72h report.
I4 – passenger injury	Slip, fall, medical emergency on board	Medical call, depot handover; formal incident report to NRSA within 7 days.
I5 – collision / derailment	Train-to-train, train-to-obstacle, train off-rail	Full network shutdown of affected line; NRSA notification within 2 hours; incident-site preservation.
I6 – mass-casualty / fire	Onboard fire not extinguished, large-scale passenger injury	Immediate evacuation protocol; emergency services; NRSA + national media escalation.

The categories map directly to `osr - event - recorder` tags, so the post-incident analysis starts from the same data the categories describe.

8. Training and recurrent qualification

- Initial driver qualification: 4 weeks classroom + 2 weeks simulator + 4 weeks supervised revenue service. Final qualification exam in written + simulator + on-road segments.
- Recurrent: 2 days refresher every 12 months + 1 simulator session per quarter.
- Degraded-mode drills: every driver runs modes M1 / M2 / M3 in the simulator every quarter. The `osr - sim` is the primary training platform.
- Dispatcher qualification: 8 weeks OCC-side classroom + simulator + supervised desk time.
- Maintenance: role-specific certification per EN 50126 / per national NRSA rules.

Training content is a deployment deliverable; this RFC fixes the minimum hours + structure.

9. Auto-gen implications

The rulebook does not directly feed the corridor-generation pipeline — but it does feed two outputs:

- Depot capacity: `osr - sim`'s fleet-sizing needs to include enough spare trains for the rulebook's degraded-mode operations. A deployment with 10 trainsets on rostered revenue needs ≥ 1 spare in depot for maintenance (M5) and ≥ 1 cold- reserve for I-category incidents. See RFC 0014 §4 for the formula.
- Workforce sizing: the driverless financial model supplies group FTE; [lib/templates/workforce.toml](#) allocates those totals into city roles, skills and tasks. The [city delivery generator](#) writes the organisation and spreadsheet. Named duty/leave cover must still be checked against the rulebook's shift structure (§4.5).

Changing this prose does not change software parameters automatically. Update the controlled model/template inputs, regenerate and check fleet and workforce evidence when operating rules change.

10. Pitfalls and decisions

- One rulebook across deployments is a big bet. Legacy metros each have their own 200-page rulebook; consolidating to one is the simplicity gain. National NRSA's will initially resist. The counter: the process of NRSA approval works on a per-document basis, not per-system basis — a shorter document approves faster. The operator carries the regulatory overlay in a thin appendix.
- Six incident categories, not more. Operators unused to this regime sometimes want a I7 or I8 for niche scenarios. We absorb niche scenarios into the existing six rather than proliferate.
- Paper written orders are allowed in mode M2. This is not anti-digital — paper is explicitly the fallback when the digital system degrades. Written orders remove ambiguity about authority when the digital system can't issue it. Rare enough that the paperwork tax is tiny.
- No "caution mode" between M1 and M2. Legacy metros have a smorgasbord of half-ATP-half-manual modes; every one is a source of operator confusion. We commit to the hard boundary.
- ≤ 60 pages is a hard target. The rulebook is reviewed on every minor-version bump; if it exceeds 60 pages, something gets cut. Adding text is easier than subtracting.

11. Rollout

Phase	Deliverable	Dependencies
v0	This RFC ratified	—
v1	docs/operations/ scaffold with per-role stub files for dispatcher, station staff, maintenance, and control-centre roles, each with scope + cross-refs + proposed rule outline (done 2026-04-22)	v0
v2	Full unattended-operations rulebook drafted across all four role families. Every rule is one sentence + a why : paragraph, cross-referenced to the relevant crate and safety-case solution. Practising-operator review is v2.1 per-deployment work.	v1
v3	NRSA submission pack for Samawah (rulebook + safety case cross-reference)	RFC 0003 §5, v2
v4	osr - sim scenario library covering every degraded-mode scenario for operator refresher training	v2

12. Relationship to existing work

- [lib/templates/workforce.toml](#) — shift-structure + headcount inputs. v3 of this RFC aligns the two documents.
- [lib/templates/service-hours.toml](#) — service hours drive the rosterable workforce; the rulebook sets the rostering envelope.
- [docs/safety-case/](#) — references specific rulebook rules as "solution" evidence for goals around fail- restrictive operation, evacuation readiness, and training competence. GSN goals for "operational fail-restrictive behaviour" land in v4.
- [crates/osr - occ](#) — the dispatcher's OCC UI embeds the dispatcher rulebook's decision-support prompts. Software change in v3.

13. Open questions

- 1 Bilingual rulebooks. Every target-region deployment operates in at least one local language plus English. Do we maintain both upstream? v2 commits on this: likely yes, with English as the canonical and translations as separate branches in docs/operations/<lang>/.
- 2 Printable-PDF vs living-document. Rail NRSA's typically want a static PDF they can archive. The project wants a live document. Compromise: every minor-version bump produces a sealed PDF, and the commit hash is the canonical reference.
- 3 Simulator-based competence testing. osr - sim can evaluate driver actions against MA envelope in software. Can NRSA's accept simulator certification in lieu of on-road supervised hours? Per-country, probably not at v1.

14. Done criteria

- [x] Table of contents per role (§4)
- [x] Degraded-mode operations (§5)
- [x] Weather rules (§6)
- [x] Incident categorisation (§7)
- [x] Training envelope (§8)
- [x] Auto-gen implications (§9)
- [x] Pitfalls (§10)
- [x] Rollout ordered (§11)
- [x] Relationship to existing work (§12)

The next session picks up at v1 — docs/operations/ scaffold, short editorial PR that seeds the per-role files. Actual rule text lands in v2.

Reproducible city operating twins

Source: <docs/operating/city-platform.md>

Each city has a reproducible business operating baseline alongside its railway engineering baseline. ERPNext provides the native operating records. The OSR integration compiles planning inputs and returns permission-filtered execution snapshots; it never issues railway release authority.

Configuration layers

Layer	Location	Purpose
Shared configuration	generic.toml	Departments, warehouses, task categories, programme tasks and planning calendar defaults
City overrides	<code>cities/catalogue/<region>/<country>/<city>/operations/erpnext.toml</code>	Identity, local calendar, organisation, operating release and additional city tasks
Compiled package	<code>build/erpnext/cities/<slug>/</code>	Effective configuration, complete task plan, dependency graph and checksummed summary
ERP execution	Native Project, Task, Department, Warehouse, Cost Center	Actual work, assignments, dates, timesheets, costing and business document workflows
Twin feedback	Private <code>var/erpnext/operating-twins.json</code>	Observed task/category/asset status, recorded hours, task costs and project-linked business document counts

All 266 catalogue cities have specific override files. The compiler derives their actual task and asset scope from their own generated operations bundle; it does not reuse Samawah's tasks for another city. See the [Samawah profile](#) and [Mosul profile](#) for local timezones, illustrative six-day construction calendars and distinct city work packages. Other profiles inherit the shared planning assumptions until a city operator supplies overrides.

The [catalogue readiness gate](#) validates all 266 configurations against their own tracked asset, manifest and project-twin evidence. Supervision can compile directly from those reviewable assets for every city. Full ERP task plans additionally need the deterministic compressed operations payload: Samawah and Mosul keep it locally in Git, while the other 264 materialise it with `./osr city <slug>` before import. A manifest is not treated as a substitute for the task records it hashes.

Tables merge recursively. Lists explicitly replace the shared list. Unknown keys fail validation. `city_programme` adds city tasks to the shared programme; each task needs a unique ID. Removing a department requires updating every task category/programme reference that used it.

The generic company, calendar start date and timezone are unset. The shared Monday–Saturday calendar and the Iraq examples are planning assumptions, not claims about an operator's actual roster or approved public holidays. No employee, supplier, bank, payroll, tax or commissioned asset data is invented.

Repeatable deployment

```
# Creates missing profiles; existing overrides are preserved.
./osr erp city init-configs
./osr erp city validate
./osr readiness --check

# The target company must already exist in ERPNext.
./osr erp city prepare samawah --company 'OpenSourceRail Evaluation'
./osr erp import build/erpnext/cities/samawah/plan.json \
  --company 'OpenSourceRail Evaluation'

./osr erp city prepare mosul --company 'OpenSourceRail Evaluation'
./osr erp import build/erpnext/cities/mosul/plan.json \
  --company 'OpenSourceRail Evaluation'
./osr erp snapshot
```

Replace either slug with any catalogue city. If its generated operations bundle is missing, run `./osr city <slug>` first, or supply a generated bundle with `prepare --bundle <path>`. Generation and preparation do not contact ERP. Review `effective-config.json`, `summary.json` and `plan.json` before import.

To make company binding permanent, add `[organisation] company = "..."` to the city profile. CLI `--company`, `--start-date` and `--release` overrides are captured in the effective configuration and its checksum. No credentials belong in tracked city profiles; use separate private site configuration.

When updating the integration code on an existing site:

```
./osr erp backup
./osr erp build
./osr erp up
./osr erp bench migrate
```

Functions supplied by the city package

- Manufacturing delivery: native Tasks with OSR manufacturing identity, task types, baseline day offsets and native predecessor relationships.
- Maintenance planning: fleet work maps to Fleet Maintenance; other assets map to Infrastructure Maintenance. Source cadences, triggers and evidence remain attached. Operators turn approved schedules and actual commissioned assets into native Asset Maintenance/Repair workflows.
- Assurance coordination: native Tasks for QA actions retain the railway hold point, asset, evidence and release-authority references. Completing them in ERPNext does not close OSR NCRs or approve railway handback.
- Procurement packages: requirements are grouped by sourcing route, with their earliest order-by and latest required-by day. All original BOM and sourcing requirements remain in the source record. Use native Material Request/Quotation/Purchase Order forms after resolving real items, quantities, suppliers, currency and approvals. The [procurement automation](#) also converts a selected requirement into a native draft Material Request once the operator provides its Item, verified stock-unit quantity, date and city warehouse.
- Programme and people: tasks for governance, business masters, workforce planning, maintenance readiness, assurance responsibilities and recovery, plus the city's own additional tasks.
- Organisation: stable native departments, warehouse locations and a city cost centre, scoped by company and city. Task Types are shared. No existing roles or user permissions are expanded by the configuration process.
- Digital-twin feedback: native Project forms gain an OpenSourceRail Operating twin action. Workbench displays matching ERP execution in the Project Twin view, and the operating page compares city baselines and asset workload. Hours/costs come from native Task costing; these are not the full general ledger or the city's generated CAPEX estimate.

Project and Task forms keep their source evidence, checksums and configuration under an OpenSourceRail reference tab, alongside the native business tabs.

Relative working days map to dates only when `calendar.start_date` is set. Weekends and explicit holidays are skipped, including negative mobilisation offsets. A non-working anchor rolls forward to the next working day. The compiler rejects missing dependencies, cycles, duplicate IDs, invalid calendars, mixed-city records and checksum mismatches. Native ERP dependencies retain ERPNext's business semantics, including its treatment of cancelled tasks.

Baselines, upgrades and operator edits

The first operating release (`release = "1"`) upgrades the earlier Project/Task import in place when source identity and hashes match. It fills empty type, department and date fields, adds dependencies only when none exist, and preserves existing task IDs, status, progress, assignments, priorities and operator-entered planning fields. It also creates the new work categories. Subsequent identical imports skip configured tasks and reuse city masters.

A changed configuration or source package cannot silently replace an applied baseline. Increment the city's `release` and prepare again to create a separate reviewable project baseline. Existing records stay intact. There is no automatic transfer of actual costs or approval of a new release. Workbench's engineering view shows execution only when exactly one exported ERP project matches the city and engineering revision; multiple candidates require explicit review on the operating page. No release is silently selected as authoritative.

Imports use native document validation, one database transaction and a final project-total recalculation. A failure rolls back new masters/tasks and metadata. The importer preloads permitted task identities and uses ERPNext's native project bulk-update flag to avoid recalculating thousands of records repeatedly. Use the bench-backed CLI for large city packages; a synchronous HTTP import can exceed a deployment's request timeout. Native task-tree updates can hold database locks during large imports, so schedule them outside active task editing.

Refreshing the integrated twin

```
# One private, atomic export. Previous valid feedback survives export failures.
./osr erp snapshot

# Linux user timer: refresh every five minutes, using the local Docker/bench site.
./osr erp feedback start
./osr erp feedback status
./osr erp feedback stop
```

The timer and service are installed under `~/ .config/systemd/user/` as `osr-erp-feedback`. The user session and container runtime must be running. The snapshot file is mode 0600 in a private directory; OSR's static server does not expose it. Only the loopback Workbench exposes the aggregate snapshot endpoint. Shared/network deployments need an authenticated integration service and their own permissions; this local bridge is not a shared-host auth scheme. The bundled single-site proxy validates browser WebSocket origins and routes ERP notification authentication through the internal Compose network. Review that proxy configuration when adding a public domain or TLS termination.

ERP callers can GET `/api/method/osr_erpnext.city_runtime.city_status?project=...`. The method checks Project access and uses native permission-filtered lists for Tasks and business documents. Unavailable document categories are marked unavailable, not reported as zero. No employee personal details are exported. The local bench export runs with the site's administrative visibility.

Snapshots show their observation time; the UI labels snapshots older than one hour. Refresh view reads the latest exported file; it does not claim to query ERP directly. ERP unavailability leaves the previous snapshot available with its original timestamp and does not affect railway control.

Verification

```
tools/automation/osr-python -m pytest -q \
  tools/automation/tests/test_erpnext_city.py \
  tools/automation/tests/test_erpnext_planning.py
./osr erp city validate
```

The tests cover configuration inheritance, every city's identity, working-day calendars, dependency rejection, deterministic packages, import fingerprints, legacy keys and company/release separation. Live validation should additionally exercise native imports, repeated imports, permission-filtered feedback and the Project/Workbench views against a private ERP site.

Upstream behaviour: [ERPNext Tasks](#), [Asset Maintenance](#).

Deployment Planning Reference

Source: [docs/deployment-planning-reference.md](https://docs.opensource-rail.org/deployment-planning-reference.md)

This page defines the common interpretation of every generated OpenSourceRail city README and national brief. Location folders contain only local inputs, calculated results, exceptions and evidence links. Shared assumptions belong here or in the linked machine-readable templates—not in hundreds of generated copies.

Generation And Review Boundary

The city catalogue is deterministic:

```
city catalogue + OSM/raster evidence + shared templates
- routed network + stations + service + fleet
- energy, civil, CAPEX and finance calculations
- engineering and simulation evidence
- concise city README and national aggregation
```

Generated READMEs are indexes and decision summaries. The authoritative local inputs and results remain `design.toml`, the expanded scenario TOML, and the JSON/GeoJSON evidence inside each city folder. Git records changes to inputs, generators and outputs together. Do not hand-edit a generated city or country README; change its source or a shared template and regenerate it.

Network And Station Planning

`osr_geo` rasterises source-locked OpenStreetMap roads, buildings, water, protected areas and demand anchors onto the planning grid. `osr - design` uses a demand-rewarded least-cost route, places stations against the demand surface, and classifies each segment as at-grade, elevated or bridge. Tunnels are not in the upstream catalogue.

The standard spacing policy is 1.6 km in central areas, 3 km in ordinary urban areas, and up to 7 km on low-demand outer approaches. Cross-line platforms inside the 600 m station-complex envelope are consolidated as one interchange. Coverage, transfer reachability, civil mix and source hits are reported in each city's `*.design-quality.yaml`; they are planning indicators rather than survey acceptance.

Service, Fleet And Capacity

The default service is GoA 4 battery-electric operation with line-specific time windows and headways. Fleet sizing covers peak revenue service, planned maintenance spares and cold reserve. Off-peak surplus performs terminal or depot inspections without creating a second service-rotation fleet unless a local schedule proves one is required.

Healthy trainsets may stable overnight at protected powered stations near their first departure. Red defects, overdue heavy maintenance, isolation failures or security failures return a set to the main depot. Station stabling requires charging, CCTV, remote traction isolation, emergency access and an OCC-assigned track slot.

Capacity uses the selected trainset family's AW2 planning capacity, the actual generated timetable and a controlled practical load factor. The annual paid-trip range is a capacity-use scenario, not a ridership forecast. Replace it with a surveyed demand model before an investment decision.

Energy Method

Each city calculation combines:

- trainset energy per car-km and the generated timetable;
- climate and non-revenue movement factors;
- opportunity charging at powered stations;
- station/depot PV and stationary LFP storage;
- a dedicated solar plant or equivalent PPA asset for the remaining annual traction-energy requirement; and
- residual grid import after those assets.

The generator checks every line's powered-stop gap, traversal energy, delivered charge, protected onboard reserve and operating margin. These are feasibility screens; grid connection, protection, harmonics, thermal, fire and local solar yield studies remain deployment work.

Civil And Cost Method

Civil geometry is generated from the parametric source and exported to IFC4.3 for Bonsai coordination. Checked quantities feed the generated [civil-cost-model.toml](#), which is then consumed by city CAPEX, finance, IFC properties and documentation. Current design targets are 2.584 M USD/route-km at-grade, 9.748 M USD/route-km elevated and 18 M USD/route-km for bridges. The retained 3/12/18 M benchmarks remain visible for audit.

These figures are unquoted planning targets. Land, utilities, tax and duty, site geotechnics, project risk, programme, independent checking and local supplier rates must be added or validated before procurement. The complete shared cost basis is in the [generated cost model](#).

Capital And Finance Method

City CAPEX includes civil works, stations, depots, rolling stock, residual train control, charging interfaces, timetable-sized dedicated solar and EPC/project services. The national trainset factory is excluded from cities and counted once in each national brief, sized to the largest single-city fleet programme.

The procurement-origin model separates imported value—the minimum foreign currency or international-capital requirement—from locally supplied labour, materials, fabrication and services that can use domestic funding. Country finance parameters define construction period, debt rates, tenor and the local bond/equity split.

For Iraq, the primary proposal is the [schedule-linked funding programme](#). It assigns proposed Chinese buyer credit only to assumed eligible imported component invoices, then splits the remainder among government appropriations, IQD bonds and IQD bank credit. The [controlled assumptions](#) remain uncommitted. Monthly native-currency debt schedules, interest on staged draws, fees, reserves, public support and downside cash gaps are explicit. Construction uses the actual resource-constrained procurement milestones under an assumed 260-working-day year; the generic five-year financing comparator does not define the Iraq construction schedule. Physical and operational release gates remain open even when all planning example artifacts are regenerated.

The foreign-turnkey comparison is an editable like-for-like sensitivity, not a vendor quotation. Both cases use the same network, fleet, service, energy scope and financing schedule. Replace the multiplier, external share and all funding terms with scope-normalised bids and signed lender terms before approval.

OPEX, revenue, NPV, IRR and DSCR outputs are planning screens. Capacity-use scenarios are not demand forecasts, and no grant, tax treatment, land-value capture or lender commitment exists unless explicit controlled evidence says otherwise.

QA, Maintenance And Assurance

Shared construction QA gates, manufacturing travellers and maintenance intervals are generated from the controlled templates. City-specific asset counts and schedules are materialised in the local operations bundle. The common system covers civil works, trackform, stations, rolling stock, passenger systems, energy, signalling/comms, depots and integrated trial running.

Generated evidence proves internal consistency and reproducibility—not railway certification. Survey, geotechnical release, structural design, fire/egress, product acceptance, calibrated operations modelling, independent safety assessment and authority approval remain deployment responsibilities.

Local Evidence Contract

Each city README links the current status of these local artifacts:

Artifact	Purpose
design.toml	Authoritative generated network, fleet and CAPEX design
<slug>.toml	Expanded simulator scenario
<slug>.design-quality.yaml	Coverage, routing and civil-quality gates
engineering/finance/summary.json	Reconciled CAPEX, OPEX and finance cases
engineering/simulation/validation-summary.json	Nominal and degraded simulation acceptance
engineering/gis/summary.json	GIS package provenance and layer counts
engineering/sumo/summary.json	Executed timetable screening
engineering/energy/summary.json	Grid, charging and solar screening
operations/	Asset, QA, manufacturing and maintenance evidence

The README generator verifies source hashes and passing status before publishing these evidence links. Repository health also regenerates city and country pages and rejects drift.

Regeneration

```
# One city
tools/automation/regenerate-city.sh samawah

# Complete catalogue, using current caches
tools/automation/regenerate-all.sh

# Documentation, links and generated drift
python3 tools/automation/check-readmes.py
python3 tools/automation/check-markdown-links.py
python3 tools/automation/repo-health.py --quiet
```

Changes to common assumptions should normally touch one template or this page, then regenerate affected outputs. Location folders should change only when their local input, calculated result, evidence or explicit exception changes.

Ballastless Slab Trackform Designs

Source: <docs/civil/slab-trackforms.md>

This note records the OSR planning-grade slab trackforms for urban at-grade and elevated sections. The parametric CAD source lives in design/component-catalogue/src/osr_mech/civil/slab.py.

The designs are reference envelopes for cost, clash, fabrication, and constructability review. Deployment partners still need local structural checks for concrete grade, reinforcement, geotechnical support, drainage, stray-current control, thermal movement, seismic load, flood/scour, and product acceptance of the direct-fixation fasteners.

At-Grade Urban Slab

Long open, machine-accessible corridors use a continuously slipformed slab by default. The table below defines the six-metre single-track ST6 product kept for constrained streets, utility crossings, short possessions, flood-prone and replaceable zones. Both methods retain the same controlled track support section and concrete quantity basis; a project method-zone schedule determines panel counts.

Item	Reference value
Trackform	OSR-ST6 single-track transportable panel; two rows form double track
Panel length	6.0 m
Overall panel width	2.9 m
Track-centre spacing	3.5 m
Base slab thickness	250 mm
Rail plinth width	380 mm
Rail plinth height above base slab	160 mm
Direct-fixation pitch	Maximum 650 mm; centred 600 mm on the 6 m reference module
Rail seats per 6 m panel	20
Nominal concrete per 6 m panel	5.08 m ³
Planning bare panel mass	12.7 t

Use ST6 where the corridor is in a constrained street, utility or transition zone, short possession, flood-prone location, station interface, depot throat, or other place where ballast dust, tamping possessions, and loose aggregate are unwanted.

The slab uses two edge troughs for drainage, low-voltage/data routing, and maintainable cable access. The direct-fixation plinths are continuous so baseplates can recover small alignment tolerances without breaking out concrete.

Elevated Deck Slab

Item	Reference value
Trackform	Local direct-fixation plinths over thin non-structural alignment layer
Panel length	6.0 m
Overall alignment-layer width	2.7 m
Deck compatibility	Fits the 2.9 m OSR-Pi20/Pi25 flange
Alignment layer thickness	40 mm; may be omitted by project design
Rail plinth width	380 mm
Rail plinth height above base slab	160 mm
Direct-fixation pitch	Maximum 650 mm; centred 600 mm on the 6 m reference module
Rail seats per 6 m panel	20
Nominal concrete per 6 m panel	1.38 m ³

Use one elevated trackform run per single-track Pi-beam. A double-track elevated section therefore uses two parallel Pi-beam/trackform assemblies.

The Pi-beam upper flange is the structural deck. The alignment layer provides local tolerance recovery only and may be omitted where surveyed casting plus adjustable baseplates can recover rail geometry. A full-width reinforced or floating slab is reserved for locations where a vibration/noise and structural study justifies its dead weight.

The cable/drainage trough is installed on the inner/non-egress side. The separate outer 1.0 m walkway/barrier cassette remains clear of service covers, cabinets, screen posts and other permanent obstructions.

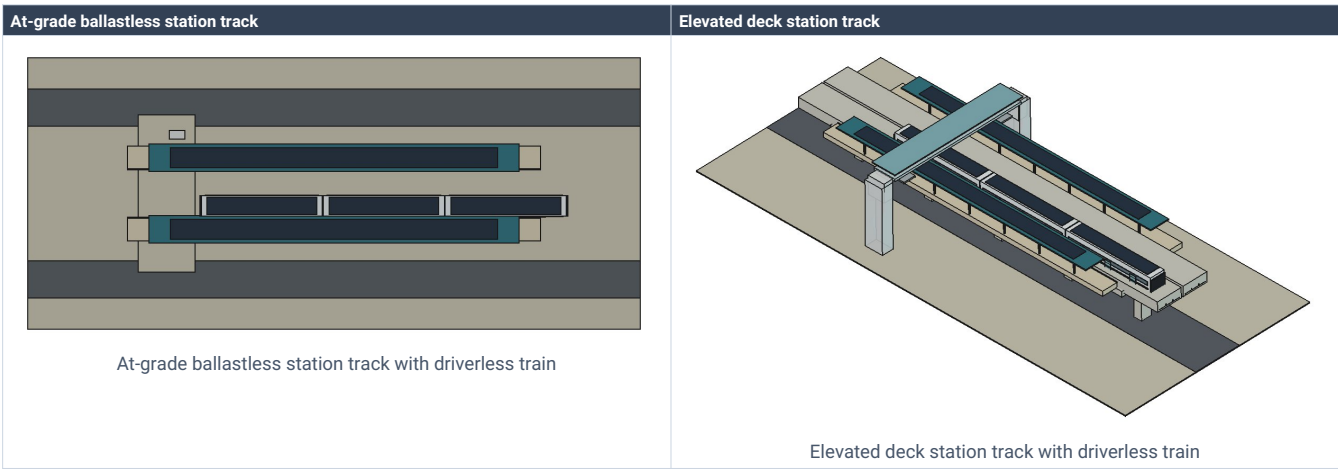
Iraqi manufacture and complete seat kits

The [Baghdad detailed engineering plan](#) adds the local slab production route, prototype inspection, lifting/bedding requirements and a complete fastener child schedule. The 6 m module has ten seats per rail centred at 300, 900, ..., 5700 mm, avoiding the former pad projection beyond the panel ends. Supplier anchor edge distances still require release. Continuous-run quantities retain the existing 650 mm planning basis; precast method zones use their own panel/seat installation schedule.

Concrete volume and bare mass cover base/plinths only. The CAD service troughs are allocation envelopes; covers, drainage sections, steel, inserts, bedding and rigging need separate schedules. Make panels, cages, moulds, ducts and suitable metalwork in Iraq; source qualified fastener/anchor systems initially and qualify any localisation through system fatigue, stiffness, insulation and process tests. The catalogue does not issue a rebar schedule, mix grade, lifting approval or supplier drill pattern.

FreeCAD Scene Renders

The slab trackforms are shown below in station context with driverless rolling stock and the station access elements that affect civil cost.



The source scene document is [design/component-catalogue/models/cad/station-scenes.FCStd](#) and is regenerated with [design/component-catalogue/scripts/freecad_station_scenes.sh](#).

Fabrication And Installation Rules

- Slipform long accessible runs over the released machine-controlled formation. Keep the 6 m ST6 module for constrained and replaceable zones; use closure pours only at turnouts, station ends, curves that require local survey correction, and transitions to bridge/special spans.
- Cast embedded inserts, earthing/bonding points, drainage scuppers, and lifting anchors in the precast yard.
- Require surveyed casting beds and a dimensional QA record for every batch.
- Use adjustable baseplates on all running rails.
- Keep prestressed elevated girders conservative; recycled aggregate and aggressive low-carbon mixes can be used first in non-prestressed slabs, piers, pile caps, and station modules.
- Treat direct-fixation pads, anchors, and baseplates as replaceable wear/interface parts, not as buried civil structure.

Cost Model Link

These slab designs feed the generated civil planning contract:

Civil class	Design-derived target	Retained benchmark
At-grade ballastless slab	2.584 M USD/route-km	3.0 M USD/route-km
Elevated guideway with local-plinth trackform	9.748 M USD/route-km	12.0 M USD/route-km
Bridge/water-crossing guideway	18.0 M USD/route-km	18.0 M USD/route-km

The active values come from [civil-cost-model.toml](#) and are planning targets, not tender or turnkey EPC estimates. Build the project estimate from [viaduct-quantity-cost-model.toml](#) plus site rates, utilities, access, risk, programme, independent checking, and contingency. Short elevated stubs, stations, soft ground, deep utilities, tight curves, or repeated

special spans carry project-specific premiums.